



**ARBA MINCH UNIVERSITY
COLLEGE OF BUSINESS AND ECONOMICS
MANAGEMENT DEPARTMENT**

Course Materials for Theme 5 in Management Program

Theme 5: Management Science and Financial Management (20%)

This module contains three courses:

- 1. Operation research***
- 2. Operation management and,***
- 3. Financial management***

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**OPERATIONAL RESEARCH (OR)
STUDENT HANDBOOK**

CHAPTER ONE

OPERATIONS RESEARCH – AN OVERVIEW

1.1. The origin and Development of operations research

The term Operations Research, was first coined in 1940 by McClosky and Trefthen in a small town, Bowdsey, of the United Kingdom. This new science came into existence in military context. Its concept was derived from the fact that “Unity is strength”. Why unity is strength? Because, when there is calamity to the nation, citizens of all shades join their opinions together to do their might to solve the problem. How this came to be proved? It was proved during World War II.

Two blocks took part in the Second World War, namely, Axes power and Allied power. Axes forces include Nazi German, Italy and Japan, in one block. Contrarily, the allied forces included America, Great Britain and Russia. During World War II, there was a natural calamity to Great Britain from Nazi German Forces. Superior weapons and strategy of Germans became a real threat to Allied Forces. Consequently, Allied Forces were threatened on **land, sea and air** using superior weapons such as submarines, U-boats and air crafts. Being challenged by this serious situation, the Government of Great Britain appealed to the people and requested talents from all walks of life to join together and find a solution to the problem in order to overcome the threatening situation. Based on an appeal made by Government of Great Britain, military management called on scientists from various disciplines and organized them into teams to assist in solving strategic and tactical problems, i.e., to discuss, evolve and suggest ways and means to improve the execution of various military projects. Accordingly, scientists reported to the military management and they were grouped into various teams and each team is given strategic problem to come up with the possible solution. These combined efforts produced fantastic results. By their joint efforts, experience and deliberations, they suggested certain approaches that showed remarkable progress. Based on suggestions provided by teams of scientists, Allied forces defeated Axes forces.

This signaled to the birth of Operations Research as a separate discipline. This new approach to systematic and scientific study of operations of the systems was called the Operations Research or Operational Research (abbreviated as O.R.)

Following the end of World War II, the success of military teams attracted the attention of industrial managers who were seeking solutions to their complex executive-type problems.

During the year 1950, O.R. achieved recognition as a subject worthy of academic study in the Universities. Since then, the subject has been gaining more and more importance for the students of **Economics, Management, Public Administration, Behavioral Sciences, Social Work, Mathematics, Commerce and Engineering**. With a view to increasing the impact of O.R. and establishing rapport between all its serious students and users, the Operations Research society of America was formed in 1950. Other countries followed suit, and in 1957 the International Federation of O.R. Societies was established.

O.R. is, therefore, a systematic method consisting of stating the problem in clear terms, collecting facts and data, analyzing them, and reaching certain conclusions in the form of solutions to the problem. The ultimate aim of it is to find out an optimum solution which is most appropriate under the given circumstances.

1.2.DEFINITIONS:

Dictionary meaning of research is “a careful investigation or inquiry especially through search for new facts”. It also means “systematized effort to gain new knowledge”.

As far as the definition of O.R. is concerned, there is no single universally accepted definition. It is so because it has been defined by many authors in different ways. However, with the consideration of your effort to refer various definitions, the definition we assume to be more comprehensive is given below.

Operations Research is a systematic analysis of a problem through scientific methods, carried out by appropriate specialists, working together as a team, constituted at the instance of management for the purpose of finding an optimum and the most appropriate solution, to meet the given objective under a given set of constraints.

There are many more definitions as various authors have defined the term differently. As it has been written by many authors, discussing all the definitions exhaustively is very impossible. The student, therefore, can consult more definitions from such books as J.K. Sharma, 1997, S.D Sharma and others.

Meaning of O.R. is evident from the concept given in the definitions here above. However, some more aspects of O.R. are given here under to get better insight into it.

1. O.R is the application of scientific methods, techniques and tools to the problem to find out an answer.
2. O.R. is a management tool, in the hands of a manager, to take a decision.

3. O.R. is a scientific approach to the decision-making process.
4. O.R. is an “*Applied Research*” which aims at finding a solution for immediate problem facing a society, industry or a business enterprise. It is not fundamental research.
5. O.R. is decision-oriented research, which provides quantitative basis to managers of the organization for taking decisions.
6. O.R. is applied decision theory. It uses scientific, mathematical and logical means to take decisions.

N.B. - *O.R. is a team effort. Team efforts are as old as mankind. What is then new in O.R. approach? O.R. is a systematic approach using only scientific methods to find a solution. This distinguishes O.R. from team efforts-past or present.*

Question: Define the meaning of Operations Research and discuss the origin and development of it.

1.3.NATURE AND CHARACTERISTIC OF OPERATIONS RESEARCH:

Some significant features of O.R. are given here under as follows:

- | | |
|---|------------------------------------|
| i) Decision-making | v) Digital computer |
| ii) Scientific approach | vi) Methodological Approach |
| iii) Objective | vii) Wholistic Approach. |
| iv) Inter-disciplinary team approach | |

It is not so easy to exhaustively list down the features of O.R. as the matter of fact that it constitutes very broad area. However, the following features can also be mentioned in addition to those already listed here above.

1.4.APPLICATIONS OF OPERATIONS RESEARCH

Operation Research is mainly concerned with the techniques of applying scientific knowledge, besides the development of science. It provides an understanding which gives the expert/manager new insights and capabilities to determine better solutions in his decision-making problems, with great speed, competence and confidence. In recent years, O.R. has successfully entered many different areas of research in Defense, Government, Service Organizations and Industry. Some applications of O.R. in the functional areas of management are briefly presented here under.

Finance, Budgeting, and Investment
Marketing
Physical Distribution
Purchasing, Procurement and

Exploration
Personnel
Production
Research and Development

Besides the above-mentioned applications of O.R. in the context of modern management, its use has now extended to a wide range of problems, such as the problems of communication and information, socio-economic fields and national planning.

1.5.OPPORTUNITIES AND SHORTCOMINGS OF OPEERATIOS RESEARCH

The use of O.R. to improve decision-making has become almost universal today. However, this remarkable achievement of O.R. is not totally free from shortcomings. Certain shortcomings result from lack of awareness on the part of managers about their roles, while others result due to avoidance of the behavioral and organizational issues which are a part of every successful application.

Opportunities of O.R.

1. Using O.R. approach, the decision-maker can determine a solution to his routine or repetitive problem. For obtaining solution of such type of problems, it is necessary to build a model so that future solution can be obtained using the model thus freeing managers to concentrate on more pressing matters.
2. O.R. requires business managers to be quite explicit about their objective, their assumptions and the way of visualizing the constraints.
3. While using O.R. approach, a manager has to consider very carefully all those variables which influence his decisions and the way these variables in a problem interact with each other. He then selects a decision which is best for the organization as whole.
4. O.R. approach allows a decision-maker to solve a complex problem involving multiple variables much more quickly than if he had to compute them using traditional methods. Sometimes it may not be possible to solve such complex problems without using O.R. methods.
5. Using O.R. approach, a decision-maker can examine a situation from various angles by simulating the model which he has constructed for the real problem. He can change various conditions under which decisions are being made, and examine the effect of these changes through appropriate experiments on the model, to determine the best or optimal solution for

the problem under consideration. All these experiments can be carried out without causing any series damage to the existing system or incurring excessive cost.

Shortcomings of O.R.

- 1) There are certain problems which a decision-maker may have to solve only once. Constructing a complex O.R. model for solving such problems is often too expensive when compared with the cost of other less sophisticated approaches available to solve them.
- 2) Sometimes O.R. specialists become too much enamored with the model they have built and forget the fact that their model does not represent the “real world problem” in which decisions have to be made.
- 3) Sometimes the basic data are subject to frequent changes. In such cases, modification of O.R. models are a costly affair.
- 4) Many O.R. models are so complex that they cannot be solved without the use of computer. Also, the solutions obtained from these models are difficult to explain to managers and hence fail to gain their support and confidence.
- 5) Magnitude of computation involved, lack of consideration for non-quantifiable factors and psychological issues involved in implementation are some of the other shortcomings of O.R.

1.6.Models and Modeling in Operations Research

The approximation or abstraction, maintaining only the essential elements of the system, which may be constructed with various forms by establishing relationships among specified variables and parameters of the system, is called a model. Models attempt to describe the essence of a solution or activity by abstracting from reality so the decision-maker can study the relationship among relevant variables in isolation. Hence, models do not, and cannot represent every aspect of reality because of the innumerable and changing characteristics of the real-life problems to be represented. Instead, they are a limited approximation of reality.

A model is constructed to analyze and understand the given system for the purpose of improving its performance. The reliability of the solution obtained from a model depends on the validity of the model in representing the system under study. The key to model building lies in abstracting only the relevant variables that affect the criteria of the measures of performance of the given

system and expressing the relationship in a situation. A model should be as simple as possible so as to give the desired result.

There are many ways to classify models. Classification schemes can also provide a useful frame of reference for modelers. There are five classification schemes for models.

1. Classification based on the structure

- a) Physical models
 - ✓ *Iconic models*
 - ✓ *Analogue models*
- b) Symbolic models
 - ✓ *Verbal models*
 - ✓ *Mathematical models*

2. Classification based on function or purpose

- a) Descriptive models
- b) Predictive models
- c) Normative (optimization) models

3. Classification based on time reference

- a. Static models
- b. Dynamic models

4. Classification based on the degree of certainty

- a) Deterministic models
- b) Probabilistic (stochastic) models

5. Classification based on the method of solution or quantification:

- a. Heuristic models
- b. Analytical models
- c. Simulation model

Chapter Review Questions

1. Discuss the concept, history and application of OR.
2. *Model building is the essence of the operations research approach? Discuss.*
3. *What is meant by a mathematical model of a real situation? Discuss the importance of models in the solution of O.R. problems.*

4. What is the purpose of a mathematical model? How does a model achieve this purpose?
Consider in your answer the concept that a model is an abstraction of reality.
5. Explain how and why operations research methods have been valuable in aiding executive decisions?
6. What are the features, strengths and weaknesses of OR?
7. Discuss the concept of modeling and the various types of models.

CHAPTER TWO

LINEAR PROGRAMMING PROBLEM

1.7. Introduction:

Linear programming (LP) is a mathematical modeling technique useful for economic allocation of *scarce or limited* resources, such as labor, material, machine, time, warehouse space, capital, energy, etc., to several competing activities, such as products, services, jobs, new equipment, projects, etc., on the basis of a given criterion of optimality. **Scarce resources** mean resources that are not infinite in availability during the planning period. The criterion of **optimality** generally is either performance, return on investment, profit, cost, utility, time, distance, etc.

What is linear programming? *“Linear programming is a versatile mathematical technique in O.R. and a plan of action to solve a given problem involving linearly related variables in order to achieve the laid down objective function under a given set of constraints.”*

1.8. Component of Linear Programming

1. The Objective Function
2. The Decision Variables
3. The constraints
4. Parameters

Example:

Maximize: $4X_1 + 7X_2 + 5X_3$ (Profit) _____ objective function

Subject to:

$2X_1 + 3X_2 + 6X_3 \leq 300$ labor hrs

$5X_1 + X_2 + 2X_3 \leq 200$ lb raw material A } System constraints

$3X_1 + 5X_2 + 2X_3 \leq 360$

$X_1 = 30$ } Individual constraints

$X_2 \geq 40$ }

$X_1, X_2, X_3 \geq 0 \rightarrow$ Non-negativity constraints.

System constraints – involve more than one decision variables.

Individual constraints – involve only one decision variable.

None-negativity constraints specify that no variable will be allowed to take on a negative value.

The non-negativity constraints typically apply in an LP model, whether they are explicitly stated or not.

1.9.Assumption of LP Models

- | | |
|----------------|-------------------|
| 1.Additivity | 4.Certainty |
| 2.Linearity | 5.Non-negativity. |
| 3.Divisibility | |

1.10. Advantages & Limitations of Linear Programming

Advantages of LP; The following are some of the advantages of LP approach:

- ✿ Scientific Approach to Problem Solving
- ✿ Evaluation of All Possible Alternatives
- ✿ Quality of Decision
- ✿ Focus on Grey-Areas
- ✿ Flexibility
- ✿ Maximum optimal Utilization of Factors of Production

Limitations of Linear Programming:

Although Linear Programming is successful and having wide applications in business and trade for solving optimization problems, yet it has certain demerits or limitations as follows:

- ✚ Linear Relationship
- ✚ Constant Value of objective & Constraint Equations

- ✚ No Scope for Fractional Value Solutions
- ✚ Degree of Complexity
- ✚ Multiplicity of Goals

1.11. Main Application Areas of Linear Programming

Some of the important application areas of Linear Programming are the following:

- Military Applications
- Agriculture.
- Environmental Protection.
- Facilities Location.
- Product-Mix.
- Production.
- Mixing or Blending.
- Job Analysis.
- Wages and Salary Administration
- Transportation & Trans-Shipment.
- Portfolio Selection.
- Profit Planning & Contract.
- Traveling Salesmen Problem.
- Media Selection/Evaluation.
- Staffing.

1.12. FORMULATION OF MATHEMATICAL MODEL OF LP

The basic steps in formulating a linear programming model are presented as follows:

Step I Identification of the decision variables.

- Express each constraint in words. For this first see whether the constraint is of the form $\geq$ (at least, as large as), or of the form $\leq$ (no larger than) or $=$ (exactly equal)
- Then express the objective function verbally.
- Step (a) and (b) should then allow you to verbally identify the decision variables.

Step II Identification of the constraints

Step III formulate the objective function

Step IV formulate the LPP

EXAMPLES

Example 1: A company has three operational departments (weaving, processing and packing) with capacity to produce three different types of clothes namely suiting, shirting and woolens yielding a profit of Birr 2, 4 and 3 respectively. One meter of suiting requires 3 minutes in weaving, 2 minutes in processing and 1 minute in packing. Similarly, one meter of shirting requires 4 minutes in weaving, 1 minute in processing and 3 minutes in packing. One meter of woolen requires 3 minutes in each department. In a week, total run time of each department is 60, 40 and 80 hours for weaving, processing and packing respectively. Formulate the linear programming problem (LPP) to find the product mix to maximize the profit.

The data can be summarized as follows before formulating an LPP model.

Departments	Products in meters			Availability (in minutes)
	Suiting	Shirting	Woolens	
Weaving	3	4	3	60 x 60 = 3600
Processing	2	1	3	40 x 60 = 2400
Packing	1	3	3	80 x 60 = 4800
Profit contr.	2	4	3	

LPP model formulation

Let x_1 , x_2 and x_3 represent the rate of production in the three departments.

Maximize $Z = 2x_1 + 4x_2 + 3x_3$

..... objective function

Subject to:

$$\begin{array}{llll} 3x_1 + 4x_2 + 3x_3 \leq 3600 & \text{.....} & \text{Weaving time constraint} \\ 2x_1 + x_2 + 3x_3 \leq 2400 & \text{.....} & \text{Processing time constraint} \\ x_1 + 3x_2 + 3x_3 \leq 4800 & \text{.....} & \text{Packing time constraint} \\ x_1, x_2, x_3 \geq 0. & \text{.....} & \text{Non-negativity condition.} \end{array}$$

Example 2: A small scale manufacturer produces two types of products A and B which give a profit margin of Birr 4 and Birr 3, respectively. There are two plants I and II, each having a capability of 72 and 48 hours per day. Cycle times of A and B are 2 hours and 1 hour in plant number I, respectively. Similar figures for plant II are 1 and 2 hours. Formulate an LPP model for maximizing the profit.

The summarized data of the above problem is as follows:

	Products		
Plant	A	B	Availability
I	2	1	72
II	1	2	48
Profit	4	3	

LPP model formulation:

$$\text{Maximize } Z = 4x + 3y$$

Subject to:

$$2x + y \leq 72$$

$$x + 2y \leq 48$$

$$x, y \geq 0.$$

1.13. SOLVING LINEAR PROGRAMMING:

There are two types of finding a solution for Linear programming problems.

- **Graphic solution and**
- **Simplex solution**

1.13.1. THE GRAPHICAL METHOD:

The Graphical method can be used to solve simple LPPs having two decision variables only. For LPPs with three or more variables, this method cannot be employed. To use the graphical method for solving linear programming problems of two decision variables, the following steps are required:

Step I *Defining the problem*

Step II *Plot the constraints graphically*

Step III *Shade the feasible region*

- a. For "greater than" & "greater than or equal to" constraints, the feasible region or the solution space is the area that lays above the constraint lines.
- b. For "Less Than" & "Less than or equal to" constraint, the feasible region or the solution space is the area that lays below the constraint lines.

Step IV *Selecting the graphic solution technique.*

1. Corner Point Method and
2. Iso-profit (or Iso-cost) method may be used; however, it is easier to generate solution by using the corner point method and we use this method to find solution.

Corner Point Method to find Solution for Graphics Method

- I. Since the solution point (x, y) always occurs at the corner point of the feasible or solution space, identify each of the extreme points or corner points of the feasible region by the method of simultaneous equations.
- II. By putting the value of the corner point's co-ordinates [e.g. (2, 3)] into the objective function, calculate the profit (or the cost) at each of the corner points.
- III. In a Maximize problem, the optimal solution occurs at that corner point which gives the highest profit. In a minimization problem, the optimal solution occurs at that corner point which gives the lowest profit.

Examples on Maximization LP problem

Example 1 Use the graphical method to solve the following LP problem.

$$\text{Maximize } Z = 15x_1 + 10x_2$$

Subject to the constraints

$$4x_1 + 6x_2 \leq 360$$

$$3x_1 + 0x_2 \leq 180$$

$$0x_1 + 5x_2 \leq 200$$

$$x_1, x_2 \geq 0.$$

Solution:

1. If the given problem is not in mathematical form, then convert the problem into mathematical form. However, in the case of above example, the given problem is already in mathematical form.

2. We shall treat x_1 as the horizontal axis and x_2 as the vertical axis. Each constraint will be plotted on the graph by treating it as a linear equation and then *appropriate inequality conditions* will be used to mark *the area of feasible solutions*.

Consider the first constraint $4x_1 + 6x_2 \leq 360$. Treat it as the equation, $4x_1 + 6x_2 = 360$. The easiest way to plot this line is to find any two points that satisfy the equation, then drawing a straight line through them. The two points are generally the points at which the line intersects the x_1 and x_2 axes. For example, when $x_1 = 0$, we get $6x_2 = 360$ or $x_2 = 60$, dividing both sides by 6. Similarly, when $x_2 = 0$, $4x_1 = 360$, then $x_1 = 90$, applying the same rule as above.

These two points are then connected by a straight line as will be depicted in the Figure 2.1 here below. But the question is: where are the points satisfying $4x_1 + 6x_2 \leq 360$. In the case of this constraint, any point above the constraint line violates the inequality condition. But any point below the line does not violate the constraint. Thus, the inequality and non-negativity condition can only be satisfied by the shaded area (feasible region) as may be shown in Figure 2.1 here below. Similarly, the constraints $3x_1 \leq 180$ and $5x_2 \leq 200$ are also plotted on the graph and are indicated by the shaded area as shown in Fig. 2.1.

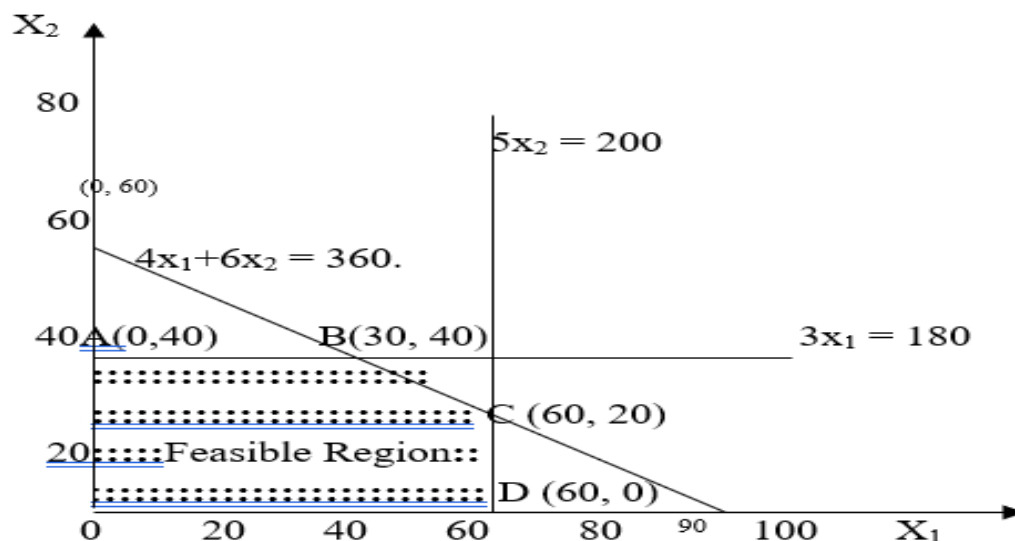


Figure 2.1 Graphical Solution of LP Problem.

Since all constraints have been graphed, the area which is bounded by all the constraint lines including all the boundary points is the feasible region (solution space). The feasible region is shown in Figure 2.1 by the shaded area OABCD.

Since the optimal value of the objective function occurs at the one of the extreme points of the feasible region, it is necessary to determine their coordinates. The coordinates of extreme points of the feasible region are: O= (0, 0), A= (0, 40), B= (30, 40), C= (60, 20), D = (60, 0).

Evaluate objective function value at each extreme point of the feasible region as shown here below, in order to arrive at decision making.

Extreme point	Coordinates		Objective function value
	x_1	x_2	$Z = 15x_1 + 10x_2$
O	0	0	$15(0) + 10(0) = 0$
A	0	40	$15(0) + 10(40) = 400$
B	30	40	$15(30) + 10(40) = 850$
C	60	20	$15(60) + 10(20) = 1100$
D	60	0	$15(60) + 10(0) = 900$

Since we desire Z to be maximum for maximization case, we conclude that maximum value of $Z = 1,100$ is achieved at the point C (60, 20). Hence, the optimal solution to the given LP problem is $x_1 = 60$, $x_2 = 20$ and $\text{Max } Z = 1,100$.

Example 2 A firm makes two products X and Y, and has a total production capacity of 9 tones per day, X and Y requiring the same production capacity. The firm has a permanent contract to supply at least 2 tons of X and at least 3 tons of Y per day to another company. Each tone of X requires 20 machine hours of production time and each tone of Y requires 50 machine hours of production time. The daily maximum possible number of machine hours is 360. All the firm's output can be sold, and the profit made is Birr 80 per ton of X and Birr 120 per ton of Y. It is required to determine the production schedule for maximum profit and to calculate this profit.

LP model formulation: Let x_1 and x_2 = number of units (in tons) of product X and Y to be manufactured, respectively. Then, the LP model of the given problem can be written as:

$$\text{Maximize (total profit) } Z = 80x_1 + 120x_2$$

Subject to the constraints:

$$\left. \begin{array}{l} x_1 + x_2 \leq 9 \\ x_1 \geq 2 \\ x_2 \geq 3 \end{array} \right\} \text{ (Production capacity constraints).}$$

$$\left. \begin{array}{l} 20x_1 + 50x_2 \leq 360 \end{array} \right\} \text{ (machine hours constraint)}$$

$$x_1, x_2 \geq 0. \quad \text{(Non-negativity restriction)}$$

Graphical Solution:

For solving this problem graphically, let us graph each constraint by first treating it as a linear equation in the same way as discussed earlier. Then use the inequality condition of each constraint to mark the feasible region as shown in Figure 2.2.

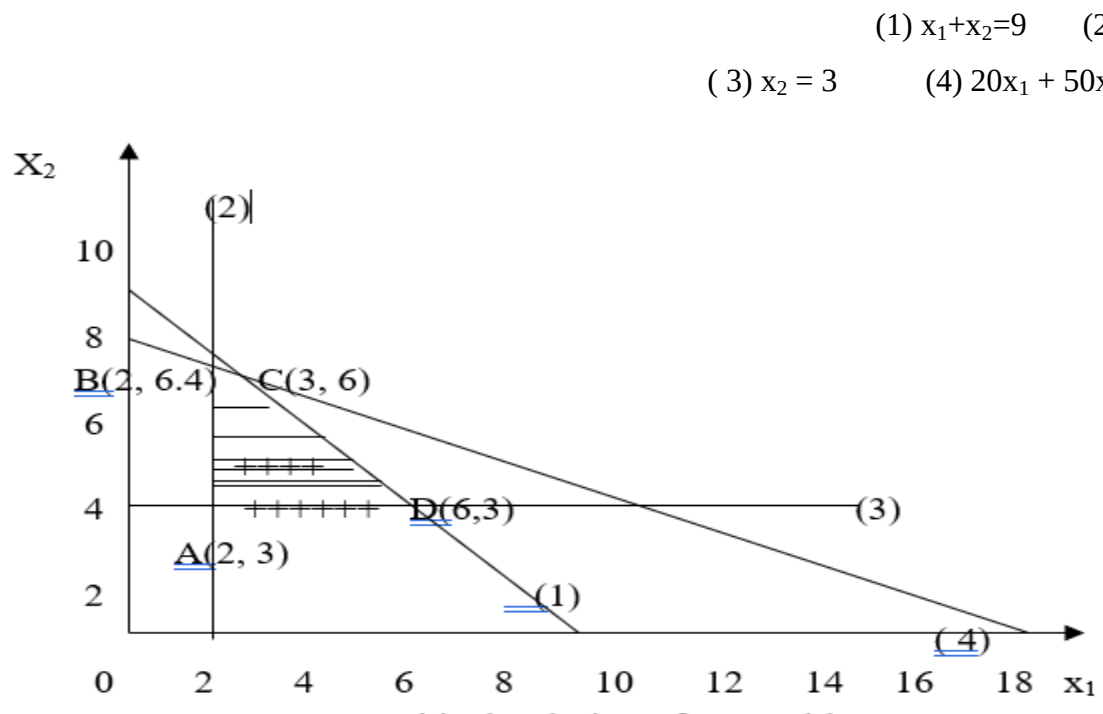


Figure 2.2 Graphical Solution of LP problem

So computed coordinates of the extreme points of the feasible region are: A= (2, 3), B= (6, 3), C= (3, 6), and D= (2, 6.4). The value of the objective function at each of these extreme points is as follows:

Extreme point	Coordinates		Objective function value $Z = 80x_1 + 120x_2$
	x_1	x_2	
A	2	3	$80(2) + 120(3) = 520$
B	2	6.4	$80(2) + 120(6.4) = 928$
C	3	6	$80(3) + 120(6) = 960$
D	6	3	$80(6) + 120(3) = 840$

Decision: The maximum value of the objective function $Z = 960$ occurs at the extreme point

C (3, 6). Hence the company should produce, $x_1 = 3$ tons of product X and $x_2 = 6$ tons of product Y in order to yield a maximum profit of Birr 960.

Minimization Case and graphical method solution

Minimization case is used to minimize the cost the organization could incur otherwise. Cost can be minimized by reducing wastages and increasing efficiency. The increase of efficiency can lead the organization to productivity and profitability.

Minimization case is used when the constraint is “greater than or equal to” which may symbolically be represented as ($\geq$). When the minimization case is employed, the solution space will be situated above the constraint lines. Thus, shading will take place above the constraints.

Example 3 Use the graphical method to solve the following LP Problem.

$$\text{Minimize } Z = 3x_1 + 2x_2$$

Subject to the constraints:

$$5x_1 + x_2 \geq 10$$

$$x_1 + x_2 \geq 6$$

$$x_1 + 4x_2 \geq 12$$

$$x_1, x_2 \geq 0$$

Solution: Plot on a graph each constraint by first treating them as a linear equation in the same way as discussed earlier. Use the inequality condition of each constraint to mark the feasible region as may be shown in Figure 2.3.

$$5x_1 + x_2 = 10$$

$$x_1 + x_2 = 6$$

$$x_1 + 4x_2 = 12$$

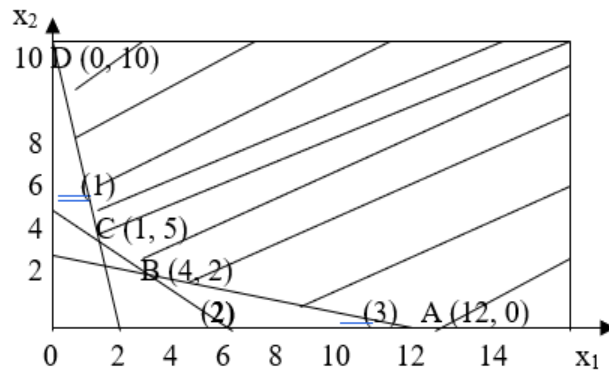


Figure 2.3 Graphical solution of LP problem.

Accordingly, the coordinates of the extreme points of the feasible region (bounded from below) are: A = 12, 0), B = (4, 2), C = (1, 5), and D = (0, 10). The value of the objective function at each of these extreme points is as follows:

Extreme point	Coordinates		Objective function value $Z = 3x_1 + 2x_2$
	X_1	X_2	
A	12	0	$3(12) + 2(0) = 36$
B	4	2	$3(4) + 2(2) = 16$
C	1	5	$3(1) + 2(5) = 13$
D	0	10	$3(0) + 2(10) = 20$

Decision: The minimum value of the objective function $Z = 13$ occurs at the extreme point C (1, 5). Hence, the optimal solution to the given LP problem is: $x_1 = 1$, $x_2 = 5$ and $\min z = 13$.

1.13.1.1. SPECIAL CASES IN LINEAR PROGRAMMING

a) Alternative (or Multiple Optimal Solutions)

‘There are two conditions that should be satisfied for an alternative optimal solution to exist:

1. The given objective function is parallel to a constraint that forms the boundary (or edge) of the feasible solution region.
2. The constraint should form a boundary on the feasible region in the direction of optimal movement of the objective function.

Example; Use the graphical method to solve the following LP problem.

$$\text{Maximize } Z = 10x_1 + 6x_2$$

Subject to the constraint:

$$5x_1 + 3x_2 \leq 30$$

$$x_1 + 2x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

Solution: 1. Convert inequalities, in the constraints, into equalities, such as

$$\text{Maximize } Z = 10x_1 + 6x_2$$

Subject to the constraint:

$$5x_1 + 3x_2 = 30$$

$$x_1 + 2x_2 = 18$$

$$x_1, x_2 \geq 0$$

2. Find x and y coordinates of each constraint computing it as follows:

$$1) 5x_1 + 3x_2 = 30$$

$$x_1 = 0, x_2 = 30/3 = 10.*$$

$$x_1 = 30/5 = 6, x_2 = 0$$

$$2) x_1 + 2x_2 = 18$$

$$x_1 = 0, x_2 = 9$$

$$x_1 = 18, x_2 = 0$$

3. Plot the graph using the coordinates of X and Y.

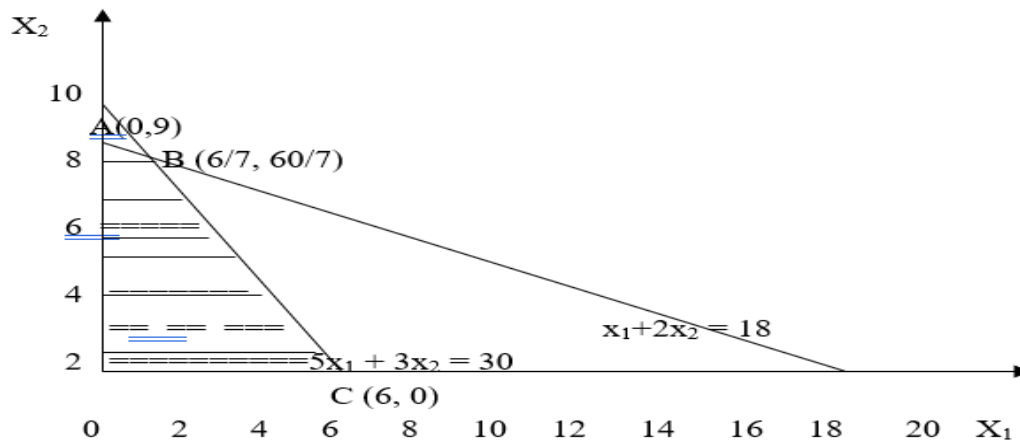


Figure 2.4 Graphical Solution of LP problem

Here we observe that the objective function is parallel to the line BC (or the first constraint), which forms the boundary of the feasible region. This is because the slope of the objective function and that of the first constraint ($5x_1 + 3x_2 = 30$) is $-5/3$. Therefore, this implies that any point including extreme points B and C on the same line between B and C is an optimal solution.

We may disregard all other solutions obtained on the line segment BC and consider only those obtained at extreme points B and C to establish that the solution to an LP problem will always lie at an extreme point of the feasible region.

The evaluation of four extreme points is presented here below:

Extreme point	Coordinates		Objective function value
	x_1	x_2	$Z = 10x_1 + 6x_2$
O	0	0	$10(0) + 6(0) = 0$
A	0	9	$10(0) + 6(9) = 54$
B	6/7	60/7	$10(6/7) + 6(60/7) = 60$
C	6	0	$10(6) + 6(0) = 60$

Since on two different extreme points B and C the value of objective function is same, i. e., $\max Z = 60$, two alternative solutions: $x_1 = 6/7$, $x_2 = 60/7$ and $x_1 = 6$, $x_2 = 0$ exist.

Remark: If a constraint to which the objective function is parallel does not form the boundary of the feasible region, the multiple solutions will not exist and such a constraint is called redundant constraint, i.e., a redundant constraint is one whose addition or removal does not change the feasible region.

b) Unbounded Solution

Sometimes an LP problem will not have a final solution. This means when one or more decision variable values and the value of the objective function (maximization case) are permitted to increase infinitely without violating the feasibility condition, then the solution is said to be unbounded.

Example Use the graphical method to solve the following *unbounded* LP problem.

$$\text{Max. } Z = 3X_1 + 2X_2$$

St:

$$X_1 - X_2 \leq 1$$

$$X_1 + X_2 \geq 3$$

$$X_1, X_2 \geq 0$$

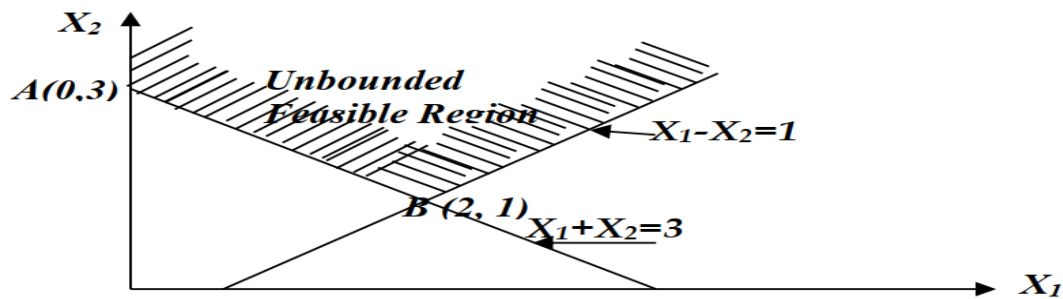


Figure 2.5 unbounded solution to an LP problem

It may be noted from Figure 2.5 that there exist an infinite number of points in the convex region for which the value of the objective function increases as we move from extreme point (origin), to the right. That is, both the variables x_1 and x_2 can be made arbitrarily large and the value of objective function Z is also increased. Thus, the problem has an unbounded solution.

c) Infeasible Solution

Infeasibility is a condition that arises when no value of the variables satisfy all of the constraints simultaneously. This means there is no unique (single) feasible region. Such a problem arises due to wrong model formulation with conflicting constraints.

Example:

$$\text{Max } Z = 20X_1 + 30X_2$$

St:

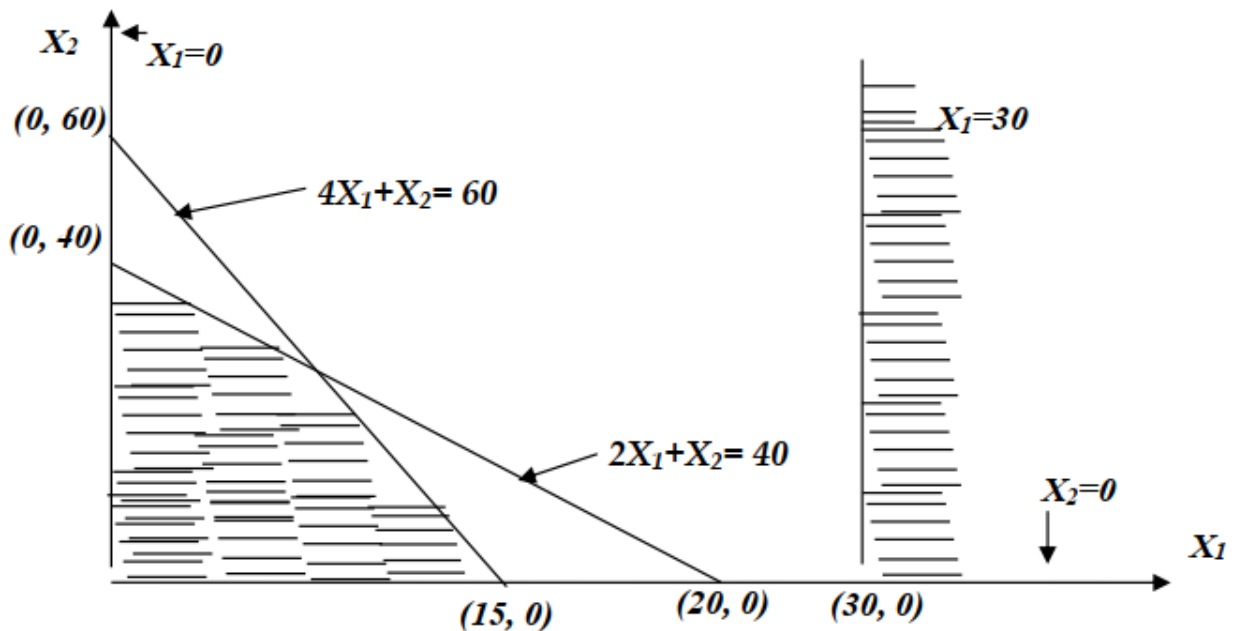
$$2X_1 + X_2 \leq 40$$

$$4X_1 + X_2 \leq 60$$

$$X_1 \geq 30$$

$$X_1, X_2 \geq 0$$

Solution:



Note: -In the above graph, there is no common point in the shaded area.

-All constraints cannot be satisfied simultaneously and there is no feasible solution to the problem.

SELF PRACTICE PROBLEMS:

Solve the following problems using graphical method and submit them to your instructor.

1. A manufacturing firm produces two machine parts: p_1 and p_2 using milling and grinding machines. The different machine times required for each part, the machining times available on different machines and the profit on each machine part are given as below:

Machine	Manufacturing time required		Maximum time available / week (min)
	P_1	P_2	

Lathe	10	5	2,500
Milling machine	4	10	2,000
Grinding machine	1	1.5	450
Profit per unit (\$)	50	100	

Determine the number of pieces of p_1 and p_2 to be manufactured per week to maximize profit.

2. A furniture manufacturer makes two products: chairs and tables. Processing of these products is done on two machines A and B. A chair requires 2 hours on machine A and 6 hours on machine B. A table requires 5 hours on machine A and no time on machine B. There are 16 hours per day on machine A and 30 hours on machine B. Profit gained by the manufacturer from a chair and table is 2 and 10, respectively. What should be the daily production of each of the two products?
3. A company produces two types of leather belts, say A and B. Belt A is of superior quality and B is inferior. Profit on the two is 40 and 30 cents per belt, respectively. Each belt of type A requires twice as much as required by a belt of type B. If all the belts were of type B, the company could produce 1,000 belts per day. But the supply of leather is sufficient only for 800 belts per day. Belt A requires a fancy buckle and only 400 of them are available per day. For belt B only 700 buckles are available per day. How should the company manufacture the two types of belts in order to have a maximum overall profit?

1.13.2. SIMPLEX METHOD

INTRODUCTION

The graphical method to solving **LPPs** provides fundamental concepts for fully understanding the LP process. However, the graphical method can handle problems involving only two decision variables (say X_1 and X_2).

In 1940's **George B. Dantzig** developed an algebraic approach called the **Simplex Method** which is an efficient approach to solve applied problems containing numerous constraints and involving many variables that cannot be solved by the graphical method.

The simplex method is an **ITERATIVE** or “**step by step**” method or repetitive algebraic approach that moves automatically from one basic feasible solution to another basic feasible solution improving the situation each time until the optimal solution is reached at.

Scope of solution of LPP by simplex method

Following types of problems are solved by simplex method:

- **Maximize Z** with inequalities of constraints in “ $\leq$ ” form.
- **Minimize Z** with inequalities of constraints in “ $\geq$ ” form.
- **Maximize Z** with inequalities of constraints in “ $\geq$ ” form.
- **Maximize Z or Minimize Z** with inequalities of constraints in “ $\leq$ ”, “ $\geq$ ” or “ $=$ ” form.

The summary of the extra variable to be added in the given LP problem to convert it in to

Types of constraint	Extra variable needed	Coefficient of extra variables In the objective function		Presence of extra variable in initial solution mix
		Max Z	Min Z	
Less than or equal to $\leq$	A slack variable is added	0	0	Yes
Greater than or equal to ($\geq$)	A surplus variable is subtracted and artificial variable is added	0	0	No
		-M	+M	Yes
Equal to ($=$)	Only an artificial variable is added	-M	+M	Yes

standard form is given in the following table.

Remark: a *slack variable* represents unused resource, either in the form of time on a machine, labor hour, money, warehouse space or any number of such resources in various business problems. Since variable yields no profit, therefore such variable are added to the original objective function with zero coefficients.

A *surplus variable* represents amount by which solution values exceed a resource. These variables are also called *negative slack variables*. Surplus variables carry a zero coefficient in the objective function.

Artificial variable (A): Artificial variable is a variable that has no meaning in a physical sense but acts as a tool to create an initial feasible LP solution.

1 MAXIMIZATION PROBLEMS

➤ Maximize Z with inequalities of constraints in " $\leq$ " form

Example:

Solve the problem using the simplex approach

$$\text{Max. } Z = 300x_1 + 250x_2$$

Subject to:

$$2x_1 + x_2 \leq 40 \quad (\text{Labor})$$

$$x_1 + 3x_2 \leq 45 \quad (\text{Machine})$$

$$x_1 \leq 12 \quad (\text{Marketing})$$

$$x_1, x_2 \geq 0$$

Solution

Step 1

Formulate LPP Model

Step 2

Standardize the problem

i.e Convert constraint inequality into equality form by introducing *Slack variable*.

Let that s_1 , s_2 , and s_3 are unused labor, machine and marketing hrs respectively.

$$\text{Max. } Z = 300x_1 + 250x_2 + 0s_1 + 0s_2 + 0s_3$$

St:

$$2x_1 + x_2 + s_1 = 40$$

$$x_1 + 3x_2 + s_2 = 45$$

$$x_1 + s_3 = 12$$

$$x_1, x_2, s_1, s_2, s_3 \geq 0$$

Standard form

Step 3

Obtain the initial simplex tableau

To represent the data, the simplex method uses a table called the *simplex table* or the *simplex matrix*.

==> In constructing the initial simplex tableau, the search for of the optimal solution begins at the origin. Indicating that nothing can be produced;

Thus, first assumption, No production implies that $x_1 = 0$ and $x_2 = 0$

$$\Rightarrow 2x_1 + x_2 + s_1 + 0s_2 + 0s_3 = 40$$

$$\Rightarrow x_1 + 3x_2 + 0s_1 + s_2 + 0s_3 = 45$$

$$2(0) + 0 + s_1 + 0 s_2 + 0 s_3 = 40$$

$$\underline{s_1 = 40} - \text{Unused labor hrs.}$$

$$0 + 3(0) + 0 s_1 + s_2 + 0 s_3 = 45$$

$$\underline{s_2 = 45} - \text{Unused machine hrs.}$$

$$\Rightarrow x_1 + 0 s_1 + 0 s_2 + s_3 = 12$$

$$0 + 0 s_1 + 0 s_2 + s_3 = 12$$

$$\underline{s_3 = 12} - \text{Unused Marketing hrs.}$$

$$\begin{aligned} \text{Therefore, } \text{Max. } Z &= 300x_1 + 250x_2 + 0 s_1 + 0 s_2 + 0 s_3 \\ &= 300(0) + 250(0) + 0(40) + 0(45) + 0(12) = \underline{0} \end{aligned}$$

Note:

In general, whenever there are n variables and m constraints (excluding the non-negativity), where m is less than n ($m < n$), $n-m$ variables must be *set equal to zero* before the solution can be solved algebraically.

a. Basic variables are variables with non-zero solution values.

Or: basic variables are variables that are in the basic solution. Basic variables have 0 values in the $C_j - Z_j$ row.

b. Non-basic variables are variables with zero solution values.

Or: non-basic variables are variables that are out of the solution.

$\Rightarrow n=5$ variables (x_1 , x_2 , s_1 , s_2 , and s_3) and $m=3$ constraints (Labor, machine and marketing constraints), excluding non-negativity.

Therefore, $n-m=5-3=2$ variables (x_1 and x_2) are set equal to zero in the 1st simplex tableau. These are non-basic variables. 3 Variables (s_1 , s_2 , and s_3) are basic variables (in the 1st simplex tableau) because they have non-zero solution values.

Step 4

Construct the initial simplex tableau

Initial simplex tableau

<i>Profit per unit column</i>	<i>Basic or Solution variable column</i>	<i>Real or decision variables column</i>					<i>Profit per unit row</i>
C_j		300	250	0	0	0	
<i>SV</i>		X_1	X_2	S_1	S_2	S_3	Q
0	S_1	2	1	1	0	0	40
0	S_2	1	3	0	1	0	45
0	S_3	1	0	0	0	1	12
Z_j		0	0	0	0	0	0
$C_j - Z_j$		300	250	0	0	0	

Constraint equation rows R_1
 R_2
 R_3

Gross Profit row
Net Profit row /Indicator row/

Step 5: Choose the “incoming” or “entering” variables

Note:

The entering variable is the variable that has *the most positive value* in the $C_j - Z_j$ row also called as *indicator row*. Or the entering variable is the variable that has *the highest contribution to profit per unit*.

- X_1 in our case is the entering variable
- The column associated with the entering variable is called *key* or *pivot column* (X_1 column in our case)

Step 6: Choose the “leaving” or “outgoing” variable

==> In this step, we determine the variable that will leave the solution for X_1 (or entering variable)

Note:

- The row with *the minimum* or *lowest positive (non-negative) replacement ratio* shows the variable to leave the solution.

$$\text{Replacement Ratio (RR)} = \frac{\text{Solution Quantity (Q)}}{\text{Corresponding values in pivot column}}$$

Note: $RR > 0$

- The variable leaving the solution is called *leaving variable* or *outgoing variable*.
- The row associated with the leaving variable is called *key or pivot row* (s_3 column in our case)
- The element that lies at the intersection of the pivot column and pivot row is called pivot element ($N^o 1$ in our case)

Step 7: Repeat step 3-5 till optimum basic feasible solution is obtained.

Or: repeat step 3-5 till no positive value occurs in the $C_j - Z_j$ row.

Note:

- Divide each element of the pivot row by the pivot element to find new values in the key or pivot row.
- Perform row operations to make *all other entries for the pivot column equal to zero*.

2nd simplex tableau

C_j	300	250	0	0	0	
SV	X_1	X_2	S_1	S_2	S_3	Q
0 S_1	0	1	1	0	-2	16
0 S_2	0	3	0	1	-1	33
300 X_1	1	0	0	0	1	12
Z_j	300	0	0	0	300	
$C_j - Z_j$	0	250	0	0	-300	

$R2 \text{ NEW} = R2 \text{ OLD} - R3 \text{ NEW}$

$R1 \text{ NEW} = R1 \text{ OLD} - 2 R3 \text{ NEW}$

$R3 \text{ NEW} = R3 \text{ OLD}$

3rd simplex tableau

C_j	300	250	0	0	0	
SV	X_1	X_2	S_1	S_2	S_3	Q
0	S_1	0	0	1	-1/3	-5/3
						5

250	X_2	0	1	0	$1/3$	$-1/3$	11
300	X_1	1	0	0	0	1	12
	Z_j	300	250	0	$250/3$	$650/3$	6350
	$C_j - Z_j$	0	0	0	$-250/3$	$-650/3$	

$$R2 \text{ NEW} = R2 \text{ OLD} / 3$$

$$R1 \text{ NEW} = R1 \text{ OLD} - R2 \text{ NEW}$$

$$R3 \text{ NEW} = R3 \text{ OLD}$$

Since all the $C_j - Z_j \leq 0$ optimal solution is reached at.

Therefore, $X_1=12$, $X_2=11$, $S_1=5$ and $Max Z=6350$

Exercise:

A Juice Company has available two kinds of food Juices: **Orange Juice** and **Grape Juice**. The company produces two types of punches: **Punch A** and **Punch B**. One bottle of punch A requires 20 liters of Orange Juice and 5 liters of Grape Juice. 1 Bottle of punch B requires 10 liters of Orange Juice and 15 liters of Grape Juice.

From each of bottle of Punch A a profit of \$4 is made and from each bottle of Punch B a profit of \$3 is made .Suppose that the company has 230 liters of Orange Juice and 120 liters of Grape Juice available

Required:

- Formulate this problem as a LPP
- How many bottles of Punch A and Punch B the company should produce in order to maximize profit? (Using the simplex method)
- What is this maximum profit?

Solution:

Juice needed for one bottle of

<i>Juice</i>	<i>Punch A</i>	<i>Punch B</i>	<i>Juice Available</i>
<i>Orange Juice (lt)</i>	20	10	230
<i>Grape Juice (lt)</i>	<u>5</u>	<u>15</u>	120
Profit per tent	<u>\$4</u>	<u>\$3</u>	

Let X_1 = the No of bottles of punch A produced.

X_2 = the No of bottles of punch B produced.

• **LPP Model**

$$\text{Max } Z = 4X_1 + 3X_2$$

St:

$$20X_1 + 10X_2 \leq 230 \quad \text{Orange Constraint}$$

$$5X_1 + 15X_2 \leq 120 \quad \text{Grape Constraint}$$

$$X_1, X_2 \geq 0 \quad \text{Non-negativity constraint}$$

• **Standard form**

$$\left. \begin{array}{l} \text{Max } Z = 4x_1 + 3x_2 + 0s_1 + 0s_2 + 0s_3 \\ \text{St:} \\ 20x_1 + 10x_2 + s_1 + 0s_2 = 230 \\ 5x_1 + 15x_2 + 0s_1 + s_2 = 120 \\ x_1, x_2, s_1, s_2 \geq 0 \end{array} \right\} \text{Standard form}$$

Where, s_1 = Unused orange juice

s_2 = Unused grape juice

Initial simplex tableau

C_j	4	3	0	0	
SV	X_1	X_2	S_1	S_2	Q
0 S_1	20	10	1	0	230
0 S_2	5	15	0	1	120
Z_j	0	0	0	0	0
$C_j - Z_j$	4	3	0	0	

$$R1 \text{ NEW} = R1 \text{ OLD} / 20$$

$$R2 \text{ NEW} = R2 \text{ OLD} - 5R1 \text{ NEW}$$

2nd simplex tableau

C_j		4	3	0	0	
	SV	X_1	X_2	S_1	S_2	Q
4	X_1	1	1/2	1/20	0	11.5
0	S_2	0	25/2	-1/4	1	62.5
	Z_j	4	2	1/5	0	46
	$C_j - Z_j$	0	1	-1/5	0	

C_j		4	3	0	0	
	SV	X_1	X_2	S_1	S_2	Q
4	X_1	1	0	3/50	-1/25	9
3	X_2	0	1	-1/50	2/25	5
	Z_j	4	3	0.18	0.08	51
	$C_j - Z_j$	0	0	-0.18	-0.08	

$R2\ NEW = R2\ OLD * 2/25$

$R1\ NEW = R1\ OLD - 1/2 * R2\ NEW$

Optimal solution with all $c_j - z_j$ values less than or equal to zero

$X_1 = 9$ bottles of punch A

$X_2 = 5$ bottles of punch B

$s_1 = 0$ $s_2 = 0$ **Max Z = \$51**

Exercise:

Solve the following LPPs using the simplex method

1. Max $Z = 3x_1 + 5x_2$

Subject to:

$$x_2 \leq 6$$

$$3x_1 + 2x_2 \leq 18$$

$$x_1, x_2 \geq 0$$

Answer:

$x_1 = 2, x_2 = 6, s_1 = 0, s_2 = 0$ and **Max Z = \$36**

Max Z = \$1,000

2. Max $Z = 20x_1 + 10x_2$

Subject to:

$$5x_1 + 4x_2 \leq 250$$

$$2x_1 + 5x_2 \leq 150$$

$$x_1, x_2 \geq 0$$

Answer:

$x_1 = 50, x_2 = 0, s_1 = 0, s_2 = 50$ and

2. MINIMIZATION PROBLEMS

➤ Minimize Z with inequalities of constraints in " $\geq$ " form

There are two methods to solve minimization LP problems:

1. Direct method/Big M-method/

➤ *Using artificial variables*

2. Conversion method

➤ *Minimization by maximizing the dual*

❖ **Surplus Variable (-s):**

➤ A variable inserted in a greater than or equal to constraint to create equality. It represents the amount of resource usage above the minimum required usage.

➤ Surplus variable is subtracted from a $\geq$ constraint in the process of converting the constraint to standard form.

Example:

1. $2x_1 + x_2 \leq 40$ ==> is a constraint inequality

$$x_1 = 12 \text{ and } x_2 = 11 \Rightarrow 2x_1 + x_2 + s = 40 \Rightarrow 2(12) + 11 + s = 40$$

$$\Rightarrow \underline{s=5} \text{ unused resource}$$

2. $5x_1 + 3x_2 \leq 45$

$$x_1 = 12 \text{ and } x_2 = 11 \Rightarrow 5x_1 + 3x_2 + s = 45 \Rightarrow 5(12) + 3(11) + s = 45$$

$$\Rightarrow \underline{s=0} \text{ unused resource (No idle resource)}$$

3. $5x_1 + 2x_2 \geq 20$

$$x_1 = 4.5 \text{ and } x_2 = 2 \Rightarrow 5x_1 + 2x_2 - s = 20 \Rightarrow 5(4.5) + 2(2) - s = 20$$

$$\Rightarrow \underline{s=6} \text{ unused resource}$$

4. $2x_1 + x_2 \geq 40$

$$x_1 = 0 \text{ and } x_2 = 0 (\text{No production}) \Rightarrow 5x_1 + 2x_2 - s = 20 \Rightarrow 5(4.5) + 2(2) - s = 20$$

$$\Rightarrow \underline{s=-6} (\text{This is mathematically unacceptable})$$

Thus, in order to avoid the mathematical contradiction, we have to add artificial variable (A)

1.13.2.1. Big M-method

/Charnes Penalty Method/

The Big-M Method is a method which is used in removing artificial variables from the basis. In this method; we assign coefficients to artificial variables, undesirable from the objective function point of view. If objective function Z is to be minimized, then a very large positive price (called **penalty**) is assigned to each artificial variable. Similarly, if Z

is to be maximized, then a very large negative price (also called *penalty*) is assigned to each of these variables.

Following are the characteristics of Big-M Method:

- a. High penalty cost (or profit) is assumed as M
- b. M is assigned to artificial variable A in the objective function Z.
- c. Big-M method can be applied to minimization as well as maximization problems with the following distinctions:
 - i. *Minimization problems*
-Assign **+M** as coefficient of artificial variable A in the objective function Z
 - ii. *Maximization problems:*
-Here **-M** is assigned as coefficient of artificial variable A in the objective function Z
- d. Coefficient of S (slack/surplus) takes zero values in the objective function Z
- e. For minimization problem, the incoming variable corresponds to the *highest negative* value of $C_j - Z_j$.
- f. Solution is optimal when there is no negative value of $C_j - Z_j$. (For minimization case)

Example:

1. Minimize $Z = 25x_1 + 30x_2$

Subject to:

$$20x_1 + 15x_2 \geq 100$$

$$2x_1 + 3x_2 \geq 15$$

$$x_1, x_2 \geq 0$$

Solution

Step 1

Standardize the problem

$$\text{Minimize } Z = 25x_1 + 30x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$$

Subject to:

$$20x_1 + 15x_2 - s_1 + A_1 = 100$$

$$2x_1 + 3x_2 - s_2 + A_2 = 15$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

Step 2

Initial simplex tableau

The initial basic feasible solution is obtained by setting $x_1 = x_2 = s_1 = s_2 = 0$

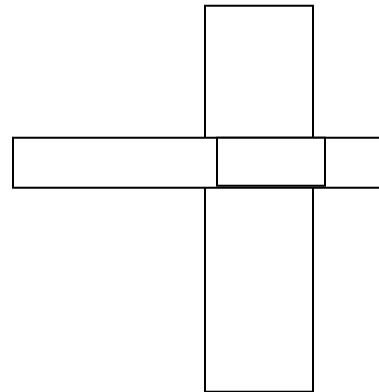
No production, $x_1 = x_2 = s_1 = 0 \Rightarrow 20(0) + 15(0) - 0 + A_1 = 100 \Rightarrow A_1 = 100$

$x_1 = x_2 = s_2 = 0 \Rightarrow 0(0) + 3(0) - 0 + A_2 = 15 \Rightarrow A_2 = 15$

C_j		25	30	0	0	M	M	
	SV	X_1	X_2	S_1	S_2	A_1	A_2	Q
M	A_1	20	15	-1	0	1	0	100
M	A_2	2	3	0	-1	0	1	15
	Z_j	22M	18M	-M	-M	M	M	115M
	$C_j - Z_j$	25 - 22M	30 - 18M	M	M	0	0	

*Initial
simplex
tableau*

$100/20=5$



$R1 \text{ NEW} = R1 \text{ OLD} / 20$

$R2 \text{ NEW} = R2 \text{ OLD} - 2R1 \text{ NEW}$

Note: Once an *artificial variable* has left the basis, it has served its purpose and can therefore be removed from the simplex tableau. An artificial variable is never considered for re-entry into the basis.

2nd Simplex Tableau

C_j		25	30	0	0	M	
	SV	X_1	X_2	S_1	S_2	A_2	Q
25	X_1	1	$3/4$	$-1/20$	0	0	5
M	A_2	0	$3/2$	$1/10$	-1	1	5
	Z_j	25	$75/4+3/2M$	$-5/4+1/10M$	$-M$	M	$125+5M$
	$C_j - Z_j$	0	$45/4-3/2M$	$5/4-1/10M$	M	0	

$$R1 \text{ NEW} = R1 \text{ OLD} - 3/4 R2 \text{ NEW}$$

$$R2 \text{ NEW} = R2 \text{ OLD} * 2/3 \quad \text{OR} \quad R2/3/2$$

3rd Simplex Tableau

C_j		25	30	0	0	
	SV	X_1	X_2	S_1	S_2	Q
25	X_1	1	0	$-1/10$	$1/2$	$5/2$
30	X_2	0	1	$1/15$	$-2/3$	$10/3$
	Z_j	25	30	$-1/2$	$-15/2$	162.5
	$C_j - Z_j$	0	0	$1/2$	$15/2$	

All $C_j - Z_j \geq 0$ Optimal solution is reached

$$X_1 = 5/2$$

$$X_2 = 10/3 \text{ and Min } Z = 162.5$$

Note:

As long as an "A" variable is available in the solution variable column, the solution is *infeasible*.

2. Use the penalty (Big-M) method to solve the following LPP

$$\text{Min } Z = 5x_1 + 3x_2$$

Subject to:

$$2x_1 + 4x_2 \leq 12$$

$$2x_1 + 2x_2 = 10$$

$$5x_1 + 2x_2 \geq 10$$

$$x_1, x_2 \geq 0$$

Solution

$$\text{Min } Z = 5x_1 + 3x_2 + 0s_1 + 0s_2 + MA_1 + MA_2$$

Subject to:

If no production

$$2x_1 + 4x_2 + s_1 = 12 \implies x_1 = x_2 = 0 \implies s_1 = 0 \text{ (Solution Value in the initial simplex tableau)}$$

$$2x_1 + 2x_2 + A_1 = 10 \implies x_1 = x_2 = 0 \implies A_1 = 10 \text{ (Solution Value in the initial simplex tableau)}$$

$$5x_1 + 2x_2 - s_2 + A_2 = 10 \implies x_1 = x_2 = s_2 = 0 \implies A_2 = 10 \text{ (Solution Value in the initial simplex tableau)}$$

$$x_1, x_2, s_1, s_2, A_1, A_2 \geq 0$$

C_j	5	3	0	0	M	M	
SV	X_1	X_2	S_1	S_2	A_1	A_2	Q
0 S_1	2	4	1	0	0	0	12
M A_1	2	2	0	0	1	0	10
M A_2	5	2	0	-1	0	1	10
Z_j	7M	4M	0	-M	M	M	20M
$C_j - Z_j$	5-7M	3-4M	0	M	0	0	

Initial
tableau

Simplex

$$R1 \text{ NEW} = R1 \text{ OLD} - 2 R3 \text{ NEW}; R2 \text{ NEW} = R2 \text{ OLD} - 2 R3 \text{ NEW}; R3 \text{ NEW} = R3 \text{ OLD} / 5$$

2nd simplex tableau

C_j	5	3	0	0	M	
SV	X_1	X_2	S_1	S_2	A_1	Q
0 S_1	0	16/5	1	2/5	0	8

M	A_1	0	$6/5$	0	$2/5$	1	6
5	X_1	1	$2/5$	0	$-1/5$	0	2
	Z_j	$5M$	$6/5M + 2$	0	$2/5M - 1$	M	$10 + 6M$
	$C_j - Z_j$	0	$-6/5M + 1$	0	$-2/5M + 1$	0	

$$R1 \text{ NEW} = R1 \text{ OLD} * 5/16$$

$$R2 \text{ NEW} = R2 \text{ OLD} - 6/5 * R1 \text{ NEW}$$

$$R3 \text{ NEW} = R3 \text{ OLD} - 2/5 * R1 \text{ NEW}$$

3rd simplex tableau

C_j		5	3	0	0	M	
	SV	X_1	X_2	S_1	S_2	A_1	Q
3	X_2	0	1	$5/16$	$1/8$	0	2.5
M	A_1	0	0	$-3/8$	$1/4$	1	3
5	X_1	1	0	$-1/8$	$-1/4$	0	1
	Z_j	$5M$	3	$-3/8M + 5/16$	$M/4 - 7/8$		$12.5 + 3M$
	$C_j - Z_j$	0	0	$3/8M - 5/16$	$-M/4 + 7/8$	0	

$$R1 \text{ NEW} = R1 \text{ OLD} - 1/8 R2 \text{ NEW}$$

$$R2 \text{ NEW} = R1 \text{ OLD} * 4$$

$$R3 \text{ NEW} = R3 \text{ OLD} + 1/4 * R2 \text{ NEW}$$

4th Simplex tableau (OPTIMAL SOLUTION)

C_j		5	3	0	0	
	SV	X_1	X_2	S_1	S_2	Q
3	X_2	0	1	1/2	0	1
0	S_2	0	0	-3/2	1	12
5	X_1	1	0	-1/2	0	4
	Z_j	5	3	-1	0	23
	$C_j - Z_j$	0	0	1	0	

$X_1=4, X_2=1, S_1=0, S_2=12$ and $MinZ=23$

Exercise

Find the optimal solution using simplex method

1. $Min Z=10x_1 + 5x_2$

Subject to:

$$2x_1 + 5x_2 \geq 150$$

$$3x_1 + x_2 \geq 120$$

$$x_1, x_2 \geq 0$$

Ans: $x_1=450/13, x_2=210/13$ and $Min Z=\$540$

2. $Min Z=4x_1 + 5x_2$

Subject to:

$$x_1 + 2x_2 \geq 80$$

$$3x_1 + x_2 \geq 75$$

$$x_1, x_2 \geq 0$$

Ans: $x_1=14, x_2=33$ and $Min Z=\$221$

1.13.2.2. SPECIAL CASES IN SIMPLEX METHOD

1. Mixed constraints

Example

$Max Z=6x_1 + 8x_2$

Subject to:

$$x_2 \leq 4$$

$$x_1 + x_2 = 9$$

$$6x_1 + 2x_2 \geq 24$$

$$x_1, x_2 \geq 0$$

• **Standard form**

$$\text{Max } Z = 6x_1 + 8x_2 + 0s_1 + 0s_2 - M A_1 - M A_2$$

St:

$$x_2 + s_1 = 4$$

$$x_1 + x_2 + A_1 = 9$$

$$6x_1 + 2x_2 - s_2 + A_2 = 24$$

$$\text{All Variables} \geq 0$$

Standard form

Initial simplex tableau

C_j		6	8	0	0	-M	-M	
	SV	X_1	X_2	S_1	S_2	A_1	A_2	Q
0	S_1	0	1	1	0	0	0	4
-M	A_1	1	1	0	0	1	0	9
-M	A_2	6	2	0	-1	0	1	4
	Z_j	-7M	-3M	0	0	+M	-M	24
	$C_j - Z_j$	7M + 6	3M + 8	0	0	-M	0	0

Ans: At the 4th tableau: $X_1 = 5$, $X_2 = 4$, $S_2 = 14$ and $\text{Max } Z = 62$

Note:

For the initial basis, use *artificial* variables for constraints that have them. Otherwise, use a constraint *slack* variable. Hence, *surplus* variables will not appear in an initial solution.

2. Two incoming variables

/ Or Tie for entering variables/

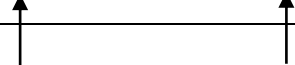
In order to break this tie, the selection for the key column (entering variable) can be made arbitrary. However; the number of solution can be minimized by adopting the following rules:

1. *If there is a tie between two decision variables, then the selection can be made arbitrary.*
2. *If there is a tie between a decision variable and a slack (or surplus) variable, then select the decision variable to enter into basis first.*
3. *If there is a tie between slack or surplus variable, then selection can be made arbitrary.*

Example:

If the equation is max Z:

C_j					Q
SV	X_1	X_2	S_1	S_3	
Z_j					
$C_j - Z_j$	5	2	5	0	



In such a case, X_1 is the entering variable

3. Degeneracy

/ Tie for leaving basic variable (key row) /

If there is a tie for the smallest ratio, this is a signal tat degeneracy exists. Degeneracy can occur right in the first (initial tableau). This normally happens when the number of constraints is less than the number of variables in the objective function. Problem can be overcome by *trial and error method*.

C_j		5	8	2	0	0	0	
	SV	X_1	X_2	X_3	S_1	S_2	S_3	Q
8	X_2	1/4	1	1	-2	0	0	10
0	S_2	4	0	1/3	-1	1	0	20
0	S_3	2	0	2	2/5	0	1	10
	Z_j	2	8	8	16	0	0	80
	$C_j - Z_j$	3	0	-6	-16	0	0	

4. Infeasibility

In it, all $C_j - Z_j$ row entries will be the proper sign to imply *optimality*, but an *artificial variable* (A) will still be in the solution mix. This called infeasible solution.

Example:

Minimization case

C_j		5	8	0	0	M	
	SV	X_1	X_2	S_1	S_2	A_2	Q
5	X_1	1	1	-2	3	0	200
8	X_2	0	1	1	2	0	100
M	A_2	0	0	0	-1	1	20
	Z_j	5	8	-2	$31-M$	M	$1,800+200M$
	$C_j - Z_j$	0	0	2	$M-31$	0	

Even though all $C_j - Z_j$ are positive or 0 (i.e the criterion for an optimal solution in a minimization case), *no feasible solution* is possible because *an artificial variable* (A_2) remains in the solution mix.

5. Unbounded Solutions

If the entire replacement ratios turn out to be *negative* or *undefined*, it indicates that the problem is unbounded.

Example:

Maximization

C_j		6	9	0	0		
	SV	X_1	X_2	S_1	S_2	Q	<u>RR</u>
9	X_2	-1	1	2	0	30	$30/-1=-30$ $10/-2=-5$ Unacceptable RRs
0	S_2	-2	0	-1	1	10	
	Z_j	-9	9	18	0	270	
	$C_j - Z_j$	15	0	-18	0		

↑
Pivot Column

The solution in the above case is not optimal because not all $C_j - Z_j$ entries are 0 or negative, as required in a maximization problem. The next variable to enter the solution should be X_1 . To determine which variable will leave the solution, we examine the ratios of the quantity column numbers to their corresponding numbers in the X_1 or pivot column. Since both pivot column numbers are negative, an unbounded solution is indicated.

- No unbounded solutions, no outgoing variable will exist.

6. Multiple Optimal Solutions

Multiple optimal solutions exist when non-basic variable contains *zero* on its $C_j - Z_j$ row.

Example:

Maximization problem

C_j		3	2	0	0	
	SV	X_1	X_2	S_1	S_2	Q
2	X_2	3/2	1	1	0	6
0	S_2	1	0	1/2	1	3
	Z_j	3	2	2	0	12
	$C_j - Z_j$	0	0	-2	0	

$$\text{Max } Z = 3X_1 + 2X_2$$

$X_1=0, X_2=6, S_2=3$ and $MaxZ=12$ or: $X_1=3, X_2=3/2$ and $MaxZ=12$

The $C_j - Z_j$ value of the Non-basic variable (X_1) is 0. Thus, there is alternative optimal solution.

Exercise:

1. Solve the following LPP by the simplex algorithm

$$\text{Min } Z = 6x_1 + 8x_2$$

Subject to:

$$x_1 + 2x_2 \geq 80$$

$$3x_1 + x_2 \geq 75$$

$$x_1, x_2 \geq 0$$

What are the values of the basic variables at each iteration?

Which are the non-basic variables at each iteration?

1. At the 3rd iteration of a particular LP maximization problem, the following tableau is established:

C_j								
	SV	X_1	X_2	X_3	S_1	S_2	S_3	Q
5	X_3	0	1	1	-2	0	0	5
6	X_1	1	-3	0	0	0	1	12
0	S_2	0	2	0	1	1	-1	10
	Z_j	6	-13	5	5	0	21	97
	$C_j - Z_j$	0	16	0	-5	0	-21	

What special condition exists as you improves the profit and move to the next iteration?

Proceed to solve the problem for optimal solution

2. Solve the following LPP to show that it has alterative optimal solutions.

a. $MaxZ = 6x_1 + 3x_2$ $x_2 \leq 3$

Subject to:

$$x_1, x_2 \geq 0$$

$$2x_1 + x_2 \leq 8$$

b $Min Z = 2x_1 + 8x_2$

$$3x_1 + 3x_2 \leq 18$$

Subject to:

$$5x_1 + x_2 \geq 10$$

$$2x_1 + 2x_2 \geq 14$$

$$x_1 + 4x_2 \geq 14$$

$$x_1, x_2 \geq 0$$

3. Solve the following LPP to show that it has unbounded solution.

a. $\text{Max} Z = -2x_1 + 3x_2$

Subject to:

$$x_1 \leq 5$$

$$2x_1 - 3x_2 \leq 6$$

$$x_1, x_2 \geq 0$$

b. $\text{Max} Z = 3x_1 + 6x_2$

Subject to:

$$3x_1 + 4x_2 \geq 12$$

$$-2x_1 + x_2 \leq 4$$

$$x_1, x_2 \geq 0$$

4. Solve the following LPP to show that it has no feasible solution.

a. $\text{Max} Z = -2x_1 + 3x_2$

Subject to:

$$x_1 - x_2 \geq 4$$

$$x_1 + x_2 \leq 6$$

$$x_1 \leq 2$$

$$x_1, x_2 \geq 0$$

b. $\text{Max} Z = 3x_1 + 3x_2$

Subject to:

$$2x_1 + x_2 \leq 2$$

$$3x_1 + 4x_2 \geq 1$$

$$x_1, x_2 \geq 0$$

EXERCISE

1. Bekas Chemicals Company manufactures two chemicals A and B which are sold to the manufacturers of soaps and detergents. On the basis of the next month's demand, the management has decided that the total production of chemicals A and B should be at least 350 kilograms. Moreover, a major customer order for 125 kilograms of product A must also be supplied. Product A requires 2 hours of processing time per kilogram and product B requires one hour of processing time per kilogram. For the coming month, 600 hours of processing time are available. The company wants to meet the above requirements at minimum total production cost. The production costs are birr 2 per kilogram for product A and birr 3 per kilogram for product B. bekas chemicals company wants to determine its optimal product mix and the total minimum cost relevant thereto.
 - A. Formulate the above as a linear programming problem
 - B. Solve the problem with the simplex method
 - C. Does the problem have multiple optimal solutions? Why?
2. A finished product must weight exactly 150 grams. The two raw materials used in manufacturing the product are A, with a cost of birr 2 per unit and B with a

cost of birr 8 per unit. At least 14 units of B and not more than 20 units of A must be used. Each unit of A and B weights 5 and 10 grams respectively. How much of each type of raw material should be used for each unit of the final product in order to minimize the cost? Use the simplex method

3. Solve the following LPP

$$\text{Maximize } Z = 8x_1 - 4x_2$$

$$\text{St. } 4x_1 + 5x_2 \leq 20$$

$$-x_1 + 3x_2 \geq -23$$

$$x_1 \geq 0, x_2 \text{ unrestricted in sign}$$

1.14. SENSATIVITY ANALYSIS AND DUALITY

1.14.1. SENSATIVITY ANALYSIS

Sensitivity Analysis is concerned with the study of 'Sensitivity' of the optimal solution of an LPP with discretion variables (changes) in parameters. The degree of sensitivity of the solution due to those variations can range from no change at all to a substantial change in the optimal solution of the given LPP. Thus, in sensitivity analysis, we determine the range over which the LP model parameters can change without affecting the current optimal solution. The process of studying the sensitivity of the optimal solution of an LPP is called *post-optimal analysis*.

The two sensitivity analysis approaches are:

I. Trial and error approach and

II. Analytical approach

Analytical approach

Five types of discrete changes in the original LP model may be investigated during the sensitivity analysis:

A. *Changes of the coefficients of the objective function (c_j)*

B. *Changes of the RHS Quantity (b_j)*

C. *Changes of the input-output coefficient*

D. *Add/delete constraints*

E. *The addition of a new variable to the problem*

A. Changes of the coefficients of the objective function (c_j)

Decision variables can be:

i. Basic (in the solution)

ii. Non-basic (out-of the solution)

● Range for the coefficients of basic decision variables

The *range of optimality* is the range over which a basic decision variable coefficient in the objective function can change without changing the optimal solution mix. However, this change will change only the optimal value of the objective function.

Example:

	X_2	S_1	S_2
$C_j - Z_j$ row	-0.842	-0.311	-0.067
X_1 row	1.053	0.067	0

Non-basic variables
Max. $Z = 5x_1 + 4.5x_2 + x_3$

Subject to:

$$15x_1 + 15.8x_2 \leq 150$$

$$5x_1 + 6.4x_2 + 15x_3 \leq 77$$

$$2.8x_2 + 11.8x_3 \leq 36$$

$$x_1, x_2, x_3 \geq 0$$

The optimal tableau for this solution is:

C_j		5	4.5	1	0	0	0	
	SV	X_1	X_2	X_3	S_1	S_2	S_3	Q
5	X_1	1	1.053	0	0.067	0	0	10
1	X_3	0	0.67	1	-0.022	0.067	0	1.8
0	S_3	0	1.924	0	0.258	-0.773	1	15.12
	Z_j	5	5.342	1	0.311	0.067	0	51.8
	$C_j - Z_j$	0	-0.842	0	-0.311	-0.067	0	

Determine the range of optimality for the coefficients of the basic-decision variables.

Solution:

Analysis of basic decision variables

The analysis will be conducted on products on X_1 and X_3 which are in the basic solution. Divide each $C_j - Z_j$ row entry for variables not in the solution (for instance, by X_2, S_1 and S_2 values) by the associated variable a_{ij} from X_1 or X_3 row.

I. Analysis of X_1

$C_j - Z_j \text{ row}$	-0.8	-4.64	$-\infty$
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Steps:

a. Take the $C_j - Z_j$ row

of the optimal solution of the non-basic variables

b. Take the X_1 row of the non-basic variables

c. $\frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$

❖ Upper Limit

The *smallest positive number* in the $C_j - Z_j$ row tells how much the profit X_1 of can be increased before the solution is changed.

$$\text{Upper Limit} = C_j \text{ (for } X_1) + \text{the smallest positive value of } \frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$$

$$= 5 + \infty = \infty$$

Note: C_j (for X_1) = 5 (Look in the OF of the LP problem)

❖ Lower Limit

The *largest negative number closest* (negative amount closest to 0)

$$\text{Lower Limit} = C_j \text{ (for } X_1) + \text{The largest negative value of } \frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$$

$$= 5 + (-0.8) = \underline{4.2}$$

Therefore, the range of optimality for the coefficient of X_1 is $4.2 \leq C_j \text{ (for } X_1) < \infty$ (The coefficient of X_1 in the objective function can change between 4.2 and ∞ without changing the optimal solution mix $X_1=10$, $X_3=1.8$ and $S_3=15.12$)

II. Analysis of X_3

$$\text{Upper Limit} = C_j \text{ (for } X_3) + \text{The smallest positive value of } \frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$$

$$= 1 + (14.13) = \underline{15.13}$$

Note: C_j (for X_3) = 5 (Look in the OF of the LP problem)

$$\text{Lower Limit} = C_j \text{ (for } X_3) + \text{The largest negative value of } \frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$$

$$= 1 + (-1) = \underline{0}$$

Therefore, the range of optimality for the coefficient of X_3 is $0 \leq C_j \text{ (for } X_3) < 15.13$ (The coefficient of X_3 in the objective function can change between 0 and 15.13 without changing the optimal solution mix $X_1=10$, $X_3=1.8$ and $S_3=15.12$). However, this change will change only the optimal value of the objective function (i.e. Max Z will change)

● The range for the non-basic variables

If there is a variable C_j , not participating in the *optimal basis*, then, in order for the variable to be included in *the optimal solution*, its coefficient in the objective function will have to change from the existing C_j to a new level $C_j(\text{new})$.

The range of insignificance is the range over which C_j rates for non-basic variables can vary without causing a change in the optimal solution mix (variable) is called the range of insignificance.

$$C_j(\text{new}) > Z_j$$

Example:

$$\text{Max. } Z = 5x_1 + 4.5x_2 + x_3$$

Subject to:

$$15x_1 + 15.8x_2 \leq 150$$

$$5x_1 + 6.4x_2 + 15x_3 \leq 77 \quad 2.8x_2 + 11.8x_3 \leq 36$$

$$x_1, x_2, x_3 \geq 0$$

The optimal tableau for this solution is:

C_j		5	4.5	1	0	0	0	
SV		X_1	X_2	X_3	S_1	S_2	S_3	Q
5	X_1	1	1.053	0	0.067	0	0	10
1	X_3	0	0.67	1	-0.022	0.067	0	1.8
0	S_3	0	1.924	0	0.258	-0.773	1	15.12
Z_j		5	5.342	1	0.311	0.067	0	51.8
$C_j - Z_j$		0	-0.842	0	-0.311	-0.067	0	

Calculate the range of insignificance for the coefficient of non-basic variable(X_2)

Solution

$$C_j(\text{for } X_2) = 4.5 \text{ and } Z_j(\text{ for } X_2) = 5.342$$

$$C_j(\text{new for } X_2) > 5.342 \implies$$

$C_j(\text{new for } X_2)$. If the profit contribution of X_2 is greater

than 5.342, then X_2 will be included in the solution.

Thus, $-\infty < C_j(\text{new for } X_2) \leq 5.342$ is the range of insignificance for X_2 .

	X_2	S_1	S_2
$C_j - Z_j \text{ row}$	-0.842	-0.311	-0.067
$X_3 \text{ row}$	0.67	-0.022	0.067
$\frac{C_j - Z_j \text{ row}}{X_1 \text{ row}}$	-1.26	14.13	-1

Non-basic variables

C_j (new for X_2) can vary within the given range without causing a change in the optimal solution mix ($X_1=10, X_2=0, X_3=1.8, S_1=S_2=0$ and $S_3=15.12$).

B. Change in the Right Hand – Side Quantity (RHS)

Or

Change in the availability of resource (Capacity) (b_j)

❖ Shadow prices:

==>How much should a firm be willing to pay to make additional resources available?

Shadow prices signify the change in the optimal value of the objective function for **1 unit increases** in the value of the RHS of the constraint that represent the availability of scarce resources.

The negative of the number of $C_j - Z_j$ row in **its slack variable columns** provide as with shadow prices. **Or:** shadow prices are found in the Z_j row of the final simplex tableau in the slack variable columns.

❖ **RHS ranging** is the range over which shadow prices remain valid.

Example:

$$\text{Max } Z=3x_1+4x_2$$

Subject to:

$$3x_1+5x_2 \leq 15$$

$$2x_1 + x_2 \leq 8$$

$$x_2 \leq 2$$

$$x_1, x_2 \geq 0$$

The optimal tableau is given as:

C_j		3	4	0	0	0	
SV		X_1	X_2	S_1	S_2	S_3	Q
3	X_1	1	0	-0.143	0.714	0	3.57
0	S_3	0	0	-0.286	0.428	1	1.143
4	X_2	0	1	0.286	-0.428	0	0.857
$C_j - Z_j$		0	0	-0.714	-0.428	0	

Required:

1. Determine the shadow price for each constraint
2. Determine the RHS ranges over which the shadow prices are valid

❖ *Analysis of the 1st constraint (S_1)*

<u>Q</u>	<u>S_1</u>	<u>Q/S_1</u>
3.57	-0.143	-24.96
1.143	-0.286	-3.99
0.857	0.286	3.00

Lower Limit= b_j -the smallest positive number in the Q/S_j column

Upper Limit= b_j -the largest negative number in the Q/S_j column

Lower Limit= b_1 -the smallest positive number in the Q/S_1 column

$$=15-3=\underline{12}$$

Upper Limit= b_1 -the largest negative number in the Q/S_1 column

$$=15-(-3.99)=\underline{18.99}$$

Therefore, $12 \leq b_1 \leq 18.99$ (The range of resource 1 over which the shadow price \$0.714 per unit is valid).

❖ *Analysis of the 2nd constraint (S_2)*

<u>Q</u>	<u>S_2</u>	<u>Q/S_2</u>
3.57	0.714	5
1.143	0.428	2.67
0.857	-0.428	-2

Lower Limit= b_2 -the smallest positive number in the Q/S_2 column

$$=8-(2.67)=\underline{5.33}$$

Upper Limit= b_2 -the largest negative number in the Q/S_2 column

$$=8-(-2)=\underline{10}$$

Therefore, $5.33 \leq b_1 \leq 10$ (The range of resource 2 over which the shadow price \$0.428 per unit is valid).

❖ *Analysis of the 3rd constraint (S_3)*

Q	S_3	Q/S_3
3.57	0	-
1.143	1	1.143
0.857	0	-

Lower Limit = b_3 - the smallest positive number in the Q/S_3 column

$$= 2 - (1.143) = \underline{0.857}$$

Upper Limit = b_3 - the largest negative number in the Q/S_3 column

$$= 2 - (-\infty) = \underline{\infty}$$

Therefore, $0.857 \leq b_3 \leq \infty$ (The range of resource 3 over which the shadow price \$0 per unit is valid)

Exercise:

1. $\text{Max. } Z = 50x_1 + 40x_2$

Subject to:

$$3x_1 + 5x_2 \leq 150 \text{ (Assembly time)}$$

$$x_2 \leq 20 \text{ (Portable display)}$$

$$8x_1 + 5x_2 \leq 300 \text{ (Warehouse space)}$$

$$x_1, x_2 \geq 0$$

The final simplex tableau is:

C_j		50	40	0	0	0	
SV		X_1	X_2	S_1	S_2	S_3	Q
40	X_2	0	1	8/25	0	-3/25	12
0	S_2	0	0	-8/25	1	3/25	8
50	X_1	1	0	-5/25	0	5/25	30
Z_j		50	40	14/5	0	26/5	1980

$C_j - Z_j$	0	0	-14/5	0	-26/5
-------------	---	---	-------	---	-------

Determine the shadow prices for the three constraints for the High Tech Co.

Answer:

The Z_j values for the three slack variables are 14/5, 0 and 26/5, respectively.

Thus, the shadow price for the assembly *time constraint* is 14/5(i.e.1 additional assembly time over the existing 150 is \$2.8)

The shadow price for the *portable display constraint* is 0.

The shadow price for the *Warehouse space constraint* is 26.5

Therefore, we see that obtaining more warehouse space will have the biggest positive impact on High Tech's profit.

Note:

With a greater than or equal to constraint, the value of the shadow price will be less than or equal to zero because a 1 unit increase in the value of the RHS cannot be helpful; it makes it more difficult to satisfy the constraint. As a result, for a maximization problem the optimal value of the objective function can be expected to decrease when the RHS side of a greater than or equal to constraint is increased.

2. Solve the following LPP using simplex method. A firm that manufactures both lawn mowers and snow blowers:

X_1 =the number of lawn mowers

X_2 =the number of snow blowers

Max. $Z = \$30x_1 + \$80x_2$

Subject to:

$2x_1 + 4x_2 \leq 1000$ Labor hours available

$6x_1 + 2x_2 \leq 1,200$ lb of steel available

$x_2 \leq 20$ snow blower engine available

$x_1, x_2 \geq 0$

a. What is the best product mix? What is the optimal profit?

Answer:

$x_1 = 100$, $x_2 = 200$ and profit = \$19,000

b. What are the shadow prices? When the optimal solution has been reached, which resource has the highest marginal value?

Answer:

The shadow price for 1 additional labor = \$15

The shadow price for 1 additional pound of steel = 0

The shadow price for 1 additional snow blowers engine made available =\$20

Thus, snow blower engine have the highest marginal value at the optimal solution.

c. Over what range in each of the RHS values are these shadows valid?

Answer: The shadow price for labor hours is valid from 800 hours to 1,066.66 hours. The shadow price for pounds of steel is valid from 1,000pounds up to an infinite number of pounds. The shadow price for snow blower engines ranges from 180 engines up to 250 engines

d. What are the ranges over which the objective function coefficients can vary for each of the two decision variables?

Answer: Without changing the current solution mix, the profit coefficient for the mowerys can range from \$0 to 40, while the coefficient for the blowers can range from \$60 to infinity.

1.14.2. DUALITY

Every LPP has another LPP associated with it, which is called its *dual*. The first way of starting a linear problem is called the *primal* of the problem. The second way of starting the same problem is called the *dual*.

Table

Primal-Dual Relationship

<i>Primal</i>	<i>Dual</i>
<i>Objective is minimization</i>	<i>Objective is maximization and vice versa</i>
<i>$\geq$ type constraints</i>	<i>$\leq$ type constraints</i>
<i>Nº of columns</i>	<i>Nº of rows</i>
<i>Nº of rows</i>	<i>Nº of columns</i>
<i>Nº of decision variables</i>	<i>Nº of constraints</i>
<i>Nº of constraints</i>	<i>Nº of decision variables</i>
<i>Coefficient of Object function</i>	<i>RHS value</i>
<i>RHS value</i>	<i>Coefficient of Object function</i>

Duality Advantage

1. The dual form provides an alternative form
2. The dual reduces the computational difficulties associated with some formulation
3. The dual provides an important economic interpretation concerning the value of scars resources used.

Example:

Write the duals to the following problems

a. $\text{Max. } Z = 5x_1 + 6x_2$

Subject to:

$$2x_1 + 3x_2 \leq 3000 \text{ (Labor constraint)}$$

$$5x_1 + 7x_2 \leq 1000 \text{ (Machine constraint)}$$

$$x_1 + x_2 \leq 5000 \text{ (Market constraint)}$$

$$x_1, x_2 \geq 0$$

Solution

Represent primal in the conventional table as follows

Dual variables	x_1	x_2	Constraints
u_1	2	3	≤ 3000
u_2	5	7	≤ 1000
u_3	1	1	≤ 500
MaxZ	5	6	

By referring the above table, dual for this can be stated as:

$$\text{Min } Z^* = 3000 u_1 + 1000 u_2 + 500 u_3$$

St:

$$2u_1 + 5u_2 + u_3 \geq 5$$

$$3u_1 + 7u_2 + u_3 \geq 6$$

$$u_1, u_2, u_3 \geq 0$$

Note:

1. For maximizing, all constraints must be brought to " $\leq$ " form
2. For minimizing, all constraints must be brought to " $\geq$ " form
3. If they are not, use multiplication factor -1
4. " $=$ " is an intersection of " $\geq$ " and " $\leq$ "

b. Obtain the dual problem of the following primal LPP

$$\text{Min. } Z = x_1 + 2x_2$$

Subject to:

$$2x_1 + 4x_2 \leq 160$$

$$x_1 - x_2 = 30$$

$$x_1 \geq 10$$

$$x_1, x_2 \geq 0$$

Solution

Rule:

1. For a maximization primal with all $\leq$ type constraints there exists a minimization dual problem with all $\geq$ type constraints and vice versa. Thus, the inequality sign is reversed in all the constraints except the non-negativity conditions.
2. Transform the $\leq$ type constraint to a $\geq$ type constraint by multiplying the constraint by -1.
3. Write the $=$ type constraint equivalent to two constraints of the type $\geq$ and $\leq$.

The standard primal LPP so obtained is:

$$\text{Min. } Z = x_1 + 2x_2$$

Subject to:

$$-2x_1 - 4x_2 \geq -160$$

$$x_1 - x_2 \geq 30$$

$$-x_1 + x_2 \geq -30$$

$$x_1 \geq 10$$

$$x_1, x_2 \geq 0$$

Let u_1, u_2, u_3 , and u_4 be the dual variables corresponding to the four constraints in given order, then the dual of the given primal problem can be formulated as follows:

$$\text{Max } Z^* = -160u_1 + 30u_2 - 30u_3 + 10u_4$$

St:

$$-2u_1 + u_2 - u_3 + u_4 \leq 1$$

$$-4u_1 - u_2 + u_3 \leq 2$$

$$u_1, u_2, u_3, u_4 \geq 0$$

Let $u = u_2 - u_3$, then the above dual problem reduces to the form:

$$\text{Max } Z^* = -160u_1 + 30u - 30u_3 + 10u_4$$

St:

$$-2u_1 + u + u_4 \leq 1$$

$$-4u_1 - u \leq 2$$

$$u_1, u_4 \geq 0, u \text{ unrestricted in sign}$$

Here it may be noted that the second constraint in the primal is equality. Therefore, the corresponding dual u_2 should be unrestricted in sign.

c. Obtain the dual problem of the following primal LPP

$$\text{Max. } Z = x_1 - 2x_2 + 3x_3$$

Subject to:

$$-2x_1 + x_2 + 3x_3 = 2$$

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_1, x_2, x_3 \geq 0$$

Solution:

Since the given constraints are both equality, both the corresponding dual variables, u_1 and u_2 will be unrestricted in sign. The dual of the given primal LPP is:

$$\text{Min } Z^* = 2u_1 + u_2$$

St:

$$-2u_1 + 2u_2 \geq 1$$

$$u_1 + 3u_2 \geq -1$$

$$3u_1 + 4u_2 \geq 3$$

u_1, u_2 , unrestricted in sign

Exercise

1. Write the dual of the following primal LP problems

A. $\text{Min. } Z = 3x_1 - 2x_2 + 4x_3$

St:

$$3x_1 + 5x_2 + 4x_3 \geq 7$$

$$6x_1 + x_2 + 3x_3 \geq 4$$

$$7x_1 - 2x_2 - x_3 \leq 10$$

$$4x_1 + 7x_2 - 2x_3 \geq 2$$

$$x_1, x_2, x_3 \geq 0$$

B. $\text{Max. } Z = 2x_1 + 5x_2 + 6x_3$

St:

$$5x_1 + 6x_2 - x_3 \leq 3$$

$$-2x_1 + x_2 + 4x_3 \leq 4$$

$$x_1 - 5x_2 + 3x_3 \leq 1$$

$$-3x_1 - 2x_2 + 7x_3 \leq 6$$

$$x_1, x_2, x_3 \geq 0$$

C. $\text{Min. } Z = 7x_1 + 3x_2 + 8x_3$

St:

$$8x_1 + 2x_2 + x_3 \geq 3$$

$$3x_1 + 6x_2 + 4x_3 \geq 4$$

$$4x_1 + x_2 + 5x_3 \geq 1$$

$$x_1 + 5x_2 + 2x_3 \geq 7$$

$$x_1, x_2, x_3 \geq 0$$

D. $\text{Max. } Z = 2x_1 + 3x_2 + x_3$

St:

$$4x_1 + 3x_2 + x_3 = 6$$

$$x_1 + 2x_2 + 5x_3 = 4$$

$$x_1, x_2, x_3 \geq 0$$

2. A manufacturer of toys makes two types of toys: A and B. Processing of these two toys is done on two machines X and Y. The toy A requires two hours on machine X and six hours on machine Y. Toy B requires four hours on machine X and five hours on machine Y. There are sixteen hours of time per day available on

machine **X** and thirty hours on machine **Y**. The profit obtained on both the toys is same, i.e. \$5 per toy.

- What should be the daily production of each of the two toys?
- Formulate the primal and the dual **LPP** model
- Find the optimal solution for the dual **LPP**
- Relate the dual optimal solution to the primal optimal solution
- Give the proper interpretation to the primal and the dual solutions.

3. A company produces **P**, **Q** and **R** from three raw materials **A**, **B** and **C**. One unit of **P** requires 2 units of **A** and 3 of **B**. Product **Q** requires 2 units of **B** and 5 of **C**. One unit of **R** requires 3 units of **A**, 2 units of **B** and 4 units of **C**. Company has 8 units of material **A**, 10 units of material **B** and 15 units of material **C** available to it. Profit per unit of products **P**, **Q** and **R** are \$3, \$5 and \$4 respectively.

- Formulate the problem mathematically
- How many units of each product should be produced to maximize profit?
- Write the **DUAL** problem

4. The 3rd and final simplex tableau for the LPP is:

$$\text{Max } Z = 200x_1 + 200x_2$$

St:

$$2x_1 + x_2 \leq 8$$

$$x_1 + 3x_2 \leq 9$$

$$x_1, x_2 \geq 0$$

C_j		\$200	\$200	\$0	\$0	
	SV	X_1	X_2	S_1	S_2	Q
\$200	X_1	1	0	3/5	-1/5	3
250	X_2	0	1	-1/5	2/5	2
	Z_j	\$200	\$200	\$80	\$40	\$1,000
	$C_j - Z_j$	0	0	-\$80	-\$40	

What are the solutions of the dual variables, u_1 , u_2 and u_3 ? What is the optimal dual cost?

CHAPTER THREE

TRANSPORTATION AND ASSIGNMENT PROBLEMS

1.15. Introduction:

One important application of linear programming has been in the area of the physical distribution (transportation) of resources, from one place to another, to meet a specific set of requirement. Therefore, this chapter describes two special purpose algorithms: the **transportation model** and the **assignment model**. Both **transportation** and **assignment problems** are members of a category of linear programming techniques called network flow problems.

1.16. TRANSPORTATION PROBLEMS

Transportation problem deals with the distribution of goods from several points of supplies (sources) to a number of points of demands (destinations). It is easy to express a transportation problem mathematically in terms of an LP model, which can be solved by the simplex method. But because it involves a large number of variables and constraints, it takes a long time to solve it. However, transportation algorithms, namely the Stepping Stone Method and the MODI (Modified Distribution) Method, have been developed for this purpose. **Note:** Cost of product is the sum of production cost and distribution cost. Hence, Transportation techniques are designed to minimize the distribution cost of the firm rather than production cost.

1.16.1. The characteristics of transportation problem

- ❖ A limited supply of one commodity is available at certain sources or origins.
- ❖ There is a demand for the commodity at several destinations
- ❖ The quantities of supply at each source and the demand at each destination are constant.
- ❖ The transportation costs per unit from each source to each destination are assumed to be constant.
- ❖ No shipments are allowed between sources or between destinations. All supply and demand quantities are given in whole number or integer values.
- ❖ The problem is to determine how many units shipped from each source to each destination so that all demands are satisfied at the minimum total shipping costs.

1.16.2. Uses of transportation model

- ♣ Reduce distribution or transportation cost.
- ♣ Improve competitiveness of product.
- ♣ Assist proper location of warehouses.
- ♣ Assist proper location of new factories or plants being planned.
- ♣ Close down warehouses which are found costly and uneconomical.

1.16.3. Objectives of transportation problem:

- ❖ To identify the optimal shipping routes-**minimum cost rout.**
- ❖ To identify the maximum amount that can be shipped over the optimum route.
- ❖ To determine the total transformation cost or the profit of transportation.

1.16.4. Mathematical Model of Transportation Problem

In order to develop a model of a transportation problem, it is necessary to have the following information:

1. Supply quantity (capacity) of each origin.
2. Demand quantity of each destination.
3. Unit transportation cost for each origin-destination route.

The transportation algorithm requires the assumption that all goods be homogeneous, so that any origin is capable of supplying any destination and the assumption that transportation costs are a direct linear function of the quantity shipped over any route. The solution algorithm to a transportation problem may be summarized into the following steps:

Step 1: Formulate the problem and set up in the matrix form.

The formulation of the problem is similar to the linear programming. Here the objective function is the total transportation cost and the constraints are the supply and demand available at each source and destination respectively.

Step 2: Obtain an initial basic feasible solution. There are **3 methods** to find the initial feasible solution.

1. North-West Corner Method (**NWCM**)
2. Least Cost Method (**LCM**)

3. Vogel's Approximation Method (VAM).

Note that: The initial solution obtained by any of the three methods must satisfy the following condition:

A. The solution must be feasible i.e. It must satisfy all the supply and demand constraints

B. The number of positive allocations must equal to $m+n-1$, where **m**=the number of rows (or origins or supply centers) and **n**= the number of columns (or destination centers or demand centers).

Any solution that satisfies the above conditions is called non-degenerate basic feasible solution, otherwise, degenerate solution.

Step 3: Test the initial solution for optimality: In this chapter, we shall discuss the Stepping-Stone and the Modified Distribution (MODI) method to test the optimality of the solution obtained in Step 2. If the current solution is optimal, then stop. Otherwise, determine a new improved solution.

Step 4: Updating the solution: Repeat Step 3 until an optimal solution is reached.

Example 1: A Company has three production facilities **F1, F2** and **F3** with production capacity of **50, 60** and **25** units (**in 100's**) per week of a product, respectively. These units are to be shipped to four warehouses **W1, W2, W3** and **W4** with requirement of **60, 40, 20** and **15** units (**in 100's**) per week, respectively. The transportation cost (in birr) per unit between the warehouses is in the table below.

Origin (Supply)	W₁	W₂	W₃	W₄	Factory Capacity =i
F₁	Br.3	2	7	6	5000
F₂	7	5	2	3	6000
F₃	2	5	4	5	2500
Demand)	6000	4000	2000	1500	13500

Based on the above information formulate the following transportation problem as an LPP?

Solution Let x_{ij} = The amount of commodity to be transported from source i ($i = 1, 2, 3$) to destination j ($j = 1, 2, 3, 4$). Then the objective function of the problem is minimization of the total transportation cost. Hence, the objective function is as follows.

$$\text{Min } Z = 3x_{11} + 2x_{12} + 7x_{13} + 6x_{14} + 7x_{21} + 5x_{22} + 2x_{23} + 3x_{24} + 2x_{31} + 5x_{32} + 4x_{33} + 5x_{34}$$

Subject to the constraints:

Supply constraints:

$$x_{11} + x_{12} + x_{13} + x_{14} = 5000 \quad F_1 \text{ supply constraint}$$

$$x_{21} + x_{22} + x_{23} + x_{24} = 6000 \quad F_2 \text{ supply constraint}$$

$$x_{31} + x_{32} + x_{33} + x_{34} = 2500 \quad F_3 \text{ supply constraint}$$

Demand constraints:

$$x_{11} + x_{21} + x_{31} = 6000 \quad W_1 \text{ demand constraint}$$

$$x_{12} + x_{22} + x_{32} = 4000 \quad W_2 \text{ demand constraint}$$

$$x_{13} + x_{23} + x_{33} = 2000 \quad W_3 \text{ demand constraint}$$

$$x_{14} + x_{24} + x_{34} = 1500 \quad W_4 \text{ demand constraint}$$

$$x_{ij} \geq 0 \text{ for all } i \& j$$

In the above LPP, there are $m \times n = 3 \times 4 = 12$ decision variables & $m + n = 3 + 4 = 7$ constraints. Thus, if this problem is solved by the simplex method, then it may take considerable computational time.

3.2.4.1. METHODS FOR FINDING INITIAL SOLUTION

There are several methods available to obtain an initial basic feasible solution. Here we shall discuss only three different methods:

1. **North-west corner method (NWCM):** It is a simple and efficient method to obtain an initial feasible solution. This method does not take into account the cost of transportation on any route of transportation. The NWCM gets its name because the starting point for the allocation process is the Upper Left-hand (Northwest) corner of the transportation table. Therefore, allocate to the Northwest corner as many units as possible.

Rules to use this method:

1. Start with the cell **upper left** (north-west) corner of the transportation matrix, and allocate as many units as possible equal to minimum of the rim values for the first row and first column, i.e. $\min(X_{11})$. Adjust the row & column quantities to reflect the allocation.
2. Subtract from the row supply and from the column demand the amount you allocated.
3. If the column demand is now zero, move to the cell **next** to the **right**, if the row supply is zero, move **down** to the cell in the **next** row.
If both are zero, move **first** to the **next** cell on the **right** then down one cell.
4. Once a cell is identified as per step (3), it becomes a **northwest cell**. Allocate to it an amount as per step (1)
5. Repeat the above steps (1) - (4) until all the remaining supply and demand is gone.

Example: Formulate the initial feasible solution and find the optimum transportation cost for the following transportation problem by using north-west corner method.

Source (Plant)	Destination (Store house)				
	Store 1	Store 2	Store 3	Store 4	Supply
Plant 1	19	30	50	10	7
Plant 2	70	30	40	60	9
Plant 3	40	8	70	20	18
Demand	5	8	7	14	34

Solution: table for initial feasible solution

To From	Store 1	Store 2	Store 3	Store 4	Supply
Plant 1	19 { 5 }	30 { 2 }	50	10	7
Plant 2	70	30 { 6 }	40 { 3 }	60	9
Plant 3	40	8	70 { 4 }	20 { 14 }	18
Demand	5	8	7	14	34

Check that the solution is feasible or not:

$\Rightarrow m + n - 1$; $m=3$ and $n=4 \Rightarrow 3+4-1=6$ cells occupied (Feasible solution).

The total transportation cost of the initial feasible solution derived by the NWCM is:

<u>Route</u>		<u>Unit</u>		<u>Per unit</u>		<u>Total</u>
<u>From</u>	<u>To</u>	<u>Shipped</u>	<i>X</i>	<u>cost (\$)</u>	=	<u>Cost (\$)</u>
Plant 1	Store 1	5		19		95
plant 1	Store 2	2		30		60
Plant 2	Store 3	6		30		180
Plant 2	Store 4	3		40		120
Plant 3	Store 4	4		70		280
Plant 3	Store 4	14		20		<u>280</u>
Total Cost=						<u>\$ 1015</u>

Note: NWCM **does not** consider the cost factor for allocation.

2. THE LEAST- COST METHOD (LARGEST- PROFIT) METHOD)

Since the objective is to minimize the total transportation cost, we must try to transport as much as possible through those routes (cells) where the unit transportation cost is lowest. This method takes into account the minimum unit cost of transportation for obtaining initial solution and can be summarized as follows:

Step 1: Select the cell with the lowest unit cost in the entire transportation table and allocate as much as possible to this cell and eliminate (line out) that row or column in which either supply or demand is exhausted. If both a row and a column are satisfied simultaneously, only one may be crossed out. In case the smallest unit cost cell is not unique, then select the cell where maximum allocation can be made.

Step 2: After adjusting the supply and demand for all uncrossed-out rows and columns repeat the procedure with the next lowest unit cost among the remaining rows and columns of the transportation table and allocate as much as possible to this cell and eliminate (line out) that row and column in which either supply or demand is exhausted.

Step 3: Repeat the procedure until the entire available supply at various sources and demand at various destinations is satisfied. The solution so obtained need not be non-degenerate.

Example 2: Suppose that a firm has three factories / *sources of supply* /& four warehouses/point of demand/. The firm's production capacity at the three factories, the demand for the four destination centers located at various regions & the cost of shipping each unit from the factories to the warehouses through each route is given as follows:

	W_1	W_2	W_3	W_4	Factory Capacity
F ₁	3	2	7	6	5000
F ₂	7	5	2	3	6000
F ₃	2	5	4	5	2500
Demand	6000	4000	2000	1500	13500

A. Formulate the initial feasible solution and find the optimum transportation cost for the following transportation problem by using least cost method (LCM).

	W_1	W_2	W_3	W_4	Factory Capacity
F ₁	3 1000	2 4000	7	6	5000
F ₂	7 2500	5	2 2000	3 1500	6000
F ₃	2 2500	5	4	5	2500
Demand	6000	4000	2000	1500	13500

Then check feasibility i.e ($m + n - 1 = \text{used cell (occupied cell)}$). Hence it is feasible.

<u>From</u>	<u>Routes To</u>	<u>Units Shipped</u>	X	<u>Unit Cost</u>	<u>Total =Cost</u>
F ₁	W ₁	1000		3	\$ 3000
F ₁	W ₂	4000		2	8000
F ₂	W ₁	2500		7	17500
F ₂	W ₃	2000		2	4000
F ₂	W ₄	1500		3	45000
F ₃	W ₁	2500		2	<u>5000</u>
				TC	<u>=\$42,000</u>

3. Vogel's Approximation Method (VAM)

Vogel's approximation (penalty or regret) method is a heuristic method and is preferred to the other two methods described above. In this method, each allocation is made on the basis of the opportunity (or penalty or extra) cost that would have been incurred if allocations in certain cells with minimum unit transportation cost were missed. In this method allocations

are made so that the penalty cost is minimized. The advantage of this method is that it gives an initial solution which is nearer to an optimal solution or is the optimal solution itself. The steps in VAM are as follows:

Step 1: Calculate penalties for each row (column) by taking the difference between the smallest and next smallest unit transportation cost in the same row (column). This difference indicates the penalty or extra cost which has to be paid if one fails to allocate to the cell with the minimum unit transportation cost.

Step 2: Select the row or column with the largest penalty and allocate as much as possible in the cell having the least cost in selected row or column satisfying the rim conditions. If there is a tie in the values of penalties, it can be broken by selecting the cell where maximum allocation can be made.

Step 3: Adjust the supply and demand and cross out the satisfied row or column. If a row and column are satisfied simultaneously, only one of them is crossed out and the remaining row (column) is assigned a zero supply (demand). Any row or column with zero supply or demand should not be used in computing future penalties.

Step 4: Repeat steps 1 to 3 until the entire available supply at various sources and demand at various destinations are satisfied.

Example 3: Formulate the initial feasible solution and find the optimum transportation cost for the following transportation problem by using VAM.

	A	B	C	D	Supply
AMU, CBE, Mgmt Dep't					

Factory	F ₁	2 5	2	0 20	4	25	Row difference or Row penalty or opportunity cost
	F ₂	5 15	9 5	8	3 5	25	
	F ₃	6	4 10	3	2	10	
	Demand	20	15	20	5	60	
							2 2 - - - -
							2 2 2 2 5 0
							1 2 2 - -

Column difference	3	2	3	1
or Column penalty	3	2	-	1
	1	5	-	1
	0	0	-	0
	0	0	-	-

m= 3, n=4 ==> 3+4-1 =6 Occupied cells (feasible). The transportation cost associated with this solution is: **Total cost= 5x2 + 20x0+15x5x9 =+95x3+10x4= \$185**

3.2.4.2.METHODS FOR TESTING OPTIMAL SOLUTION

Once an initial solution is obtained, the next step is to check its optimality. An optimal solution is one where there is no other set of transportation routes (allocations) that will further reduce the total transportation cost. Thus, we have to evaluate each unoccupied cell (representing unused route) in the transportation table in term of an opportunity of reducing total transportation cost.

An unoccupied cell with the largest negative opportunity cost is selected to include in the new set of transportation routes (allocations). This is also known as an *incoming variable*. The outgoing variable in the current solution is the occupied cell (*basic variable*) in the unique closed path (loop) whose allocation will become zero first as more units are allocated to the unoccupied cell with largest negative opportunity cost. Such an exchange reduces total transportation cost. **The process is continued until there is no negative opportunity cost.** That is, the current solution cannot be improved further. This is the optimal solution. Optimum solution to a transportation problem can be obtained by following two methods (i.e Stepping stone Method and Modified Distribution method (MODI)).

1. Stepping-stone method: is an iterative technique for moving from an initial feasible solution to an optimal solution in transportation problems. For the **stopping- stone method** to be

applied to a transportation problem, one rule about the n^0 of shipping routes being used must be observed. The rule is: **“The N^0 of occupied routes (or squares) must always be equal to one less than the sum of the n^0 of rows plus the n^0 of columns.”** i.e. **Occupied shipping routes (squares) = N^0 of rows + N^0 of columns - 1 = Non degenerate solution.**

❖ **Rules for drawing each closed loop:**

1. Select an unused square (cell) to be evaluates.
2. Beginning at this cell, trace a closed loop going **clockwise** draw an arrow to an **occupied cell** in the same row (or column).
3. Move **vertically** or **horizontally** (but **never diagonally**) to another **occupied cell** “stepping –over” unoccupied or occupied cells (if necessary) without changing them. Follow the same procedure to other occupied cells until returning to the original empty cell.
4. Begin with a plus (+) sign at the unused cell, place alternative (-) signs and plus signs on each corner square of the closed path just traced. i.e. At each turn of the loop (the **loop may cross over itself at times**), plus and minus signs are alternately placed in the cells, starting with a + sign in an empty cell.
5. **There must be exactly one cell** with a + sign and **exactly one cell** with a - sign in any row or column in which the loop turns.
6. **An even n^0 of at least four cells** must participate in a loop and the **occupied cells can be visited once and only once**.
7. Repeat steps 1 to 4 until an improvement index has been calculated for all unused squares (cells). If all indices computed are greater than or equal to zero, an optimal solution has been reached. If not, it is possible to improve the current solution and decrease total shipping costs. **Note:** In a non-degenerate problem, there is only one possible way of drawing the loop for each empty cell.

➤ **How to find the value of a cell evaluator?**

The value of a cell evaluator is the sum of the per unit shipping costs in the gaining cells less the sum of the per unit shipping costs in the losing cells of the closed loop. This evaluation process must be extended to **all** unoccupied cells.

1. **The test of optimality for a minimization (cost) problem:** If one or more of the cell evaluators is **negative**, the existing solution is **not optimal**. i.e: For minimization (cost) problems, all the cell evaluators must be **positive** for optimality.
 - **Analysis of test:** Check **all** the empty cells and select for improvement the one with the **largest** improvement potential. If the solution is **not optimal**, the next step in the transportation method is to find a better solution. The operations in this step are:

A. Identify the “**incoming**” cell (the empty cell to be occupied. In a minimization case, the incoming cell is located by identifying the **most negative cell** evaluator.

B. Design an improved solution by shifting units form cell to cell.

2. The test of optimality for a maximization (profit) case: Is the reversed of minimization case.

If one or more of the cell evaluators is **positive**, the existing solution is **not optimal**.

i.e: for a maximization (profit) case, all the cell evaluators must be **negative** for optimality. If any cell evaluation is **positive**, the **solution is not optimal**.

Note:

- ❖ A cell evaluator of 0 indicates the existence of another solution just as good as the current solution. Thus, in the final solution, if cell evaluators of 0 exist, this indicates the existence of multiple optimal solutions.
- ❖ If two or more cells have the same value, then either may be selected.
- ❖ If two or more of the "losing" cells contain the same no of units, both will become empty simultaneously and a “degenerate” solution will result.
- ❖ For the minimization case; when one or more cell evaluators are negatives, the cell with the largest negative should be brought into solution because that route has the largest potential for improvement per unit.
- ❖ The loop starts and ends at the selected unoccupied cell. Every corner element of the loop must be an occupied cell.

Example 1: Use **NWCM** to find initial feasible solution and test the solution for optimality.

	Project A	Project B	Project C	SS
F₁	4	2	8	100
F₂	5	1	9	200
F₃	7	6	3	200
DD	50	150	300	500

Initial feasible solution by using NWCM is as follow as:

	Project A	Project B	Project C	SS
F₁	4 50	2 50	8	100
F₂	5	1 100	9 100	200

F_3	7	6	3	200
			200	
DD	50	150	300	500

$m=3, n=3 \Rightarrow 3+3-1=5$ (Non-degenerate solution)

Total transportation cost = $\$[50 \times 4 + 50 \times 2 + 100 \times 1 + 100 \times 9 + 200 \times 3] = \underline{\$1900}$

Evaluation path for the unoccupied cells

Table: Test of optimality

Unoccupied cells	Cell evaluators
(F_2, A)	$+5 - 4 + 2 - 1 = +2$
(F_1, C)	$+8 - 9 + 1 - 2 = -2$
(F_3, A)	$+7 - 4 + 2 - 1 + 9 - 3 = +10$
(F_3, B)	$+6 - 1 + 9 - 3 = +11$

The negative value for cell (F_1, C) indicates an improved solution is possible. For each unit we can shift into that cell, the total cost will decrease by \$2. The next question is how many units can be reallocated into that cell while retaining the balance of supply and demand for that table?

The Stepping- stone path for cell (F_1, C) is:

The + Signs in the path indicate units to be added, the - signs indicate units to be subtracted. The limit on subtraction is the **smallest quantity in a negative position** along the cell path. There are two quantities in negative positions, 50 and 100.

Because 50 is the smaller quantity, that amount will be shifted in the following manner: Subtract 50 units from each cell on the path with a minus sign and add 50 units to the quantity of each cell with a plus sign in it. With each iteration (new solution), it is necessary to evaluate the empty cells to see if further improvements is possible.

The distribution plan after reallocation of 50 units is:

	A	B	C	SS
F₁	4 50	2	8 50	100
F₂	5	1 150	9 50	200
F₃	7	6	3	200
DD	50	150	300	500

Table: Test of optimality

Unoccupied cells		Cell evaluators
(F₁,B)		+2 -8+9-1 =+2

	Project A	Project B	Project C	SS
F₁	4 50	2 50	8 50	100
F₂	5	1 100 +	9 100-	200
F₃	7	6	3 200	200
DD	50	150	300	500

(F₂,A)	+5 -4+8-9 =0
(F₃,A)	+7-4+8-3 =+8
(F₃,B)	+6-1+9-3= +11

Because none of these n^0 is negative, this is an optimal solution. Therefore, the total cost for the distribution plan is: The total transportation cost = \$ (50x4 +50x8 150x1+50x9 +200x3) = **\$1,800**

Self-test exercise: Consider the following transportation problem and answer questions?

Warehouses		Store 1	Store 2	Store 3	Supply
	A	12	20	15	50
	B	9	11	4	15
	C	20	14	8	55
	Demand	25	50	45	120

- A. Develop an initial feasible solution using the NWCM. And compute the total cost for this solution.
- B. Evaluate the solution using the stepping-stone method. Is the solution optimal? Explain.
- C. What is the total cost for the optimal solution?

2. Modified Distribution Method (MODI Method)

The steps to evaluate unoccupied cells are as follows:

Step 1 for an initial feasible solution with $m+n-1$ occupied cells, calculate u_i and v_j for rows and columns. The initial solution can be obtained by any of the three methods discussed earlier.

To start with, any one of u_i 's and v_j 's is assigned the value zero. It is better to assign zero for a particular u_i or v_j where there are maximum number of allocations in a row or column respectively, as it will reduce arithmetic work considerably. Then complete the calculation of u_i 's and v_j 's for other rows and columns by using the relation.

$c_{ij}=u_i+v_j$ for all occupied cells (i, j).

Step 2 for unoccupied cells; calculate opportunity cost by using the relationship

$d_{ij}=c_{ij}-(u_i+v_j)$ for all i and j.

Step 3 examine sign of each d_{ij}

- I. If $d_{ij}>0$, then current basic feasible solution is optimal
- II. If $d_{ij}=0$, then current basic feasible solution will remain unaffected but an alternative solution exists
- III. If one or more $d_{ij} < 0$, then an improved solution can be obtained by entering unoccupied cell (i,j) in the basis. An unoccupied cell having the largest negative value of d_{ij} is chosen for entering in to the solution mix(new transportation schedule)

Steps 4 construct a closed path (or loop) for the unoccupied cell with largest negative opportunity cost. Start the closed path with the selected unoccupied cell and mark a plus sign(+) in this cell, trace a path along the rows(or columns) to an unoccupied cell, mark the corner with minus sign(-) and continue down the column (or row) to an occupied cell and mark the corner with plus sign(+) and minus(-) alternatively. Close the path back to the selected unoccupied cell.

Step 5 select the smallest quantity amongst the cells marked with minus sign on the corners of closed loop. Allocate this value to the selected unoccupied cell and add it to other occupied cells marked with plus signs and subtract it from the occupied cells marked with minus signs.

Step 6 obtain a new improved solution by allocating units to the unoccupied cell according to step 5 and calculate the new total transportation cost.

Step 7 tests the revised solution further for optimality. The procedure terminates when all $d_{ij} \geq 0$, for unoccupied cells.

Remarks: the loop starts and ends at the selected unoccupied cell. It consists of successive horizontal and vertical (connected) lines whose end points must be occupied cells, except for a end point associated with entering unoccupied cell. This means that every corner element of the loop must be an occupied cell. It is immaterial whether the loop is traced in a clockwise or anti clock wise direction. However, for a given solution only one loop can be constructed for each unoccupied cell.

Example1: Obtain an optimal solution to the transportation problem by **MODI method** given below:

Farm	Project N ^o 1	Project N ^o 2	Project N ^o 3	Supply
Farm 1	4	2	8	100
Farm 2	5	1	9	200
Farm 3	7	6	3	200
Demand	50	150	300	500

Solution

Note: Both the **MODI** and the **stepping - stone method** will yields the same values.

Remark:

Conventionally, we **begin** by assigning a value of **zero** as the index for row 1 ($U_1=0$). Once row index has been established, it will enable us to compute column index numbers for all occupied cells in that row. Similarly, once a column index number has been determined, index numbers for all rows corresponding to occupied cells in that column can be determined.

Consider the initial feasible solution of the given example by NWCM as shown below:

	Proj ect	Proje ct	Proje ct	SS	U _i
Farm 1	4 50	2 50	8	100	U ₁ =0
Farm 2	5	1	9	200	U ₂ =-1

		100	100		
Farm 3	7	6	3	200	$U_3 = -7$
DD	50	150	300	500	
V_i	$V_1 = 4$	$V_2 = 2$	$V_3 = 1$		

Then, determine C_{ij} , use the occupied cells. For instance, $C_{11}=4$, $C_{12}=2$, $C_{22}=1$, $C_{23}=9$, and $C_{33}=3$

$$C_{ij} = U_i + V_j$$

$$\Rightarrow C_{11} = U_1 + V_1 \Rightarrow 4 = 0 + V_1 \Rightarrow \underline{V_1 = 4}, U_1 = 0 \text{ by convention}$$

$$\Rightarrow C_{12} = U_1 + V_2 \Rightarrow 2 = 0 + V_2 \Rightarrow \underline{V_2 = 2}$$

$$\Rightarrow C_{22} = U_2 + V_2 \Rightarrow 1 = U_2 + 0 \Rightarrow \underline{U_2 = -1}$$

$$\Rightarrow C_{23} = U_2 + V_3 \Rightarrow 9 = -1 + V_3 \Rightarrow \underline{V_3 = 10}$$

$$\Rightarrow C_{33} = U_3 + V_3 \Rightarrow 3 = U_3 + 10 \Rightarrow \underline{U_3 = -7}$$

Note: $C_{ij} \neq U_i + V_j$ (For unoccupied cells). For instance, from the above information, $C_{32} \neq U_3 + V_2 \Rightarrow 6 \neq -7 + 2$.

Table: Test of optimality

Unoccupied cells	Cell evaluators $D_{ij} = C_{ij} - (U_i + V_j)$
(1,3)	$C_{13} - (U_1 + V_3) = 8 - (0 + 10) = -2$
(2,1)	$C_{21} - (U_2 + V_1) = 5 - (-1 + 4) = +2$
(3,1)	$C_{31} - (U_3 + V_1) = 7 - (-7 + 4) = 10$
(3,2)	$C_{32} - (U_3 + V_2) = 6 - (-7 + 2) = +11$

In this case, we found that cell (1, 3) had an evaluation of -2, which represented an improvement potential of and \$ **2 per unit**. Hence, an improved solution is possible. The stepping-stone path for call (1, 3) is:

Farm	Proj N ^o 1	Proj N ^o 2	Proj N ^o 3	Supply
Farm 1	4 50	2 -50	8 +	100

Farm 2	5	1 100	9 100	200
Farm 3	7	6	3 200	200
Demand	50	150	300	500

The distribution plan after reallocation of 50 units is:

Farm	Project	Project	Project N° 3	Supply	U_i
Farm 1	4 50	2	8 50	100	U₁=8
Farm 2	5	1 150	9 50	200	U₂=9
Farm 3	7	6	3 200	200	U₃=3
Demand	50	150	300	500	
V_j	V₁=-4	V₂=-8	V₃=0		

Table: Test of optimality

Unoccupied cells	Cell evaluators
	$D_{ij} = C_{ij} - (U_i + V_j)$
(1,2)	$C_{12} - (U_1 + V_2) = 2 - (-8 + 8) = 2$
(2,1)	$C_{21} - (U_2 + V_1) = 5 - (-4 + 9) = 0$
(3,1)	$C_{31} - (U_3 + V_1) = 7 - (-4 + 3) = 8$
(3,2)	$C_{32} - (U_3 + V_2) = 6 - (-8 + 3) = +11$

Because none of the cell evaluators is negative, this is an optimal solution. Thus, the total cost for the distribution plan = **\$1650**

3.2.5. Special case of transportation problem

1. Unbalanced transportation problem: This is a condition that occurs when the total demand is different from total supply. Therefore, to solve this problem we should convert the unbalanced TP into a balanced TP by using dummy destination/dummy source.

- ♣ If total Supply > Total demand, then create a fictitious or artificial destination called **dummy destination**. i.e: total Supply > Total demand====> **Add dummy demand**
- ♣ Excess demand (Supply < demand) -----**Add a dummy supply**

Note: the cost of “**shipments**” to the dummy is usually set at zero ==> No real cost.

Example: Develop an initial feasible solution using NWCM

	R	S	T	Supply
A	1	2	3	100
B	4	1	5	110
Demand	80	120	60	210 260

Hence, this is unbalanced transportation problem because, **SS (210) < DD (260)**.

Therefore, to convert this unbalanced TP to a balanced TP use the dummy supply (dummy row).

i.e

	R	S	T	Supply
A	1 80	2 20	3	100
B	4	1 100	5 10	110
Dummy	0	0	0 50	50
Demand	80	120	60	260

Then $TC = X_{11}=80, X_{12}=20, X_{22}=100, X_{23}=10, X_{33}=50$ Total cost = \$270

2. Degeneracy and Its Resolution: A condition that occurs when the N^0 of occupied cells in any solutions less than the N^0 of rows plus the N^0 of columns minus 1 in a transportation table.

i.e. $N^0 \text{ of occupied cells} < m + n - 1$ **Degeneracy**

If the N^0 of occupied cells = $m + n - 1$, then the solution is non-

The degeneracy in the transportation problems may occur at two stages:

1. **when obtaining an initial solution**
2. **During improvement (or at any stage while moving towards optimal solution.**

To resolve degeneracy, we processed by allocating a **very small quantity close to zero** to one or more **unoccupied cell** so as to get **$m+n-1$** number of occupied cells. This amount is denoted by a Greek letter ϵ (epsilon) or Δ (delta). This quantity **would not affect the total cost** as well as **supply** and **demand** values.

$\epsilon = \text{Almost zero}$

- ♣ In a minimization transportation problem, allocate the smallest transportation cost to ϵ .
- ♣ In a maximization transportation problem, it should be allocated to a cell that has a high pay off value.
- ♣ Insert ϵ when it is able to create a **closed loop** for each occupied cell.

The purpose of epsilon/delta is to enable evaluation of the remaining empty cells. The choice of location for the epsilon/delta can be somewhat tricky: some empty cells may be unsuitable if they do not enable evaluations of remaining empty cells. Not all choices would be acceptable.

Actually, the N^o of epsilon/deltas needed will equal the difference between the N^o of completed cells and **$m+n-1$** . However; you will only be exposed to the most common case in which one more completed cell is needed.

The epsilon/delta cannot be placed in a cell which later turns out to be in a **negative position** of a cell path involved in reallocation because epsilon/delta will be the “**smallest quantity a negative position**” and shifting that minute quantity around the cell path will leave the solution virtually unchanged. Consequently, a certain amount of trial and error may be necessary before a satisfactory location can be identified for epsilon/delta.

Example: Consider the following transportation table and test optimality by using MODM.

	1	2	Supply	U_i
1	3 50	3 ϵ	50	$U_1=0$
2	4	6 30	30	$U_2=3$
Demand	50	30	80	
V_j	$V_1=3$	$V_2=3$		

Solution: The above problem is degeneracy (i.e $m + n - 1 = \text{used cell}$ hence, we use epsilon).

$$C_{ij} = U_i + V_j$$

$$\Rightarrow C_{11} = U_1 + V_1 \Rightarrow 3 = 0 + V_1 \Rightarrow \underline{V_1 = 3}, U_1 = 0 \text{ by convention}$$

$$\Rightarrow C_{12} = U_1 + V_2 \Rightarrow 3 = 0 + V_2 \Rightarrow \underline{V_2 = 3}$$

$$\Rightarrow C_{22} = U_2 + V_2 \Rightarrow 6 = U_2 + 3 \Rightarrow \underline{U_2 = 3}$$

$$\Rightarrow C_{33} = U_3 + V_3 \Rightarrow 3 = U_3 + 8 \Rightarrow \underline{U_3 = -5}$$

Note: $m=2$ and $n=2 \Rightarrow 2+2-1=3 \Rightarrow$ Occupied cells $= 2 < 3$ (Degeneracy)

Table: Test of optimality

<i>Unoccupied cell</i>	<i>Cell evaluator</i>
	$K_{ij} = C_{ij} - (U_i + V_j)$
(2,1)	$C_{21} - (U_2 + V_1) = 4 - (3 + 3) = -2$

	1	2	Supply	U_i
1	³ 20	³ 30	50	$U_1 = 0$
2	⁴ 30	⁶	30	$U_2 = 1$
Demand	50	30	80	
V_j	$V_1 = 3$	$V_2 = 3$		

The optimal solution is

$$C_{ij} = U_i + V_j$$

$$\Rightarrow C_{11} = U_1 + V_1 \Rightarrow 3 = 0 + V_1 \Rightarrow \underline{V_1 = 3}, U_1 = 0 \text{ by convention}$$

$$\Rightarrow C_{21} = U_2 + V_1 \Rightarrow 4 = U_2 + 3 \Rightarrow \underline{U_2 = 1}$$

$$\Rightarrow C_{12} = U_1 + V_2 \Rightarrow 3 = 0 + V_2 \Rightarrow \underline{V_2 = 3}$$

Table: Test of optimality

<i>Unoccupied cells</i>	<i>Cell evaluators</i>
	$K_{ij} = C_{ij} - (U_i + V_j)$
(2,2)	$C_{22} - (U_2 + V_2) = 6 - (1 + 3) = +2$

The total cost = $\$(20 \times 3 + 30 \times 3 + 30 \times 4) = \underline{\underline{\$270}}$

2. Alternative Optimal solutions: The existence of alternative optimal solution can be determined by an inspection of the opportunity costs, D_{ij} for the unoccupied cells. If an unoccupied cell in an optimal solution has opportunity cost of **zero**, then an alternative optimal solution can be formed with another set of allocations without increasing the total transportation cost.

3. MAXIMIZATION TRANSPORTATION PROBLEM

In general, transportation model is used for cost minimization problems. However, it is also used

to solve problems in which the objective is to **maximize total value or benefit**. That is, instead of

unit cost c_{ij} , the unit profit or payoff p_{ij} associated with each route, (i, j) is given. Then, the

objective function in terms of total profit or payoff is stated as follows:

$$\text{Maximize } Z = \sum p_{ij} X_{ij}$$

The algorithm for solving this problem is same as that for the minimization problem.

However,

since we are given profits instead of costs, therefore, few adjustments in Vogel's approximation

method (**VAM**) for finding initial solution and in the **MODI** optimality test are required.

For finding initial solution by **VAM**, the penalties are computed as difference between the

largest and next largest payoff in each row or column. In this case, row and column

differences

represent payoffs. Allocations are made in those cells where the payoff is largest corresponding

to the highest row or column difference. Since it is a maximization problem, the criterion of optimality is the converse of the rule for minimization. The rule is: ***A solution is optimal if all opportunity costs d_{ij} for the unoccupied cells are zero or negative.***

3.3. Assignment problems

The Assignment Problem (AP) refers to the class of **LPPs** that involves determining the most efficient **assignment of people to projects, sales people to territories, contracts to bidders, jobs to machines**, and so on. The objective is to assign a number of resources to an equal number of activities so as to minimize total costs or total time or maximize total profit of allocation.

The problem of assignment arises because available resources such as men, machines, etc have varying degrees of efficiency for performing different activities such as job. Therefore, cost, profit or time of performing the different activities is different.

3.3.1. Assumptions:

The AP is a special case of TP under the condition that the number of origins is equal to the number of destinations. Viz. **$m=n$** . Hence assignment is made on the basis of 1:1.

Following are the assumptions:

- ◆ *Number of jobs is equal to the number of machines or persons*
- ◆ *Each man or machine is loaded with one and only one job.*
- ◆ *Each man or machine is independently capable of handling any of the job being presented.*
- ◆ *Loading criteria must be clearly specified such as “**minimizing operating time**” or “**maximizing profit**”, or “**minimizing production cost**” or “**minimizing throughout (production cycle) time**” etc.*

Remark:

- *The AP is considered as a special TP in which the supply at each source and the demand at each destination are always one unit.*

- Since the supply and demand are always equal to one unit in each row and column, there is no need to write them in the assignment table.

Example: Service costs of different team assignment (\$ in thousands)

	Z_1	Z_2	Z_3	Supply
S_1	20	15	31	1
S_2	17	16	33	1
S_3	18	19	27	1
Demand	1	1	1	

The above problem can be presented as a LPP as follows:

$$\text{Min} Z = 20x_{11} + 15x_{12} + 31x_{13} + 17x_{21} + 16x_{22} + 33x_{23} + 18x_{31} + 19x_{32} + 27x_{33}$$

Subject to the constraints

a. **Supply constraints:**

$$x_{11} + x_{12} + x_{13} = 1 \quad S_1 \text{ constraint}$$

$$x_{21} + x_{22} + x_{23} = 1 \quad S_2 \text{ constraint}$$

$$x_{31} + x_{32} + x_{33} = 1 \quad S_3 \text{ supply constraint}$$

b. **Demand constraints**

$$x_{11} + x_{21} + x_{31} = 1 \quad Z_1 \text{ constraint}$$

$$x_{12} + x_{22} + x_{32} = 1 \quad Z_2 \text{ constraint}$$

$$x_{13} + x_{23} + x_{33} = 1 \quad Z_3 \text{ constraint}$$

$$x_{ij} \text{ either } 0 \text{ or } 1 \text{ for all } i, j$$

Since all x_{ij} can be either 0 or 1, there will be one assignment in each supply constraint and one assignment in each demand constraint.

As in the transportation problem, assignment problems can be balanced or not. In a balanced case, the number of objects to be assigned equals the number of objects to which they are assigned. Unbalanced problem can be balanced by adding a dummy (dummies) with zero cost coefficients.

3.3.2. **Methods of solving assignment problems**

An assignment problem can be solved by the following methods:

1. **Enumeration method**
2. **Simplex method**
3. **Transportation method**
4. **Hungarian method**

3.3.2.1. **Hungarian Method/Flood's Technique/**

The Hungarian Method (developed by Hungarian Mathematician **D. Konig**) of assignment provides us with efficient method of finding the optimal solution without having to make a direct comparison of every solution. It works on **the principle of reducing the given cost matrix** (the **principle of matrix reduction**) to a **matrix of opportunity costs**, which means that by subtracting and adding appropriate numbers in the cost table or matrix, we can reduce the problem to a **matrix of opportunity costs**.

Opportunity costs show the relative penalties associated with assigning resource to an activity as opposed to making the best or least-cost assignment. If we can reduce the cost matrix to the extent of having at least one zero in each row and column, then it will be possible to make optimal assignments.

Steps in solving assignment problems:

Step.1. Develop the cost table from the given problem

If the number of rows does not equal the number of columns and vice versa, then a dummy row or dummy column must be added. The assignment costs for dummy cells are always zero.

Step 2. Find the opportunity cost table

i.e. The transformation of the cost matrix to what is termed as a total-opportunity cost matrix.

It involves two operations:

a. Perform row operation

I.e. Locate the smallest element in each row of the given cost table and then subtract that the given cost table from each element of the respective row.

b. Perform column operation

I.e. In the reduced matrix obtained from 2(a), locate the smallest element in each column and then subtract that from each element of the respective column.

Notice that: each row and column now have at least one zero value.

Step 3 make assignment in the opportunity cost matrix

- a) Examine rows successively until a row with exactly one unmarked zero is obtained. Make an assignment to this single zero by making a square around it
- b) For each zero value that becomes assigned, eliminate (strike off) all other zeros in the same row and/or column
- c) Repeat steps 3(a) and 3(b) for each column also with exactly single zero value cell that has not been assigned
- d) If a row and/or column has two or more unmarked zeros and one cannot be chosen by inspection, then choose the assigned zero cell arbitrarily
- e) Continue this process until all zeros in rows/columns are either enclosed(assigned) or strike off (x)

Step 4 optimality criteria

If the number of assigned cells is equal to the number of rows/columns, then it is optimal solution. The total cost associated with this solution is obtained by adding original cost figures in the occupied cells.

Steps 5 revise the opportunity cost table

Draw a set of horizontal and vertical lines to cover all the zeros in the revised cost table obtained from step 3, by using the following procedure

- a) For each row in which no assignment was made, mark a tick
- b) Examine the marked rows, if any zero cell occurs in those rows, mark a tick to the respective columns that contain those zeros
- c) Examine marked columns. If any assigned zero occurs in those columns, tick the respective rows that contain those assigned zeros
- d) Repeat this process until no more rows or columns can be marked
- e) Draw a straight line through each marked column and each unmarked row

If the number of lines drawn (or total assignment) is equal to the number of rows (or columns), the current solution is the optimal solution, otherwise go to step 6

Step 6 develop the new revised opportunity cost table

- a) From among the cells are not covered by any line, choose the smallest element. Call this value k
- b) Subtract k from every element in the cell not covered by a line
- c) Add k to every element in the cell covered by the two lines, i.e. intersection of two lines

d) Elements in cells not covered by one line remain unchanged

Step 7 repeat steps

Repeat steps 3 to 6 until optimal is obtained

Example:

1. A computer center has three programmers. The center wants three application programs to be developed. The head of the computer center, after studying carefully the programs to be developed, estimate the computer time in minutes required by the experts for the application programs as follows; then, assign the programmers to the programs in such a way that the total computer time is minimum?

Programmers		Programs (Estimated time in		
		A	B	C
	1	120	100	80
	2	80	90	110
	3	110	140	120

Solution:

Steps 1 and 2:

A. Perform row reduction

The minimum time element in row 1, 2, and 3 is 80, 80 and 110 respectively. Subtract those elements from all elements in their respective row. The reduced time matrix is:

Row value		A	B	C
-80	1	40	20	0
-80	2	0	10	30
-110	3	0	30	10

B. Column reduction

Since column B has no one '0', perform also column reduction. The minimum time element in columns A, B and C is 0, 10 and 0 respectively. Subtract these elements from all elements in their respective column to get the reduced time matrix.

Table: After column reduction

	A	B	C
1	40	10	0
2	0	0	30

3	0	20	10
---	---	----	----

Step 3: make assignment in the opportunity cost matrix

A. *In 1 assignment, start with row/column having one zero and cancel the alternative zeros(x)*

Table: Test of optimal assignment

	A	B	C
1	40	10	0
2	0	0	30
3	0	20	10

B. Count the number of occupied cells. If the number occupied cells are equal to the number of rows/columns, the optimal solution is obtained. The pattern of assignment among programmers and programs with their respective time (in minute) is given below:

<u>Programmer</u>	<u>Program</u>	<u>Time (in minutes)</u>
1	C	80
2	B	90
3	A	110
		Total time=280 minutes

3. A department has five employees with five jobs to be performed. The time (in hours) each man will take to perform each job is given in the effectiveness matrix.

		<u>Employees</u>				
		<u>I</u>	<u>II</u>	<u>III</u>	<u>IV</u>	<u>V</u>
<u>Jobs</u>	A	10	5	13	15	16
	B	3	9	18	13	6
	C	10	7	2	2	2
	D	7	11	9	7	12
	E	7	9	10	4	12

How should the jobs be allocated, one per employees, so as to minimize the total man-hours?

Solution:

	<u>I</u>	<u>II</u>	<u>III</u>	<u>IV</u>	<u>V</u>
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Table: After row reduction

A	5	0	8	10	11
B	0	6	15	10	3
C	8	5	0	0	0
D	0	4	2	0	5
E	3	5	6	0	8

Since the number of lines less than the number of rows/columns, an improvement is possible.

Step4. Improve the present opportunity cost table. This is done by the following operations;

- Select the **smallest entry (element)** among all **uncovered elements** by **the lines** and **subtract it** from **all entries** in the **uncovered cells**. This cell is 2
- Add the same smallest entry to those cells in which lines intersect (cells with two lines them).
- Cells with one line through them are unchanged to the improved table.

		I	II	III	IV	V
	A	7	0	8	12	11
	B	0	4	13	10	1
	C	10	5	0	2	0
	D	0	2	0	0	3
	E	3	3	4	0	6

Since the number of assigned cells equals to the number of rows/columns, the solution is optimal.

<u>Job</u>	<u>Employees</u>	<u>Time (in minutes)</u>
A	II	5
B	I	3

C	V	2
D	III	9
E	IV	4
		Total time=23 Hours

Has prepared the following table, which shows the costs for various combinations of job-machine assignments:

		<i>Machine (Cost in '000s)</i>		
		A	B	C
	1	20	15	31
Job	2	17	16	33
	3	18	19	27

- A. What is the optimal (minimum-cost) assignment for this problem?
 B. What is the total cost for the optimum assignment?

Solution:

Table: After row reduction

		A	B	C
-15	1	5	0	16
-16	2	1	0	17
-18	3	0	1	9

Table: After column reduction

		A	B	C
	1	5	0	7
	2	1	0	8
	3	0	1	0

Table: optimal

		A	B	C
	1	4	0	6
	2	0	0	7
				0

	3	0	2	0
--	---	---	---	---

<u>Job</u>	<u>Machine</u>	<u>Cost(in \$)</u>
1	B	15000
2	A	17000
3	C	27000
		Total optimal assignment= <u>\$59000</u>

3.3.3. SPECIAL CASES IN THE ASSIGNMENT PROBLEMS

Certain situations can arise in which the model deviates slightly from that previously described. Among those situations are the following:

A. Multiple Optimal Solutions

While making an assignment in the reduced assignment matrix, it is possible to have two or more ways to strike off a number of zeros. Such situation indicates multiple optimal solutions with the **same optimal value of objective function**. In such cases the more suitable solution may be considered by the decision-maker. In multiple optimal solutions, **no unique 0 will exist at some point**, resulting in more than one choice for assignment and hence, more than one optimal solution. It should be noted that all optimal solutions will yield the same value of the objective function.

Example:

1. Given this final assignment table, identify two optimal solutions.

		<u>Machine</u> (Estimated time in minute)		
		1	2	3
	A	4	0	0
Job	B	0	3	2
	C	1	0	0

Solution

The first assignment must be **B-1**, because **B-1** is the only **0** that appears in a single row or column. Having made that assignment, there are two choices for the remaining two rows, and two choices for the remaining two columns. This results in two possible solutions, as shown:

Alternative 1

	<i>Machine</i> (Estimated time in minute)			
	1	2	3	
	A	4	[0]	0
Job	B	[0]	3	2
	C	1	0	[0]

Alternative 2

		<i>Machine</i> (Estimated time in minute)		
		1	2	3
	A	4	0	[0]
Job	B	[0]	3	2
	C	1	[0]	0

B. Maximization case in assignment problems

There may situations when the assignment problem calls for maximization of profit, revenue, etc as the objective function. Such problem may be solved by converting the given maximization problem into a minimization problem by the following procedure:

- Find the largest profit coefficient in the entire.*
- Subtract each entry in the original table from the largest profit coefficient.*

The transformed assignment problem so obtained can be solved by using the Hungarian method.

Example

A company has four territories open, and four salesmen available for an assignment. The territories are not equally rich in their sales potential. Based on the past performance, the following table shows the annual sales (in \$) that can be generated by each salesman in each territory. Find the optimal assignment and the maximum expected total sales.

Salesmen	Territory				
	I	II	III	IV	
	A	42	35	28	21
	B	30	25	20	15
	C	30	25	20	15
	D	24	20	16	12

Solution:

Convert maximization problem into minimization problem by subtracting all elements from the highest element (i.e 42)

Thus, the equivalent cost table is:

Sales men	Territory			
	I	II	III	IV
A	0	7	14	21
B	12	17	22	27
C	12	17	22	27
D	18	22	26	30

Thus, after improvement of the table, the optimal assignment is:

Sales men	Territory			
	I	II	III	IV
A	0	7	14	21
B	0	5	10	15
C	0	5	10	15
D	0	4	8	12

Column reduction value 0 4 8 12

Thus, the improved assignment is.

Sales men	Territory			
	I	II	III	IV
A	0	2	6	9
B	0	1	2	3
C	0	1	2	3

D	0	0	0	0
----------	---	---	---	---

After this we use covered line approach i.e

Sales men	Territory			
	I	II	III	IV
A	0	2	6	9
B	0	1	2	3
C	0	1	2	3
D	0	0	0	0

Then after use the following rules i.e

- ♣ Identify the smallest uncovered number from the table.
- ♣ Add that smallest number from the number covered by two lines.
- ♣ Subtract that smallest number from the number not covered by line.
- ♣ Neither adds nor subtracts that number from the number covered by single line.

Note: When the number of covered line is equals to either the number of row or columns, the problem is reach optimal solution.

Hence, for the above example the smallest uncovered line is 1. Then;

Sales men	Territory			
	I	II	III	IV
A	0	1	5	8
B	0	0	1	2
C	0	0	1	2
D	1	0	0	0

The new improved assignment will be;

Sales men	Territory			
-----------	-----------	--	--	--

	I	II	III	IV
A	0	1	5	8
B	0	0	1	2
C	0	0	1	2
D	1	0	0	0

This doesn't reach optimal solution because, based on the above assumption. Therefore, move to the next table. i.e the smallest uncovered number is **1** again.

Sales men	Territory			
	I	II	III	IV
A	0	1	4	7
B	0	0	0	1
C	0	0	0	1
D	2	1	0	0

Now we reach at optimal solution because, the number of column is equal to covered lines. Hence,

The optimum assignment value indicates there is multiple solution i.e

Alternative 1	Sales men	Territory	Total profit
	A	I	42
	B	IV	25
	C	II	20
	D	III	12
	Total		99 br.

Sales men	Territory	Total profit
A	I	42
B	II	25

Alternative 2

C	IV	20
D	III	15
Total		99 br.

C. Unbalanced Assignment problem

The Hungarian method of assignment requires that the number of columns and rows in the assignment matrix be equal. However, when the given cost matrix is not a square matrix, the assignment problem is called an ***unbalanced problem***. In such cases a dummy row(s) or column(s) are added in the matrix (with ***zeros as the cost elements***) to make it a square matrix. After making the given cost matrix a square matrix, the Hungarian method may be used to solve the problem.

D. Restrictions on Assignments

In certain instances, it may happen that a particular match or pairing may be either ***undesirable*** or otherwise unacceptable. For example, an employee may not have the skills necessary to perform a particular job or a machine may not be equipped to handle a particular operation. In such cases, the cost of performing that particular activity by a particular resource is considered to be very large (written as ***M*** or ∞) so as to prohibit the entry of this pair of employee-job into the final solution. When such a restriction is present, a letter (***M***) is often placed in the table in the position that would represent a pairing. Analysis is performed as usual except the ***M*** is ignored throughout the analysis. That is, ***M*** is not used in any reductions, nor is any value added to it or subtracted from it during the course of the analysis.

Chapter four

Decision theory

4.1.Introduction

The success or failure that an individual or organizations experiences, depends to a large extent on the ability of making appropriate decisions. Making of a decision requires an enumeration of feasible and viable alternatives (courses of action or strategies), the projection of consequences associated with different alternatives and a measure of effectiveness (or an objective) by which the most preferred alternative is identified. Decision theory provides an analytical and systematic approach to the study of the decision making in other words, decision theory provide a method of natural decision making where in data concerning the occurrence of different outcomes (consequences') may be evaluated to enable the decision maker to identify alternatives.

Decision models useful in helping decision makers to make the best possible decisions are classified according to the degree of certainty. The scale of certainty can range from complete certainty to complete uncertainty. The region which falls between these two extreme points corresponds to the decision making under risk. Irrespective of the type of decision model, there are certain essential characteristics which are common to all as listed below.

1. **Decision alternatives:** these are a finite number of decision alternatives available with the decision maker at each point at a time when a decision is made. The number and type of such alternatives may depend on the previous decision made and on what has happened

subsequent to those decisions. These alternatives are also called course of action (actions, act or strategies) and are under control and known to the decision maker.

2. **State of nature:** a possible future condition (consequence or event) resulting from the choice of a decision alternative depends up on certain factors which are beyond the control of a decision maker. These factors are called state of nature (future). For example, if the decision is to carry an umbrella or not, the consequence (get wet or do not) depends on what action nature takes. The states of natures are mutually exclusive and collectively exhaustive with respect to any decision problem. The state of nature may be described numerically such as, demand of 100 units of an item or non-numerically such as, employee strike, etc.
3. **Payoff:** a numerical value resulting from each possible combination of alternatives and state off nature is called payoff. The payoff values are always conditional values because of unknown state of nature. A tabular arrangement of these conditional outcome (pay off) values is called **pay off matrix**.

4.2.Steps in decision theory approach

The decision-making process involves the following steps:

- I. Identify and define the problem
- II. Listing of all possible future events called state of nature, which can occur in the context of the decision problem. Such events are not under the control of decision maker because they are erratic in nature.
- III. Identification of all the course of action (alternatives or decision choices) which are available to the decision maker. The decision maker has control over this course of action.
- IV. Expressing the payoffs resulting from each pair of course of action and state of nature. These pay offs are normally expressed in a monetary value.
- V. Apply an appropriate mathematical decision theory model to select best course of action from the given list on the basis of some criterion (measure of effectiveness) that results in the optimal (desired) pay off.

Example: a firm manufactures three types of products. The fixed and variable costs are given below.

	Fixed cost in birr	Variable cost per unit
Product A	25,000	12
Product B	35,000	9
Product C	53,000	7

The likely demand (units) of the products is given below:

Poor demand	3000
Moderate demand	7000
High demand	11,000

If the sale price of each type of product is birr 25, then prepare the payoff matrix

Solution: let D1, D2 and D3 be the poor, moderate and high demand, respectively. Then payoff is given by;

Payoff= sales revenue-cost

The calculations for payoff (in thousands) for each pair of alternative demand and the type of product (state of nature) are shown below:

$$D1A = 3 \times 25 - 25 - 3 \times 12 = 14 \quad D2A = 7 \times 25 - 25 - 7 \times 12 = 66$$

$$D1B = 3 \times 25 - 35 - 3 \times 9 = 13 \quad D2B = 7 \times 25 - 35 - 7 \times 9 = 77$$

$$D1C = 3 \times 25 - 53 - 3 \times 7 = 1 \quad D2C = 7 \times 25 - 53 - 7 \times 7 = 73$$

$$D3A = 11 \times 25 - 25 - 11 \times 12 = 118$$

$$D3B = 11 \times 25 - 35 - 11 \times 9 = 141$$

$$D3C = 11 \times 25 - 53 - 11 \times 7 = 145$$

The payoff values are shown below:

Product type	D1	D2	D3
A	14	66	118
B	13	77	141
C	1	73	145

4.3.Types of decision-making environments

Decisions are made based up on the data available about the occurrence of events as well as the decision situation (or environment). There are three types of decision-making environment: certainty, uncertainty and risk.

Type1: decision making under certainty

In this case the decision maker has the complete knowledge (perfect information) of consequence of every decision choice (course of action) with certainty. Obviously, he/she will select an alternative that yields the largest return (payoff) for the known future (state of nature).

Type 2: decision making under risk

In this case the decision maker has less than completes knowledge with certainty of the consequence of every decision choice (course of action). This means there is more than one state of nature (future) and for which he makes an assumption of the probability with which each state of nature will occur. For example, probability of getting head in the toss of a coin is 0.5.

Type 3 decision making under uncertainty

In this case the decision maker is unable to specify the probabilities with which the various state of nature (futures events) will occur. Thus, decision under uncertainty are taken with even less information than decision under risk. For example, the probability that Mr X will be the prime minister of the country 15 years from now is not known.

4.4.Decision making under uncertainty

In the absence of knowledge about the probability of any state of nature (future event) to be occurred, the decision maker must arrive at a decision only on the actual conditional pay off values together with a policy (attitude). There are several different criteria of decision making in this situation and these are:

- I. Maximax or Minimin (optimistic criterion)
- II. Maximin or Minimax (pessimistic criterion)
- III. Equally likely (Laplace criterion)
- IV. Criterion of realism (Hurwicz criterion)
- V. Criterion of regret (Savage criterion)
- I. Criterion of optimism (Maximax or Minimin)**

In this criterion the decision maker ensures that he/she should not miss the opportunity to achieve the largest possible profit (Maximax) or lowest possible cost (Minimin). Thus, he/she selects the alternative that represents the maximum of maximum (or minimum of minimum) pay off. The working method is summarized as follows:

- a) Locate the maximum (or minimum) pay off values corresponding to each alternative
- b) Select an alternative with the best anticipated pay off value (maximum for profit and minimum for cost)

II. Criterion of pessimism (Maximin or Minimax)

In this criterion the decision maker ensures that he would earn no less (or pay no more) than some specified amount. Thus, he selects the alternative that represents the maximum of the minima for profit (or minimum of the maxima in case of loss) payoff. The working method is summarized as follows:

- a) Locate the minimum (in case of profit) or maximum (in case of loss) payoff value corresponding to each alternative, then
- b) Select an alternative with the best anticipated payoff value (maximum for profit and minimum for loss or cost)

Since in this criterion the decision maker is conservative about the future and always anticipates worst possible outcome (minimum for profit and maximum for cost or loss), it is called a pessimistic decision criterion. This criterion is also known as Wald's criterion.

III. Equally likely decision (laplace) criterion

Since the probabilities of each state of nature are not known, it is assumed that all state of nature will occur with equal probability, i.e. each state of nature is assigned an equal probability. As states of nature are mutually exclusive and collectively exhaustive, so the probability of each of these must be $1/(\text{number of states of nature})$. The working method is summarized as follows:

- a) Assign equal probability value to each state of nature by using the formula:
$$1/(\text{number of states of nature})$$
- b) Compute the expected (or average) payoff for each alternative (course of action) by adding all the payoffs and dividing by the number of possible states of nature or by applying the formula:

(Probability of state of nature j) * (payoff value for the combination of alternative 'I's and state of nature 'J's).

- c) Select the best expected payoff value (maximum for profit and minimum for cost from the given average values in each strategy). This criterion is also known as criterion of insufficient reason because, except in few cases, some information of the likelihood of occurrence of state of nature is available.

IV. Criterion of realism (Hurwicz criterion)

This criterion suggests that a rational decision maker should be neither completely optimistic nor pessimistic and therefore, must display a mixture of both. Hurwicz, who suggested this criterion, introduced the idea of a sufficient of optimism (denoted by α) to measure the decision maker's degree of optimism. This coefficient lies between 0 and 1, where 0 represents a complete pessimistic attitude about the future and 1 a complete optimistic attitude about the future. Thus, α is the coefficient of optimism, then $(1-\alpha)$ will represent the coefficient of pessimism.

The Hurwicz approach that the decision maker must select an alternative that maximizes

H (criterion of realism) = α (maximum in column) + $(1-\alpha)$ (minimum in column).

The working method is summarized as follows:

- a) Decide the coefficient of optimism α and then coefficient of pessimism $(1-\alpha)$
- b) For each alternative select the largest and lowest payoff value and multiply these with α and $(1-\alpha)$ values, respectively. Then calculate the weighted average by using the formula
- c) Select an alternative with best anticipated weighted average payoff value.

V. Criterion of regret (savage criterion)

This criterion is also known as opportunity loss decision criterion or Minimax regret decision criterion because the decision maker feels regret after adopting a wrong course of action (or alternative) resulting in an opportunity loss of payoff. Thus, he/ she always intend to minimize this regret. The working method is summarized as follows:

- a) From the given payoff matrix, develop an opportunity loss (or regret) matrix as follows
 - I. Find the best payoff corresponding to each state of nature, and
 - II. Subtract all other entries (payoff) in that row from this value
- b) For each course of action (strategy or alternative) identify the worst or maximum regret value. Record this number in a new row

c) Select the course of action (alternative) with the smallest anticipated opportunity –loss value

Example: Let there be a situation in which a decision-maker has three possible alternatives A_1 , A_2 and A_3 , where the outcome of each of them can be affected by the occurrence of any one of the four possible events S_1 , S_2 , S_3 and S_4 . The monetary payoffs of each combination of A_i and S_j are given in the following table:

	State of natures			
Alternatives	S1	S2	S3	S4
A1	27	12	14	26
A2	45	17	35	20
A3	52	36	29	15

Opportunity loss table for the above table

	State of natures				
Alternatives	S1	S2	S3	S4	Maximum regret
A1	25	24	21	0	25
A2	7	19	0	6	19
A3	0	0	6	11	11

Solution: Since 11 are Minimum out of the maximum regret values in each alternative, the optimal action is A_3 .

a. Maximax A_3 (52).

b. Maximin A_1 (12)

c. Regret Criterion A_3 (11)

d. Herzwicz Criterion: Assume that the index of optimism $\alpha = 0.7$.

<i>Action</i>	<i>Max. Payoff</i>	<i>Min. Payoff</i>	<i>Weighted Average</i>
A_1	27	12	$27 \times 0.7 + 12 \times 0.3 = 22.5$
A_2	45	17	$45 \times 0.7 + 17 \times 0.3 = 36.6$
A_3	52	15	$52 \times 0.7 + 15 \times 0.3 = 40.9$

Since the average for A_3 is Maximum with 40.9, it is optimal.

Exercises

- The following matrix gives the payoff (in birr) of different strategies (alternatives) S1, S2a and S3 against conditions (events) N1, N2, N3 and N4.

State of nature

Strategy	N1	N2	N3	N4
S1	4000	-100	6000	18,000
S2	20,000	5000	400	0
S3	20,000	15,000	-2000	1000

Indicate the decision taken under the following approaches:

- Pessimistic
 - Optimistic
 - Equal probability
 - Realism
 - Regret
- In the toy manufacturing company, suppose the product acceptance probabilities are not know, but the following data is known

Product acceptance	Full	Partial	Minimal
Good	8	70	50
Fair	50	45	40
Poor	-25	-10	0

Determine the optimal decision under each of the following decision criterion and show how you arrive at it

- A. Maximax
 - B. Maximin
 - C. Equally likely
 - D. Herzwicz
 - E. Minimax regret
3. The following is a payoff(in birr's) table for three strategies and two state of nature

Strategy	State of nature	
	N1	N2
S1	40	60
S2	10	-20
S3	-40	150

Select a strategy using each of the following decision criterions:

- A. Maximax
- B. Minimax regret
- C. Maximin
- D. Equally likely
- E. Herzwicz

4.5.Decision making under risk

Decision making under risk is a probabilistic decision situation, in which more than one state of nature exists and the decision maker has sufficient information to assign probability values to the likely occurrence of each of these states. Knowing the probability distribution of the state of nature, the best decision is to select that course of action which has the largest expected payoff value. The most widely criterion for evaluating various course of action under risk is the expected monetary value (EMV) or expected utility

1. Expected monetary value

The expected monetary value (EMV) for a given course of action is the weighted average payoff which is the sum of the payoffs for each course of action multiplied by the probabilities associated with each state of nature

Steps for calculating EMV

- a) Construct a payoff matrix listing all possible courses of action and state of nature. Enter the conditional payoff values associated with each possible combination of course of action and state of nature along with the probabilities of the occurrence of each state of nature
- b) Calculate the EMV for each course of action by multiplying the conditional payoffs by the associated probabilities and add these weighted values for each course of action
- c) Select the course of action that yields the optimal EMV

2. Expected monetary loss(EML)

An alternative approach to minimizing expected monetary value (EMV) is to minimize the expected opportunity loss (EOL), also called expected value of regret. The EOL is defined as the difference between the highest profit (payoff) for a state of nature and the actual profit obtained for the particular course of action taken. In other words, EOL is the amount of payoff that is lost by not selecting the course of action that has the greatest payoff for the state of nature that actually occurs. The course of action which has minimum EOL from the weighted average value of each opportunity loss alternatives is recommended. Since EOL is an alternative decision criterion for reaching decision making under risk, therefore, the result will always be the same as those obtained by EMV criterion discussed earlier. Thus, only one of the two methods should be applied to reach a decision.

Steps for calculating EOL

- a) Prepare a conditional profit table for each course of action and state of nature combination along with the associated probabilities
- b) For each state of nature calculate the conditional opportunity loss (COL), values by subtracting each payoff from the maximum payoff for that outcome.
- c) Calculate EOL for each course of action by multiplying the probability of each state of nature with the EOL value and then adding the values
- d) Select a course of action for which the EOL value is minimum

3. Expected value of perfect information (EVPI)

In decision making under risk each state of nature is associated with the probability of its occurrence. However, if the decision maker can acquire perfect (complete and accurate) information about the occurrence of various state of nature, then he/she will be able to select a course of action that yields the desired payoff for whatever state of nature that actually occurs.

We have seen that the EMV or EOL criterion helps the decision maker select a particular course of action that optimizes the expected payoff without any additional information. Expected value of perfect information (EVPI) represents the maximum amount of money the decision maker has to pay to get this additional informational about the occurrence of various state of nature before a decision has to made.

Steps to calculate the maximum amount of money that the decision maker can pay to get perfect information is as follows

- Find expected value with perfect information (EVwPI) by multiplying the maximum values in each state of natures with corresponding probabilities and add the weighted values.
- Then, subtract EMV from EVwPI (EVwPI-EMV)

Example;

State of natures	Alternatives		
	A1	A2	A3
N1	20	12	25
N2	25	15	30
N3	30	20	22

The associated probabilities for each state of natures are 0.5, 0.4 & 0.1.

EMV; $A1 = 20(0.5) + 25(0.4) + 30(0.1) = 23$

$A2 = 12(0.5) + 15(0.4) + 20(0.1) = 14$

$A3 = 25(0.5) + 30(0.4) + 22(0.1) = 26.7$

Therefore, A3 is the best strategy with the value of 26.7.

EOL;

State of natures	Alternatives		
	A1	A2	A3
N1	5	13	0
N2	5	15	0
N3	0	10	8

$A1 = 5(0.5) + 5(0.4) + 0(0.1) = 4.5$

$$A2=13(0.5) +15(0.4) +10(0.1) =13.5$$

$$A3=0(0.5) +0(0.4) +8(0.1) =0.8$$

The best strategy is A3 with a minimum opportunity loss of 0.8.

$$EVPI=EV_{wPI}-EMV$$

$$EV_{wPI}=25(0.5) +30(0.4) +30(0.1) =27.5$$

$$EMV=26.7$$

$$EVPI=27.5-26.7=0.8$$

Therefore, the maximum amount of money that the decision maker can incur to get perfect information is 0.8 Br.

CHAPTER FIVE

5. Project management

5.1.PERT (project evaluation and review technique) and CPM (critical path method)

A project is a well-defined set of jobs or tasks, of which must be completed to finish the project. Examples of project includes, construction of a bridge, highway, power plant, repair and maintenance of an oil refinery or an airplane; design, development and marketing of a new product, research and development work, etc. such a project involve large number of interrelated activities or tasks which must be completed in a specified tome, in a specified sequence (or order) and requires resources such as personnel, money, materials, facilities and/ or space. The main objective before starting any project is to schedule the required activities in an efficient manner so as to complete it on or before specified time limit at minimum cost of its completion.

The technique of operation research used for planning, scheduling, and controlling large and complex project are often referred to as network analysis, network planning, or network planning and scheduling techniques. All these techniques are based on the representation of the project as a network of activities. A network is a graphical plan consisting of a certain configuration of arrows and nodes for showing the logical sequence of various activities to be performed to achieve project objectives. It is perhaps the reason that the above nomenclature is given to these techniques. In this chapter, we shall discuss two of the well-known techniques, PERT and CPM belong to this family.

5.1.1. THE BASIC DIFFERENCE BETWEEN PERT AND CPM

Although there are no essential differences between PERT and CPM as both of them share in common the determination of a critical path and are based on the network representation of activities and their scheduling that determines the most critical activities to be controlled so as to meet the completion date of the project. However, following are some other major differences.

PERT

1. It assumes a probability distribution for the duration of each activity. Thus, completion time estimates for all of the activities are needed
2. To perform PERT analysis on the project, the emphasis is given on the completion of a task rather than the activities required to be performed to reach particular event or task. Thus, it is also called event-oriented technique
3. It is used for one time projects involving activities of non-repetitive nature (i.e. activities which may never have been performed before) in which time estimates are uncertain, such as redesigning an assembly line or installing a new information system
4. It helps in identifying critical areas in a project so that necessary adjustment can be made to meet the scheduled completion date of the project

CPM

1. This technique was developed in connection with a construction and maintenance project in which duration of each activity was known with certainty
2. It is suitable for establishing a tradeoff for optimum balancing between schedule time and cost of the project
3. It is used for completion of the project involving activities of repetitive nature

5.2.SIGNIFICANCE OF USING PERT/ CPM

1. A network diagram helps translation of highly complex project in to a set of simple and logical arranged activities and therefore
 - Helps in clarity of thoughts and actions
 - Helps in clear and unambiguous communication developing from top to bottom and vice versa among the people responsible for executing the project
2. Detailed analysis of network help project in charge to peep in to future because
 - Difficulties and problems that can be reasonably expected to crop up during of execution, being foreseen well ahead of actual execution
 - Delays and hold ups during course of execution get minimized. Corrective action can also be taken well in time
3. Isolates activities which control the project completion and therefore, results in expeditious completion of the project
4. Helps in the division of responsibilities and therefore, enhance effective coordination among different departments involved
5. Helps in timely allocation of resources to various activities to achieve optimum utilization of resources

5.3.PERT/CPM NETWORK COMPONENTS AND PRECEDENE RELATIONSHIP

PERT/CPM network consists of two major components a discussed below

Events

- I. MERGE EVENT: an event which represents the joint completion of more than one activity is known as merge event

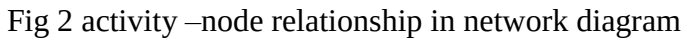


Fig 1 types of events

- II. BURST EVENT: an event which represents the initiation (beginning) of more than one activity is known as burst event

Activities

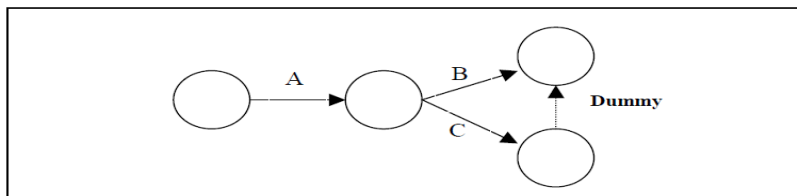
Activities in network diagram represent project operation or tasks to be conducted. As such each activity except dummy consumes time and resource and incurs costs.



- a) Predecessor activity
- b) Successor activity



- Two or more parallel activities in a project have same head and tail events or
- Two or more activities have some (but not at all) of their immediate predecessor activities in common. A dummy activity is depicted by dotted line in the network diagram



Activity on arrow (AOA) network: in this type of precedence network at each end of the activity arrow is node (or circle). These nodes represent points in time or instants, when an activity is starting or ending. Three important advantage of using AOA are as follows:

- i. Many computer programs are based on AOA network
- ii. AON diagrams can be superimposed on a time scale with the arrows drawn, the correct length to indicate the time requirement
- iii. AOA diagrams give a better sense of the flow of time throughout a project

In this chapter, only AOA network diagram will be used

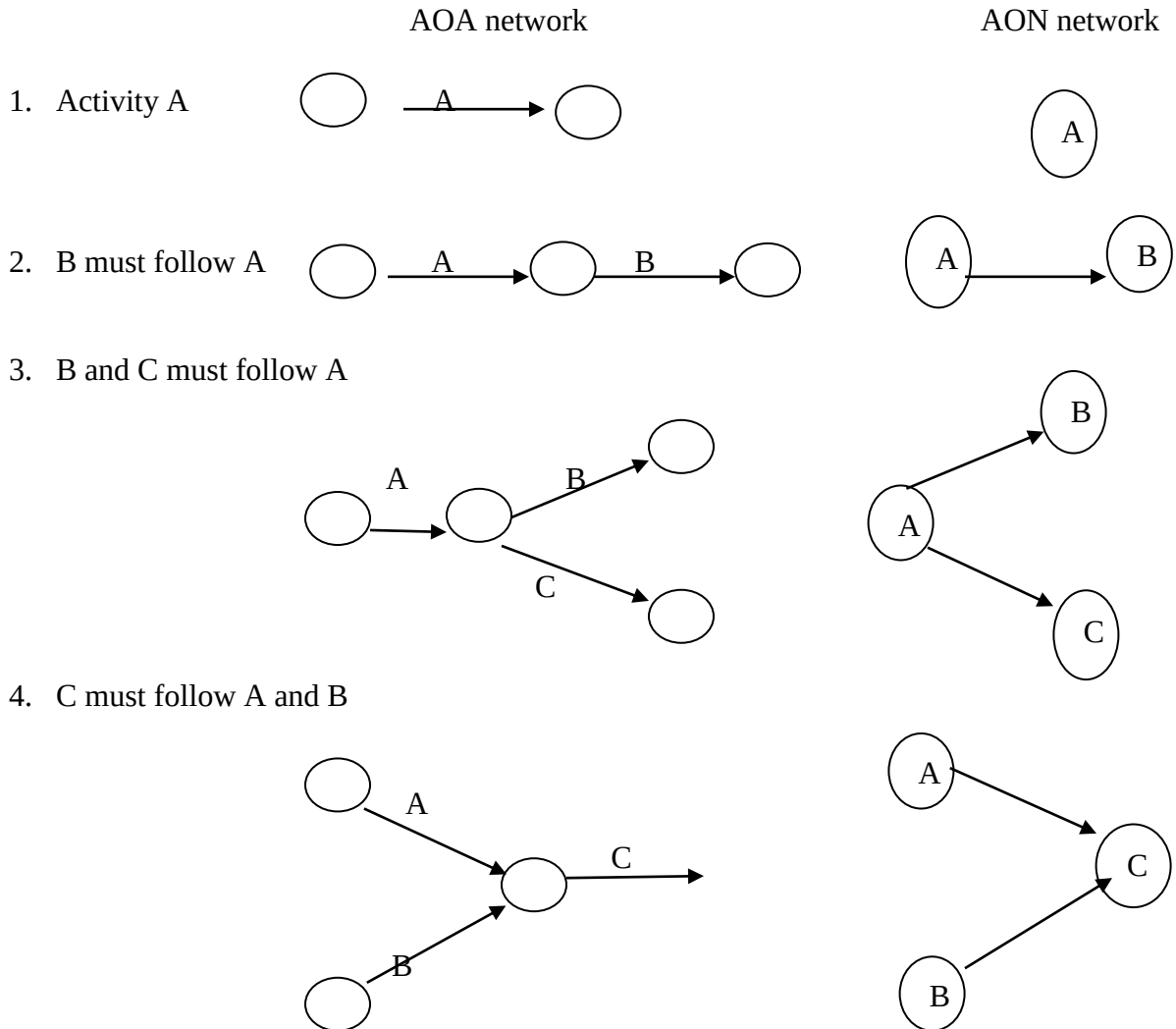


Fig 3 AOA and AON diagramming

Errors and dummies in network

1. **Looping and dangling:** looping (cycling) and dangling are considered to as fault in a network. Therefore must be avoided
 - i) A case of endless loop in a network diagram which is also known as looping is shown below where activities A, B, and C form a cycle

Due to precedence relationship, it appears from fig 4 that every activity in looping (or cycle) is a predecessor of itself. In this case it is difficult to number three events associated with activity A, B and C so as to satisfy rule 6 of constructing a network

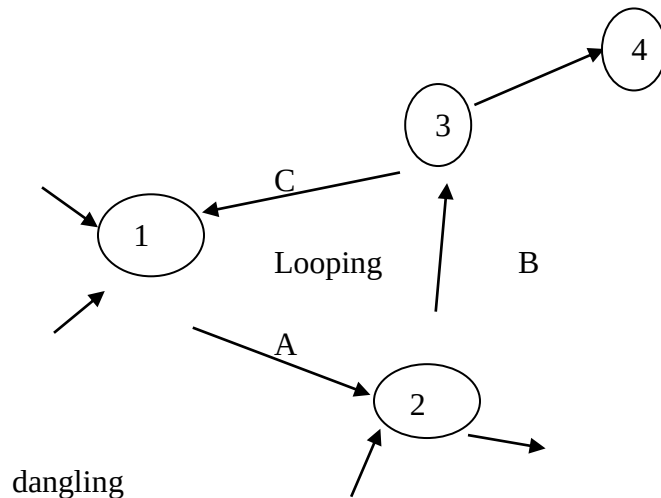


Fig 4.1 Looping in network diagram

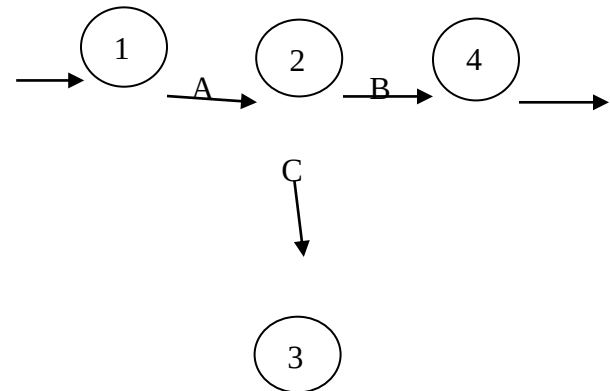


fig 4.2 Dangling in network diagram

- ii) A case of disconnect activity before the completion of all activities which is also known as dangling is shown on fig 4.2. In this case, activity C does not give any result as per rules of the network. The dangling is avoided by adopting rule 5 of constructing the network

Dummy (or redundant) activity: the following are the two cases in which use of dummy activity may help in drawing the network correctly as per the various rules.

- i) When two or more parallel activities in a project have the same head and tail events

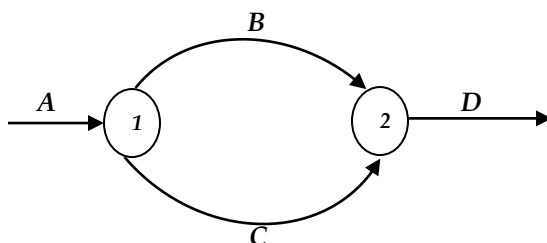


Fig: a) Parallel Activities

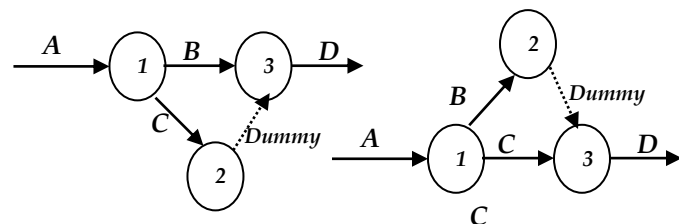


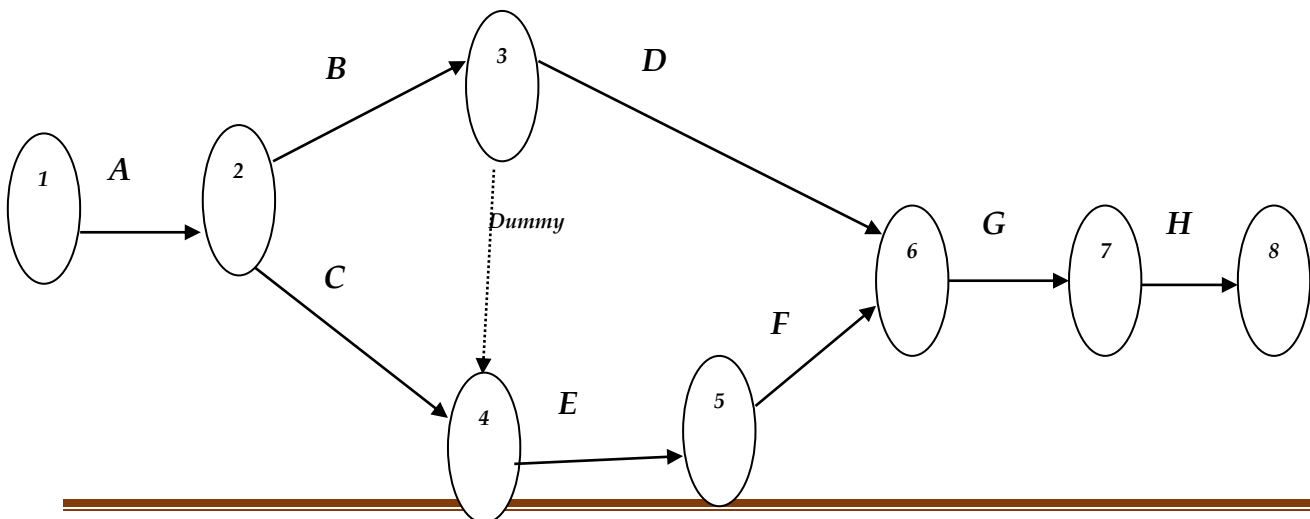
Fig: b) Dummy Activity

Example 1 an assembly is to be made from two parts X and Y. both parts must be turned on a lathe and Y must be polished whereas X need to be polished. The sequence of activities together with their predecessors is give below

Activity	Description	Predecessor activity
A	Open work order	-
B	Get material for X	A
C	Get material for Y	A
D	Turn X on lathe	B
E	Turn Y on lathe	B,C
F	Polish Y	E
G	Assemble X and Y	D,F
H	Pack	G

Draw the network diagram for the project

Solution:

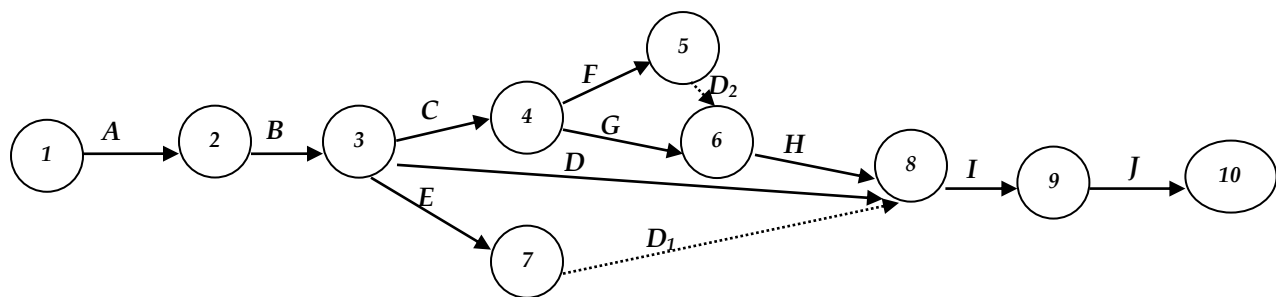


example 2: listed in the table are the activities and sequencing necessary for a maintenance job on the heat exchangers in a refinery

Activity	Description	Predecessor activity
A	Dismantle pipe connections	-
B	Dismantle heater, closure and floating front	A
C	Remove tube bundle	B
D	Clean bolts	B
E	Clean heater and floating head front	B
F	Clean tube bundle	C
G	Clean shell	C
H	Replace tube bundle	F,G
I	Prepare shell pressure test	D,E,H
J	Prepare tube pressure test and reassemble	I

Draw the network diagram for the project

Solution



5.4.CRITICAL PATH ANALYSIS

The objective of critical path analysis is to estimate the total project duration and assign a starting and finishing times to all activities involved in the project. This helps in checking actual progress against the scheduled duration of the project. The duration of individual activities may be uniquely determined (in case of CPM) or may involve the three time estimates (in case of PERT) out of which the expected duration of the activity is computed. Having done this, the following factors should be known to prepare project scheduling.

- i) Total completion time of the project
- ii) Earlier and latest start time of each activity
- iii) Float for each activity, i.e. the amount of time by which the completion of an activity can be delayed without delaying the total project completion time
- iv) Critical path and critical activities

5.4.1. FORWARD PASS METHOD (FOR EARLIEST EVENT TIME)

In this method, calculating begin from the initial even 1, proceed through the events in an increasing order of event numbers and end at the final event, say N. at each event we calculate its earliest occurrence time(E) and earliest start and finish time for each activity that begins at that event. When calculations end at the final event N, its earliest occurrence time gives the earliest possible completion time of the entire project. The method may be summarized as follows:

1. Set the earliest occurrence time of initial event 1 to zero. That is $E_1=0; i=1$
2. Calculate earliest start time for each activity that begins at event $i(=1)$. This is equal to earliest occurrence time of event (tail event). That is,
 $ES_{ij}=E_i$, for all activities (i,j) starting at event i
3. Calculate the earliest finish time of each activity that begins at event i. this is equal to the earliest start time of the activity plus the duration of the activity. That is,
 $EF_{ij}=ES_{ij}+t_{ij}=E_i+T_{ij}$, for all activities (i,j) beginning at event i.
4. Proceed to the next event, say $j>i$.
5. Calculate the earliest occurrence time for the event j. this is the maximum of the earliest finish times of all activities ending into that event, that is,
 $E_j=\max(EF_{ij})=\max(E_i+t_{ij})$, for all immediate predecessor activities
6. If $j=N$ (final event), then earliest finish time for the project, that is, the earliest occurrence time $E_N=\max(EF_{ij})=\max(E_{N-1}+t_{ij})$, for all terminal activities

5.4.2. BACKWARD PASS METHOD (FOR LATEST ALLOWABLE EVENT TIME)

In this method calculations begin from final event N. proceed through the events in the decreasing order of event numbers and end at the initial event 1. At each event, we calculate latest occurrence time (L) and latest finish and start time for each activity that is terminating at that event and the procedure continue till the initial event. The method may be summarized as follows:

1. Set the latest occurrence of last event, N equal to its earliest occurrence time (known from forward pass method). That is, $L_N = E_N, j=N$
2. Calculate latest finish time of each activity which ends at event j. this is equal to latest occurrence time of final event. That is,
 $LF_{ij} = L_j$; for all activities (i,j) ending at event j
3. Calculate the latest start times of all activities at j. it is obtained by subtracting the duration of the activity from the latest finish time of the activity. That is,
 $LS_{ij} = LF_{ij} - t_{ij} = L_j - t_{ij}$, for all activity (i,j) ending at event j
4. Proceed backward to the event in the sequence, that decrease j by 1
5. Calculate the latest occurrence time of event i(i<j). This is the minimum of the latest start times of all activities from the event. That is,
 $L_i = \text{Min}(LS_{ij}) = \text{Min}(L_j - t_{ij})$, for all immediate successor activities.
6. If j=1(initial event), then the latest time for project i.e. latest occurrence time L_1 for the initial event is given by
 $L_1 = \text{Min}(LS_{ij}) = \text{Min}(L_{j-1} - t_{ij})$

5.4.3. FLOAT (SLACK) OF AN ACTIVITY AND EVENT

The float (slack) or free time is the length of time to which a non-critical activity and/ or an event can be delayed or extended without delaying the total project completion time

EVENT FLOAT

The float (also called slack) of an event is the difference between its latest occurrence time (L_i) and its earliest occurrence time (E_i). That is,

$$\text{Event float} = L_i - E_i$$

It is a measure of how much later than expected a particular event could occur without delaying the completion of the entire project

ACTIVITY FLOAT

As mentioned earlier, float is the length of free time available within the estimated times of the non-critical activities. The computation of activity float provides a measure of the extent to which non-critical activities may be extended or delayed. Mainly three types of floats are defined for each non-critical activity of the project

Total float: it is the amount of time by which an activity can be delayed if all its preceding activities take place at their earliest possible times and following activities can be allowed to wait until their latest permissible times. The times within which an activity must be scheduled is computed from LS and ES values for each activity's start event and event. That is, for each activity (i,j) the total float is equal to the latest permissible end event time minus the earliest possible start event time minus the activity duration. That is,

$$\text{Total float } (TF_{ij}) = (L_j - E_i) - t_{ij}$$

$$= LS_{ij} - ES_{ij} = LF_{ij} - EF_{ij}$$

Remark

1. Latest occurrence time of an event is always greater than or equal to its earliest occurrence time(i.e. $L_i \geq E_i$),

$$TF_{ij} \geq (E_j - E_i) - t_{ij} \text{ Or } TF_{ij} \geq FF_{ij}$$

This implies that the value of free float may range from zero to total float and can never exceed total float value. That is, independent float $\leq$ free float $\leq$ total float

2. The calculation of various floats can help the decision maker in identifying the underutilized resources, flexibility in the total schedule and possibilities of redeployment of resources
3. Once the float of an activity is distributed, float of all other activities of the project is changed and should be recalculated
4. Few conclusions which can be drawn from the total float value of an activity are as follows
 - a) Negative total float time($L-S < 0$)
 - Resources are not adequate and activity may not finish in time. This needs extra resources or certain activities need crashing to reduce negative float value
 - b) Zero total float time($(L-S=0)$)

- Resources are just sufficient to complete the activity. Any delay in activity execution will necessary increase the project cost

c) Positive((L-S>0)

- Resources are surplus. These resources can be deployed elsewhere or execution of the activity can be delayed

CRITICAL PATH

Certain activities in a network diagram of a project are called critical activities because delay in their execution will cause further delay in the project completion time. Thus, all activities having zero total float value are identified as critical activities. It is the longest path starting from first to the last event and is shown by thick line or double lines in the network diagram

The critical path on a network diagram can be identified as:

- For all activities (i,j) lying on the critical path the E- values and L- values for tail and head events are equal. That is, $E_j=L_j$ and $E_i=L_i$
- On critical path $E_j-E_i=L_j-L_i=t_{ij}$

Finding the critical path is important for directing decision maker's attention and effort to critical activities where improvement will pay the largest dividend.

Example

- An established company has decided to add a new product to its line. It will buy the product form manufacturing concern, package is & sell it to a number of distributes selected on a geographical basis. Market research has identified the volume expected & the size of sales force required. The steps

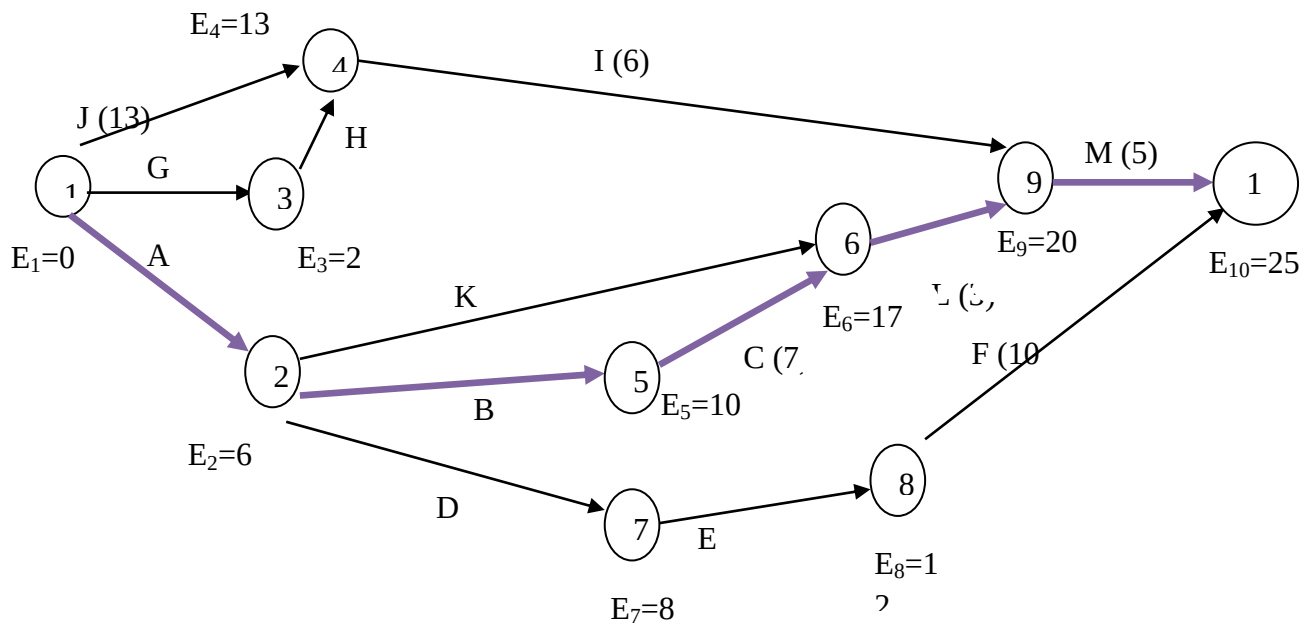
<u>Activity</u>	<u>Description</u>	<u>Predecessors</u>	<u>Time (Week)</u>
A	Organize sales office	—	6
B	Hire Salesman	A	4
C	Train salesman	B	7
D	Select advertising agency	A	2
E	Plan advertising campaign	D	4

F	Conduct	E	10
G	Design package	-	2
H	Setup packaging facilities	G	10
I	Package initial Stocks	J,H	6
J	Order Stock from manufacture	-	13
K	Select distributors	A	9
L	Sell to distributors	C,K	3
M	Ship stocks	I,L	5

Required:

- Draw the an arrow diagram
- Indicate the critical path; the critical activities & the total project time
- For each non-critical activity, find the total float

Solution: (a). the arrow diagram for the given project along with E-values and L-values is shown in the following diagram.



Forward pass method

Determine the earliest start time E_i and latest finish L_j for each event by processing as follows.

$$E_1=0$$

$$E_2= E_1+t_{1,2}=0+6$$

$$E_3= E_1+t_{1,3}=0+2=2$$

$$E_4=\text{Max} \{E_1+t_{1,4}; E_3+t_{3,4}\} \text{ } i=1,3$$

$$E_5= E_2+t_{2,5}=6+4=10$$

$$E_6=\text{Max} \{E_i + t_{i,6}\} =\text{Max} \{E_2+t_{2,6}; E_5+t_{5,6}\}$$

$$\text{Max} \{0+13, 2+10\}=13$$

$$i = 2, 5$$

$$E_8= E_7+t_{7,8}=8+4=12$$

$$\text{Max} \{6+9; 10+7\}=17$$

$$E_{10}=\text{Max}\{E_i+t_{i,10}\}$$

$$E_7= E_2+t_{2,7}=6+2=8$$

$$i=8,9$$

$$E_9=\text{Max} \{E_i+t_{i,9}\} =\text{Max} \{E_4+t_{4,9}; E_6+t_{6,9}\}$$

$$=\text{Max} \{E_8+t_{8,10}; E_9+t_{9,10}\}$$

$$i = 4, 6$$

$$=\text{Max} \{12+10; 20+5\}=25$$

$$=\text{Max} \{13+6; 17+3\} =20$$

Backward pass method

$$L_{10}=E_{10}=25$$

$$L_9=L_{10}-t_{9,10}=25-5=20$$

$$L_8=L_{10}-t_{8,10}=25-10=15$$

$$L_7=L_8-t_{7,8}=15-4=11$$

$$L_6=L_9-t_{6,9}=20-3=17$$

$$L_5=L_6-t_{5,6}=17-7=10$$

$$L_4=L_9-t_{4,9}=20-6=14$$

$$L_3=L_4-t_{3,4}=14-10=4$$

$$L_2=\min \{L_j- t_{2,j}\}$$

$$L_2=\min \{L_j- t_{1,j}\}$$

$$J=5, 6, 7$$

$$j=2, 3, 4$$

$$=\min \{L_5, - t_{2,5}; L_6, t_{2,6}; L_7-t_{2,7}\}$$

$$=\min \{L_2, - t_{1,2}; L_3, t_{1,3}; L_4-t_{1,4}\}$$

$$=\min \{10-4; 17-9; 11-2\} =6$$

$$=\min \{6-6, 5; 4-2; 14-13\}=0$$

- B. The vertical path in the network diagram above has been shown by thick lines by joining all those events where two values E_i & L_j are equal. The critical path of the project is: 1-2-5-6-9-10 the critical activities are A, B, C, L&M. the total project time is 25 weeks

- C. For each non-critical activity, the total float is:

Activity (i,j)	Duration (t _{ii})	Earliest time		Last time		Float	
		Start (E _i)	Finish (E _i +t _{ij})	Start (L _j -t _{ij})	Finish (L _j)	Total [(L _j -t _{ij})-E _i]	Free (E _j -E _i)-t _{ij}

1-3	2	0	2	2	4	2	0
1-4	13	0	13	1	14	1	0
2-6	9	6	15	8	17	2	2
2-7	2	6	8	9	11	3	0
3-4	10	2	12	4	14	2	1
4-9	6	13	19	14	20	1	1
7-8	4	8	13	11	15	3	0
8-10	10	12	22	15	25	3	3

Remark:

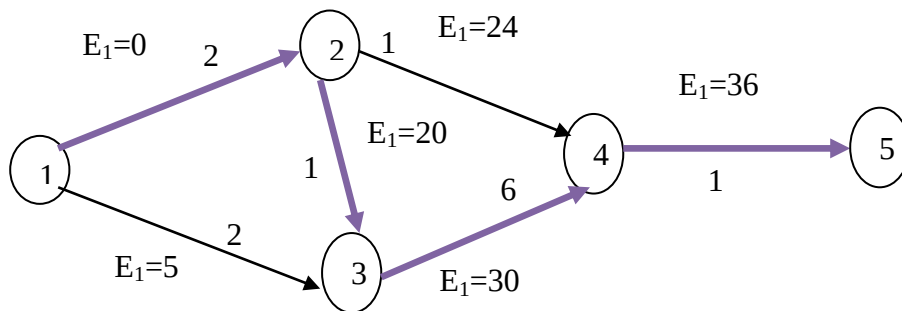
- What does the critical path really mean?
 - The critical path shows the shortest time in which a project can be completed. Even though the critical path is the longest path, it represents the shortest time it takes to complete a project.
 - There can be more than one critical path on a project. Project managers should closely monitor performance of activities on the critical path to avoid late project completion. If there is more than one critical path, project managers must keep their eyes on all of them

Example2. The following table gives the activities of a construction project and duration.

Activity	1-2	1-3	2-3	2-4	3-4	4-5
Duration (days)	20	25	10	12	6	10

- (i) Draw the network for the project.
- (ii) Find the critical path.
- (iii) Find the total, free and independent floats of each activity.

Solution: The first step is to draw the network and fix early start and early finish schedule and then late start-late finish schedule.



To find the connect activities and we get 1-2-3-path.

Check paths.

$$1-2-4-5 = 42$$

$$1-2-3-4-5 = 46^* \text{ (critical path)} \quad 1-3-4-5 = 41$$

Activity	Total Slack	Slack	Independent Slack
1-2	0	0	0
1-3	5	5	5
2-3	0	0	0
2-4	4	4	4
3-4	0	0	0
4-5	0	0	0

critical path, with 0 total slack 4-5 as the critical with alternate

5.5.PERT Networks

Program Evaluation and Review Technique (PERT) - is a network analysis technique used to estimate project duration when there is a high degree of uncertainty about the individual activity duration estimates.

The three-time estimates are:

1. Optimistic time (t_o or a): This is the shortest possible time to perform an activity, assuming that everything goes well.
2. Pessimistic time (t_p or b): This is the maximum (longest) time that is required to perform an activity, under extremely bad conditions.
3. Most likely time (t_m or m): This is the most realistic time to complete the activity.

The β distribution is not necessarily symmetric; the degree of skewness depends on the location of t_m to t_o and t_p . Thus the range specified by the optimistic time (t_o) and pessimistic time (t_p) estimates is assumed to enclose every possible estimates of the duration of the activity. The most likely time (t_m) estimate may not coincide with the midpoint ($(t_o + t_p)/2$) and may occur to its left or its right as shown in the following figure.

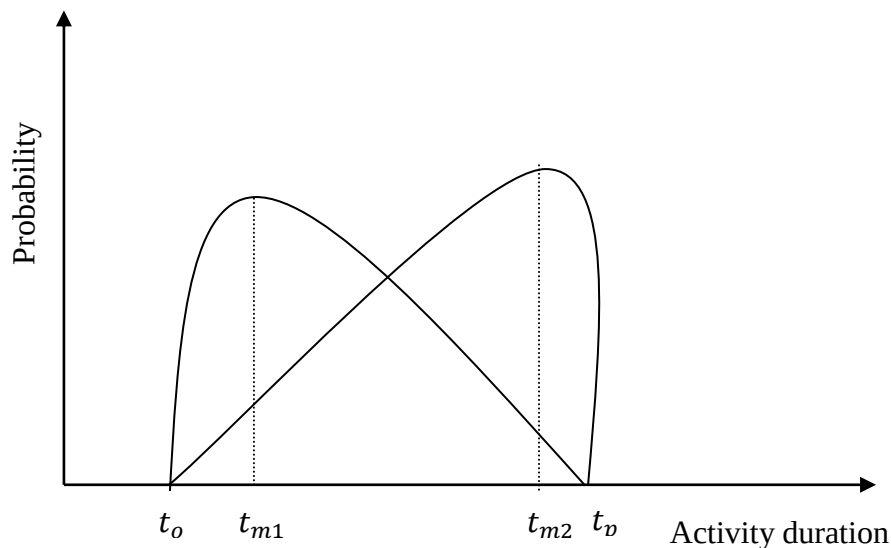


Fig an example of a β distribution

In beta distribution the midpoint $(t_o + t_p)/2$ is assumed to weight half as much as the most likely point (t_m). Thus the expected or mean (t_e) or μ) value of the captivity duration can be approximated as the arithmetic mean of $(t_o + t_p)/2$ and $2t_m$. That is

$$\text{Expected activity time } (t_e) = ((t_o + t_p)/2 + 2t_m)/3 = ((t_o + 4t_m) + t_p)/6$$

Estimating the variance is apparently based on an analogy to the normal distribution where 99% of the area under normal curve is within $\pm 3\sigma$ from the mean or fall within the range approximately 6 standard deviation in length, therefore to, tp or range $tp - to$ enclose about 6 standard deviation of a symmetric distribution, thus, if σ denotes standard deviation, then

$$6\sigma \equiv tp - to \text{ or } \sigma = tp - to / 6$$

Variance of activity time, $\sigma^2 = [1/6(tp - to)]^2$

Standard deviation, $\sigma = \sqrt{\text{Variance}}$

Estimation of project completion time

The probability distribution of time for completing an event can be approximated by the normal distribution due to central limit theorem. Thus, the probability of completing the project by schedule time (T_s) is given by

$\text{Prob}(Z = t_e / \sigma_e)$

Where t_e = expected completion time of the project

Z = number so standard deviations the scheduled time or target date lies away from the mean or expected date.

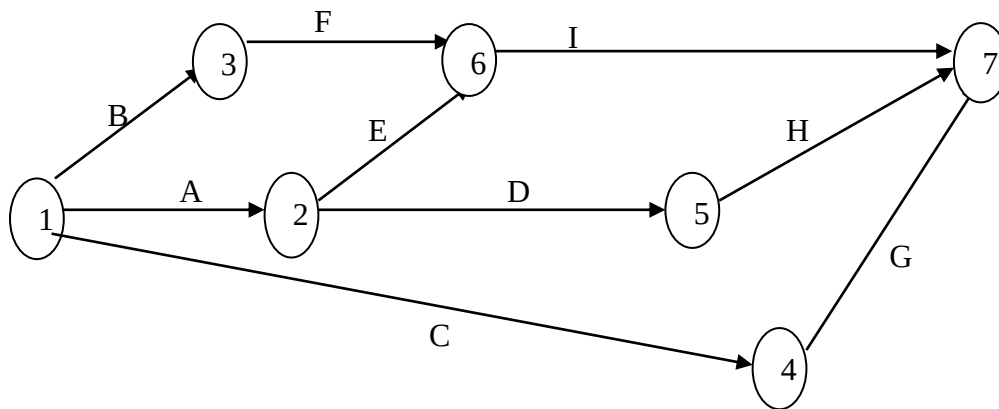
Example1.

1. A project is represented by a network shown below & has the following data

Task:	A	B	C	D	E	F	G	H	I
Predecessors	–	–	–	A	A	B	C	D	E,F
Optimistic time	5	18	26	16	15	6	7	7	3
Pessimistic time:	10	22	40	20	25	12	12	9	5
Most Likely time:	8	20	33	18	20	9	10	8	4

Determine the following

- Expected task time & their variance
- The earliest & latest expected times to reach each event
- The critical path



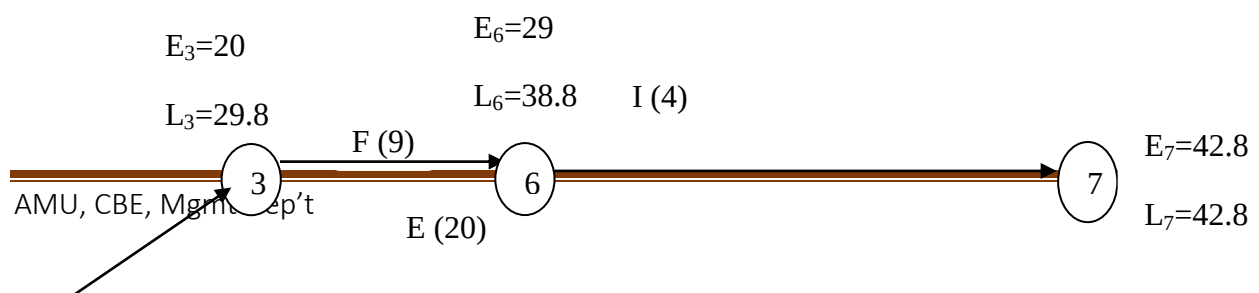
- d. The probability of an event occurring at the expected completion date if the original scheduled time of completing the project is 41.5 weeks.
- e. The duration of the project that will have 95% chance of being completed.

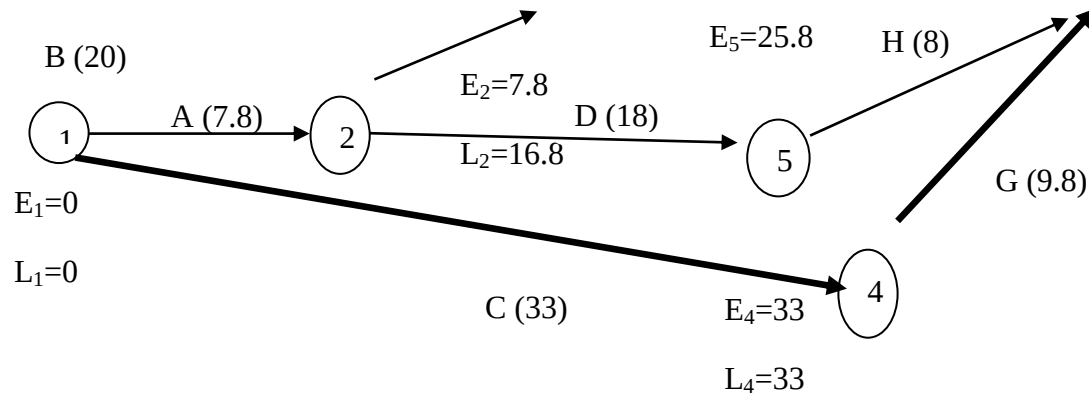
Solution (a) using the following formula, the expected activity time (t_e) or μ) and variance (σ^2) is given in the following table.

$$t_e = ((t_o + 4t_m) + t_p)/6 \text{ and } \sigma^2 = [1/6(t_p - t_o)]^2$$

Activity	t_o	T_p	t_m	$t_e = ((t_o + 4t_m) + t_p)/6$	$\sigma^2 = [1/6(t_p - t_o)]^2$
1-2	5	10	8	7.8	0.694
1-3	18	22	20	20.0	0.444
1-4	26	40	33	33.0	5.444
2-5	16	20	18	18.0	0.444
2-6	15	25	20	20.0	2.778
3-6	6	12	9	9.0	1.000
4-7	7	12	10	9.8	0.694
5-7	7	9	8	8.0	0.111
6-7	3	5	4	4.0	0.111

(b) The earliest and latest estimated time presented in the following graph.





(c). the critical path is shown by thick line in the above figure where E values and L values are the same. The critical path is 1-4-7 and the critical completion time for the project is 42.8 week.

(d) the last event 7 will occur only after 42.8 weeks. For this, we require only the duration of critical activity. This will help us in calculating the standard of the duration of the last event.

$$\text{Expected length of critical path} = 33 + 9.8 = 42.8$$

$$\text{Variance of critical path length} = 5.429 + 0.694 = 6.123$$

It is given that $T_s = 41.5$, $T_e = 42.8$ and $\sigma_e = \sqrt{6.123} = 2.474$. Therefore probability of meeting the scheduled time is given by $\text{Prob}\left(Z \leq \frac{t_s - t_e}{\sigma_e}\right) = \text{prob}(Z \leq -0.5) = 0.3$ from normal distribution table

Thus, the probability that the project can be completed in less than or equal to 41.5 weeks is 0.30. In other words, the probability that the project will be delayed beyond 41.5 weeks is 70%

(e) Given that $\text{Prob}(Z \leq t_s - t_e / \sigma_e) = 0.95$

But $Z_{0.95} = 1.46$, from normal distribution table. Thus

$$1.64 = (T_s - 42.8) / 2.47 \text{ or } T_s = 1.64 \times 2.474 + 42.8 = 46.85 \text{ weeks.}$$

CHAPTER SIX

GAME THEORY

6.1.Introduction

The concepts of game theory provide a common language to formulate structure, analyze and eventually understand different strategical scenarios. Generally, game theory investigates conflict situations, the interaction between the agents and their decisions. A game in the sense of game theory is given by a (mostly finite) number of players, who interact according to given rules. Those players might be individuals, groups, companies, associations and so on.

6.2. Objectives

At the end of this chapter, you should be able to:

- ☞ Understand the concept of game theory
- ☞ Aware the concept of two-person zero-sum game
- ☞ Make a decision under the condition of pure strategy (game with saddle point)
- ☞ Understand the concept of mixed strategy (game without saddle point) differentiate it with pure strategy.
- ☞ Define and understand the rule of dominance

6.3.The Concept of Game Theory

As far as games have defined in mathematical or numerical values, let us describe the ingredients and concepts of the game.

Game: A game is a formal description of a strategic situation.

Game theory: Game theory is the formal study of decision-making where several players must make choices that potentially affect the interests of the other players.

Rules: Mathematical games have strict rules. They specify what is allowed and what isn't.

Outcomes and payoffs: The payoffs may be monetary, or they may express satisfaction.

Uncertainty of the Outcome: A mathematical game is “thrilling” in that its outcome cannot be predicted in advance.

Decision making: A game with no decisions might be boring, at least for the mind.

No cheating: In real-life games cheating is possible. Cheating means; not playing by the rules.

Common knowledge: A fact is common knowledge if all players know it, and know that they all know it, and so on. The structure of the game is often assumed to be common knowledge among the players.

Dominating strategy: A strategy dominates another strategy of a player if it always gives a better payoff to that player, regardless of what the other players are doing.

Extensive game: An extensive game (or extensive form game) describes with a tree how a game is played.

Mixed strategy: A mixed strategy is an active randomization, with given probabilities that determine the player’s decision.

Nash equilibrium: Nash equilibrium, also called strategic equilibrium, is a list of strategies, one for each player, which has the property that no player can unilaterally change his strategy and get a better payoff.

Payoff: A payoff is a number, also called utility that reflects the desirability of an outcome to the player, for whatever reason.

Perfect information: A game has perfect information when at any point in time only one player makes a move, and knows all the actions that have been made until then.

Player: A player is an agent who makes decisions in a game.

Rationality: A player is said to be rational if he seeks to play in a manner which maximizes his own

Strategic form: A game in strategic form, also called normal form, is a compact representation of a game in which players simultaneously choose their strategies.

Strategy: In a game in strategic form, a strategy is one of the given possible actions of a player.

Zero-sum game: A game is said to be zero-sum if for any outcome, the sum of the payoffs to all players is zero.

6.4. Two Person Zero-Sum Game

A game is called zero-sum if the sum of payoffs equals zero for any outcome. That means that the winnings of the winning players are paid by the losses of the losing players.

Example 1: Assume we are playing ROCK-SCISSORS-PAPER for one dollar. Then the payoff matrix is;

	Rock	Scissor	Paper
Rock	0	1	-1
Scissor	-1	0	1
Paper	1	-1	0

The first cell says “0”, which stands for “0; 0” a payoff of 0 for both players. The second cell entry of “1” should be read as “1; -1”, a payoff of 1 for Ann which has to be paid by Beth, therefore a payoff of -1 for Beth.

6.5.Pure strategies (Game with saddle point)

In a zero-sum game, the pure strategies of two players constitute a saddle point if the corresponding entry of the payoff matrix is simultaneously a maximum of row minima and a minimum of column maxima. This decision-making is referred to as the **maximin-minimax principle** to obtain the best possible selection of a strategy for the players.

The game theory for saddle point problem is shown below

Example 1: Consider the example to solve the game whose pay-off matrix is given in the following table as follows:

Game Problem

		Player B	
		1	2
Player A	1	1	3
	2	-1	6

Maximin-Minimax Procedure

		Player B		
		1	2	Row Min
Player A →	1	1	3	1
	2	-1	6	-1
Col Max →		1	6	

If maximum value in row is equal to the minimum value in column, then saddle (equilibrium) point exists. **Maximin = Minimax (1 = 1)**

Therefore, there is a saddle point and player A plays strategy A1 and player B plays strategy B1 and the value of the game is 1.

Example 2: Check whether the following game is given in Table, determinable and fair.

Game Problem

		Player B	
		1	2
Player A	1	7	0
	2	0	8

Solution: The game is solved using maximin-minimax criteria as shown in Table.

Maximin-minimax Procedure

		Player B		
		1	2	Row Min
Player A	1	7	0	0
	2	0	8	0
Column Max		7	8	

The game is strictly neither determinable nor fair.

6.6.Mixed Strategies: Game without Saddle Point

In certain cases, no pure strategy solutions exist for the game. In other words, saddle point does not exist. In all such game, both players may adopt an optimal blend of the strategies called Mixed Strategy to find a saddle point. The optimal mix for each player may be determined by

assigning each strategy a probability of it being chosen. The probabilistic mixed strategy games without saddle points are commonly solved by any of the following methods.

No	Methods	Application to
1	Analytical method	2x2 games
2	Graphics method	2x2, mx2 and 2xn
3	Simplex method	2x2, mx2, 2xn and mxn games

(a) Analytical Method: A 2 x 2 payoff matrix where there is no saddle point can be solved by analytical method. Given the matrix;

$$\begin{matrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{matrix}$$

value of the game is;

$$V = \frac{a_{11}a_{22} - a_{21}a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

With the coordinates,

$$X_1 = \frac{a_{22} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$X_2 = \frac{a_{11} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$Y_1 = \frac{a_{22} - a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$Y_2 = \frac{a_{11} - a_{21}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

Example 1;

saddle point
analytical

A	B	
	5	1
	3	4

It is a 2 x 2 matrix and no
exists. We can solve by
method

$$A \begin{matrix} & B \\ \begin{bmatrix} 5 & 1 \\ 3 & 4 \end{bmatrix} & \begin{matrix} 1 \\ 4 \end{matrix} \\ \begin{matrix} 3 \\ 2 \end{matrix} & \end{matrix}$$

$$V = \frac{a_{11}a_{22} - a_{21}a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})}$$

$$V = \frac{5(4) - 3(1)}{(5+4) - (3+1)}$$

$$V = \frac{20 - 3}{9 - 4}$$

$$V=17/5 \quad S_A = (x_1, x_2) = (1/5, 4/5) \quad S_B = (y_1, y_2) = (3/5, 2/5).$$

(b) Graphical method

The graphical method is used to solve the games whose payoff matrix has

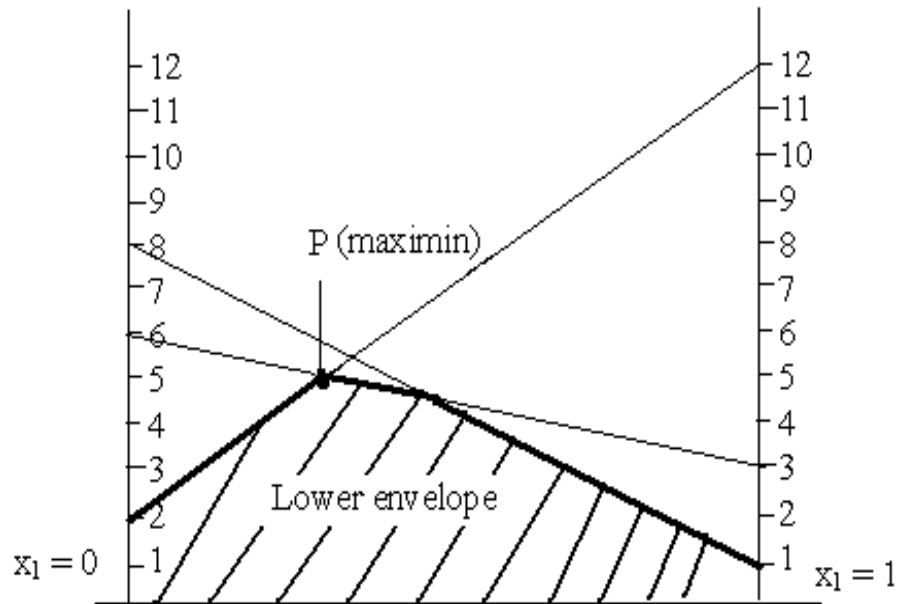
- ✓ Two rows and n columns (2 x n)
- ✓ m rows and two columns (m x 2)
- ☞ Draw two vertical axes 1 unit apart. The two lines are $x_1 = 0$, $x_1 = 1$
- ☞ Take the points of the first row in the payoff matrix on the vertical line $x_1 = 1$ and the points of the second row in the payoff matrix on the vertical line $x_1 = 0$.
- ☞ The point a_{1j} on axis $x_1 = 1$ is then joined to the point a_{2j} on the axis $x_1 = 0$ to give a straight line.
- ☞ Draw 'n' straight lines for $j=1, 2, \dots, n$ and determine the highest point of the lower envelope obtained. This will be the **maximin point**.
- ☞ The two or more lines passing through the maximin point determines the required 2 x 2 payoff matrix. This in turn gives the optimum solution by making use of analytical method.

Example 2;

Solve by graphical method

	B1	B2	B3
A1	1	3	12
A2	8	6	2

Solution



	B2	B3	
A1	3	12	4
A2	6	2	9
	10	3	

$$V = \frac{a_{11}a_{22} - a_{21}a_{12}}{(a_{11} + a_{22}) - (a_{12} + a_{21})} = \frac{6 - 72}{5 - 18}$$

$$V = 66/13$$

$$SA = (4/13, 9/13)$$

$$SB = (0, 10/13, 3/13)$$

6.7.The Rule of Dominance

The principle of dominance in Game Theory (also known as dominant strategy **or** dominance method) states that if one strategy of a player dominates over the other strategy in all conditions then, the later strategy can be ignored.

general, the following rules are used to reduce the size of payoff matrix.

Rules (principles) of dominance

- ✚ Rule 1: If all the elements in a row (say i^{th} row) of a payoff matrix are less than or equal to the corresponding elements of the other row (say j^{th} row) then, the player A will never choose the i^{th} strategy and we say i^{th} strategy is dominated by j^{th} strategy and will delete the i^{th} row.
- ✚ Rule 2: If all the elements in a column (say r^{th} column) of a payoff matrix are greater than or equal to the corresponding elements of the other column (say s^{th} column) then, the player B will never choose the r^{th} strategy or in the other words the r^{th} strategy is dominated by the s^{th} strategy and we delete r^{th} column .
- ✚ Rule 3: A pure strategy may be dominated if it is inferior to average of two or more other pure strategies. Now, consider some simple examples.

Example 1: Given the payoff matrix for player A, obtain the optimum strategies for both the players and determine the value of the game.

A	B		
	6	-3	7
	-3	0	4

Solution

A	B		
	B1	B2	B3
A1	6	-3	7
A2	-3	0	4

When A chooses strategy A1 or A2, B will never go to strategy B3. Hence strategy B3 is redundant.

A	B			Raw minima
	B1	B2	B3	-3
A1	6	-3	7	-3

A2	-3	0	4
Column maxima	6	0	

Minimax (=0), maximin (= -3). Hence, this is not a pure strategy with a saddle point. Let the probability of mixed strategy of A for choosing A1 and A2 strategies are p_1 and $1 - p_1$ respectively. We get $6 p_1 - 3 (1 - p_1) = -3 p_1 + 0 (1 - p_1)$ or $p_1 = 1/4$. Again, q_1 and $1 - q_1$ being probabilities of strategy B, we get $6 q_1 - 3 (1 - q_1) = -3 q_1 + 0 (1 - q_1)$ or $q_1 = 1/4$.

Hence, optimum strategies for players A and B will be as follows:

$$S_A = (A_1=1/4 \text{ \& } A_2=3/4)$$

$$S_B = (B_1=1/4, B_2=3/4 \text{ \& } B_3=0)$$

$$\text{Expected value of the game} = q_1 (6 p_1 - 3(1 - p_1)) + (1 - q_1) (-3 q_1 + 0(1 - q_1)) = 3/4.$$

CHAPTER SEVEN

7. QUEUING THEORY

7.1.Introduction

The queuing theory also called the waiting line theory, owes its development to A K Erlang's effort to analyze telephone traffic congestion with a view to satisfying the randomly arising demand for the services of the Copenhagen automatic telephone system, in year 1909. Queuing Theory is a collection of mathematical models of various queuing systems. It is used extensively to analyze production and service processes exhibiting random variability in market demand (arrival times) and service times.

Queues or waiting lines arise when the demand for a service facility exceeds the capacity of that facility, that is, the customers do not get service immediately upon request but must wait, or the service facilities stand idle and wait for customers.

Queuing theory can be applied to a wide variety of operational situations where imperfect matching between the customers and the service facilities is caused by one's inability to predict accurately the arrival and service time of customers. In particular, it can balance the following two conflicting costs:

- a) Cost of offering the service
- b) Cost incurred due to delay in offering service

Figure1.1. Illustrates both types of costs as a function of level of service. The optimum service level is one that minimizes the sum of the two costs.

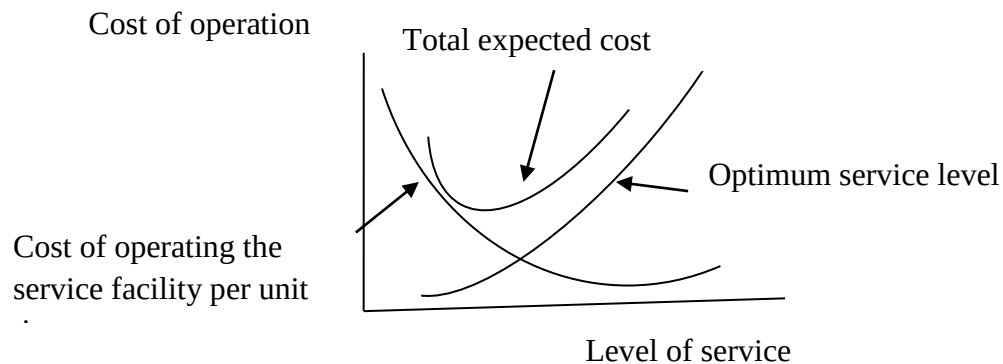


Fig.6.1. Queuing costs vs. level of service

Of the two types of costs mentioned above, the cost of waiting is difficult to estimate. However, it is measured in terms of loss of sales or goodwill when the customer is a human being who has no sympathy with the service. But, if the customer is a machine waiting for repair, then cost of waiting is measured in terms of the cost of lost production. The waiting line is developed because the service to a customer may not be rendered immediately as the customer reaches to the service facility.

7.2.Essential Features of queuing system

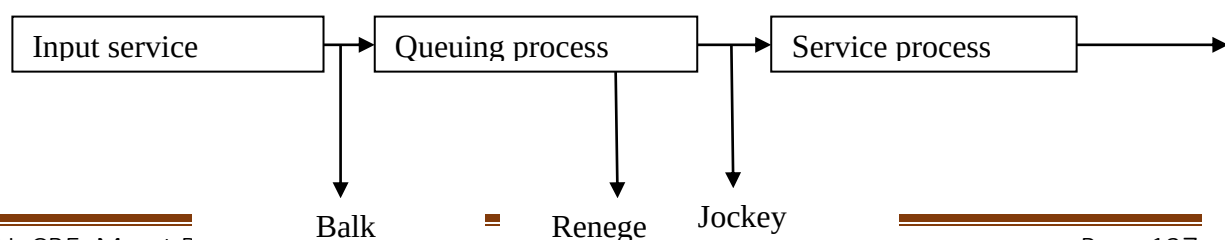
The essential features or characteristics of queuing system are:

- | | |
|--------------------|----------------------------------|
| 1. Input source | 3. Queue discipline |
| 2. Queuing process | 4. Service process(or mechanism) |

The customer's entry into the service system depends upon the queue conditions. If at the time of customer's arrival, the server is idle, then the customer is served immediately. Otherwise, the customer is asked to join the queues, which can have several configurations.

Input source of queue

Service system



An input source is characterized by

- Size of the calling population
- Pattern of arrivals at the system
- Behavior of the arrivals

An input source need not be homogeneous population but may consist of several sub-populations.

- ❖ *Size of calling population*
- ❖ *Pattern of arrivals*
- ❖ *Dynamic arrival process*
- ❖ *Behavior of arrivals*
- ❖ *Static arrival process*

The arrival time distribution can be approximated by one of the following probability distributions;

- ✱ Poisson distribution
- ✱ Exponential distribution
- ✱ Erlang distribution

Frequently in queuing problems, the number of arrivals per unit of time can be estimated by a probability distribution known as the Poisson distribution, as it adequately supports many real-world situations.

The Poisson distribution is a discrete probability distribution of the number of customers (events) arriving in some time interval. The Poisson probability mass function is

$$P(X=n/P_n=\lambda t) = \frac{(\lambda t)^n e^{-\lambda t}}{n!}, n=0, 1, 2, \dots$$

Now, if we are interested in the inter arrival time probability distribution, then the time interval from 0 to t, the probability of no arrival is given by

$$P(X=0/P_n=\lambda t) = \frac{(\lambda t)^0 e^{-\lambda t}}{0!} = e^{-\lambda t}$$

Let us define the random variable T as the time between successive arrivals. Since a customer can arrive at any time, T must be a continuous random variable. The probability of no arrival in the time interval from 0 to t will be equal to the probability that T exceeds t, so we have

$$P(T > t) = p(X = p_n \lambda t) = e^{-\lambda t}$$

The probability that there is an arrival in the time interval from 0 to t is given by

$$P(T \leq t) = 1 - p(T > t) = 1 - e^{-\lambda t}; t \geq 0$$

This probability is also called the cumulative probability distribution function, F(t) of T. Also the distribution of the random variable T is referred to as the exponential distribution. Its probability density function (pdf) is

$$f(t) = \begin{cases} \lambda e^{-\lambda t} & ; t \geq 0 \\ 0 & ; t < 0 \end{cases}$$

In many practical situations, the inter-arrival time is approximated by an exponential distribution (sometimes called the negative exponential distribution).

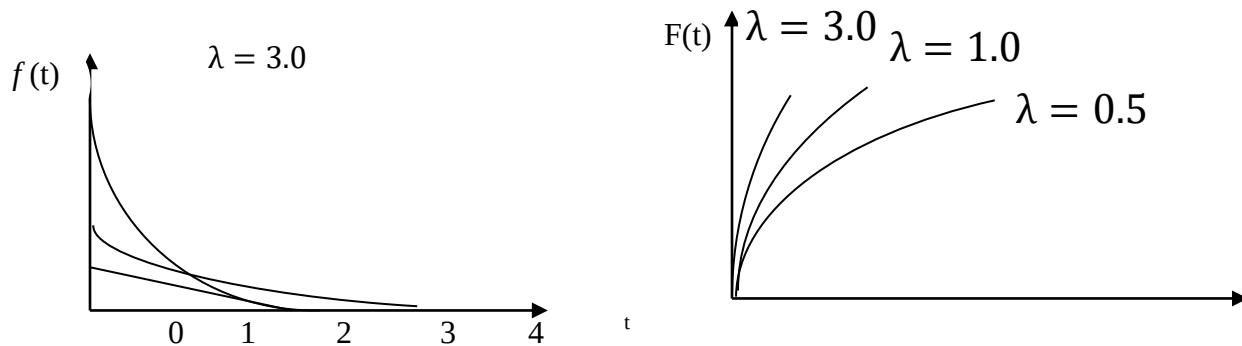


Fig 6.2. _ Exponential probability distribution function fig---Exponential cumulative probability density function

The mean of exponential distribution is the expected or average time between arrivals $E(T)$.

Thus with λ arrivals per unit time, $E(T) = \frac{1}{\lambda}$

Hence from the above discussion we may conclude that the Poisson distribution of arrivals with arrival rate λ is equal to the negative exponential distribution of inter-arrival times with mean inter arrival times $\frac{1}{\lambda}$.

The probability density function of exponential distribution can also be used to compute the probability that the next customer arrives within time, T of the previous arrival.

Example if $\lambda = 1.2$ and $t = 1.5$ $\lambda t = 1.8$ so

$$F(1.5) = P(T \leq 1.5) = 1 - (x=0/p_n=1.8) \\ = 1 - 0.165 = 0.835$$

Also, if customers arrive at the service system at an average rate of 24 customers per hour, and a customer has already arrived, then probability of another customer arriving in the next 5 minutes (i.e. $t = 1/12$ hr) is

$$P(\text{inter-arrival time} \leq 1/12) = 1 - e^{-24(1/12)} \\ = 1 - e^{-2} = 1 - 0.1353 = 0.8647$$

Further if $\lambda = 24$ customers per hour, $t = 1/12$ hour and $\lambda t = 24 \times (1/12) = 2$, the probability of n customers arriving within the next 10 minutes is:

$$P(x = 2/p_n = 2) = \frac{e^{-24(\frac{1}{12})} \left(\frac{24}{12}\right)^2}{2!} = 0.27$$

7.3. Queuing process

The queuing process refers to the number of queues, and their respective lengths. The number of queues depends upon the layout of the service system. Thus, there may be single or multiple queues. Certain service systems adopt “take a number” policy to avoid a formal queue to form; the length (or size) of the queue depends upon the operational situation such as physical space, legal restrictions, and attitude of the customers.

A multiple queue system at a service system can also be finite or infinite. But this has certain advantages such as:

- I. The service provided to the customers can be differentiated by adopting different service rules called queuing discipline
- II. Division of manpower is possible
- III. Customers has the option of joining any queue and can also switch to the end of any other queues
- IV. Balking behavior of the customers can be controlled

Queue discipline

In the queue structure, the important thing to know is the queue discipline. The queue discipline is the order or manner in which customers from the queue are selected for service.

There are a number of ways in which customers in the queue are served. Some of these are:

a) Static queue disciplines

- i. First-come, first-served (FCFS)
- ii. Last-come-first-served (LCFS)

b) Dynamic queue disciplines

- i. Service in Random Order (SIRO).
- ii. Priority Service
- iii. Pri-emptive priority
- iv. Non-pre-emptive priority

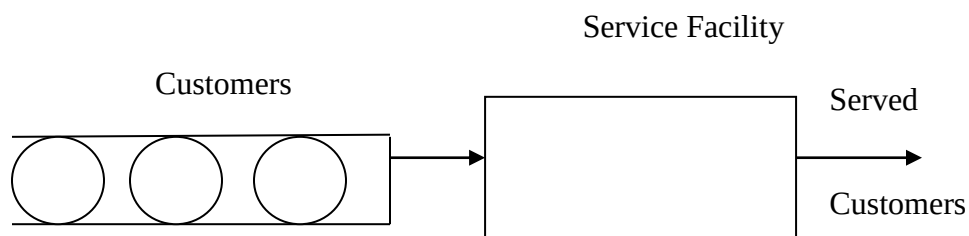
For the queuing models that we shall consider, the assumption would be that the **customers are serviced on the first-come-first-served basis.**

Service process (mechanism)

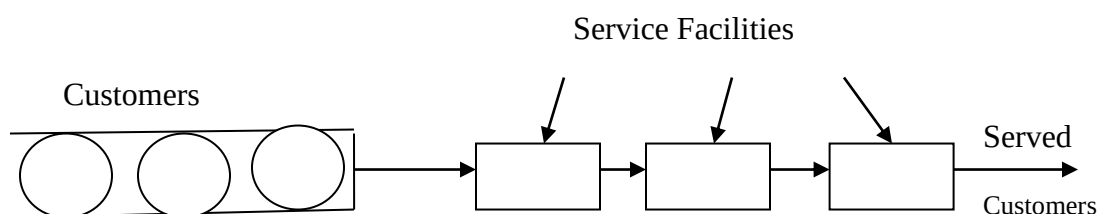
The service process is concerned with the manner in which customers are serviced and leave the service system. It is characterized by

- The arrangement of the service system
- The distribution of the service lines
- Server's behavior
- Management policies

Arrangement of service facilities the capacity of the service mechanism is measured in terms of customers that can be served simultaneously and/or effectively. The service facilities (or servers), commonly known as service channels may be series or in parallel or mixed (partly in series and partly in parallel). The series arrangement consists of a sequence of a number of service facilities such that a customer must go through one facility after another in a particular sequence before the whole service is completed. Each service facility may however, work independently of the others, having its own rules of service.

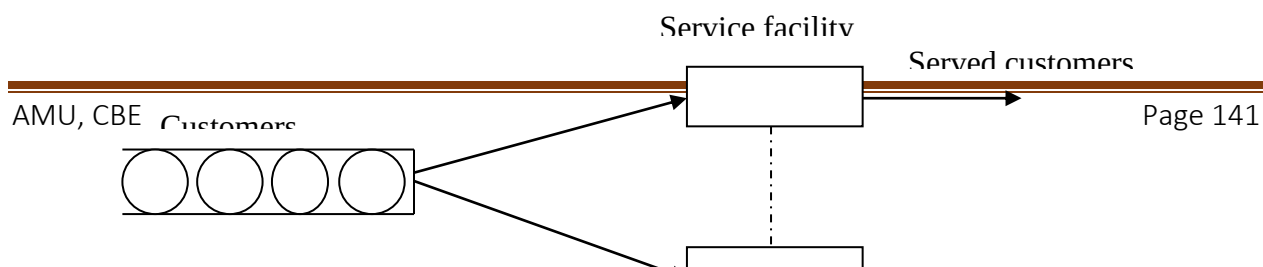


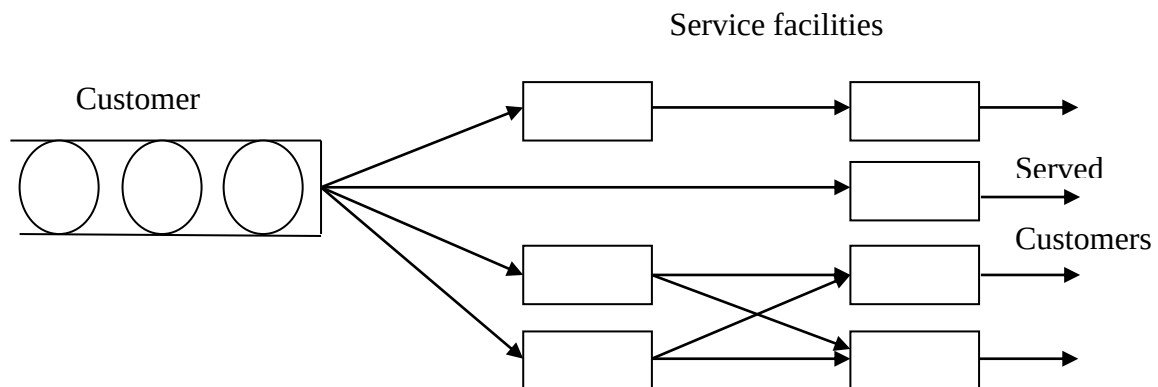
a) Single queue, single server



b) Single queue

In the parallel arrangement the incoming customer may join the queue of his choice in front of service facilities or may be served by any service facilities.





Mixed Arrangement of Service Facilities

Distribution or service time-the time taken by the server from the commencement of service to the completion of service for a customer is known as the **service time**. If μ is the average service rate, then the expected number of customers served during time interval t will be μt . Thus, if we consider zero time at the start of service, the probability that services is not completed by time t is given by

$$p(x = 0/p_n = \mu t)e^{-\mu t}$$

If the random variable T represents the service time, then the probability of service completion within time t is given by

$$p(T \leq t) = 1 - e^{-\mu t}, t \geq 0$$

7.4.Probability distribution in queuing system

It is assumed that customers joining a queuing system arrive in random manner and follow a Poisson distribution or equivalently the inter-arrival times follow exponential distribution.

In most of the cases, service times are also assumed to be exponentially distributed.

The number of arrivals and departures (those served) during an interval of time in a queuing system is controlled by the following assumptions (also called axioms).

- The probability of an event (arrival or departure) occurring during the time interval $(t, t+\Delta t)$ depends on the length of time interval Δt .

- b. The probability of more than one event occurring during the time interval $(t, t + \Delta t)$ is negligible. It is denoted by $0(\Delta t)$.
- c. At most one event (arrival or departure) can occur during a small time interval Δt . the probability of an arrival during the time interval $(t, t + \Delta t)$ is given by

$$p_1(\Delta t) = \lambda \Delta t + 0(\Delta t)$$

Where λ is a constant and independent of the total number of arrivals up to time t ; Δt is a small-time interval and $0(\Delta t)$ represents the quantity that becomes negligible when compared to

Δt as $\Delta t \rightarrow 0$, i.e.

$$\lim_{\Delta t \rightarrow 0} \{ 0(\Delta t) / \Delta t \} = 0$$

Distribution of arrivals (pure birth process)

The arrival process assumes that the customers arrive at the queuing at the queuing system and never leave it. Such a process is called pure birth process. The aim is to derive an expression for the probability $P_n(t)$ of n arrivals during time interval $(t, t + \Delta t)$. The terms commonly used in the development of various queuing models are the following:

Δt = a time interval so small that the probability of more than one customer's arrival is negligible, i.e. during any given small interval of time Δt only one customer can arrive.

$\lambda \Delta t$ = probability that a customer will arrive in the system during time Δt .

$1 - \lambda \Delta t$ = probability that no customer will arrive in the system during time Δt .

If the arrivals are completely random, then the probability distribution of a number of arrivals in a fixed time interval follows Poisson distribution.

Distribution of inter-arrival times (exponential process)

If the number of arrivals, n , in time t follows the Poisson distribution, then

$$p_n(t) = \frac{(\lambda t)^n}{n!} e^{-\lambda t}, n = 0, 1, 2, \dots$$

Is an associated random variable defined as the inter-arrival T follows the exponential distribution $f(t) = \lambda e^{-\lambda t}$ and vice versa.

Markovian property of inter- arrival times

The Markovian property of inter arrival times states that the probability that a customer currently in service is completed at some time t is independent of how long he has already been in service. That is

$$\text{prob.} \{T \geq t_1 / T \geq t_0 \mid T \geq t_0\} = \text{Prob.} = \{T \geq t_1 \mid T \geq t_0\} = \text{prob.} \{0 \leq T \leq t_1 - t_0\}$$

Where T is the time between successive arrivals

Distribution of departures (pure death process)

The departure process assumes that no customer joins the system while service is continued for those who are already in the system. Let, at time $t=0$ (starting time) there be $N \geq 1$ customers in the system. Since service is being provided at the rate of μ .

Basic axioms

- i. Probability of the departure during time Δt is $\mu \Delta t$.
- ii. Probability of more than one departure between time t and $t + \Delta t$ is negligible.
- iii. The number of departures in non-overlapping intervals are statistically independent.

The following terms are used in the development of various queuing models.

$\mu \Delta t$ = probability that a customer in-service at time t will complete service during time Δt .

$1 - \mu \Delta t$ = probability that the customer in-service at time t will not complete service during time Δt .

The probability density function $s(t)$ of service time is given by

$$s(t) = \begin{cases} \mu e^{-\mu t}; & 0 \leq t \leq \infty \\ 0 & ; t < 0 \end{cases}$$

This shows that service times follows negative exponential distribution with mean $1/\mu$ and variance $1/\mu^2$.

Distribution of service times

The service time of short duration have the highest probability of occurrence. As the service time increases, the probability of occurrence tails off exponentially towards zero probability.

The area under the negative exponential distribution curve is determined as:

$$\begin{aligned} F(T) &= \int_0^T \mu e^{-\mu t} dt = [-\mu e^{-\mu t}]_0^T \\ &= -e^{-\mu T} + e^0 = 1 - e^{-\mu T} \end{aligned}$$

It is also described as:

$$F(T) = f(t \leq T) = 1 - e^{-\mu T}$$

Where $F(T)$ is the area under the curve to the left of T . Thus

$1 - F(T) = f(t > T) = e^{-\mu T}$ is the area under the curve to the right of T .

7.5. Single Server Queuing Models

Model I {M/M1}: (∞ /FCFS)} Exponential service-unlimited Queue

The derivation of this model is based on certain assumptions about the queuing system:

- I. Exponential distribution of inter-arrival time or Poisson distribution of arrivals
- II. Single waiting line with no restriction on length of queue (i.e infinite capacity) and no balking or reneging
- III. Queue discipline is “first-come, first served”
- IV. Single server with exponential distribution of service time

The following events (possibilities) may occur during a small interval of time, Δt just before time t . it is assumed that the system is in state n (number of customers) at time t .

- 1) The system is in state n (number of customers) and no arrival and no departure, leaving the total to n customer.
- 2) The system is in state $n+1$ (number of customers) and no arrival and one departure, reducing the total to n customers.
- 3) The system is in state $n-1$ (number of customers) and one arrival and no departure, bringing the total to n customers.

We will begin with determining probability, P_n of n customers in the system at time t and value of its various operating characteristics is summarized as follows:

Figure XX illustrates the process of determining P_n by considering each possible number of customers either waiting or receiving service at each state which may be entered by the arrival of new customer or left by the completion of the leading customer's service.



Step 1 obtain system of differential difference equation

If $p_n(t)$ is the probability of a customers at time t in the system, then the probability that the system will contain n customers at time $(t + \Delta t)$ can be expressed as the sum of the joint probabilities of the three mutually exclusive and collectively exhaustive cases as mentioned above. That is

For $n \geq 1$ and $t \geq 0$.

$p_n(t + \Delta t) = p_n(t) - \text{prob. (no arrival in } \Delta t \text{ and on departure in } \Delta t)$

$+ p_{n+1}(t) \cdot \text{Prob. (no arrival in } \Delta t \text{ and one departure in } \Delta t)$
 $+ p_{n-1}(t) \cdot \text{Prob. (no arrival in } \Delta t \text{ and no departure in } \Delta t)$
 $p_n(t) \{ \text{prob. (no arrival in } \Delta t) \times \text{prob. (no departure in } \Delta t) \}$
 $p_{n+1}(t) \cdot \text{Prob. (no arrival in } \Delta t) \times \text{prob. (one departure in } \Delta t)$
 $+ p_{n-1}(t) \cdot \text{Prob. (one arrival in } \Delta t) \times \text{prob. (one departure in } \Delta t)$
 $p_n(t) \{ 1 - \lambda \Delta t \} (1 - \mu \Delta t) + p_{n+1}(t) \{ 1 - \lambda \Delta t \} \mu \Delta t + p_{n-1}(t) \{ \lambda \Delta t \} \{ 1 - \mu \Delta t \}$
 $p_n(t) \{ 1 - (\lambda + \mu) \Delta t \} + p_{n+1}(t) \mu \Delta t + p_{n-1}(t) \lambda \Delta t$ Terms involving $(\Delta t)^2$ since Δt is very small, therefore terms involving $(\Delta t)^2$ can be neglected. Subtracting $p_n(t)$ from both sides and dividing by Δt , we got

$$\frac{p_n(t + \Delta t) - p_n(t)}{\Delta t} = \lambda p_{n+1}(t) - (\lambda + \mu) p_n(t)$$

Taking limit on both sides as $\Delta t \rightarrow 0$, then above equation reduces to

$$p'_{n(t)} = \frac{d}{dt} \{ p_{n(t)} \} = \lambda p_{n-1}(t) + \mu p_{n+1}(t) - (\lambda + \mu) p_n(t); n \geq 1 \quad (1)$$

Similarly, if there is no customer in the system at time $(t + \Delta t)$, when there will be no service completion during Δt . Thus for $n=0$ and $t \geq 0$, we have only two probabilities instead of three.

The resulting equation is

$$\frac{p_n(t + \Delta t) - p_0(t)}{\Delta t} = \mu p_1(t) - \lambda p_0(t) + \frac{0(\Delta t)}{\Delta t}$$

Taking limit on both sides as $\Delta t \rightarrow 0$, we get

$$p'_{0(t)} = \frac{d}{dt} \{ p_{0(t)} \} = \mu p_1(t) - \lambda p_0(t); n = 0 \quad (2)$$

Step 2 obtaining the system of steady state equations

In the steady-state, $p_n(t)$ is independent of time t , and the rate of change $p_n(t)$ can be considered to be zero. That is

$$\lim_{n \rightarrow \infty} p_n(t) = p_n$$

$$\text{And } \lim_{n \rightarrow \infty} \frac{d}{dt} \{ p_{n(t)} \} = 0; n = 0, 1, 2, \dots$$

Consequently, equations (1) and (2) may be written as:

$$\lambda p_{n-1} + \mu p_{n+1} - (\lambda + \mu) p_n = 0; n \geq 1 \quad (3)$$

$$\mu p_1 - \lambda p_0 = 0; n = 0 \quad (4)$$

Thus these equations constitute the system of steady-state difference equations. The solution of these equations can be obtained either by (i) iterative method, (ii) generating functions method, or (iii) linear operator method. Here we shall be using the iterative method to find the values of $P_1, P_2, \dots$ in terms of P_0, λ and μ .

Step 3 Solve the system of difference equations

From equation 4 we get

$$\lambda p_0 = \mu p_1 \text{ or } p_1 = \left(\frac{\lambda}{\mu}\right) p_0; n = 0$$

Putting $n=1$ in equation (3), we get

$$0 = -(\lambda + \mu)p_1 + \lambda p_0 + \mu p_2$$

$$\text{Or } p_2 = \left(\frac{\lambda + \mu}{\mu}\right) p_1 - \frac{\lambda}{\mu} p_0 = \left(\frac{\lambda + \mu}{\mu}\right) \frac{\lambda}{\mu} p_0 - \frac{\lambda}{\mu} p_0 = \left(\frac{\lambda}{\mu}\right)^2 p_0$$

$$p_3 = \left(\frac{\lambda + \mu}{\mu}\right) p_2 - \left(\frac{\lambda}{\mu}\right) p_1$$

$$= \left(\frac{\lambda + \mu}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^2 p_0 - \left(\frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right) p_0 = \left(\frac{\lambda}{\mu}\right)^3 p_0; n=2$$

In general, by using the inductive principle, we get

p_n = probability of being in state n (i.e. n customers in the system)

$$= \left(\frac{\lambda}{\mu}\right)^n p_0; n=1,2,\dots$$

To obtain the value of p_0 , we make use of the fact that (sum of all possibilities)

$$\sum_{n=0}^{\infty} p_n = p_0 + p_1 + \dots + p_n + \dots = 1$$

Or

$$\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n p_0 = 1 \text{ or } p_0 = \frac{1}{\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n}; \frac{\lambda}{\mu} < 1$$

The denominator of this expression is an infinite geometric series, whose sum is:

$$\sum_{n=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^n = \frac{1}{1 - (\lambda/\mu)}$$

$$\text{And hence } p_0 = 1 - \frac{\lambda}{\mu} = 1 - p$$

$$\text{And } p_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right) = p^n (1-p); p < 1, n=0,1,2,\dots$$

p_w = Probability that an arriving customer has to wait = $1 - p_0 = \lambda/\mu$

This expression gives the required probability distribution for exactly n customers in the queuing system. Hence p_0 denotes the probability of the system being empty (no customer).

Step 4 obtain probability density function of waiting time excluding service time distribution

The waiting time distribution of each customer in the steady-state is same, and it is a continuous random variable except that there is a non-zero probability that the delay will be zero probability that the delay will be zero, i.e. waiting time is zero. Let w be the time required by the server to serve all the customers present in the system at a particular time in the steady state. Let $\phi_w(t)$ be the probability distribution function of w , i.e. $\phi_w(t) = p(w \leq t), 0 \leq t \leq \infty$.

If an arriving customer find $n(\geq 1)$ customers already in the system, then in order for a customer to get service at time between 0 and t, all the customers must have been serviced by time t. let $s_1, s_2, \dots, s_n$ denote service of n customers, respectively. Thus

$$\begin{cases} 0 & ; n = 0 \\ \sum_{i=1}^n s_i & ; n \geq 1 \end{cases}$$

The distribution function of waiting time w for a customer who has to wait is given by

$$p\{w \leq t\} = \begin{cases} p_0 = 1-p & ; n=0, t=0 \\ p \left\{ \sum_{i=1}^n s_i \leq t \right\} & ; n \geq 1, t > 0 \end{cases}$$

Since the service time for each customer is independent and identically distributed, therefore its probability function density is $\phi_s(t) = \mu e^{-\mu t}$, $t > 0$, where μ the mean service rate is. Thus

$$\begin{aligned} \phi_n(t) &= \sum_{i=1}^n [p_n \text{ prob}\{(n-1) \text{ customers got service at time } t\}] \\ &\times \text{prob}\{\text{one customer is under service during toime } \Delta t\} \\ &= \sum_{i=1}^n \left(1 - \frac{\lambda}{\mu}\right) \left(\frac{\lambda}{\mu}\right)^n \left[\frac{(\mu t)^{n-1} e^{-\mu t}}{(n-1)!} \right] \mu \Delta t \end{aligned}$$

Thus the expression for $\phi_w(t)$ may be written as

$$\begin{aligned} \phi_w(t) &= p\{w \leq t\} = \begin{cases} 1-p & ; t=0 \\ \sum_{n=1}^{\infty} p_n \int_0^t \phi_n(t) dt \end{cases} \\ &= \begin{cases} 1-p & ; t=0 \\ \sum_{n=1}^{\infty} p^n (1-p) \int_0^t \frac{(\mu t)^{n-1} \mu e^{-\mu t} dt}{(n-1)!} & ; t > 0 \end{cases} \\ &= \begin{cases} 1-p & ; t=0 \\ \int_0^t p(1-p) \mu e^{-\mu t} dt \sum_{n=1}^{\infty} \frac{(\mu p t)^{n-1}}{(n-1)!} & ; t > 0 \end{cases} \\ &= \begin{cases} 1-p & ; t=0 \\ p(1-p) \int_0^t \mu e^{-\mu(1-p)t} dt \end{cases} \end{aligned}$$

This shows that the waiting time distribution is discontinuous at $t=0$ and continuous in the range $0 < t < \infty$. thus expression for $\phi_w(t)$ may also be written as

$$\begin{aligned} \phi_w(0) &= 1-p & ; t=0 \\ \phi'_w(t) &= \lambda(1-p)e^{-\mu(1-p)t} dt \\ &= \lambda\left(1 - \frac{\lambda}{\mu}\right)e^{-(\mu-\lambda)t} dt & ; t > 0 \end{aligned}$$

Step 5 calculate busy period distribution

For the busy period distribution, let the random variable w denote the total time (waiting and service) that a customer spends in the system. Then the probability density function for the distribution is given by

$$\begin{aligned}\phi(w: w > 0) &= \frac{\phi(w)}{\text{prob}(w > 0)}; \phi(w) = \frac{d}{dt} \{\phi_w(t)\} \\ &= \frac{\phi(w)}{\int_0^\infty \phi(t) dt} = \frac{\lambda(1 - \frac{\lambda}{\mu})e^{-(\mu-\lambda)t}}{\int_0^\infty \lambda(1 - \frac{\lambda}{\mu})e^{-(\mu-\lambda)t} dt} \\ &= \frac{\lambda(1 - \frac{\lambda}{\mu})e^{-(\mu-\lambda)t}}{\frac{\lambda}{\mu}} = (\mu - \lambda)e^{-(\mu-\lambda)t}; t = 0\end{aligned}$$

Now $\int_0^\infty \phi(w: w > 0) dt = \int_0^\infty (\mu - \lambda) e^{-(\mu-\lambda)t} dt = 1$

Which is required distribution of busy period

Remark if λ is viewed as a measurement of demand for service, and μ as the capacity of the service facility, then $\mu - \lambda$ represents the ‘excess capacity’ of the system to fill the demand.

Performance measure for Model I

1. a) Expected number of customers in the system (customers in the line plus the customer being served)

$$\begin{aligned}L_s &= \sum_{n=0}^{\infty} n P_n = \sum_{n=0}^{\infty} n(1-p)p^n, 0 < p < 1 \\ &= (1-p) \sum_{n=0}^{\infty} n p^n = p(1-p) \sum_{n=1}^{\infty} n p^{n-1} \\ &= p(1-p) \{1 + 2p + 3p^2 + \dots\} \\ &= (1-p) \left\{ \frac{p}{(1-p)^2} \right\} [\text{sum of an arithmetico-geometric series}]\end{aligned}$$

Or $L_s = \frac{p}{1-p} = \frac{\lambda}{\mu-\lambda}; p = \frac{\lambda}{\mu}$

- b) Expected number of customers waiting in the queue (i.e. queue length)

$$\begin{aligned}L_q &= \sum_{n=1}^{\infty} (n-1) p_n = \sum_{n=1}^{\infty} n P_n - \sum_{n=1}^{\infty} p_n \\ &= \sum_{n=0}^{\infty} n P_n - \left[\sum_{n=0}^{\infty} p_n - p_0 \right] \\ L_s - (1 - p_0) &= \frac{\lambda}{\lambda - \mu} - \frac{\lambda}{\mu}; 1 - p_0 = \frac{\lambda}{\mu}\end{aligned}$$

$$L_q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

2. a) expected waiting time for a customer in the queue

$$\begin{aligned} W_q &= \int_0^{\infty} t \cdot \left\{ \frac{d}{dt} \phi_w(t) \right\} dt \\ &= \int_0^{\infty} t \cdot \lambda (1 - p) e^{-\mu(1-p)t} dt \end{aligned}$$

Integrating by parts $W_q = 1(1 - p) \left[\frac{te^{-\mu(1-p)t}}{-\mu(1-p)} - \frac{e^{-\mu(1-p)t}}{\mu^2(1-p)^2} \right]_0^{\infty}$

$$= \lambda \left(1 - \frac{\lambda}{\mu} \right) \frac{1}{(\mu - \lambda)^2}$$

Or $W_q = \frac{\lambda}{\mu(\mu - \lambda)} = \frac{L_q}{\lambda}$

- b) Expected waiting time for a customer in the system (waiting and service)

$$W_s = \text{Expected waiting time in queue} + \text{Expected service time}$$

$$= W_q + \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)} + \frac{1}{\mu}$$

Or $= \frac{1}{\mu - \lambda} = \frac{L_s}{\lambda}$

3. probability that the number of customers in the system is greater than or equal to k

$$p(n \geq k) = \left(\frac{\lambda}{\mu} \right)^k; = \left(\frac{\lambda}{\mu} \right)^{k+1}$$

4. the variance (fluctuation) of queue length

$$\begin{aligned} Var(n) &= \sum_{n=1}^{\infty} n^2 p_n - \left(\sum_{n=1}^{\infty} n P_n \right)^2 \\ &= \sum_{n=1}^{\infty} n^2 p_n - (L_s)^2 = \sum_{n=1}^{\infty} n^2 (1 - p) p^n - \left(\frac{p}{1 - p} \right)^2 \\ &= (1 - p) [1 \cdot p^2 + 2^2 \cdot p^2 + 3^2 \cdot p^3 + \dots] - \left(\frac{p}{1 - p} \right)^2 \\ Var(n) &= \frac{p}{(1 - p)^2} = \frac{\lambda \mu}{(\mu - \lambda)^2} \end{aligned}$$

probability that the queue is non-empty

$$\begin{aligned} p(n > 1) &= 1 - p_0 - p_1 \\ &= 1 - \left(1 - \frac{\lambda}{\mu} \right) - \left(1 - \frac{\lambda}{\mu} \right) \left(\frac{\lambda}{\mu} \right) = \left(\frac{\lambda}{\mu} \right)^2 \end{aligned}$$

5. probability of k or more customers in the system

$$\begin{aligned} p(n \geq k) &= \sum_{n=k}^{\infty} p_n = \sum_{n=k}^{\infty} (1 - p) p^k \\ &= (1 - p) p^k \sum_{n=k}^{\infty} p^{n-k} \end{aligned}$$

$$= (1-p)p^k[1 + p + p^2 + \dots] = \frac{(1-p)p^k}{(1-p)} = p^k$$

$$\text{Or } p(n \geq k) = \left(\frac{\lambda}{\mu}\right)^k; p(n > k) = \left(\frac{\lambda}{\mu}\right)^{k+1}$$

6. expected length of non-empty queue

$$L_b = \frac{\text{Expected length of waiting line}}{p(n > 1)}$$

$$= \frac{L_b}{p(n > 1)} = \frac{\frac{\lambda^2}{\mu}(\mu - \lambda)}{(\lambda/\mu)^2} = \frac{\mu}{\mu - \lambda}$$

7. probability of an arrival during the service time when system contains r customers

$$p(n = r) = \int_0^\infty p_r(t)s(t).dt = \int_0^\infty \frac{(\lambda t)^r e^{-\lambda t}}{r!} \cdot \mu e^{-\mu t} dt$$

$$= \frac{\lambda^r \mu}{r!} \int_0^\infty e^{-(\lambda + \mu)t} \cdot t^r dt = \frac{\lambda^r \mu}{r!} \frac{p(r+1)}{(\lambda + \mu)^{r+1}}$$

$$= \left(\frac{\lambda}{\lambda + \mu}\right)^r \left(\frac{\mu}{\lambda + \mu}\right); \text{ since } \int_0^\infty e^{-xt} t^n dt = \frac{p(n+1)}{x^{n+1}}; p(r+1) = r!$$

Example

A television repairman finds that the time spent on his jobs has an exponential distribution with a mean of 30 minutes. If he repairs sets in the order in which they come in, and if the arrival of sets follows a Poisson distribution approximately with an average rate of 10 per 8-hour day, what is the repairman's expected idle time each day? How many jobs are headed of the average set just brought in?

Solution

From the data of the problem, we have

$$\lambda = 10/8 = 5/4 \text{ sets per hour; and } \mu = (1/30)60 = 2 \text{ sets per hour}$$

a) Expected idle time of repairman each day

Number of hours for which the repairman remains busy in an 8 hour day (traffic intensity) is given by

$$(8) \left(\frac{\lambda}{\mu}\right) = (8) \left(\frac{5}{8}\right) = 5 \text{ hours}$$

Hence the idle time for a repairman in an 8-hours day will be: (8-5) = 3 hours

b) Expected (or average) number of TV sets in the system

$$L_s = \frac{\lambda}{\mu - \lambda} = \frac{5/4}{2 - (5/4)} = \frac{5}{3} = 2(\text{approximately}) \text{ TV sets}$$

Example 2

Arrivals at telephone booth are considered to be Poisson with an average time of 10 minutes between one arrival and the next. The length of phone call is assumed to be distributed exponentially with mean three minutes.

- What is the probability that a person arriving at the booth will have to wait?
- The telephone department will install a second booth when convinced that an arrival would expect waiting for at least 3 minutes for a phone call. By how much should the flow of arrivals increase in order to justify a second booth?
- What is the average length of queue that forms from time to time?
- What is the probability that it will take him more than 10 minutes altogether to wait for the phone and complete his call?

Solution

From the data of the problem we have

$\lambda = 1/10 = 0.10$ person per minute and $\mu = 1/3 = 0.33$ person per minute

- Probability that a person has to wait at the booth

$$p(n > 0) = 1 - p_0 = \lambda / \mu \\ = 0.10 / 0.33 = 0.3$$

- The installation of second booth will be justified only if the arrival rate is more than the waiting time. Let λ' be the increased arrival rate. Then expected waiting time in the queue will be

$$W_q = \frac{\lambda'}{\mu(\mu - \lambda')} \\ 3 = \frac{\lambda'}{0.33(0.33 - \lambda')} \text{ or } \lambda' = 0.16$$

Where $W_q = (3 \text{ given})$ and $\lambda = \lambda'$ (say) for the second booth

Hence, the increase in the arrival rate is $0.60 - 0.10 = 0.06$ arrivals per minute

- Average length of non-empty queues

$$L_b = \frac{\mu}{\mu - \lambda} = \frac{0.33}{0.23} = 2 \text{ customers approximately}$$

- Probability of waiting for 10 minutes or more is given by

$$p(t \geq 10) = \int_0^{\infty} \frac{\lambda}{\mu} (\mu - \lambda) e^{-(\mu - \lambda)t} dt \\ = \int_{10}^{\infty} (0.3) (0.23) e^{0.23t} dt = 0.069 \left[\frac{e^{-0.23t}}{-0.23} \right]$$

This shows that 3 percent of the arrivals on average will have to wait for 10 minutes or more before they can use the phone.

Model II $\{(M/M/1):(\infty/SIRO)\}$

This model is identical to the model I with a difference only in queue discipline. Since the derivation of P_n is independent of any specific queue discipline, therefore in this model also we have

$$P_n = (1 - p)p^n; n = 1, 2, \dots$$

Consequently, other results will also remain unchanged in any queuing system as long as P_n remain unchanged.

Model III $\{(M/M/1) : (N/FCFS)\}$ exponential service, Finite (or limited) Queue

The model is different from model I in respect of the capacity of the system. Suppose that no more than N customers can be accommodated at any time in the system due to certain reasons. Thus any customer arriving when the system is already contains N customers does not enter the system and is lost.

The difference equations derived in model I will also be same for this model as long as $n < N$. The systems of steady-state difference equations for this model are:

$$\begin{aligned} \lambda p_0 &= \mu p_1 & ; n = 0 \\ (\lambda + \mu) p_n &= \lambda p_{n-1} + \mu p_{n+1} & ; n = 1, 2, \dots, N-1 \\ \lambda p_{N-1} &= \mu p_N & ; n = N \end{aligned}$$

Assumptions of this model are same as that of Model I except that the length of the queue is limited. Moreover, in this case the service rate does not have to exceed arrival rate ($\mu > \lambda$) in order to obtain steady state conditions.

Using the usual procedure from first two difference equations the probability of a customer in the system for $n = 0, 1, 2, \dots, N$ are obtained as follows.

$$P_n = (\lambda/\mu)^n p_0; n \leq N$$

Now in order to obtain the value of p_0 use the fact that

$$\begin{aligned} 1 &= \sum_{n=0}^N p_n = \sum_{n=0}^N (\lambda/\mu)^n p_0 \\ &= p_0 \sum_{n=0}^N (\lambda/\mu)^n = p_0 \sum_{n=0}^N p^n \\ &= p_0 [1 + p + p^2 + \dots + p^N] = p_0 \left[\frac{1-p^{N+1}}{1-p} \right] \end{aligned}$$

$$\text{And consequently, } p_0 = \frac{1-p}{1-p^{N+1}}; p \neq 1 \text{ and } p < 1; p = \lambda/\mu$$

$$\begin{cases} \left(\frac{1-p}{1-p^{N+1}} \right) p^n ; n \leq N; p \neq 1 (\lambda \neq \mu) \\ \frac{1}{N+1} ; p = 1 (\lambda = \mu) \end{cases}$$

The steady-state solution in this case exists even for $p > 1$. this is due to the limited capacity of the system which controls the arrivals by the queue length ($= N - 1$) not by the relative rates of arrivals and departure, λ and μ . if $\lambda < \mu$ and $N \rightarrow \infty$, then $p_n = \left(1 - \frac{\lambda}{\mu}\right) (\lambda/n)^n$, which is same as in model I.

Performance Measure for Model III

1. Expected number of customers in the system

$$\begin{aligned} L_s &= \sum_{n=1}^N np_n = \sum_{n=1}^N n \left(\frac{1-p}{1-p^{N+1}} \right) p^n \\ &= \frac{1-p}{1-p^{N+1}} \sum_{n=0}^N np^n = \frac{1-p}{1-p^{N+1}} (p + 2p^2 + 3p^3 + \dots + Np^N) \\ &\begin{cases} \frac{p}{1-p} - \frac{(N+1)p^{N+1}}{1-p^{N+1}} ; p \neq 1 (\lambda \neq \mu) \\ \frac{N}{2} ; p = 1 (\lambda = \mu) \end{cases} \end{aligned}$$

2. Expected Queue length or expected number of customers waiting in the system

$$L_q = L_s - \frac{\lambda}{\mu} = L_s - \frac{\lambda(1-p_N)}{N}$$

3. Expected waiting time of a customer in the system (waiting + service)

$$W_s = \frac{L_s}{\lambda(1-p_N)}$$

4. Expected waiting time of a customer in the queue

$$W_q = W_s - \frac{1}{\mu} \text{ or } \frac{L_q}{\lambda(1-p_N)}$$

5. Fraction of potential customers lost (=fraction of time system is full)

$$P_N = p_0 p^N$$

Effective arrival rate, $\lambda_e = \lambda(1-p_N)$

Effective traffic intensity, $\rho_e = \lambda_e/\mu$

Example

Consider a single server queuing system with Poisson input, exponential service times. Suppose the mean arrival rate is 3 calling units per hour, the expected service time is 0.25 hour and the

maximum permissible calling units in the system is two. Derive the steady-state probability distribution of the distribution of the number of calling units in the system, and then calculate the expected number in the system.

Solution

From the data of the problem, we have

$\lambda = 3$ units per hour; $\mu = 4$ per hour, and $N=2$

Then traffic intensity, $p = \lambda/\mu = 3/4 = 0.75$

The steady-state probability distribution of the number of n customers (calling units) in the system is

$$p_n = \frac{(1-p)p^n}{1-p^{N+1}}; p \neq 1$$

$$= \frac{(1-0.75)(0.75)^n}{1-(0.75)^{2+1}} = (0.43)(0.75)^n$$

$$p_0 = \frac{(1-p)}{1-p^{N+1}} = \frac{1-0.75}{1-(0.75)^{2+1}} = \frac{0.25}{1-(0.75)^3} = 0.431$$

The expected number of calling units in the system is given by

$$L_s = \sum_{n=1}^N np_n = \sum_{n=1}^2 n(0.43)(0.75)^n$$

$$= 0.43 \sum_{n=1}^2 n(0.75)^n = 0.43\{(0.75) + 2(0.75)^2\} = 0.81$$

REVIEW QUESTIONS FOR ALL CHAPTERS

Part I: Multiple choice

1. Operations research analysts do not
 - A. Predict future operation
 - B. Build more than one model
 - C. Collect relevant data
 - D. Recommend decision and accept
2. Decision variables are
 - A. Controllable
 - B. Uncontrollable
 - C. Parameters
 - D. None of the above
3. A model is
 - A. An essence of reality
 - B. An approximation
 - C. An idealization
 - D. All of the above
4. A physical model is example of
 - A. An iconic model
 - B. An analogue model

- C. A verbal model
D. A mathematical model
5. Operations research approach is
- A. Multi-disciplinary
B. Scientific
C. Intuitive
D. All of the above
6. Minimization of objective function in LP model means
- A. Value occurs at allowable set of decisions
B. Lowest value is chosen among allowable decisions
C. Highest value is chosen among allowable decisions
D. Both A&B
7. LP is a?
- A. Constrained optimization technique
B. Technique for economic allocation of limited resources
C. Mathematical technique
D. All
8. The role of artificial variable in the simplex table is?
- A. To aid in finding an initial solution
B. To obtain optimal dual prices in the final simplex table
C. To start phases of simplex method
D. All

9. Mathematical model of LPP is important because:
- It helps in converting the verbal description and numerical idea in to mathematical expression
 - Decision makers prefer to work with formal models
 - It captures the relevant relationship among decision factors
 - It enables the use of algebraic technique
10. Which of the following is not a characteristics of Linear programming model
- Alternative course of action
 - Limited objective function
 - Limited amount of constraints
 - Work when all the information is known
 - Non negativity condition on the value of decision variables
 - All
11. Constraints in an LP model represents
- Limitations
 - Requirements
 - Balancing limitations and requirements
 - All of the above
12. One of the following is not the distinguishing feature of an LP models
- Relationship among all variables is linear
 - It has single objective and constraints
 - Value of decision variables is nonnegative
 - All except A
13. Each constraint in an LP model is expressed as an
- Inequality with $\geq$ sign
 - Inequality with $\leq$ sign
 - Equation with $=$ sign
 - All of the above
14. Which of the following is limitation of LP
- Linear Relationship
 - Constant Value of objective & Constraint Equations
 - No Scope for Fractional Value Solutions.
 - Degree of Complexity.
 - Multiplicity of Goals.
 - All
15. For a minimization problem, the objective function coefficient for an artificial variable is?
- $-M$
 - $+M$
 - Zero
 - None
16. A variable which does not appear in the basic variable column of simplex table is ?
- Never equal to zero
 - Always equal to zero
 - Called a basic variable
 - None
17. We must add artificial variable to write the standard form of LPP is?
- Equality constraint
 - For greater than constraint
 - Both A and B
 - None

- 18.** If a negative value appears in the RHS value then
- The solution is optimal
 - The solution is infeasible
 - The solution is unbounded
 - All
- 19.** If dual has an unbounded solution, primal has
- No feasible solution
 - feasible solution
 - unbounded solution
 - None
- 20.** the RHS constant of a constraint in a primal problem appears in the corresponding dual as
- a coefficient in the objective function
 - a RHS constant of constraint
 - an input- output coefficient
 - none
- 21.** shadow price indicates how much one unit change in the resource value will change the
- optimality range of an objective function
 - optimal value of the objective function
 - value of basic variables in the optimal solution
 - none
- 22.** the dual of a dual is
- A primal
 - Both A and B
 - A dual
 - None
- 23.** For any primal and its dual
- Optimal value of the objective function is same
 - Primal will have an optimal solution if and only if dual does not too
 - Both dual and primal cannot be infeasible
 - All of the above
- 24.** To ensure best marginal increase in the objective function value, a resource value may be increased whose shadow p[rise is comparatively
- Large
 - Neither A nor B
 - Small
 - Both A and B
- 25.** the entering variable in the sensitivity analysis of objective function coefficient is always
- Decision variable
 - Slack variable
 - Non basic variable
 - None
 - Basic variable
- 26.** Which of the following criteria is not applicable to decision making under risk
- Maximize expected return
 - Maximize return
 - Minimize expected regret
 - Knowledge of likely hood occurrence of each state of nature
- 27.** while performing sensitivity analysis, the upper bound infinity on the value of the RHS of the constraint means that

- A. there is slack in constraint
- B. the constraint is redundant
- C. the shadow price for that constraint is zero
- D. None

- 28.** A type of decision making environment is
- A. Certainty
 - B. Uncertainty
 - C. Risk
 - D. All
- 29.** The concept of utility is used to
- A. Measure the utility of money
 - B. Take in to account a version of risk
 - C. Both A and B
 - D. None
- 30.** Which of the following criteria is not used for decision making under uncertainty
- A. Maximin
 - B. Maximax
 - C. Minimax
 - D. Minimize expected loss
- 31.** The expected value of perfect information (EVPI) is
- A. Equal to expected regret of the optimal decision under risk
 - B. The utility of additional information
 - C. Maximum expected opportunity loss
- 32.** None The objective of network analysis is to
- A. Minimize total project duration
 - B. Minimize total project cost
 - C. Minimize production delays, interruption and conflicts
 - D. All of the above
- 33.** In time cost tradeoff analysis
- A. Cost decrease linearly as time increases
 - B. Cost at normal time is zero
 - C. Cost increases linearly as time increases
 - D. None
- 34.** In PERT network each activity time assume a Beta distribution because
- A. It is uni-modal distribution that provides information regarding the uncertainty of time estimates of activities.
 - B. It has got finite non negative error
 - C. It need not be symmetrical about model value
 - D. All

35. The activity which can be delayed without affecting the execution of the immediate succeeding activity is determined by

- A. Total float
- B. Free float
- C. Independent float
- D. A and B
- E. None

36. Generally pert technique deals with the project of

- A. Reiterative nature
- B. Non-Reiterative nature
- C. Deterministic nature
- D. None

Part II: Solve the following problems

37. Maximize $Z=10X_1+15X_2$

2. Min. $Z=3X_1+2X_2$

Subject to:

Subject to:

$$2X_1+X_2 \leq 26$$

$$5X_1+X_2 \geq 10$$

$$2X_1+4X_2 \leq 56$$

$$X_1 + X_2 \geq 6$$

$$-X_1+X_2 \leq 5$$

$$X_1 + 4 X_2 \geq 12$$

$$X_1, X_2 \geq 0$$

$$X_1, X_2 \geq 0$$

38. A manufacturer produces two different models; **X** and **Y**, of the same product .The raw materials r_1 and r_2 are required for production. At least 18 Kg of r_1 and 12 Kg of r_2 must be used daily. Almost at most 34 hours of labor are to be utilized .2Kg of r_1 are needed for each model **X** and 1Kg of r_1 for each model **Y**. For each model of **X** and **Y**, 1Kg of r_2 is required. It takes 3 hours to manufacture a model **X** and 2 hours to manufacture a model **Y**. The profit realized is \$50 per unit from model **X** and \$30 per unit from model **Y**. How many units of each model should be produced to maximize the profit?

39. A manufacturing firm produces two machine parts **P₁** and **P₂** using milling and grinding machines .The different machining times required for each part, the machining times available on different machines and the profit on each machine part are as given below:

Manufacturing time

Maximum time

Machine	Required (min)	available per week (minimum)	
	P_1	P_2	
<i>Lathe</i>	<i>10</i>	<i>5</i>	<i>25,000</i>
<i>Milling Machine</i>	<i>4</i>	<i>10</i>	<i>2000</i>
<i>Grinding Machine</i>	<i><u>1</u></i>	<i><u>1.5</u></i>	<i>450</i>
<i>Profit per unit (\$)</i>	<i><u>\$50</u></i>	<i><u>\$100</u></i>	

Determine the number of pieces of P_1 and P_2 to be manufactured per week to maximize profit.

40. Consider the following 'Primal' LP formulation

Maximize $12x_1 + 10x_2$

Subject to $2x_1 + 3x_2 \leq 18$

$2x_1 + x_2 \leq 14$

$x_1, x_2 \geq 0$

Formulate the 'Dual' for this problem

41. Suppose that a firm has three factories / *sources of supply* / & four warehouses/point of demand/. The firm's production capacity at the three factories, the demand for the four destination centers located at various regions & the cost of shipping each unit from the factories to the warehouses through each route is given as follows:

Destinations

	W_1	W_2	W_3	W_4	<i>Factory Capacity</i>
F_1	3	2	7	6	5000
F_2	7	5	2	3	6000

F_3	2	5	4	5	2500
Demand	6000	4000	2000	1500	13500

Required:

- Develop an initial feasible solution using NWCM & Compute the total cost
- Develop an initial feasible solution using least-cost method & compute the total cost.

- 42.** Three garment plants are available for monthly production of four styles of men's shirts. The capacities of the three plants are 45,000, 93,000 and 60,000 shirts. The number of shirts required in style "a" through "d" is 28,000, 65,000, 35,000 & 70,000, respectively. The profits, in \$ per shirt, at each plant for each style are shown below.

Table: The garment plants' profit.

STYLE PLANT	A	b	c	D
1	8	12	-2	6
2	13	4	3	10
3	0	7	11	8

How many shirts of each type to produce in each plant so that profit is maximized?

- 43.** A car rental company has one car at each of five depots **a, b, c, d** and **e**. A customer in each of the five towns **A, B, C, D** and **E** requires a car. The distance in (in kilometers) between the depots and towns where the customers are, is given in the following distance matrix:

Depots

	A	B	c	d	e
A	160	130	175	190	200
B	135	120	130	160	175
C	140	110	155	170	185

<i>D</i>	50	50	90	80	110
<i>E</i>	55	35	70	80	105

☞ How should the cars be assigned to the customers so as to minimize the distance traveled?

44. An investor is given the following investment alternatives and percentage rate of return.

	State of nature (market condition)		
	Low (%)	Medium (%)	High (%)
Regular sharing	2	5	8
Risky sharing	-5	7	15
Property	-10	10	20

Over the past 300 days, 150 days have been medium market conditions and 60 days have had high market increases.

On the basis of these data, state the optimal investment strategy for the investor. According to the f given information the probabilities of low medium and high market conditions would be 90/300 or 0.30, 150/300 or 0.50 and 60/300 or 0.20, respectively.

A. Calculate the expected payoffs for each of the alternatives

B. Calculate the expected return

C. In which alternative the investor invest his money?

Part II Solve the following question

45. Listed in the table are the activities & sequencing necessary for the completion of a recruitment procedure for management trainees (MT) in a firm.

<u>Activity</u>	<u>Predecessor Activity</u>
A	-
B	A
C	A
D	C
E	B
F	C
G	D, E
H	G
I	H
J	F
K	I, J
L	K
M	L
N	M, N.

Draw a network diagram for the project Answer:

- 46.** The owner of a chain of fast food restaurants is considering a new computer system for accounting and inventory control. A computer company sent the following information about the computer system installation.

Activity	Description	Predecessors	Expected time (days)		
			Optimistic	Most likely	Pessimistic
A	Select the computer model	-	4	6	8
B	Design input output system	A	5	7	15
C	Design monitoring system	A	4	8	12
D	Assemble computer hardware	B	15	20	25
E	Develop the main programs	B	10	18	26
F	Develop input-output routines	C	8	9	16
G	Create database	E	4	8	12
H	Install the system	D,F	1	2	3
I	Test and implement	G,H	6	7	8

Required:

- Construct an arrow diagram for this problem
 - Determine the critical path and compute the expected completion time
 - Determine the probability of completing the project in 55 days
- 47.** A company distributes its products by trucks loaded at its only loading station. Both, company's trucks and constructor's trucks, are used for this purpose. It was found out that on average every five minutes, one truck arrived and the average loading time was 3 minutes. 50% of the trucks belong to the constructor. Find out

- A. The probability that a truck has to Wait
 - B. The waiting time of truck that waits and
 - C. The expected waiting time of constructor truck per day, assuming a 24 shift
- 48.** Customers arrive at a one window drive according to a poison distribution with mean of 10 minutes and service time per customer is Solve exponential with mean of 6 minutes. The space in front of the window can accommodate only 3 vehicles including the serviced one. Other vehicles have to wait outside this space. Calculate
- A. probably that an arriving customer can drive directly to the space in front of the window
 - B. Probability that an arriving customer will have to wait outside the directed space.
 - C. how long an arriving customer is expected to wait before gating the service
- 49.** A branch of Dashen Bank has only one typist. Since the typing work varies in length (number of pages to the typed) the typing rate is randomly distributed approximating a poison distribution with mean service rate of 8 letters per hour. The letter arrives at a rate of 5 per hour during the entire 8 hours work day. If the type writer is values at birr 1.50 per hour, determine;
- A. Equipment utilization
 - B. Average system time
 - C. The percent time that an arriving letter has to wait
 - D. Average cost due to waiting on the part of type writer, i.e. it remaining idle.

For further information, refer the course module, handouts, and reference books mentioned in the course outline which is provided by the course instructor. Any material related to this course can be useful to refer.

Operation management

STUDENT HANDBOOK

CHAPTER ONE

INTRODUCTION TO OPERATIONS MANAGEMENT

1.1. Introduction

Every business is managed through three major functions: finance, marketing, and operations management. Other business functions (such as accounting, purchasing, human resources, and engineering) support these three major functions. **Finance** is the function responsible for managing cash flow, current assets, and capital investments. **Marketing** is responsible for sales, generating customer demand, and understanding customer wants and needs. **Operation is** that part of a business organization that is responsible for producing goods and/or services. Goods are physical items that include raw materials, parts, subassemblies such as motherboards that go into computers, and final products such as cell phones and automobiles. Services are activities that provide some combination of time, location, form, or psychological value. Examples of goods and services are found all around you. Every book you read, every video you watch, every e-mail or text message you send, every telephone conversation you have, and every medical treatment you receive involves the operations function of one or more organizations. So, does everything you wear, eat, travel in, sit on, and access the Internet with. The operations function in business can also be viewed from a more far-reaching perspective: The collective success or failure of companies' operations functions has an impact on the ability of a nation to compete with other nations, and on the nation's economy.



Figure 1; Organizational chart showing the three major business functions

What is Operation Management?

- Operations management is the activity of managing the resources which produce and deliver products and services. The administration of process that transforms inputs of labor, capital and materials in to output bundles of products and services that are valued by customers.
- **Operation Management:** is the management of systems or processes that create goods and/or provide services. Or **Operation Management-** is the design, operation, and improvement of those systems that create and deliver the firms primary products and services, like marketing and finance..

Every organization has an operations function because every organization produces some type of products and/or services. However, not all types of organization will necessarily call the operations function by this name. Operations managers are the people who have particular responsibility for managing some, or all, of the resources which compose the operations function. Again, in some organizations the operations manager could be called by some other name. For example, he or she might be called the ‘fleet manager’ in a distribution company, the ‘administrative manager’ in a hospital, or the ‘store manager’ in a supermarket.

1.2. System Concepts in Operations Management

A system may be defined as “a purpose full collection of people, objects and procedures for operating within an environment”. Thus, every organization can be represented as a system consisting of interacting sub systems. The basic process of the system converts (transforms) the resource inputs (such as people, plant, part processes, planning and controlling systems) in to some useful form of outputs. Using the generalized concepts of operations management, we can call such system as operations system and the operation management has the prime responsibility for transforming input in to output.



1. **Inputs:** Inputs to the system may be labor, material, machine, facilities, energy, information, technology etc. Inputs in operating system can be: customer in a bank, patient in a hospital,

commuters to public transport system, files and papers to an office situations and program to be run in a computer center. The types of inputs used will vary from one industry to another.

2. **Components:** are machines, persons, tools, in the system.
3. **Outputs:** similarly output from the system may be in terms of finished products, transported goods, delivered massages, cured patients, serviced customers etc.
4. **Transformation;** all operation functions are essentially a part of the conversion process which transforms entities in shape, size, form, location, space, time, and state. The essence of the operations function is to add value and efficiency during the transformation process: **Value-added** is the term used to describe the difference between the cost of inputs and the value or price of outputs. The greater the **value added**, the more productive a business is. An obvious way to add value is to reduce the cost of activities in the transformation process. Activities that do not add value are considered a waste; these include certain jobs, equipment, and processes. **Efficiency:** it means being able to perform activities well, and at the lowest possible cost. An important role of operations is to analyze all activities, eliminate those that do not add value, and restructure processes and jobs to achieve greater efficiency. Today's business environment is more competitive than ever, and the role of operations management has become the focal point of efforts to increase competitiveness by improving value added and efficiency.

<i>System</i>	<i>Inputs</i>	<i>Components</i>	<i>Transformations</i>	<i>Output</i>
Hospital	Patients	Doctors, Nurses, Staff, Equipment, Facility, Energy, etc.	Physiological, Examination, Monitoring, Surgery and Therapy etc.	Cured person
Bank	Clients	Teller, Staff, Facility etc.	Exchange	Loan/deposit etc.
College	High school graduate	Teacher, Class room, Books etc.	Imparting knowledge	Educated person

1.3. Objective of Operations Management

Operation Management involves management of the entire process responsible for converting inputs into outputs. The following are the objectives of Operations Management.

- ✚ **To provide customer service;** The main objective of any operating management systems is to utilize resources judiciously for the satisfaction of customer needs and wants.
- ✚ **Effective utilization of resources;** Resources that are used in the business organization must be carefully utilized. It aims at obtaining maximum output from the available resources with minimum cost.
- ✚ **To reduce cost of production;** Operation management aims at reduction in the cost of production of goods and services.
- ✚ **To improve product quality;** Quality control and maintenance are the two important objectives of operations management. Quality control consists of all those activities, which are designed to define, maintain and control specific quality of products within reasonable limits.
- ✚ **To fix time schedule;** Another important objective of operation management is to establish time schedule for various operation activities. The schedule fixation includes the operating cycle time, inventory turnover rate, machine utilization rate, capacity utilization etc.
- ✚ **Proper utilization of Machinery;** Operation management has to take number of decisions with regard to machinery and equipment.

1.4. Production Management Vs Operations Management

There are two points of distinction between production management and operations management. First, the term production management is more used for a system where tangible goods are produced. Whereas, operations management is more frequently used where various inputs are transformed into intangible services. Viewed from this perspective, operations management will cover such service organizations as banks, airlines, utilities, pollution control agencies, super bazaars, educational institutions, libraries, consultancy firms and police departments, in addition, of course, to manufacturing enterprises. The second distinction relates to the evolution of the subject. Operations management is the term that is used nowadays. Production management precedes operations management in the historical growth of the subject.

1.5. Why Study Operations Management?

You may be wondering why you need to study operations management. Actually, there are a number of very good reasons.

- ✓ One is that operations management activities are at the Core of all business organizations, regardless of what business they are in.
- ✓ We want (and need) to know how goods and services are produced.
- ✓ Second, 50 percent or more of all jobs are in operations management-related areas-such areas as customer service, quality assurance, production planning and control, scheduling, job design, inventory management, and many more.
- ✓ Third, activities in all of the other areas of business organizations, such as finance, accounting, human resources, logistics, management information systems (MIS), marketing, purchasing, as well as others are all interrelated with operations management activities. So, it is essential for people to work in these areas to have a basic understanding of operations management.

1.6. Manufacturing Operation Vs Service Operation

Production of goods results in a tangible output, such as an automobile, a clock radio, a golf ball, a refrigerator-anything that we can see or touch. It may take place in a factory, but can occur elsewhere. Service, on the other hand, generally implies an act. A physician's examination, TV and auto repair, lawn care, and projecting a film in a theater are examples of services. The majority of service jobs fall into these categories:

1.6.1. Similarities between Manufacturing and Services

- ◆ Every organization whether manufacturing or service giving, has processes (transform input in to output) that must be designed and managed effectively.
- ◆ Every organization is concerned about quality, productivity, and the timely response to customers. A service provider, like a manufacturer, must make choices about the capacity, location, and lay out of its facility.
- ◆ Ever organization deals with suppliers of outside services and materials, as well as scheduling problems.
- ◆ Finally, matching staffing levels and capacities with forecasted demand is universal problems.

1.6.2. Difference between Manufacturing and Service

Manufacturing and service organizations differ chiefly because manufacturing is goods-oriented and service is act-oriented. The differences involve the following

Physical nature of the product: manufacturing goods are tangible and durable products whereas services are intangible, perishable products often being ideas, concepts or information. Moreover, manufactured goods are output that can be produced, stored, and transported in anticipation of future demand. By contrast, service cannot be pre-produced.

Customer contact: Most customers for manufactured products have little or no contact with the production system. Primary customer contact is left to distributors and retailers. However, in many service organizations the customers themselves are inputs and active participants in the process. At a college, for example, the student studies, attends lectures, takes exams, and finally receives a Diploma.

Response time to customer demand: While manufacturers generally have days or weeks to meet customer demand, many services must be offered within minutes of customer arrival.

Location and size of operation: Manufacturing facilities often serve regional, national, or even international markets and therefore, generally requires larger facilities, more automations and greater capital investment. In general, service cannot be shifted to distant locations. Thus, service organization requires direct customer contact and must locate relatively near their customers.

Measurement of quality and productivity: As manufacturing system tends to have tangible products quality and productivity is relatively easy to measure.

Labor content of jobs: Services often have a higher degree of labor content than manufacturing jobs do, although automated services are an exception.

Uniformity of inputs: Service operations are often subject to a higher degree of variability of inputs. Each client, patient, customer, repair job, and so on presents a somewhat unique situation that requires assessment and flexibility.

Inventory: Unlike manufactured goods, services cannot be stored. Instead, they must be provided “on demand.”

Wages: Manufacturing jobs are often well paid, and have less wage variation than service jobs, which can range from highly paid professional services to minimum-wage workers.

Ability to patent: Product designs are often easier to patent than service designs.

Typical differences between manufacturing and services (Normann, 1991)

	Manufacturing	Service
Tangibility	Product is generally concrete	Service is intangible
Ownership	Transferred when sold	Not generally transferred
Resale	Can be resold	Cannot generally be resold
Demonstrability	Can be demonstrated	Does not exist before Purchase
Storability	Can be stored by providers and customers	Cannot be stored
Simultaneity	Production precedes consumption	Generally, coincide
Location	Production/selling/ consumption locally differentiable	Spatially united
Transportability	Can be transported	Generally, not (but producers might be)
Production	Seller alone produces	Buyer/client takes part
Contact	Indirect contact possible between client and provider	Direct contact usually Necessary
Internationalization	Can be exported	Service usually cannot (but delivery system often can)
Output	can be inventoried	Cannot be inventoried
Inventory	Use much inventory	Use fewer inventory
Facility	Require large facility	Small facilities
Uniformity of input and output	High	Low

Quality and productivity	Easy to measure	Difficult to measure
Automation	Easy	Difficult
Response time	Long	Short

1.7. Operations Management Decision Making Areas

Operations management is essentially function concerning decision making with respect to operation system so as to render the necessary customer satisfaction at lowest cost. Operation managers manage all activities in the transformation of input in to output by making three broad types of decisions.

1.7.1. Strategic (long term) decisions:

A decision is said to be strategic if it has a long-term impact influence a large part of the system and is difficult to undo once implemented. Some examples of such decisions are:

- ✓ *Product selection and design*
- ✓ *Process selection and planning*
- ✓ *Facilities location*
- ✓ *Facilities lay out and materials handling*
- ✓ *Capacity planning*

1.7.2. Operational (short term) decisions

Operational level decisions deal with short term planning and control problems. Some of these are:

- ✓ *Production planning, scheduling and control*
- ✓ *Inventory planning and control*
- ✓ *Quality assurance*
- ✓ *Work and job design*
- ✓ *Cost reduction and control*

1.7.3. Controlling decisions

In every system, the actual accomplishment of objectives may not be as planned for various reasons. It is there for very important to monitor the actual performance by measuring the actual output or some performance indicators. The decision in control system should consider cost

benefit aspects of control in mind. If cost of control exceeds its benefits, it becomes counterproductive. Thus, selective control must be exercised employing the exception principle or Pareto's law. While taking these decisions, one must bear in mind the general objective of Operation Management.

1.8. Operation Management and Its Environment

A production system is only one of the major components of an organization. Production is affected by and has an impact on decision in other functional areas of the firm. In addition, various external influences affect the overall objectives and policies of a company and have important implications for production. These important relationships for a typical firm are illustrated below.

At the outermost level, external to the firm itself, are several environment factors that influence the overall policies and objectives of the company. Four of the most important external factors are: economic condition, government regulation, competition and technology.

- *An economic factor* includes interest rate, availability of capital, general economic conditions, tax regulation, and economies of scale. For example, a sudden rise in oil prices, coupled with an increasingly energy conscious public, has had a dramatic effect on the product lines of automobile manufacturers.
- *Government regulation*: -compliance with government regulation on pollution controls and environmental impacts has had an impact on production.
- *Competition*: - the nature of competition, market share, and how the firm reacts to competitive strategies have a significant influence on product lines and other strategic decisions. This has been especially evident in the fast food industry. When a fast food franchise opens on a well-traveled highway, a variety of other similar operations almost immediately spring up nearby.
- *Technology*: - new technology in manufacturing processes can drastically affect product design and production methods. The company is forced to incorporate the new technology in order to stay in the business i.e. production must react quickly to strategic changes in a firm's business plan.

Operation function Vs other functional areas

Managing operations well requires effective interactions with the other functional areas of business. When people from the various functions do not work together well, it is very difficult to satisfy customers or even make minor improvements in operations. In most organizations, operation is an internal function that is buffered from the external environment by other organizational function. Within an organization, the production function is influenced by other functional areas such as:

- 1. Marketing:** is responsible for generating and maintaining demand for the firm's products, insuring customer satisfaction, and developing new markets and product potentials. Coordination of production and marketing is important in order to use demand forecast effectively, to project workloads, and to ensure sufficient work capacity to handle the demand and delivered finished products on time.
- 2. Finance:** is responsible for obtaining funds, controlling their uses, analyzing investment opportunities, and ensuring that the firm operates on a cost-effective basis and in most cases at profit. Financial decision affects the choices of manufacturing equipment's, use of overtime, cost control policies, price volume decisions, and in fact, nearly all facets of the organizations.
- 3. Accounting:** keeps records on cost and price that relates to such factors as financial decisions, purchasing and payroll.
- 4. Engineering:** determine guidelines for product quality, production methods, and other technical specifications.
- 5. Personnel and labor relation:** recruit and train employees and are responsible for employee's morale, wage administration, union negotiation, and so on. Because peoples are the most important entity in any organization, this function is vital in helping a production system run smoothly.
- 6. Research and development:** are responsible to investigate new ideas and their potential uses as consumer products.
- 7. Purchasing and traffic:** are responsible for the acquisition of materials and supplies necessary for production and the distribution of the finished goods to customers respectively.

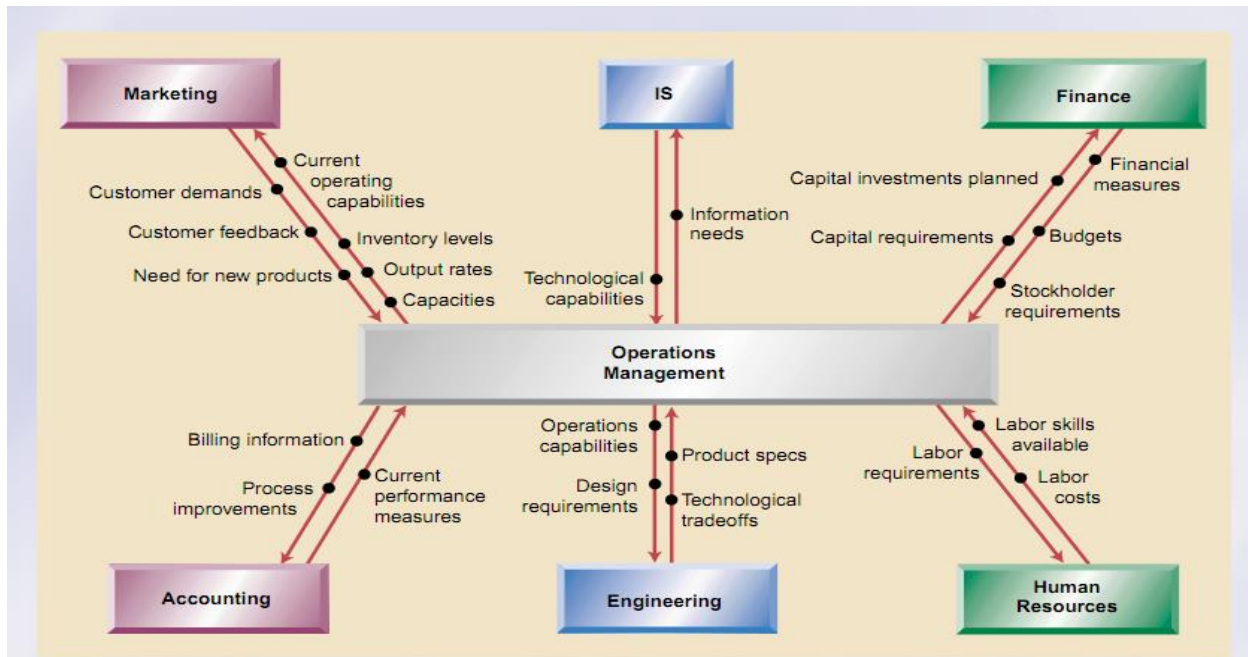


Figure 2; Information flow between operations and other business functions

1.9. Productivity (Measuring Productivity)

Productivity is defined in terms of utilization of resources, like material and labor. In simple terms, productivity is the ratio of output to input. It has been said, “If you can’t measure it, you can’t manage it”. This is particularly true of productivity. Now we come to the main question of how we know that we are managing our operation system well. This concerns the efficiency with which we are converting the input in to outputs. This conversion efficiency can be roughly gauged by the ratio of: **Output /inputs** (a term which generally known as productivity of the system).

Productivity is used for making comparison or to measure improvement. Productivity is a relative term i.e. it gives sense only when we compare it with: company’s previous performance, with other similar company’s performance or with the performance of leader of the industry.

Productivity may be expressed as:

- ❖ *Total factor productivity measure:* -is the ratio of all output to all input i.e. total outputs/total inputs. Total inputs include all resources used in the production of goods and services: labor, capital, raw materials, and energy.

- ❖ *Multi factor productivity*: - measures only a sub set of these inputs. i.e. **output/** (*labor + capital*), **output/** (*labor+ capital + materials*), **output/** (*materials + energy*).
- ❖ *Single factor (partial productivity) measure*: -is the ratio of output to a single resource (inputs). i.e. *output/labor, output/capital, output/material, output/energy* etc.

Numerical examples

Example 1. Three employees process 600 insurance policies in a week. They work 8 hrs. per day, 5 days a week. Required calculate labor productivity?

Solutions

Labor productivity=output/inputs.

Where, output is number of insurance policies processed and input is hours worked.

$$\text{Labor productivity} = \frac{600 \text{ polices}}{(3 \text{ employees}) \times (40 \text{ hrs/employees})} = 5 \text{ polices/hours}$$

Total or multifactor productivity measures are generally preferable than partial productivity. The reason for this is that focusing on productivity improvement in a narrow portion of an organization may actually decrease total productivity. See the following example.

Example 2: ABC Furniture Company produced 10,000 chairs, with annual labor and equipment cost of \$50,000 and \$25000 respectively. Total productivity can be calculated as:

$$\text{Productivity: chairs/dollar input} = \frac{10000 \text{ chairs}}{\$50000 + \$25000} \Rightarrow 0.133 \text{ chairs/dollar input}$$

Labor productivity, however, is measured for this example as:

$$10,000/50,000 = 0.20 \text{ chairs/dollar inputs} - \text{labor}.$$

Suppose that a \$10,000 reduction in labor can be achieved by investing in a more advanced machine. Labor productivity will increase to: $10,000/40,000 = 0.25$ chairs/labor- dollar. Thus, from a productivity perspective, it would appear that this investment is attractive. If, however, that annual cost with the new equipment increases to \$40,000 then, total productivity will be:

$$\frac{10,000}{\$40,000 + 40000} = 0.125 \text{ chairs/dollar inputs}$$

And hence overall productivity would decrease. It is necessary, therefore to examine the simultaneous effects of all changes on productivity.

Productivity is more difficult to measure in service organization than in manufacturing organization. In manufacturing, physical inputs and outputs are easy to identify and the values of outputs are easy to measure. For service, the value of an output is often intangible. For instance, how does one assess the value of healthy patient?

Productivity measures in service organization are often stated as: benefit/cost ratio.

1.9.1. Factors that affect Productivity

Productivity stands tall on four important pillars of Capital, Quality, Management and Technology. These pillars are also responsible for positively as well as negatively affecting the productivity of the organization. Example: Capital, Quality, Management Technology and Caution.

Other Factors Affecting Productivity

Standardization: We live in a world where for the sake of convenience, reliability and safety, majority of the products and services have been standardized. If for a moment any process whether it relates to manufacturing or services is made standard less, the vital concept of compatibility would be lost.

Use of Internet: Use of Internet/Extranet especially for the services side, even though there are knowledge base applications available for the manufacturing side as well but primarily it has been the services side which has been able to exploit the resourcefulness of the Internet.

Computer viruses: A lot of time IT based services industry have fallen a prey to computer viruses and hackers.

Searching for lost or misplaced items: This speaks low about the coordinating activities and can lead to loss in production time and increase in idle time. Often this also leads to increase in replacement costs.

Scrap rates: Any aberration in the raw materials or processed product can lead to increase in scrap. The increase in scrap rate in fact can decrease the utilization of resources in general and raw material.

New workers: Organizations spend millions of Dollars every year to train their employees. A trained workforce is not only reliable and dependable but also ensures productivity.

1.9.2. Improving Productivity

A company or a department can take a number of key steps toward improving productivity:

1. Develop productivity measures for all operations; measurement is the first step in managing and controlling an operation.
2. Look at the system as a whole in deciding which operations are most critical; it is overall productivity that is important.
3. Develop methods for achieving productivity improvements, such as soliciting ideas from workers (perhaps organizing teams of workers, engineers, and managers), studying how other firms have increased productivity, and reexamining the way work is done.
4. Establish reasonable goals for improvement.
5. Make it clear that management supports and encourages productivity improvement.
6. Measure improvements and publicize them

There are several ways in which operations manager can improve productivity. These may be classified as:

- *increasing output while keeping input constant*
- *decreasing input while keeping output constant*
- *increasing output at higher rate than increasing input*
- *decreasing output at lower rate than decreasing input*
- *Increasing output while decreasing input (Most challenging but effective).*

1.10 Efficiency and Effectiveness

While many businesses might wonder how to produce on a greater scale, using less can be just as important.

- ✚ Efficiency is determined by the relationship between resources expended and the benefits gained in achieving a goal. One system is more efficient than another if its operating costs are lower for the same or better quality product, or if the product's quality is greater for the same or lower costs.
- ✚ Operational efficiency is a term that describes a business's ability to reduce inputs while producing the same number of outputs. Prioritizing this metric can help a

company achieve desirable profit margins and produce high-quality products that satisfy consumer demands. Operational efficiency is about doing things right.

✚ **Effectiveness** defines the degree to which a goal is achieved. Thus, a system is more or less effective depending on how much of a particular goal it achieves and the degree to which it achieves better outcomes than other systems do.

✚ Operational effectiveness is about doing the right things. It revolves around making sure that all the core work done by the organization creates value to their end customer; that the core Value Stream is correctly designed. Organizational effectiveness denotes the concept of how effectively an organization achieves the outcomes it intends to produce. In general, effectiveness the extent to which stated objectives are met the policy achieves what it intended to achieve.

Exercise; A furniture company has provided the following data for the year 2000 and 2001:

	<u>2000</u>	<u>2001</u>
Output	22000	35000
Inputs		
- Labor	10000	15000
- RM & supplies	8000	12500
-Depreciation	700	1200
-Others	2200	4800

Required:

1. Compute total productivity for year 2000 and 2001
2. Compute multi factor productivity for labor, and RM & supplies, and labor, capital and others for the year 2001.
3. Compute partial productivity for labor, and RM and supplies for the year 2000

Chapter two

Operations strategy for competitive advantage

2.1 Identifying Missions and Strategies

An effective operations management effort must have a mission so it knows where it is going and a strategy so it knows how to get there. This is the case for a small or domestic organization, as well as a large international organization.

2.1.1 What is Mission?

Economic success, indeed survival, is the result of identifying missions to satisfy a customer's needs and wants. We define the organization's mission as its purpose – what it will contribute to society. Mission statements provide boundaries and focus for organizations and the concept around which the firm can rally. The mission states the rationale for the organization's existence. Developing a good strategy is difficult, but it is much easier if the mission has been well defined. Once an organization's mission has been decided, each functional area within the firm determines its supporting mission. By functional area we mean the major disciplines required by the firm, such as marketing, finance/accounting, and production/operations. Missions for each function are developed to support the firm's overall mission. Then within that function lower – level supporting missions are established for the OM functions. The following are some of sample missions.

. Example of some selected company mission statements

- I. **Arba Minch University:** Promoting democratic thinking, offering high quality education and training, conducting demand driven and problem solving research and consultancy, and rendering community service in order to contribute to the development endeavors of the country.
- II. **Microsoft;** to help people and businesses throughout the world to realize their full potential.
- III. **U.S. Dept. of Education;** to promote student achievement and preparation for global competitiveness and fostering educational excellence and ensuring equal access.

IV. **Sample Operations Management Mission:** to produce products consistent with the company's mission as the worldwide low-cost manufacturer.

2.1.2 What is Strategy?

To maintain a competitive position in the marketplace, a company must have a long-range plan. This plan needs to include the company's long-term goals, an understanding of the marketplace, and a way to differentiate itself from its competitors. All other decisions made by the company must support this long-range plan. Otherwise, each person in the company would pursue goals that he or she considered important, and the company would quickly fall apart. Strategy can be defined as follows:

'Strategy is the direction and scope of an organization over the long term: ideally which matches its resource to its changing environment and in particular its markets, customers or clients so as to meet stakeholders' expectations.' (Johnson et. Al, 2008).

With the mission established, strategy and its implementation can begin. Strategy is an organization's action plan to achieve the mission. Each functional area has a strategy for achieving its mission and for helping the organization reach the overall mission. These strategies exploit opportunities and strengths, neutralize threats, and avoid weaknesses. Operations strategy is a long term plan for the production of a company's products/rendering services and it provides a road map for what the production/operations function must do if business strategies are to be achieved. Operations strategy is the decisions which shape the long-term capabilities of the company's operations and their contribution to overall strategy through the on-going reconciliation of market requirements and operations resources.

Firms achieve missions in three conceptual ways: 1). Differentiation, 2) cost leadership, and 3). Response. This means operations managers are called on to deliver goods and services that are (1) better, or at least different, (2) cheaper, and (3) more responsive. Operations managers translate these strategic concepts into tangible tasks to be accomplished. Any one or combination of these three strategic concepts can generate a system that has a unique advantage over competitors. Operations strategy consists of goals, plans and a direction for the operations function that are linked to the business strategy and other functional strategies, leading to a competitive advantage for the firm. The operations manager's job is to implement an OM strategy, provide competitive advantage, and increase productivity.

Strategy can be seen to exist at 3 main levels of corporate, operation and functional

Corporate strategy

- ☞ Defines long range plan for the organization.

Operations strategy

- ☞ Develop a plan for the operations function focusing on specific competitive priorities.

Functional level strategy

- ☞ Developed to focus on the identified competitive priorities.

2.2. Strategy Formulation

Strategy formulation is the process by which an organization chooses the most appropriate courses of action to achieve its defined goals. This process is essential to an organization's success, because it provides a framework for the actions that will lead to the anticipated results. Strategic plans should be communicated to all employees so that they are aware of the organization's objectives, mission, and purpose. Strategy formulation forces an organization to carefully look at the changing environment and to be prepared for the possible changes that may occur. A strategic plan also enables an organization to evaluate its resources, allocate budgets, and determine the most effective plan for maximizing ROI (return on investment).

A company that has not taken the time to develop a strategic plan will not be able to provide its employees with direction or focus. Rather than being proactive in the face of business conditions, an organization that does not have a set strategy will find that it is being reactive; the organization will be addressing unanticipated pressures as they arise; and the organization will be at a competitive disadvantage.

Strategy formulation requires a defined set of six steps for effective implementation. Those steps are:

1. define the organization,
 2. define the strategic mission,
 3. define the strategic objectives,
 4. define the competitive strategy,
 5. implement strategies, and
 6. evaluate progress.
- Defining the organization requires a company to identify its customers by end benefits sought, by specific target markets, or by technology.

- Defining the strategic mission ensures that the company is able to identify its values, the nature of its business, its competitive advantage, and its vision for the future.
- Strategic objectives should be defined based on performance targets and may include increases in market share, customer service improvements, corporate expansion, sales increases, production methods, etc.
- Competitive strategy includes an evaluation of the overall industry and marketplace, the nature of the competition's position, and the company's internal strengths and weaknesses.
- A company must implement its strategic plan in order to achieve success. It must develop appropriate tactics, which are the action steps for meeting the strategies directives.
- Strategies must be evaluated and revised on a regular basis in order to meet the changing needs and challenges of the marketplace and business environment.

2.3. Achieving Competitive Advantage through Operations

Competitive priorities are the capabilities that the operations function can develop in order to give a company a competitive advantage in its market. Each of the three strategies provides an opportunity for operations manager to achieve competitive advantage. Competitive advantage implies the creation of a system that has a unique advantage over competitors. The idea is to create customer value in an efficient and sustainable way. Pure forms of these strategies may exist, but operations managers will more likely be called on to implement some combination of them.

A. Competing on Differentiation

Differentiation is concerned with providing uniqueness. A firm's opportunities for creating uniqueness are not located with a particular function or activity but can arise in virtually everything the firm does. Moreover, because most products include some service, and most services include some product, the opportunities for creating this uniqueness are limited only imagination. Indeed, differentiation should be thought of as going beyond both physical characteristics and service attributes to encompass everything about the product or service that influences the value that the customers derive from it. Therefore, effective operations managers

assist in defining everything about a product or service that will influence the potential value of the customer. E.g. Convenience of product features or convenience of location distribution...

In the service sector, one option for extending products differentiation is through an experience. Differentiation by experience in services is a manifestation of the growing “experience economy.” The idea of experience differentiation is to engage the customer – to use people’s five senses so they become immersed, or even an active participant, in the product.

B. Competing on Cost

Low – cost leadership entails achieving maximum value as defined by your customer. To the companies which compete directly on price, cost will clearly be their major operations objective. The lower the cost of producing their goods and services, the lower can be the price to their customers. Even those companies which compete on things other than price, however, will be interested in keeping their costs low. Every euro or dollar removed from an operation’s cost base is a further euro or dollar added to its profits. Provide the maximum value as perceived by customer. Low-cost does not imply low quality.

C. Competing on Response

The third strategy option is response. Response is often thought of as flexible response, but it also refers to reliable and quick response. Indeed, we define response as including the entire range of values related to timely product development and delivery, as well as reliable scheduling and flexible performance.

Flexible performance: Flexibility means being able to change the operation in some way. Flexibility is matching market changes in design innovation and volumes.

Product/service flexibility: the operation’s ability to introduce new or modified products and services.

Mix Flexibility: the operation’s ability to produce a wide range or mix of products and services.

Volume Flexibility: the operation’s ability to change its level of output or activity to produce different quantities or volumes of products and services over time.

Delivery Flexibility: the operation’s ability to change the timing of the delivery of its services or products.

The second aspect of response is the **reliability of scheduling**. Doing things in time for customers to receive their goods or services exactly when they are needed, or at least when they were promised. The third aspect of response is quickness. It is quickness in design, production,

and delivery. It is the elapsed time between customers requesting products or services and their receipt of them.

The Need for Trade-Offs

You may be wondering why the operations function needs to give special focus to some priorities but not all. As more resources are dedicated to one priority, fewer resources are left for others. The operations functions must place emphasis on those priorities that directly support the business strategy.

- Trade-offs is the need to focus more on one competitive priority than on others.
- Therefore, it needs to make trade-offs between the different priorities.
- For example, consider a company that competes on using the highest quality component parts in its products. Due to the high quality of parts, the company may not be able to offer the final product at the lowest price. In this case, the company has made a trade-off between quality and price.
- Similarly, a company that competes on making each product individually based on customer specifications will likely not be able to compete on speed. Here, the trade-off has been made between flexibility and speed.
- It is important to know that every business must achieve a basic level of each of the priorities, even though its primary focus is only on some.
- For example, even though a company is not competing on low price, it still cannot offer its products at such a high price that customers would not want to pay for them.
- Similarly, even though a company is not competing on time, it still has to produce its product within a reasonable amount of time; otherwise, customers will not be willing to wait for it.
- One way that large facilities with multiple products can address the issue of trade-offs is using the concept of plant-within-a-plant (PWP), introduced by well-known Harvard professor Wickham Skinner.
- The PWP concept suggests that different areas of a facility be dedicated to different products with different competitive priorities.

- These areas should be physically separated from one another and should even have their own separate workforce. As the term suggests, there are multiple plants within one plant, allowing a company to produce different products that compete on different priorities.
- For example, hospitals use PWP to achieve specialization or focus in a particular area, such as the cardiac unit, oncology, radiology, surgery, or pharmacy.
- Similarly, department stores use PWP to isolate departments, such as the Sears auto service department versus its optometry centre.

Order winners and order qualifiers

To help a company decide which competitive priorities to focus on, it is important to distinguish between *order winners* and *order qualifiers*, which are concepts, developed by Terry Hill, a professor at Oxford University.

Order qualifiers are those competitive priorities that a company has to meet if it wants to do business in a particular market.

Order winners, on the other hand, are the competitive priorities that help a company win orders in the market.

Example: Consider a simple restaurant that makes and delivers pizzas. Order qualifiers might be low price (say, less than \$10.00) and quick delivery (say, under 15 minutes) because this is a standard that has been set by competing pizza restaurants. The order winners may be “fresh ingredients” and “home-made taste.” These characteristics may differentiate the restaurant from all the other pizza restaurants. However, regardless of how good the pizza, the restaurant will not succeed if it does not meet the minimum standard for order qualifiers. Knowing the order winners and order qualifiers in a particular market is critical to focusing on the right competitive priorities.

2.4. Focused Operations

Focused Operations Management shows how to do much more with existing resources in terms of throughput, response time and quality. It provides a system view and will touch upon performance measures, operations management, quality, cost-accounting, pricing, and above all, value creation and value enhancement. Recently, Slack and Lewis (2002, pp. 98-106) summarized the discussion on focus. They argue that just as there are many ways of segmenting

markets, there are also several approaches to focusing operations where the organization of process technologies, staff and processes can be based on several criteria. They distinguish six approaches to focusing operations: a performance objective focus (to satisfy the performance requirements of a particular market), a product/service specification focus (to produce a range of products for a target market), a geographic focus (an operation for a specific geographic market), a variety focus (e.g. a division between standard and non-standard products), volume focus (high vs low), and a process requirements focus (based on similarity in process technology).

CHAPTER THREE

PRODUCT AND SERVICE DESIGN

Introduction

The essence of any organization is the product and service it offers. There is an obvious link between the design of those products and services and the success of the organization. Organizations that have well-designed products or services are successful than those with poorly designed products or services. Hence, organizations have a vital stake in achieving good product and service design.

Product and service design plays a strategic role in the degree to which an organization is able to achieve its goal. It is a major factor in customer satisfaction, product and service quality, and production cost. The customer connection is obvious: The main concern of the customer is the organization's products and services, which became the ultimate basis for judging the organization.

3.1. Reasons for Product or Service Development

Product development is the process through which companies react to market signals, respond to changes in customer demand, adopt new technologies, foray into new areas, and ensure continuous growth. It is a core process in achieving strategic objectives, renewal of the company business model and deterring competition from displacing the company from its market position.

New Product Development is a comprehensive set of multi-disciplinary processes that turns an opportunity into a product to satisfy a customer. Intense global competition, short product and technology lifecycles, unpredictable consumer buying patterns and possible market stagnation makes new product development a critical activity in most businesses. There are different reasons of developing new product or service.

Technological Reasons /Obsolete Technology

A company may need to discontinue production of a product since technology has become obsolete and is beyond maintenance. New technology requires development of new products.

Product Target Gap

Sometimes a company wants to deliver a promised benefit to its customers but, it delivers less than what is promised. A “Product Target Gap” is there. The company must change the product and come up with something new, unless it decides to make its product “Price Advantage Product” and targets relatively more bargain seekers.

Economic Reasons

1. Price and/or availability of Raw Materials & Other Resources

Direct and/or indirect resources (both raw materials and skilled human resources may become unavailable over time. For instance, leather products, platinum, palladium, natural silk, black pearl, Matura diamond, etc. forced many companies to change their products. . Moreover, prices of resources may increase that may, in turn, increase the cost of production higher than/equal to choke price or close to it. In addition, prices of energy, raw materials, and wages may increase to a level to make the product unprofitable; the company goes for something new.

2. An Unutilized Resource

When a company cannot fully utilize a resource, it tries to develop a new product to better utilize that resource.

3. A New Resource

When a company acquires a new resource (a new formula, a new design, a new technology, a newer invention, a new raw material), it tries to make something new.

Changing Customer Tastes & Preferences Customers do not want to buy same thing again and again

They want something different over time. It is a common sense that over time our needs change as we want to enhance or to maintain our lifestyles. In our Product Inventory Basket (PIB), we add many new things and drop many old things. Consumer tastes & preferences change with age, marital status, occupational change, migration, and many other things.

1. Socio-legal Reasons

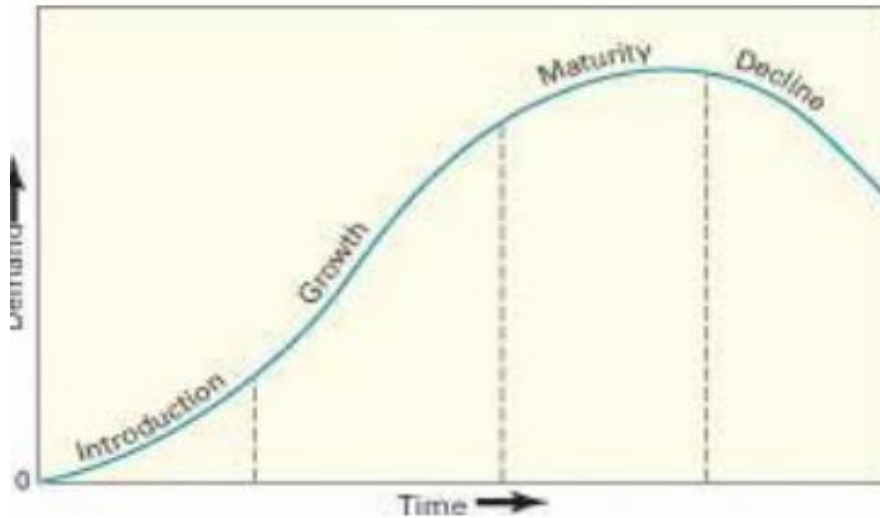
A company's one or more products may become illegal, immoral, or opposed to public policy. When Products or their production and/or disposal are Unfriendly for ecological balance and/or physical environment (called environmental outcomes of a product) must be changed and new ones must be developed.

2. Marketing Reason

- (a) A New Market Opportunity - When a company finds a new market for its product; it changes the product to better fulfil market-specific requirements.
- (b) Contestable Market Effect - If a product is bringing profit to its producers, others also start developing similar substitutes which are new products for them.
- (c) Competitive Moves: When a company's product is deleted from the distribution channel, it may need to change its product and/or develop a new product to better keep the channel members satisfied.

3.2. Strategies for New Product Introduction

Another factor in product design is the stage of the life cycle of the product. Most products go through a series of stages of changing product demand called the product life cycle. There are typically four stages of the product life cycle: introduction, growth, maturity, and decline. The product life cycle can be quite short for certain products, as seen in the computer industry. For other products it can be extremely long, as in the aircraft industry. A few products, such as paper, pencils, nails, milk, sugar, and flour, do not go through a life cycle. However, almost all products do, and some may spend a long time in one stage.



Introduction

- Weigh trade-offs between eliminating 'defects' and getting the product or service to the market at an advantageous time
- Accurate demand forecasts are important to ensuring adequate capacity availability

Introduction Stage of the PLC



3.3. New Product Development Process

To get to a final design of a product or service, the design activity must pass through several key stages. These form an approximate sequence, although in practice designers will often recycle or backtrack through the stages. The basic stages of product/service design are the following

1. Idea generation
2. Feasibility analysis

- | | |
|---|-------------------------|
| 3. Initial (Preliminary) Product Design | 6. Final Product Design |
| 4. Prototype development | 7. Product introduction |
| 5. Prototype Testing | 8. Follow-up evaluation |

1. Concept (idea) Generation: The ideas for new product or service concepts can come from sources outside the organization, such as customers or competitors, and from sources within the organization, such as staff (for example, from sales staff and front-of-house staff) or from the R&D department.

a) Ideas from customers: Marketing, the function generally responsible for identifying new product or service opportunities, may use many market research tools for gathering data from customers in a formal and structured way, including questionnaires and interviews. Listening to the customer, in a less structured way, is sometimes seen as a better means of generating new ideas. Focus groups, for example, are one formal but unstructured way of collecting ideas and suggestions from customers.

b) Ideas from competitor activity: All market-aware organizations follow the activities of their competitors. A new idea may give a competitor an edge in the marketplace, even if it is only a temporary one, then competing organizations will have to decide whether to imitate or alternatively to come up with a better or different idea. Sometimes this involves reverse engineering that is, taking apart a product to understand how a competing organization has made it.

c) Ideas from staff: The contact staff in a service organization or the salesperson in a product-oriented organization could meet customers every day. These staff may have good ideas about what customers like and do not like.

d) Ideas from research and development: One formal function found in some organizations is research and development. Research usually means attempting to develop new knowledge and ideas in order to solve a particular problem or to grasp an opportunity. Development is the attempt to utilize and operationalize the ideas that come from research.

2. Concept Screening (Feasibility Analysis)

Not all concepts which are generated will necessarily be capable of further development into products and services. The purpose of the concept-screening stage is to take the flow of concepts and evaluate them. Evaluation in design means assessing the worth or value of each design option, so that a choice can be made between them. This involves assessing each concept or option against a number of design criteria. Major criteria are the following:

- **Market feasibility:** evaluate the new ideas in terms of whether the product has market or not,
- **Financial (economic) feasibility:** how well the product quality performance and costs confirm to the design objectives and,
- **Technical feasibility:** availability of technology and skilled labor. At last, the most feasible and viable product concept is selected for the next stages.

3. Initial (Preliminary) Product Design

This stage of the product-design process is concerned with developing the best design for the new product ideas. It is the translation of ideas in to products. Preliminary product design must specify the product completely. At the end of the product design phase, the firm has a set of product specifications and engineering drawing (or computer image) specified in sufficient detail that production prototype can be built and tested. In the preliminary design tradeoffs between cost, quality and product performances are considered.

4. Prototype Construction

If the preliminary product design is approved, a prototype/s may be built for further testing and analysis. Several prototypes which closely resembles with the final products may be made by hand from some artificial materials such as plastics, mud, clay, wood etc. For example, the auto industry regularly makes clay models of new automobiles.

5. Prototype Testing:

A model is tested for its physical properties or uses under actual operating conditions. Testing of prototypes is aimed at verifying marketing and technical performance. The purpose of a test market is to gather quantitative data on customer acceptance of the new product. Prototype is

also tested for technical product performance. In general, such testing is important in uncovering any problems and correcting them prior to full scale production. For example, auto manufacturer perform extensive road tests on new models; similar experiments are performed on tires, airplanes, and sports equipment.

6. Final Product Design

Prototype testing may indicate certain changes in the preliminary product design. If changes are made the product may be tested further to ensure final product performance. Market test is used to determine the extent of consumer acceptance. If this market test is unsuccessful, return to the design review phase (this phase is handled by marketing). During the final phase these changes are incorporated in to the design specification. Drawing and specifications for the product are to be developed at this stage. And go for full scale operations.

7. Product Introduction: this stage is used to promote the product; handled by marketing.

8. Follow-up Evaluation: it helps to determines if changes are needed & refine forecasts; handled by marketing.

3.4. Value Analysis

Value analysis is defined as “an organized creative approach which has its objective, the efficient identification of unnecessary cost-cost which provides neither quality nor use nor life nor appearance nor customer features.” Value analysis focuses engineering, manufacturing and purchasing attention to one objective-equivalent performance at a lower cost. Value analysis refers to an examination of the *function* of parts and materials in an effort to reduce the cost and/or improve the performance of a product. Typical questions that would be asked as part of the analysis include: Could a cheaper part or material be used? Is the function necessary? Can the function of two or more parts or components be performed by a single part for a lower cost? Can a part be simplified? Could product specifications be relaxed, and would this result in a lower price? Could standard parts be substituted for nonstandard parts?

Value analysis is concerned with the costs added due to inefficient or unnecessary specifications and features.

Value is not inherent in a product, it is a relative term, and value can change with time and place. It can be measured only by comparison with other products which perform the same function. Value is the relationship between what someone wants and what he is willing to pay for it. In fact, the heart of value analysis technique is the functional approach. It relates to cost of function whereas others relate cost to product. It is denoted by the ratio between function and cost.

$$\text{Value} = \frac{\text{Function}}{\text{Cost}}.$$

3.4.1 Steps in Value Analysis

In order to answer the above questions, three basic steps are necessary:

1. Identifying the function: Any useful product has some primary function which must be identified—a bulb to give light, a refrigerator to preserve food, etc. In addition it may have secondary functions such as withstanding shock, etc. These two must be identified.

2. Evaluation of the function by comparison: Value being a relative term, the comparison approach must be used to evaluate functions. The basic question is, ‘Does the function accomplish reliability at the best cost’ and can be answered only comparison.

3. Develop alternatives: Realistic situations must be faced, objections should overcome and effective engineering manufacturing and other alternatives must be developed. In order to develop effective alternatives and identify unnecessary cost the following thirteen value analysis principles must be used:

- Avoid generalities.
- Get all available costs.
- Use information only from the best source.
- Brain-storming sessions.
- Blast, create and refine: In the blast stage, alternative productive products, materials, processes or ideas are generated. In the ‘create’ stage the ideas generated in the blast stage are used to generate alternatives which accomplish the function almost totally. In the refining stage the alternatives generated are sifted and refined so as to arrive at the final alternative to be implemented.

- Identify and overcome road blocks.
- Use industry specialists to extend specialized knowledge.
- Key tolerance not to be too light.
- Utilize the pay for vendors' skills techniques.
- Utilize vendors' available functional products.
- Utilize specialty processes.
- Utilize applicable standards.
- Use the criterion 'Would I spend my money this way?'

3.5. Product Design

Product design brings together marketing analysts, art directors, sales forecasters, engineers, finance experts, and other members of a company to think and plan strategically. It is exciting and creative, and it can spell success or disaster. Product design is the process of defining all the features and characteristics of just about anything you can think of, for both products and services.

Consumers respond to a product's appearance, color, texture, performance. All of its features, summed up, are the product's design. Someone came up with the idea of what this product will look like, taste like, or feel like so that it will appeal to you. This is the purpose of product design.

Product design: defines a product's characteristics, such as its appearance, the materials it is made of, its dimensions and tolerances, and its performance standards.

3.5.1. The product design process

Certain steps are common in the development of most product designs. *They are idea generation, product screening, preliminary design and testing, and final design.*

Step 1: Idea Generation

- ➡ All product designs begin with an idea. The idea might come from a product manager who spends time with customers and has a sense of what customers want, from an engineer with a flair for inventions, or from anyone else in the company.

- ➡ To remain competitive, companies must be innovative and bring out new products regularly.

Sources of product ideas

➤ Ideas from Customers, Competitors, and Suppliers

The first sources of ideas are **customers**, the driving force in the design of goods and services.

- ➡ Marketing is a vital link between customers and product design.
- ➡ Market researchers collect customer information by studying customer buying patterns and using tools such as customer surveys and focus groups.
- ➡ Management may love an idea, but if market analysis shows that customers do not like it, the idea is not viable.
- ➡ Analysing customer preferences is an on-going process.
- ➡ Customer preferences next year may be quite different from what they are today. For this reason, the related process of forecasting future consumer preferences is important, though difficult.

Competitors are another source of ideas. A company learns by observing its competitors' products and their success rate.

- ➡ This includes looking at product design, pricing strategy, and other aspects of the operation.
- ➡ Studying the practices of companies considered “best in class” and comparing the performance of our company against theirs is called **benchmarking**.
- ➡ **Reverse Engineering**: Another way of using competitors' ideas is to buy a competitor's new product and study its design features.
- ➡ Using a process called **reverse engineering**; a company's engineers carefully disassemble the product and analyse its parts and features.
- ➡ Product design ideas are also generated by a company's R & D (research and development) department, whose role is to develop product and process innovation.

Suppliers are another source of product design ideas.

- ➡ To remain competitive more companies are developing partnering relationships with their suppliers, to jointly satisfy the end customer.
- ➡ Suppliers participate in a program called **early supplier involvement (ESI)** where suppliers are involved in the early stages of product design.

Step 2: Screening

- ☞ After a product idea has been developed it is evaluated to determine its likelihood of success this is called **product screening**.
- ☞ The company's product screening team evaluates the product design idea according to the needs of the major business functions. In their evaluation, executives from each function area may explore issues such as the following:
 - ➡ **Operations** What are the production needs of the proposed new product and how do they match our existing resources? Will we need new facilities and equipment? Do we have the labour skills to make the product? Can the material for production be readily obtained?
 - ➡ **Marketing** What is the potential size of the market for the proposed new product? How much effort will be needed to develop a market for the product and what is the long-term product potential?
 - ➡ **Finance** The production of a new product is a financial investment like any other. What are the proposed new product's financial potential, cost, and return on investment?

To assist in product analysis, several methods have been developed.

- I. **Screening by check list:** One is a check list scoring method that involves developing a list of factors along with a weight for each. If the total score is above a certain minimum level the new product idea may be selected for further development. Alternatively, the method may be used to rank product ideas in priority order for selection.
- II. **Break-Even Analysis: A Tool for Product Screening**
Break-even analysis is a technique that can be useful when evaluating a new product. This technique computes the quantity of goods a company needs to sell just to cover its costs, or break even, called the "break-even" point. When evaluating an idea for a new product it is helpful to compute its break-even quantity. An assessment can then be made as to how

difficult or easy it will be to cover costs and make a profit. A product with a break-even quantity that is hard to attain might not be a good product choice to pursue. Next we look at how to compute the break-even quantity.

The total cost of producing a product or service is the sum of its fixed and variable costs. A company incurs **fixed costs** regardless of how much it produces. Fixed costs include overhead, taxes, and insurance. For example, a company must pay for overhead even if it produces nothing. **Variable costs**, on the other hand, are costs that vary directly with the amount of units produced, and include items such as direct materials and labor. Together, fixed and variable costs add up to total cost:

$$\text{Total cost} = \text{FC} + \text{VC}(Q)$$

Where F = fixed cost

VC = variable cost per unit

Q = number of units sold

Figure 3.1 shows a graphical representation of these costs as well as the break-even quantity. Fixed cost is represented by a horizontal line as this cost is the same regardless of how much is produced. Adding variable cost to fixed cost creates *total cost*, represented by the diagonal line above fixed cost. When $Q = 0$, total cost is only equal to fixed cost. As Q increases, total cost increases through the variable cost component. The blue diagonal in the figure is revenue, the amount of money brought in from sales:

$$\text{Revenue} = (SP) Q$$

Where SP = selling price per unit

When $Q = 0$, revenue is zero. As sales increase, so does revenue. Remember, however, that to cover all costs we have to sell the break-even amount. This is the quantity Q_{BE} , where revenue equals total cost. If we sell below the break-even point we incur a loss, since costs exceed revenue. To make a profit, we have to sell above the break-even point. Since revenue equals total cost at the break-even point, we can use the previous equations to compute the value of the break-even quantity:

Total cost= total revenue

$$F + (VC) Q = (SP) Q$$

Solving for Q, we get the following equation:

$$BEQ = \frac{\text{Fixed cost}}{\text{selling price/unit} - \text{variable cost/unit}}$$

Note that we could also find the break-even point by drawing the graph and finding where the total cost and revenue lines cross.

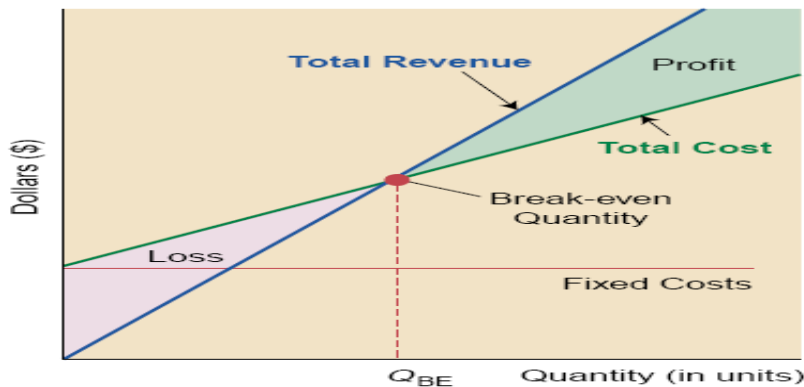


Figure 3.1: Graphical approach to break-even analysis

Break-even analysis is useful for more than just deciding between different products. It can be used to make other decisions, such as evaluating different processes or deciding whether the company should make or buy a product.

Step 3: Initial (Preliminary) Product Design and Testing

Once a product idea has passed the screening stage, it is time to begin preliminary design and testing. At this stage, design engineers translate general performance specifications into technical specifications. Prototypes are built and tested. Changes are made based on test results, and the process of revising, rebuilding a prototype, and testing continues.

Step 4: Final Product Design

Following extensive design testing the product moves to the final design stage. This is where final product specifications are drawn up. The final specifications are then translated into specific processing instructions to manufacture the product, which include selecting equipment, outlining jobs that need to be performed, identifying specific materials needed and suppliers that will be used, and all the other aspects of organizing the process of product production.

3.5.2. Factors to be Considered in Product Design

This section provides an overview of various approaches to product design, including product life cycles, manufacturing design, remanufacturing, robust design, concurrent engineering, computer-aided design, and modular design.

Product Life Cycle

Many new products go through a product life cycle in terms of demand. In the last stage of a life cycle, some firms adopt a defensive research posture whereby they attempt to prolong the useful life of a product or service by improving its reliability, reducing costs of producing it (and, hence, the price), redesigning it, or changing the packaging.

Manufacturing Design

The term design for manufacturing (DFM) is also used to indicate the designing of products that are compatible with an organization's capabilities. A related concept in manufacturing is design for assembly (DFA). A good design must take into account not only how a product will be fabricated, but also how it will be assembled. Design for assembly focuses on reducing the number of parts in an assembly, as well as the assembly methods and sequence that will be employed.

Environmental regulations and recycling have given rise to another concern for designers, design for recycling (DFR). Here the focus is on designing products to allow for disassembly of used products for the purpose of recovering components and materials for reuse.

Remanufacturing

Remanufacturing refers to removing some of the components of old products and reusing them in new products. This can be done by the original manufacturer, or another company.

Among the products that have remanufactured components are automobiles, printers, copiers, cameras, computers, and telephones.

Robust Design

Some products will perform as designed only within a narrow range of conditions, while other products will perform as designed over a much broader range of conditions. The latter have robust design. The more robust a product, the less likely it will fail due to a change in the environment in which it is used or in which it is performed.

Concurrent Engineering

To achieve a smoother transition from product design to production, and to decrease product development time, many companies are using simultaneous development, or concurrent engineering. In its narrowest sense, concurrent engineering means bringing design and manufacturing engineering people together early in the design phase to simultaneously develop the product and the processes for creating the product.

Computer-Aided Design (CAD)

Computers are increasingly used for product design. Computer-aided design (CAD) uses computer graphics for product design. The designer can modify an existing design or create a new one by means of a light pen, keyboard, a joystick, or a similar device. Once the design is entered into the computer, the designer can maneuver it on the screen. The designer can obtain a printed version of the completed design and file it electronically, making it accessible to people in the firm who need this information (e.g., marketing). A major benefit of CAD is the increased productivity of designers. No longer is it necessary to laboriously prepare mechanical drawings of products or parts and revise them repeatedly to correct errors or incorporate revisions.

Product Variety

The issue of product variety must be considered from both marketing and an operations point of view. From a marketing point of view, the advantage of a large number of products is the ability to offer customer more choices. Sales may drop if the firm does not offer as many products as its competitors. From an operations point of view, high product variety is seen as

leading to higher cost, greater complexity, and more difficulty in specializing equipment and people.

The ideal operations situation is often seen as a few high volume products with stabilized production configurations. Operations managers often prefer less product variety. There is an optimum amount of product variety which results in maximum profits. Both too little and too much product variety will lead to low profits.

Modular Design

Modular design makes it possible to have relatively high product variety and low component variety at the same time. The basic idea is to develop a series of basic product components, or modules that can be assembled into a large number of different products. To the customer, it appears there are a great number of different products. To operations, there are only a limited number of basic components and processes.

Controlling the number of different components that go into products is of great importance to operations, since this makes it possible to produce more efficiently for large volumes while also allowing standardization of processes and equipment. A large number of product variations greatly increase the complexity and cost of operations.

Modular design offers a fundamental way to change product design thinking. Instead of designing each product separately, the company designs products around standard component modules and standard processes. Common modules should be developed that can serve more than one product line, and unnecessary product frills should be eliminated. This approach will still allow for a great deal of product varieties but the number of unnecessary product variations will be reduced.

The usual way to develop products is to design each one separately without much attention to the other products in the line. Each product is optimized, but the product line as a whole is not. Modular design requires a broader view of product lines, and it may call for changes in individual products to optimize the product line in its entirety.

3.6. Service Design

Service is an act, something that is done to or for a customer (client, patient, etc.). It is provided by a service delivery system, which includes the facilities, processes, and skills needed to provide the service.

Service design: is the process of establishing all the characteristics of service including physical, sensual and psychological benefits.

- ◆ Physical (facilitating good)
- ◆ Sensual (explicit service)
- ◆ Psychological(implicit service)

3.7.1. Phases in Service Design

1. **Conceptualize:** Idea generation, assessment of customer wants /needs, assessment of demand potential
2. Identify **service package** components
3. Determine **performance specifications**
4. Translate performance specifications into **design specifications**
5. Translate design specifications into delivery specifications

3.7.2. Service Blueprinting

A useful tool for conceptualizing a service delivery system is the service blueprint, which is a method for describing and analyzing a service process. A service blueprint is much like an architectural drawing, but instead of showing building dimensions and other construction features, a service blueprint shows the basic customer and service actions involved in a service operation.

Generally, design of the service is the specifications of how the service should be delivered. A service blueprint is “a picture or map that accurately portrays the service system so that the different people involved in providing it can understand and deal with it objectively regardless of their roles or their individual points of view. It depicts the process of service delivery, roles of

customers, roles of service employees, and visible elements of service. As a design tools, they enable management to study and analyze services prior to their actual implementation.

3.7.3. How are Services different from Manufacturing

A. Intangible Product

Service organizations produce an intangible product, which cannot be touched or seen. It cannot be stored in inventory for later use or traded in for another model. The service produced is *experienced* by the customer. The design of the service needs to specify exactly what the customer is supposed to experience. For example, it may be relaxation, comfort, and pampering or it may be efficiency and speed.

Defining the customer experience is part of the service design. It requires identifying precisely what the customer is going to feel and think, and consequently how he or she is going to behave. This is not always as easy as it might seem. The experience of the customer is directly related to customer expectations.

B. High Degree of Customer Contact

Service organizations typically have a high degree of customer contact. The customer is often present while the service is being delivered, such as at a theater, restaurant, or bank. Also, the contact between the customer and service provider is often the service itself, such as what you experience at a doctor's office. For a service to be successful this contact needs to be a positive experience for the customer, and this depends greatly on the service provider.

Chapter four

Process selection and capacity planning

4.1 Process Selection

In Chapter Three, we have discussed issues involved in product design. Though product design is important for a company, it cannot be considered separately from the selection of the process. In this section, we will look at issues involved in process design.

Process is a method used in an industry for doing or making something. Process transforms inputs into outputs. Consequently, they are at the core of operations management. Not surprisingly, process planning plays a key role in the ability of an organization to achieve its mission. Process selection refers to the way an organization chooses to produce its goods or provide its services.

- ▶ Essentially it involves the choice of technology and related issues, and it has major implications for capacity planning, layout of facilities, equipment, and design of work systems. Process selection occurs as a matter of course when new products or services are being planned. However, it also occurs periodically due to technological changes in equipment. How an organization approaches process selection is determined by the organization's process strategy.

4.1.1. Types of Processes

One of the first decisions a manager makes in designing a well-functioning process is to choose a process type that best achieves the relative importance placed on quality, time flexibility and cost. The manager has five process types, which form a continuum, to choose from:

- | | |
|---|--|
|  Project |  Line, and |
|  Job |  Continuous |
|  Batch | |

1. Project Process

A project process deals with one-of-a kind products that are tailored to the unique requirement of each customer. Projects are set up to handle complex jobs that involve unique sets of activities. Examples of a project process are building a shopping center, planning a major event, running a political campaign, doing management consulting work, or developing a new technology or product. A project process is characterized by a high degree of job customization, the large scope

of each project, and the release of substantial resources once a project is completed. A project process lies at the high customization low volume end of the process choice continuum. The sequence of operations and the process involved in each are unique to the project, creating one-of-a-kind products or services made specifically to customer orders.

2. Job process

Next in the continuum of process choices is the job process. It is appropriate for manufacturers of small batches of many different products, each of which is custom designed and, consequently, requires its own unique set of processing steps, or routing, through the production process. Examples are providing emergency room care, handling special delivery mail or making customized cabinets. Customization is relatively high and volume for any one product or service is low. However, volumes are not as low as for a project process, which by definition does not produce in quantity. The work force and equipment are flexible and handle various tasks.

As with a project process, companies choosing a job process often bid for work. Typically, they make products to order and do not produce them ahead of time. A job process primarily organizes all like resources around itself; equipment and workers capable of certain types of work are located together. These resources process all jobs requiring that type of work. Because customization is high and most jobs have a different sequence of processing steps, this process choice creates jumbled flows through the operations rather than a line flow.

Characteristics of Job process

- | | |
|---|--|
| ✓ Small production runs | ✓ Highly skilled labor |
| ✓ Discontinuous flow of materials | ✓ Highly competent knowledgeable supervision |
| ✓ Disproportionate manufacturing cycle time | ✓ Large work-in-progress |
| ✓ General purpose machines and process layout | ✓ Limited functions of production planning and control |

3. Batch Process

Batch process is a step up from job process in terms of product standardization, but it is not as standardized as line process. Example of a batch process are scheduling air travel for a group, making components that feed an assembly line, processing mortgage loans, and manufacturing capital equipment. A batch process differs from the job process with respect to volume, variety and quantity.

- ◆ The primary difference is that volumes are higher because the same or similar products or services are provided repeatedly.
- ◆ Another difference is that a narrow range of products and services is provided. Variety is achieved more through an assemble-to-order strategy than the job process's make-to-order or customized services strategy. Some of the components going into the final product or service may be processed in advance.
- ◆ A third difference is that production lots or customer groups are handled in larger quantities (or batches) than they are with job processes. A batch of one product or customer grouping is processed, and then production is switched to the next one. Eventually, the first product or service is produced again. A batch process has average or moderate volumes, but variety is still too great to warrant dedicating a separate process for each product or service.

Characteristics of Batch process

- Short runs
- Skilled labor in specific trades
- Supervisor to possess knowledge of a specific process
- Limited span of control
- General purpose machines and process type of layout
- Manual materials handling
- Manufacturing cycle time affected due to queues
- Large work-in-progress • Flexibility of production schedules
- Need to have production planning and control

4. Line Process

Line Process is characterized by a linear sequence of operations used to make the product or service. Products created by a line process include automobiles, appliances and toys. Services based on a line process are fast food restaurants and cafeterias. A line process lies between the batch and continuous processes on the continuum, volumes are high, and products or services are standardized, which allows resources to be organized around a product or service. There are line flows, with little inventory held between operations. Each operation performs the same process over and over, with little variability in the products or services provided. Production orders are not directly linked to customer orders, as is the case with project and job process.

Characteristics of Line process

- Continuous flow of material
- Special purpose machines and product type layout
- Mechanized materials handling
- Low skilled labor
- Short manufacturing cycle time
- Easy supervision
- Limited work-in-progress
- Lesser flexibility in production schedules

5. Continuous Process

Continuous processes systems are sometimes referred to as flow systems because of the rapid rate at which items move through the system. This form of processing is used when highly standardized products are involved. Examples are petroleum refineries, chemical plants, and plants making beer, steel, and food. Firms with such facilities are also referred to as the process industry. A continuous process is the extreme end of high volume, standardized production with rigid line flows. Its name derives from the way materials move through the process.

Characteristics of Continuous Process

- Special purpose machines with built-in controls
- Highly mechanized/automated materials handling

- Virtually zero manufacturing cycle time
- Low skilled labor
- Supervisor to be processes specialist
- Negligible work-in-progress
- Limited production planning and control functions

4.2. Capacity Planning

Capacity issues are important for all organizations, and at all levels of an organization. **Capacity** refers to an upper limit or ceiling on the load that an operating unit can handle. The operating unit might be a plant, department, machine, store, or worker. The capacity of an operating unit is an important piece of information for planning purposes: It enables managers to quantify production capability in terms of inputs or outputs, and thereby make other decisions or plans related to those quantities. **Capacity** can also be defined as the maximum output rate that can be achieved by a facility. The facility may be an entire organization, a division, or only one machine.

Capacity planning is the process of establishing the output rate that can be achieved by a facility. If a company does not plan its capacity correctly, it may find that it either does not have enough output capability to meet customer demands or has too much capacity sitting idle.

For many companies, measuring capacity can be straight forward. It is the maximum number of units that can be produced in a specific time. However, for some organization, determining capacity can be more difficult. Capacity can be measured in terms of beds (a hospital) or, active members (a church). Other organizations use total work time available as a measure of overall capacity. Different people have different interpretations of what capacity means, and the units of measurement are often very different.

Although it is functional, it can be refined into two useful definitions of capacity:

1. **Design capacity** is the maximum output rate that can be achieved by a facility under ideal conditions. Design capacity can be sustained only for a relatively short period of time. A company achieves this output rate by using many temporary measures, such as overtime, overstaffing, maximum use of equipment, and subcontracting.

- 2. Effective capacity** is the maximum output rate that can be sustained under normal conditions. These conditions include realistic work schedules and breaks, regular staff levels, scheduled machine maintenance, and none of the temporary measures that are used to achieve design capacity. Note that effective capacity is usually lower than design capacity.

Measuring Effectiveness of Capacity Use Regardless of how much capacity we have, we also need to measure how well we are utilizing it. **Capacity utilization** simply tells us how much of our capacity we are actually using. Capacity utilization can simply be computed as the ratio of actual output over capacity:

Design capacity is the maximum rate of output achieved under ideal conditions. Effective capacity is usually less than design capacity (it cannot exceed design capacity) owing to realities of changing product mix, the need for periodic maintenance of equipment, lunch breaks, coffee breaks, problems in scheduling and balancing operations and similar circumstances. Actual output cannot exceed effective capacity and is often less because of machine breakdown, absenteeism, shortage of materials and quality problems as well as factors that are outside the control of operations managers.

These different measures of capacity are useful in defining two measures of system effectiveness: *efficiency* and *utilization*. Efficiency is the ratio of actual output to effective capacity. Utilization is the ratio of actual output to design capacity

$$\text{Efficiency} = \frac{\text{actual output}}{\text{effective capacity}} \times 100\%$$

$$\text{Utilization} = \frac{\text{actual output}}{\text{design capacity}} \times 100\%$$

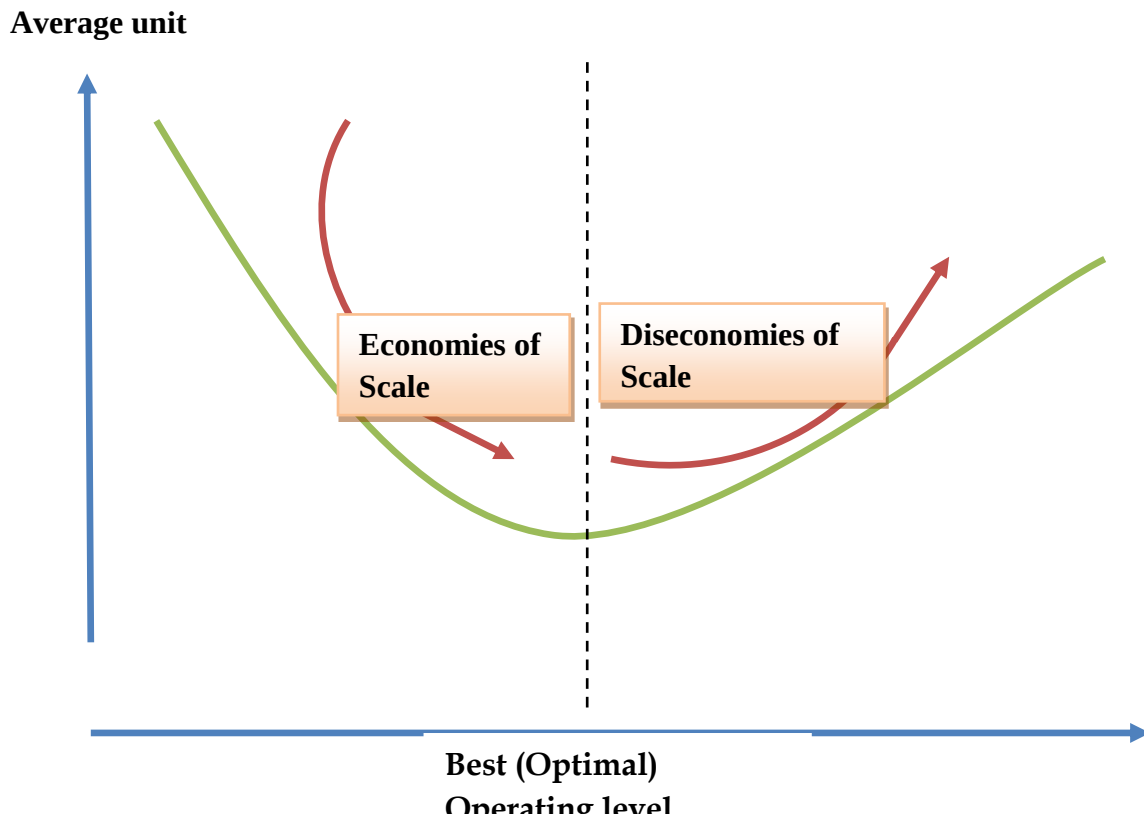
4.3.Developing Capacity Alternatives

Aside from the general considerations about the development of alternatives (i.e., conduct a reasonable search for possible alternatives, consider doing nothing, take care not to overlook non-quantitative factors), some specific considerations are relevant to developing capacity alternatives. The considerations to be discussed in this section include the following:

A. Qualitatively

1. **Design flexibility into systems.** The long-term nature of many capacity decisions and the risks inherent in long-term forecasts suggest potential benefits from designing flexible systems. For example, provision for future expansion in the original design of a structure frequently can be obtained at a small price compared to what it would cost to remodel an existing structure that did not have such a provision.
2. **Take a "big picture" approach to capacity changes.** A consideration for managers contemplating capacity increases is whether the capacity is for a new product or service, or a mature one. Mature products or services tend to be more predictable in terms of capacity requirements, and they may have limited life spans. The predictable demand pattern means less risk of choosing an incorrect capacity, but the possible limited life span of the product or service may necessitate finding an alternate use for the additional capacity at the end of the life span. New products tend to carry higher risk because of the uncertainty often associated with predicting the quantity and duration of demand. That makes flexibility appealing to managers.
3. **Prepare to deal with capacity "chunks."** Capacity increases are often acquired in fairly large chunks rather than smooth increments, making it difficult to achieve a match between desired capacity and feasible capacity. For instance, the desired capacity of a certain operation may be 55 units per hour; but suppose that machines used for this operation are able to produce 40 units per hour each. One machine by itself would cause capacity to be 15 units per hour short of what is needed, but two machines would result in an excess capacity of 25 units per hour.
4. **Identify the optimal operating level.** Production units typically have an ideal or optimal level of operation in terms of unit cost of output. At the ideal level, cost per unit is the lowest for that production unit; larger or smaller rates of output will result in a higher unit

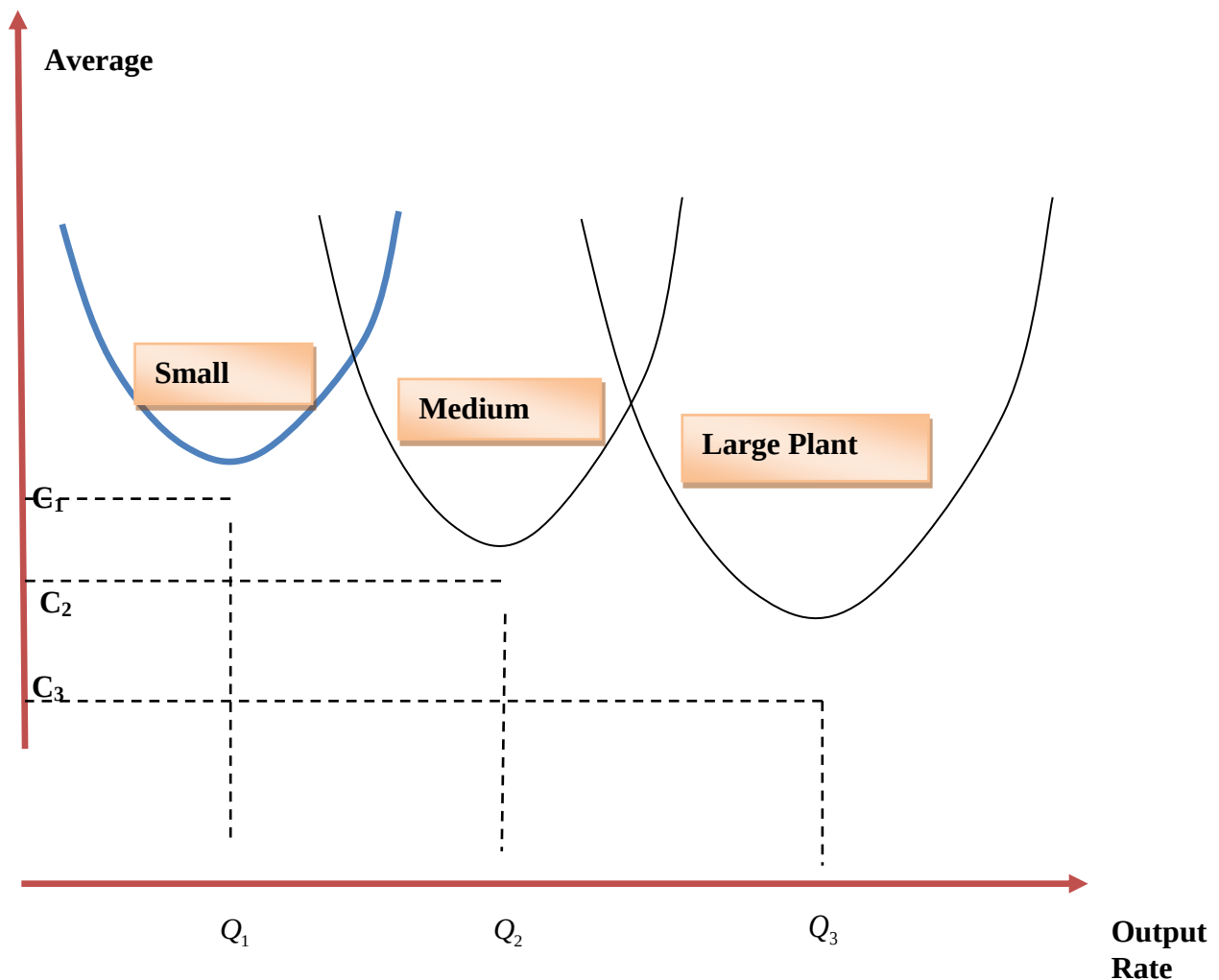
cost. The figure below illustrates this concept.



The explanation for the shape of the cost curve is that at low levels of output, the costs of facilities and equipment must be absorbed (paid for) by very few units. Hence, the cost per unit is high. As output is increased, there are more units to absorb the "fixed" cost of facilities and equipment, so unit costs decrease. A firm realizes economies of scale as long as each successive unit is produced at a lower cost than it preceding unit.

However, beyond a certain point (the optimal operating level), unit costs will start to rise even though the fixed costs are spread over even more and more units. The very causes for such average cost increment include worker fatigue and reduced morale, equipment breakdowns, the loss of flexibility, which leaves less of a margin for error, greater difficulty in coordinating operations, difficulty in scheduling, damaged goods, and increased use of overtime. Production of each successive unit beyond the optimum operating unit at a higher cost than the cost at which each preceding unit is produced is also what we call diseconomies of scale.

Both optimal operating rate and the amount of the minimum cost tend to be a function of the general capacity of the operating unit. For example, as the general capacity of a plant increases, the optimal output rate increases and the minimum cost for the optimal rate decreases. Thus, larger plants tend to have higher optimal output rates and lower minimum costs than smaller plants. Figure 4-2 illustrates these points.



The larger the size of the plant, the greater is also the optimum operating output rate and the smaller is the average cost at this output rate i.e. $Q_1 < Q_2 < Q_3$ and $C_1 > C_2 > C_3$ which are the average cost per units at the optimum operating levels of small plant, medium plant, and large plant respectively.

B. Quantitatively

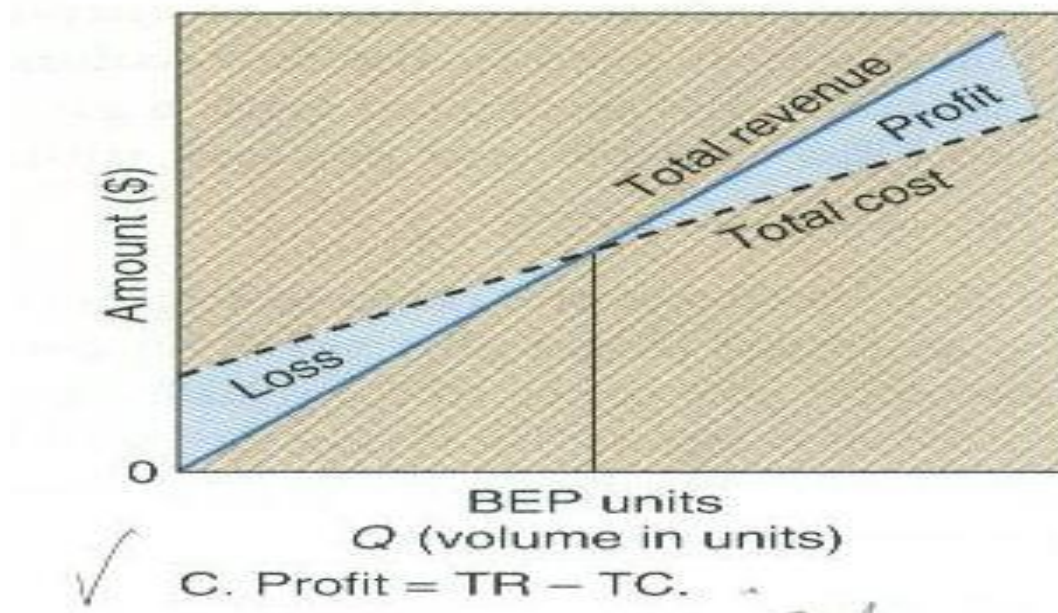
An organization needs to examine alternatives for future capacity from a number of different perspectives. Most obvious are economic considerations: Will an alternative be economically feasible? How much will it cost? How soon can we have it? What will operating and maintenance costs be? What will its useful life be? Will it be compatible with present personnel and present operations?

A number of techniques are useful for evaluating capacity alternatives from an economic standpoint. Some of the more common are cost-volume analysis, financial analysis, decision theory, and waiting-line analysis.

1. Cost-Volume Analysis

Cost-volume analysis, also called Break-Even Analysis, focuses on relationships between cost, revenue, and volume of output. The purpose of cost-volume analysis is to estimate the income of an organization under different operating conditions. It is particularly useful as a tool for comparing capacity alternatives.

Use of the technique requires identification of all costs related to the production of a given product. These costs are then assigned to fixed costs or variable costs. *Fixed costs* tend to remain constant regardless of volume of output. Examples include rental costs, property taxes, equipment costs, heating and cooling expenses, and certain administrative costs. *Variable costs* vary directly with volume of output. The major components of variable costs are generally materials and labor costs. We will assume that variable cost per unit remains the same regardless of volume of output.



The variable assuming the horizontal axis is volume of output in units and the vertical line represents corresponding dollar value of sales.

BEP units = break-even units

Assumptions of Cost-Volume Analysis:

1. One product is involved
2. Everything produced can be sold
3. Variable cost per unit is the same regardless of volume
4. Fixed costs do not change with volume
5. Revenue per unit is constant with volume
6. Revenue per unit exceeds variable cost per unit

Example

Publishing company intends to publish a book in residential landscaping. Fixed cost is \$125,000 per year, variable cost per unit is \$32, and selling price per unit is \$42.

- A) How many units must be sold per year to break even?
- B) How much annual revenue is required to break even?
- C) If annual sales are 20,000 units, what are the annual profits?

D) What variable cost per unit would result in \$100,000 annual profits if annual sales are 20,000 units?

2. Decision Trees Approach

Decision tree is a quantitative technique evaluating alternatives and structures a complex and multiphase decisions by showing:

- The decisions that must be made (the candidate alternatives available)
- The sequence in which the decisions must occur
- The interdependence among the decisions (shows main branches and sub-branches)

It allows objective evaluation of alternatives incorporates uncertainty and develops expected (anticipated) values or payoffs.

Example:

Rafi Café is about to build a new restaurant. An architect has developed three building designs, each with a different seating capacity. Rafi estimates that the average number of customers per hour will be 80, 100, or 120 with respective probabilities of 0.4, 0.2, and 0.4. The payoff table showing the profits for the three designs is depicted as below:

Pay off table:

Design Capacity	Anticipated Volume of profits for the Average Number of Customers Per Hour of:			
	C ₁ =80	C ₂ =100	C ₃ = 120	
A	\$10,000	\$15,000	\$15,000	
B	\$18,000	\$18,000	\$12,000	
C	\$6,000	\$16,000	\$21,000	
T	o	t	a	l

Required:

Depending on the information provided above, and using a decision tree approach, calculate the different facility alternatives' expected values and select the best alternative

3. Financial Analysis

- I. **Net Present Value (NPV):** is the difference between the sum of the present value of all the series of future cash flows of an investment proposal of a specific facility with a specific production capacity and the present value of the investment on it. Net present value summarizes the initial cost of an investment, its estimated annual cash flows, and any expected salvage value in a single value called the *equivalent current value*, taking into account the time value of money.

Under the net present value method, cash inflows are discounted to their present value and then compared with the capital outlay required by the investment. The interest rate used in discounting the future cash inflows is the required minimum rate of return. A proposal is acceptable when its NPV is zero or positive and the higher the positive NPV, the more attractive is the investment. In situations where there are two or more capacity alternatives, all with positive NPV, the one with the highest net present value is selected provided that they are exclusive to each other.

$$PV = \sum_{t=1}^n \frac{FV}{(1+r)^t}$$

Where $t=1, 2, 3, 4 \dots n$

And n is the number of years

FV = the amount of cash flow at the end of the i^{th} year.

r = the required rate of return or simply the discount rate.

PV = the present value of all the series of annual cash inflows for n years

Exercise

Assuming that the company requires a minimum return of 12% and depending on the information given in the table below, evaluate the two capacity alternatives using the NPV method and comment which one is economically better.

Capacity Alternative-I				Capacity Alternative-II		
Cash Flow	Year	PV factor (PV of Birr 1)	Present Value	Cash Flow	Year	Present Value =
(130,000)	0	1	(\$130,000)	(\$130,000)	0	$PV = R \left[\frac{1 - (1 + i)^{-n}}{i} \right]$ Where R is the amount of the annuity i.e. the regular amount of the income (\$26,000 in this case).
36,000	1	0.893	32,148	26,000	1	
32,000	2	0.797	25,504	26,000	2	
29,000	3	0.712	20,648	26,000	3	
27,000	4	0.636	17,172	26,000	4	
26,000	5	0.567	14,742	26,000	5	
24,000	6	0.507	12,168	26,000	6	
23,000	7	0.452	10,396	26,000	7	
22,000	8	0.404	8,888	26,000	8	
21,000	9	0.361	7,581	26,000	9	
20,000	10	0.322	6,440	26,000	10	
PV			\$155,687			\$146,906
NPV			\$25,687			\$16,906

The net value of the first alternative is greater than that of the second alternative and, thus, the former one is better.

Chapter five

Facility Location

5.1 The Need for Location Decisions

Location decisions arise for a variety of reasons:

- As part of a marketing strategy to expand markets
- Addition of new facilities
- Growth in demand that cannot be satisfied by expanding existing facilities
- Depletion of basic inputs/resources (e.g. mining & fishing)
- Cost advantage

5.2 The Nature of Location Decisions

Facility location is determining the best geographic location for a company's facility.

► *Facility location decisions are particularly important for two reasons.*

- First, they require long-term commitments in buildings and facilities, which means that mistakes can be difficult to correct.
- Second, these decisions require sizable financial investment and can have a large impact on operating costs and revenues.

The location decision often depends on the type of business. For industrial location decisions, the strategy is usually minimizing costs, although innovation and creativity may also be critical. For retail and professional service organizations, the strategy focuses on maximizing revenue. Warehouse location strategy, however, may be driven by a combination of cost and speed of delivery. A poor choice of location might result in excessive transportation costs, a shortage of qualified labor, loss of competitive advantage, inadequate supplies of raw materials, or some similar condition that is detrimental to operations. For service, a poor location could result in lack of customers and/or high operating costs. For both manufacturing and services, location decisions can have a significant impact on competitive advantage. Businesses therefore have to think long and hard about where to locate a new facility.

- ➡ In most cases, there is no one best location for a facility. Rather, there are a number of acceptable locations.
- ➡ One location may satisfy some factors whereas another location may be better for others. If a new location is being considered in order to provide more capacity, the company needs to consider options such as expanding the current facility if the current location is satisfactory.

5.3 Factors affecting location decisions

Many factors influence location decisions. However, it often happens that one or a few factors are so important that they dominate the decision. For example, in manufacturing, the potentially dominating factors usually include availability of an abundant energy and water supply and proximity to raw materials. In service organizations, possible dominating factors are market related and include traffic patterns, convenience, and competitors' locations, as well as proximity to the market. This section presents you a brief description of some of these important factors.

The main factors that affect location decisions are presented as follows:

- i. **Global- international:** is the highest level in the location decision hierarchy. Decision makers who are considering expanding in to a new country must consider macroeconomic, demographic, and political issues of long-term significance. They must consider international trade issues, such as
 - International trade issues (currency exchange risk, balance of trade, quotas, tariffs etc.)
 - market access issues (such as free trade agreement, consumer sentiment towards imported goods)
 - labor issues (availability, wages, skill and training, and regulations)
 - Political concerns (stability of current regime, risk of asset nationalization, local ownership laws etc.)
 - Cultural issues (compatibility of business practices and products with local culture)
 - Legal issues (environmental regulations, accounting & reporting requirements etc)

ii. **Regional considerations:** once the decision has been made to locate a facility in a particular country, decision makers must choose a region based on regional issues like:

- *Supply issues (availability of material inputs):* select the region endowed with abundant or sufficient inputs.
- *proximity to market:* select the region which is near to the major market if product is perishable, fragile, if the product needs large transportation space, transportation cost is expensive, and if further processing increases the volume, bulk, fragility and perish ability.
- *proximity to material:* a possible reason to select regions which is near resources includes: when further processing reduces the bulk (e.g. sugar processing), perish ability (e.g., freezing, canning, pasteurizing etc)
- *Transportation facility:* transformation facilities are essential for the economic operations of production systems. Operations manager must study the characteristics of the new materials and finished products in regard to their transportation (water, railroad, road, pipelines, and airline transport) need and search for the location (region) that provides facilities of transportation with a reasonable cost.
- *Climate:* select the region with favorable climate which is important in order to acquire and maintain productive work forces. Certain industries such as agricultural business require specific climatic conditions.

iii. **Community considerations:** choice of community depends on the following factors:

- *Community attitudes:* in order to ensure the long term existences in that community, it is mandatory to win the interest, enthusiasms, and cooperation of the society. Otherwise, poor relation with community will result in putting the survival of the organization under question mark. So select the community with positive attitudes towards the company.
- *Community government and financial incentives:* it is important to assess the current situation and attempt to predict the future situations in regard to the policies of local government. In general stable, competent, honest, and cooperative government officials are great asset to a newly located company. Local government may be evaluated in terms of financial incentives they offer, taxation policies, peace and securities etc.

- *Community facilities*: is concerned with the availability of schools, churches, medical facilities, residential housings, recreational opportunities, highways etc.
- iv. **Site considerations**: after identifying the community in the already selected region, the final step in location selection decision is to screen out the best site out of the possible ones. The followings are the basic factors that influence site selection.
- *Size and cost of the site*: while selecting exact site, consider the size of the site and its cost (building and construction costs). Generally, the size of the site must be large enough to satisfy requirements such as employees parking requirement & future expansion plans and the cost of the land must be reasonable.
 - *Drainage and soil condition*: poor drainage leads to accumulations of water around the plant which may be harmful for some organization. Similarly, if the load bearing capacity of the soil is low, it will be difficult to establish sound building foundations.
 - *Land development cost*: cost related to excavation, grading, filling, construction of road, sidings, etc must also be taken in to consideration while selecting the site.
 - *Utilities*: selection of site is influenced by cost of acquiring and using utilities like electricity, natural gas, water as well as disposal facilities. For example all enterprises need safe and pure water. Some organizations like breweries need water even with some extra ordinary quality. So select the site which is better furnished with utilities.
 - *Access concerns*: select the site which is easily accessible for customers, suppliers, utilities and other concerned bodies.

Evaluating Location Alternatives:

There are a number of techniques that are helpful in evaluating location alternatives: factor rating method, centre of gravity method, Locational cost-profit-volume analysis and transportation model.

1. Factor Rating Method:

A typical location decision involves both qualitative and quantitative inputs, which tend to vary from situation depending on the needs of each organization. Factor rating is a general approach that is useful forevaluating a given alternative and comparing alternatives. The process of selecting a new facility location involves a series of following steps:

- ✚ Determine which factors are relevant (e.g., location of market, water supply, parking, revenue potential)
- ✚ Assign a weight to each factor that indicates its relative importance compared with all other factors. Typically, weights sum to 1.00.
- ✚ Decide on a common scale for all factors (e.g., 0 to 100)
- ✚ Score each location alternative
- ✚ Multiply the factor weight by the score for each factor, and sum the results for each location alternative
- ✚ Choose the alternative that has the highest composite score

Example: Assume that a small manufacturing firm has recently decided to expand its operations to include several new lines which it hopes to produce in a separate location because of space limitations in its existing plant. The following rating sheet illustrates relevant factors and factor weightings for two alternative locations.

		Weighted score (out of 100)		Weighted score	
Factor/criteria	Weight	Site A	Site B	Site A	Site B
Proximity to the existing store	0.1	100	60	$0.1 \times 100 = 10$	$0.1 \times 60 = 6$
Traffic volume	0.05	80	80	$0.05 \times 80 = 4$	$0.05 \times 80 = 4$
Rental cost	0.40	70	90	$0.40 \times 70 = 28$	$0.40 \times 90 = 36$
Size	0.10	86	92	$0.10 \times 86 = 8.6$	$0.10 \times 92 = 9.2$
Layout	0.20	40	70	$0.20 \times 40 = 8$	$0.20 \times 70 = 14$
Operating cost	0.15	80	90	$0.15 \times 80 = 12$	$0.15 \times 90 = 13.5$
TOTAL	1.00			70.6	82.7

At the weighted score of the two alternatives indicate, alternative A=70 and alternative B= 85.7. Therefore, based on factor rating method, alternative B is selected which has the higher weighted score value.

2. Location Break Even Analysis:

Break-Even Analysis - is a technique used to compute the amount of goods that must be sold just to cover costs. The break-even point is precisely the quantity of goods a company needs to sell to break even. Break-even analysis is the use of cost-volume analysis to make an economic comparison of location alternatives. Whatever is sold above that point will bring a profit. At the break-even point, total cost and total revenue are equal and the equation will use those to solve for Q, which is the break-even quantity:

$$PQ = F + VQ \quad \text{where, } Q = F / P - V$$

Step 1: For Each Location, determine Fixed and Variable Costs. Fixed costs are incurred regardless of how many units are produced and include items such as overhead, taxes, and insurance. Variable costs are costs that vary directly with the number of units produced and include items such as materials and labour. Total cost is the sum of fixed and variable costs.

Step 2: Plot the Total Costs for each Location on one Graph. To plot any straight line we need two points. One point is $Q = 0$, which is the y intercept. Another point can be selected arbitrarily, but it is best to use the expected volume of sales in the future.

Step 3: Identify ranges of output for which each location has the lowest total cost.

Step 4: Solve Algebraically for the Break-Even Points over the Identified Ranges. Select the location that gives the lowest cost for the range of output required by the new facility

Example: Fixed and variable costs for three potential plant locations are shown below: If the selling price is = Br. 120 Expected volume = 2,000 units

Location	Fixed cost per year	Variable cost per unit	Total cost
Hawassa	Br. 30,000	Br. 75	Br. 180,000
Dredawa	60,000	45	150,000
Addis Ababa	110,000	25	160,000

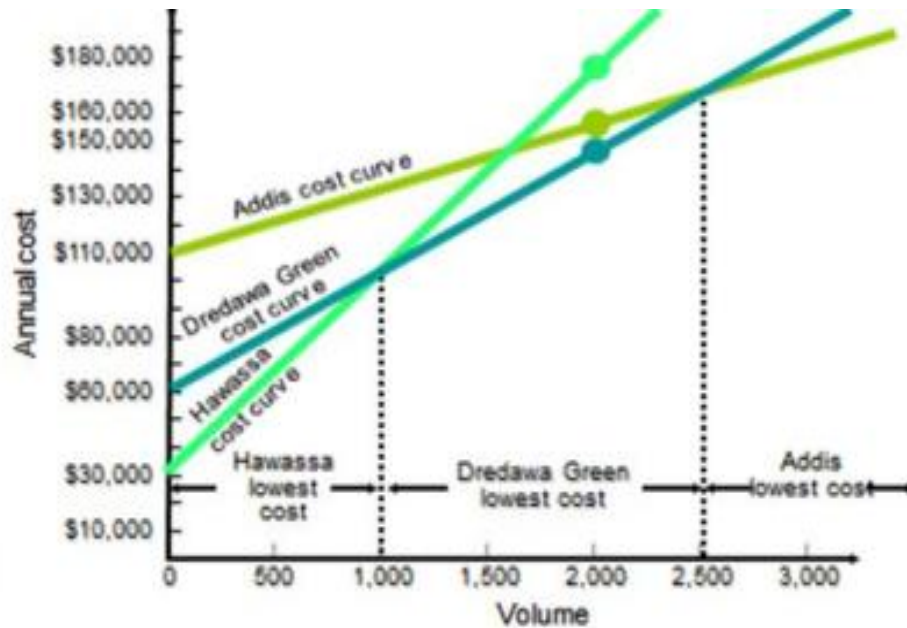


Figure 5.1 From the figure 2 we can understand that up to 1,000 units the best alternative is Hawassa; while from 1,000 to 2,500 units the best alternative is Dredawa and beyond 2, 500 is Addis.

3. The Centre of Gravity Method:

The center of gravity method is a method to determine the location of a distribution center that will minimize distribution costs. It treats distribution cost as a linear function of the distance and the quantity shipped. The center of gravity method takes into account the locations of plants and markets, the volume of goods moved, and transportation costs in arriving at the best location for a single intermediate warehouse. The center of gravity is defined to be the location that minimizes the weighted distance between the warehouse and its supply and distribution points, where the distance is weighted by the number of tones supplied or consumed. The first step in this procedure is to place the locations on a coordinate system. The origin of the coordinate system and scale used are arbitrary, just as long as the relative distances are correctly represented. This can be easily done by placing a grid over an ordinary map. The center of gravity is determined by the formula.

$$Cx = \frac{\sum d_{ix} V_i}{\sum V_i} \quad Cy = \frac{\sum d_{iy} V_i}{\sum V_i}$$

Where, Cx is X coordinate of the centre of gravity

CY is Y coordinate of the centre of gravity

Dix is X Coordinate of the i^{th} location

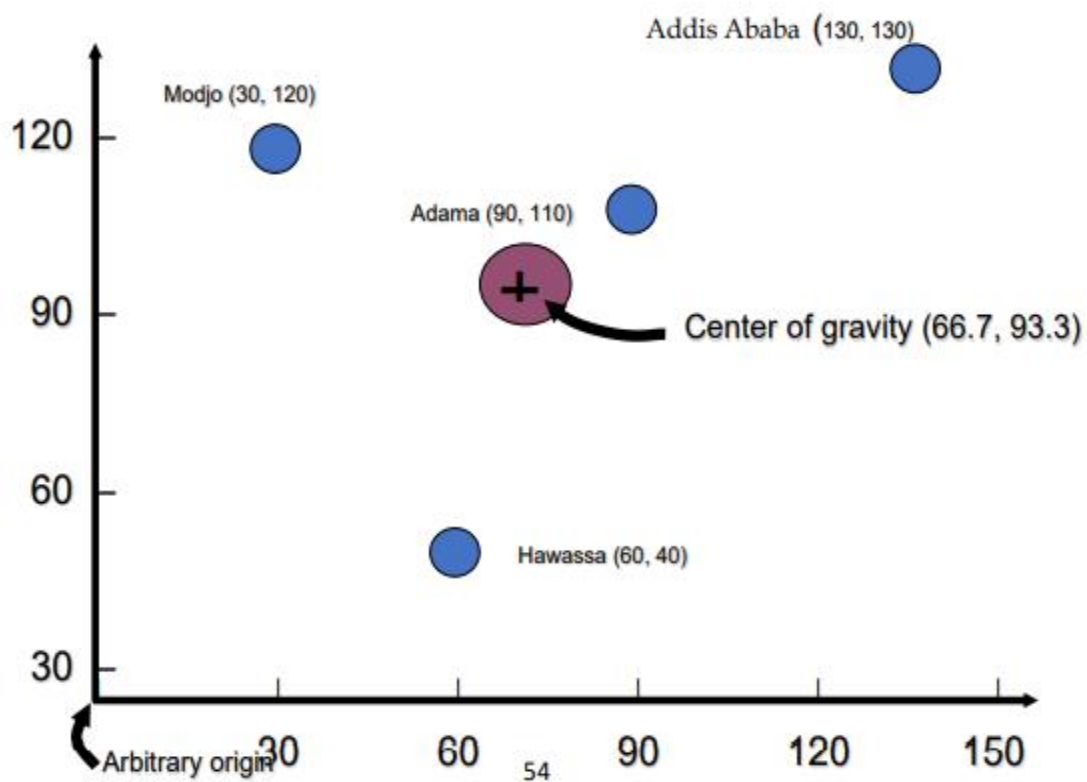
Diy is Y Coordinates of the i^{th} location

Vi is volume of goods transported to the i^{th} location

Example: A small manufacturing facility is being planned that will feed parts to three heavy manufacturing facilities. The locations of current plants with their coordinates and volume requirements are given in the following table.

Plant Location	Coordinates (X, Y)	Volume (Parts per month)
Modjo	30, 120	2,000
Adama	90, 110	1000
Addis Ababa	130, 130	1,000
Hawassa	60, 40	2,000

$$\begin{aligned}
 \text{x-coordinate} &= \frac{(30)(2000) + (90)(1000) + (130)(1000) + (60)(2000)}{2000 + 1000 + 1000 + 2000} \\
 &= 66.7 \\
 \text{y-coordinate} &= \frac{(120)(2000) + (110)(1000) + (130)(1000) + (40)(2000)}{2000 + 1000 + 1000 + 2000} \\
 &= 93.3
 \end{aligned}$$



4.

Chapter six

Facility layout

6.1. Layout Types

Facility layout decision entails determining the placement of departments, workstations, machines, and stockholding points within a productive facility.

The general objective of layout decision is to ensure a smooth flow of work, material, people, and information through the system.

Layout decisions are important for three basic reasons:

- (1) They require substantial investments of money and effort,
- (2) They involve long-term commitments, which make mistakes difficult to overcome, and
- (3) They have a significant impact on the cost and efficiency of short-term operations.

There are four basic layout types: *process*, *product*, *hybrid*, and *fixed position*. In this section we look at the basic characteristics of each of these types. Then we examine the details of designing some of the main types.

A. Process layouts

Process layouts: are layouts that group resources based on similar processes or functions. This type of layout is seen in companies with intermittent processing systems. You would see a process layout in environments in which a large variety of items are produced in a low volume.

Since many different items are produced, each with unique processing requirements, it is not possible to dedicate an entire facility to each item. It is more efficient to group resources based on their function. The products are then moved from one resource to another, based on their unique needs. The challenge in process layouts is to arrange resources to maximize efficiency and minimize waste of movement. If the process layout has not been designed properly, many products will have to be moved long distances, often on a daily basis.

Process layouts are very common. A hospital is an example of process layout. Departments are grouped based on their function, such as cardiology, radiology, laboratory, oncology, and pediatrics. The patient, the product in this case, is moved between departments based on his or her individual needs. A university is another example. Colleges and departments are grouped

based on their function. You, the student, move between departments based on the unique program you have chosen.

Advantage of process layout

- ✓ Flexibility of equipment and labor assignment
- ✓ Breakdown of one machine does not stop the whole process

Disadvantage of process layout

- ✓ Orders take more time and money
- ✓ High labor skill increases required level of training and experience

Recall that process layouts are designed to produce many different items, often to customer specifications. To achieve this goal they have certain unique characteristics:

1. *Resources used are general purpose.* The resources in a process layout need to be capable of producing many different products.
2. *Facilities are less capital intensive.* Process layouts have less automation, which is typically devoted to the production of one product.
3. *Facilities are more labor intensive.* Process layouts typically rely on higher-skilled workers who can perform different functions.
4. *Resources have greater flexibility.* Process layouts need to have the ability to easily add or delete products from their existing product line, depending on market demands.
5. *Processing rates are slower.* Process layouts produce many different products, and there is greater movement between workstations. Consequently, it takes longer to produce a product.
6. *Material handling costs are higher.* It costs more to move goods from one process to another.
7. *Scheduling resources is more challenging.* Scheduling equipment and machines is particularly important in this environment. If it is not done properly, long waiting lines can form in front of some work centers while others remain idle.
8. *Space requirements are higher.* This type of layout needs more space due to higher inventory storage needs. Improper design of process layouts can result in costly inefficiencies, such as high material handling costs. A good design can help bring order to an environment that might otherwise be very chaotic.

B. Product Layouts

Product layouts are layouts that arrange resources in a straight-line fashion to promote efficient production. They are called product layouts because all resources are arranged to meet the production needs of the product. This type of layout is used by companies that have repetitive processing systems and produce one or a few standardized products in large volume. Examples of product layouts are seen on assembly lines, in cafeterias, or even at a car wash. In product layouts the material moves continuously and uniformly through a series of workstations until the product is completed.

The challenge in designing product layouts is to arrange workstations in sequence and designate the jobs that will be performed by each station in order to produce the product in the most efficient way possible.

Advantage of product layout

- ✓ High rate of output
- ✓ Low unit due to high volume
- ✓ Labor specialization reduces training costs and time
- ✓ High utilization of labor and equipment

Disadvantage of product layout

- ✓ Poorly skilled workers exhibit little interests in maintaining equipment or in the quality of output
- ✓ The system is inflexible in response to changes in the volume of output or changes in product or process design.

Remember that product layouts are designed to produce one type or just a few types of products in high volume. Product layouts have the following characteristics:

1. *Resources are specialized.* Product layouts use specialized resources designed to produce large quantities of a product.
2. *Facilities are capital intensive.* Product layouts make heavy use of automation, which is specifically designed to increase production.
3. *Processing rates are faster.* Processing rates are fast, as all resources are arranged in sequence for efficient production.

4. *Material handling costs are lower.* Due to the arrangement of work centers in close proximity to one another, material handling costs are significantly lower than for process layouts.
5. *Space requirements for inventory storage are lower.* Product layouts have much faster processing rates and less need for inventory storage.
6. *Flexibility is low relative to the market.* Because all facilities and resources are specialized, product layouts are locked into producing one type of product. They cannot easily add or delete products from the existing product line.

The differences between process and product layouts

Process layouts	Product layout
◆ Able to produce a large number of different product	◆ Able to produce a small number of products efficiently
◆ Resource used are general purpose	◆ Resource used are specialized
◆ Facilities are more labour intensive	◆ Facilities are more capital intensive
◆ Greater flexibility relative to the market	◆ Low flexibility relative to the market
◆ Slower processing rate	◆ Processing rates are faster
◆ High material handling cost	◆ Lower material handling cost
◆ Higher space requirements	◆ Lower space requirements

C. Combination (hybrid) layout

Hybrid layouts combine aspects of both process and product layouts. This is the case in facilities where part of the operation is performed using an intermittent processing system and another part is performed using a continuous processing system.

D. Fixed-position layout

Fixed-position layout is used when the product is large and cannot be moved due to its size. All the resources for producing the product including equipment, labor, tools, and all other resources have to be brought to the site where the product is located. Examples of fixed-position layouts include building construction, dam or bridge construction, shipbuilding, or large aircraft manufacture. The challenge with a fixed-position layout is scheduling different work crews and jobs and managing the project.

6.2. Service Layouts

The major factors considered for service providers, is an impact of location on sales and customer satisfaction. Service facility layout will be designed based on degree of customer contact and the service needed by a customer. These service layouts follow conventional layouts as required. For example, for car service station, product layout is adopted, where the activities for servicing a car follows a sequence of operation irrespective of the type of car. Hospital service is the best example for adaptation of process layout. Here, the service required for a customer will follow an independent path.

6.3. Line Balancing

Line balancing is the process of assigning tasks to workstations in a product layout in order to achieve a desired output and balance the workload among stations. Recall that product layouts arrange resources in sequence so that the product can be made as efficiently as possible. This type of layout is used in repetitive processing systems that produce a large volume of one standardized product.

Steps in assembly line balancing

Step 1 Specify the sequential relationships among tasks using a precedence diagram. The diagram consists of circles and arrows. Circles represent individual tasks; arrows indicate the order of task performance.

Step 2 Determine output rate

$$\text{Maximum production} = \frac{\text{Production time per day}}{\text{Bottleneck time}}$$

Step 3 Determine the required cycle time (C), using the following formula:

$$C = \frac{\text{Production time per day}}{\text{Output per day (in units)}}$$

Step 4 Determine the theoretical minimum number of workstations (N_t) required to satisfy the cycle time constraint, using the following formula:

$$N_t = \frac{\text{Sum of task times (T)}}{\text{Cycle time (C)}}$$

Step 5 Evaluate the efficiency of the balance derived using the formula:

$$\text{Efficiency} = \frac{\text{Sum of task times (T)}}{\text{Actual number of workstations (N}_a\text{) x Cycle time (C)}}$$

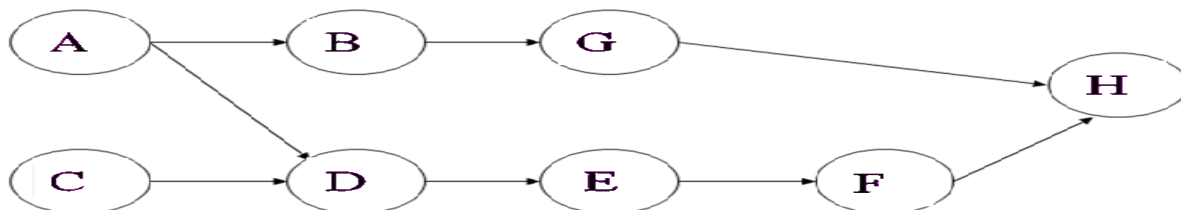
Example

An assembly line must be designed to produce 100 units. The tasks and required time are provided as follows:

Task	Time (minute)	Predecessors
A	2	None
B	1	A
C	3.25	None
D	1.2	A,C
E	0.5	D
F	1	E
G	1	B
H	1.4	F,G

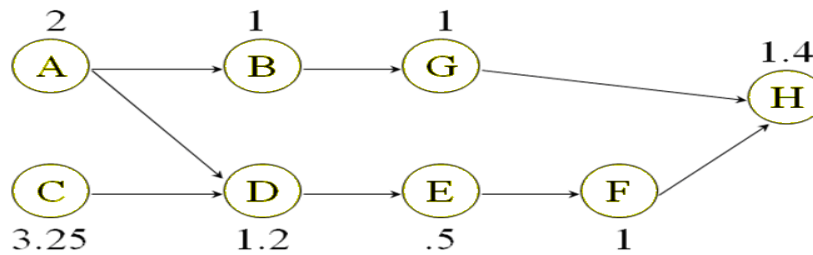
Solution

Step 1 Draw the precedence diagram



Step 2 Determine Maximum output rate

Question: Which process step defines the maximum rate of production?



Answer: Task C is the cycle time of the line and therefore, the maximum rate of production.

$$\text{Maximum production} = \frac{\text{Production time per day}}{\text{Bottleneck time}}$$

$$\text{Production time per day} = 7 \text{ hrs} \times 60 \text{ minute} = \underline{420 \text{ minutes}}$$

Bottleneck time: the longest task in the process i.e 3.25 (task C)

$$\text{Maximum production} = \frac{420 \text{ minute}}{3.25 \text{ minute/unit}}$$

Step 3 Determine the cycle time

$$C = \frac{\text{Production time per day}}{\text{Output per day (in units)}} \quad C = \frac{420 \text{ minute}}{100 \text{ unit/day}} = 4.2 \text{ minute/unit}$$

Step 4 Determine the theoretical minimum number of workstations (N_t)

$$N_t = \frac{\text{Sum of task times (T)}}{\text{Cycle time (C)}} \quad N_t = \frac{11.35 \text{ mins/unit}}{4.2 \text{ mins/unit}} = 2.702 \text{ round up to } 3$$

Step 5 Evaluate the efficiency of the balance derived using the formula:

$$\text{Efficiency} = \frac{\text{Sum of task times (T)}}{\text{Actual number of workstations (N}_a\text{) } \times \text{ Cycle time (C)}} = \frac{11.35 \text{ mins/unit}}{3 \times 4.2 \text{ mins/unit}} = 0.901$$

or 90 % of efficiency.

workout:

A firm must produce 40 units/day during an 8-hour workday. Tasks, times, and predecessor activities are given below.

Task	Time (Minutes)	Predecessor(s)
A	2	-
B	2	A
C	8	-
D	6	C
E	3	B
F	10	D, E
G	4	F
H	3	G
Total	38 minutes	

Determine the cycle time and the appropriate number of workstations to produce the 40 units per day.

- 1 Draw the precedence diagram
2. Determine the theoretical minimum number of workstations (N_t)
3. Evaluate the efficiency of the balance

Chapter seven

Quality management and control

7.1. Meaning of Quality

Broadly defined, **quality** refers to the ability of a product or service to consistently meet or exceed customer requirements or expectations. However, different customers will have different requirements, so a working definition of quality is customer-dependent.

One way to think about quality is the degree to which performance of a product or service meets or exceeds customer expectations. The difference between these two, that is Performance Expectations, is of great interest. If these two measures are equal, the difference is zero, and expectations have been met. If the difference is negative, expectations have not been met, whereas if the difference is positive, performance has exceeded customer expectations.

Customer expectations can be broken down into a number of categories, or *dimensions*, that customers use to judge the quality of a product or service. Understanding these helps organizations in their efforts to meet or exceed customer expectations. The dimensions used for goods are somewhat different from those used for services.

Product Quality: - **Product** quality is often judged on nine dimensions of quality (David Garvin, 1987)

- ✓ **Performance** —main characteristics of the product.
- ✓ **Aesthetics** —appearance, feel, smell, taste.
- ✓ **Special features** —extra characteristics.
- ✓ **Conformance** —how well a product corresponds to design specifications?
- ✓ **Reliability** —dependable performance.
- ✓ **Durability** —ability to perform over time.
- ✓ **Perceived quality** —indirect evaluation of quality (e.g., reputation).
- ✓ **Serviceability** —handling of complaints or repairs.
- ✓ **Consistency** —quality doesn't vary.

Service Quality: - The dimensions of product quality don't adequately describe service quality. Instead, service quality is often described using the following dimensions (Valerie, Zeithaml, Parasuraman and Leonard, 1990)

- ✚ **Convenience** —the availability and accessibility of the service.
- ✚ **Reliability** —the ability to perform a service dependably, consistently, and accurately.
- ✚ **Responsiveness** —the willingness of service providers to help customers in unusual situations and to deal with problems.
- ✚ **Time** —the speed with which service is delivered.
- ✚ **Assurance** —the knowledge exhibited by personnel who come into contact with a customer and their ability to convey trust and confidence.
- ✚ **Courtesy** —the way customers are treated by employees who come into contact with them.
- ✚ **Tangibles** —the physical appearance of facilities, equipment, personnel, and communication materials.
- ✚ **Consistency** —The ability to provide the same level of good quality repeatedly.
- ✚ **Expectations** —Meet (or exceed) customer expectations.

Today; there is no single universal definition of quality. Some people view quality as “performance to standards. “Others view it as “meeting the customer’s needs” or “satisfying the customer.” So, **what is Quality?** Quality can be defined in a number of deferent ways:

1. **Conformance to specifications:** measures how well the product or service meets the targets and tolerances determined by its designers. For example, the dimensions of a machine part may be specified by its design engineers as 3 ± 0.05 inches. This would mean that the target dimension is 3 inches but the dimensions can vary between 2.95 and 3.05 inches. Also, consider the amount of light delivered by a 60-watt light bulb. If the bulb delivers 50 watts it does not conform to specifications.
2. **Fitness for use:** focuses on evaluates how well the product performs its intended function or use. For example, a Mercedes Benz and a Jeep Cherokee both meet a fitness for use definition if one considers transportation as the intended function. However, if the definition becomes more specific and assumes that the intended use is for transportation on mountain roads and carrying fishing gear, the Jeep Cherokee has a greater fitness for use.
3. **Value for price paid:** Quality defined in terms of product or service usefulness for the price paid. Is a definition of quality that consumers often use for product or service usefulness. This is the only definition that combines economics with consumer criteria; it assumes that

the definition of quality is price sensitive. For example, suppose that you wish to sign up for a personal finance seminar and discover that the same class is being taught at two different colleges at significantly different tuition rates. If you take the less expensive seminar, you will feel that you have received greater value for the price.

4. **Support services:** Quality defined in terms of the support provided after the product or service is purchased. Quality does not apply only to the product or service itself; it also applies to the people, processes, and organizational environment associated with it. For example, the quality of a university is judged not only by the quality of staff and course offerings, but also by the efficiency and accuracy of processing paperwork.
5. **Psychological criteria:** is a subjective definition that focuses on the judgmental evaluation of what constitutes product or service quality. Different factors contribute to the evaluation, such as the atmosphere of the environment or the perceived prestige of the product. For example, a hospital patient may receive average health care, but a very friendly staff may leave the impression of high quality. Similarly, we commonly associate certain products with excellence because of their reputation; Rolex watches and Mercedes-Benz automobiles are examples.

7.2. The Determinants of Quality

The degree to which a product or a service successfully satisfies its intended purpose has four primary determinants:

- | | |
|------------------------------------|----------------------------|
| 1. Design. | 3. Ease of use. |
| 2. Products conformance to design. | 4. Service after delivery. |

The design phase is the starting point for the level of quality eventually achieved. Design involves decisions about the specific characteristics of a product or service such as size, shape, and location. **Quality of design** refers to the intention of designers to include or exclude certain features in a product or service. For example, many different models of automobiles are on the market today. They differ in size, appearance, roominess, fuel economy, comfort, and materials used. These differences reflect choices made by designers that determine the quality of design.

Design decisions must take into account customer wants, production or service capabilities, safety and liability (both during production and after delivery), costs, and other similar

considerations. Designers may determine customer wants from information provided by marketing, perhaps through the use of consumer surveys or other market research. Marketing may organize focus groups of consumers to express their views on a product or service (what they like and don't like, and what they would like to have).

Designers must work closely with representatives of operations to ascertain that designs can be produced; that is, that production or service has the equipment, capacity, and skills necessary to produce or provide a particular design. A poor design can result in difficulties in production or service. For example, materials might be difficult to obtain, specifications difficult to meet, or procedures difficult to follow. Moreover, if a design is inadequate or inappropriate for the circumstances, the best workmanship in the world may not be enough to achieve the desired quality. Also, we cannot expect a worker to achieve good results if the given tools or procedures are inadequate. Similarly, a superior design usually cannot offset poor workmanship.

Quality of conformance refers to the degree to which goods and services conform to (i.e., *achieve*) the intent of the designers. This is affected by factors such as the capability of equipment used; the skills, training, and motivation of workers; the extent to which the design lends itself to production; the monitoring process to assess conformance; and the taking of corrective action (e.g., through problem solving) when necessary. One important key to quality is reducing the variability in process outputs (i.e., reducing the degree to which individual items or individual service acts vary from one another).

The determination of quality does not stop once the product or service has been sold or delivered. **Ease of use and user instructions** are important. They increase the chances, but do not guarantee, that a product will be used for its intended purposes and in such a way that it will continue to function properly and safely. Customers, patients, clients, or other users must be clearly informed on what they should or should not do; otherwise, there is the danger that they will take some action that will adversely affect quality. Some examples include the doctor who fails to specify that a medication should be taken *before* meals and *not* with orange juice and the attorney who neglects to inform a client of a deadline for filing a claim.

7.3. The Costs of Quality

Any serious attempt to deal with quality issues must take into account the costs associated with quality. Those costs can be classified into three categories: appraisal, prevention, and failure.

Appraisal costs relate to inspection, testing, and other activities intended to uncover defective products or services, or to assure that there are none. They include the cost of inspectors, testing, test equipment, labs, quality audits, and field testing.

Prevention costs relate to attempts to prevent defects from occurring. They include costs such as planning and administration systems, working with vendors, training, quality control procedures, and extra attention in both the design and production phases to decrease the probability of defective workmanship.

Failure costs are incurred by defective parts or products or by faulty services.

- **Internal failures** are those discovered during the production process; internal failures occur for a variety of reasons, including defective material from vendors, incorrect machine settings, faulty equipment, incorrect methods, incorrect processing, carelessness, and faulty or improper material handling procedures. The costs of internal failures include lost production time, scrap and rework, investigation costs, possible equipment damage, and possible employee injury. Rework costs involve the salaries of workers and the additional resources needed to perform the rework (e.g., equipment, energy, raw materials). Beyond those costs are items such as inspection of reworked parts, disruption of schedules, the added costs of parts and materials in inventory waiting for reworked parts, and the paperwork needed to keep track of the items until they can be reintegrated into the process.

External failures are those discovered after delivery to the customer. External failures are defective products or poor service that goes undetected by the producer. Resulting costs include warranty work, handling of complaints, replacements, and liability/litigation, payments to customers or discounts used to offset the inferior quality, loss of customer goodwill, and opportunity costs related to lost sales.

7.4. Quality Circles and Quality Improvement Teams

7.4.1. Quality Circles

One of the first team-based approaches to quality improvement was quality circles, called quality-control circles in Japan when they originated during the 1960s; they were introduced in the United States in the 1970s. A quality circle is a small, voluntary group of employees and their supervisor(s), comprising a team of about 8 to 10 members from the same work area or department. The supervisor is typically the circle moderator, promoting group discussion but not directing the group or making decision; decisions result from group consensus. A circle meets about once a week during company time in a room designated especially for that purpose, where the team works on problems and projects of their own choice. These problems may not always relate to quality issues; instead, they focus on productivity, costs, safety, or other work-related issues in the circle's area. Quality circles follow an established procedure for identifying, analyzing, and solving quality related (or other) problems.

A group technique for identifying and solving problems is brainstorming to generate ideas. Free expression is encouraged, and criticism is not allowed. Only after brainstorming is completed are ideas evaluated.

7.4.2. Quality Improvement Teams

Quality improvement teams (QIT), also called process improvement teams, focus attention on business processes rather than separate company functions. It was noted previously that quality circles are generally composed of employees and supervisors from the same work area or department, whereas process improvement teams tend to be cross functional or even cross-business between suppliers and their customers. A process improvement team would include members from the various interrelated functions or departments that constitute a process. For example, a process improvement team for customer service might include members from distribution, packing, manufacturing and human resources. A key objective of a process improvement team is to understand the process the team is addressing in terms of how all the parts (functions and departments) work together. The process is then measured and evaluated, with the goal of improving the process to make it more efficient and the product or service better.

7.5. International Quality Documentation Standards

7.5.1 The Malcolm Baldrige National Quality Award (MBNQA)

The Malcolm Baldrige National Quality Award was established in 1987, when Congress passed the Malcolm Baldrige National Quality Improvement Act. The award is named after the former Secretary of Commerce, Malcolm Baldrige, and is intended to reward and stimulate quality initiatives. It is designed to recognize companies that establish and demonstrate high quality standards. The award is given to no more than two companies in each of three categories: manufacturing, service, and small business. Past winners include Motorola Corporation, Xerox, FedEx, 3M, IBM, and the Ritz-Carlton.

To compete for the Baldrige Award, companies must submit a lengthy application, which is followed by an initial screening. Companies that pass this screening move to the next step, in which they undergo a rigorous evaluation process, conducted by certified Baldrige examiners. The examiners conduct site visits and examine numerous company documents. They base their evaluation on seven categories, which are shown in table below.

Categories	Points
Leadership	120
Strategic planning	85
Customer and market focus	85
Information and analysis	90
Human resource focus	85
Process management	85
Business result	400
Total point	1000

7.5.2 The Deming Prize

The Deming Prize is a Japanese award given to companies to recognize their efforts in quality improvement. The award is named after W. Edwards Deming, who visited Japan after World War II upon the request of Japanese industrial leaders and engineers. While there, he gave a series of lectures on quality. The Japanese considered him such an important quality guru that they named the quality award after him. The award has been given by the Union of Japanese Scientists and Engineers (JUSE) since 1951. Competition for the Deming Prize was opened to foreign companies in 1984. In 1989 Florida Power & Light was the first U.S. Company to receive the award.

The **European Quality Award** is Europe's most prestigious award for organizational excellence. The European Quality Award sits at the top of regional and national quality awards, and applicants have often won one or more of those awards prior to applying for the European Quality Award.

7.5.3 ISO 9000 Standards

Increases in international trade during the 1980s created a need for the development of universal standards of quality. Universal standards were seen as necessary in order for companies to be able to objectively document their quality practices around the world. Then in 1987 the International Organization for Standardization (ISO) published its first set of standards for quality management called ISO 9000. The International Organization for Standardization (ISO) is an international organization whose purpose is to establish agreement on international quality standards. It currently has members from 91 countries, including the United States. To develop and promote international quality standards, ISO 9000 has been created. ISO 9000 consists of a set of standards and a certification process for companies. By receiving ISO 9000 certification, companies demonstrate that they have met the standards specified by the ISO. The standards are applicable to all types of companies and have gained global acceptance. In many industries ISO certification has become a requirement for doing business. Also, ISO 9000 standards have been adopted by the European Community as a standard for companies doing business in Europe.

In December 2000 the first major changes to ISO 9000 were made, introducing the following three new standards:

- ISO 9000:2000–Quality Management Systems–Fundamentals and Standards: Provides the terminology and definitions used in the standards. It is the starting point for understanding the system of standards.
- ISO 9001:2000–Quality Management Systems–Requirements: This is the standard used for the certification of a firm’s quality management system. It is used to demonstrate the conformity of quality management systems to meet customer requirements.
- ISO 9004:2000–Quality Management Systems–Guidelines for Performance: Provides guidelines for establishing a quality management system. It focuses not only on meeting customer requirements but also on improving performance.

These three standards are the most widely used and apply to the majority of companies. However, ten more published standards and guidelines exist as part of the ISO 9000 family of standards.

To receive ISO certification, a company must provide extensive documentation of its quality processes. This includes methods used to monitor quality, methods and frequency of worker training, job descriptions, inspection programs, and statistical process-control tools used. High-quality documentation of all processes is critical. The company is then audited by an ISO 9000 registrar who visits the facility to make sure the company has a well-documented quality management system and that the process meets the standards. If the registrar finds that all is in order, certification is received. Once a company is certified, it is registered in an ISO directory that lists certified companies. The entire process can take 18 to 24 months and can cost anywhere from \$10,000 to \$30,000. Companies have to be re-certified by ISO every three years.

One of the shortcomings of ISO certification is that it focuses only on the process used and conformance to specification. In contrast to the Baldrige criteria, ISO certification does not address questions about the product itself and whether it meets customer and market requirements. Today there are over 40,000 companies that are ISO certified. In fact, certification has become a requirement for conducting business in many industries.

7.5.4 ISO 14000 Standards

The need for standardization of quality created an impetus for the development of other standards. In 1996 the International Standards Organization introduced standards for evaluating a company's environmental responsibility. These standards, termed ISO 14000, focus on three major areas:

- Management systems standards measure systems development and integration of environmental responsibility into the overall business.
- Operations standards include the measurement of consumption of natural resources and energy.
- Environmental systems standards measure emissions, effluents, and other waste systems. With greater interest in green manufacturing and more awareness of environmental concerns, ISO 14000 may become an important set of standards for promoting environmental responsibility.

7.6. Total Quality Management

The term **total quality management (TQM)** refers to a quest for quality in an organization. There are three key philosophies in this approach. One is a never-ending push to improve, which is referred to as *continuous improvement*; the second is the *involvement of everyone* in the organization; and the third is a goal of *customer satisfaction*, which means meeting or exceeding customer expectations.

TQM expands the traditional view of quality—looking only at the quality of the final product or services—to *looking at the quality of every aspect of the process* that produces the product or service. TQM systems are intended to prevent poor quality from occurring.

We can describe the TQM approach as follows:

1. Find out what customers want. This might involve the use of surveys, focus groups, interviews, or some other technique that integrates the customer's voice in the decision-making process. Be sure to include the *internal customer* (the next person in the process) as well as the *external customer* (the final customer).
2. Design a product or service that will meet (or exceed) what customers want. Make it easy to use and easy to produce.

3. Design processes that facilitate doing the job right the first time. Determine where mistakes are likely to occur and try to prevent them. When mistakes do occur, find out why so that they are less likely to occur again. Strive to make the process “mistake-proof.”
4. Keep track of results, and use them to guide improvement in the system. Never stop trying to improve.
5. Extend these concepts throughout the supply chain.
6. Top management must be involved and committed. Otherwise, TQM will just be another fad that fails and fades away.

It would be incorrect to think of TQM as merely a collection of techniques. Rather, TQM reflects a whole new attitude toward quality. It is about the *culture* of an organization. To truly reap the benefits of TQM, the organization must change its culture.

7.7. Quality Control and Improvement

Quality control (QC) is a component of process control, and is an essential element of the quality management system. It monitors the processes related to the examination phase of testing and allows for detecting errors in the testing system. These errors may be due to test system failure, adverse environmental conditions or operator performance. QC gives the laboratory confidence that test results are accurate and reliable before patient results are reported.

Kaizen is the Japanese term for continuous improvement, not only in the workplace but also in one’s personal life, home life, and social life. In the workplace, kaizen means involving everyone in a process of gradual, organized, and continuous improvement. Every employee within an organization should be involved in working together to make improvements. If an improvement is not part of a continuous, ongoing process, it is not considered kaizen.

Employees are most directly involved in kaizen when they are determining solutions to their own problems. Employees are the real experts in their immediate workspace. In its most basic form, kaizen is a system in which employees identify many small improvements on a continual basis and implement these improvements themselves. Employees identify a problem, come up with a solution, check with their supervisor, and then implement it. This works to involve all employees in the improvement process and gives them a feeling that they are really participating in quality

improvement, which in turn keeps them excited about their jobs. Nothing motivates someone more than when he/she comes up with a solution to his/her own problem. Small individual changes have a cumulative effect in improving entire process, and with this level of participation improvement occurs across the entire organization. No companywide TQM program can succeed without this level of total employee involvement in continuous improvement.

Chapter eight

Aggregate planning

Aggregate-planning is intermediate-range capacity planning that typically covers a time horizon of 2 to 12 months, although in some companies it may extend to as much as 18 months. It is particularly useful for organizations that experience seasonal or other fluctuations in demand or capacity. The goal of aggregate planning is to achieve a production plan that will effectively utilize the organization's resources to satisfy expected demand. Planners must make decisions on output rates, employment levels and changes, inventory levels and changes, back orders, and subcontracting.

Organizations make capacity decisions on three levels: long term, intermediate term, and short term. **Long-term decisions** relate to product and service selection (i.e., determining which products or services to offer), facility size and location, equipment decisions, and layout of facilities. These long-term decisions essentially define the capacity constraints within which intermediate planning must function. **Intermediate decisions**, as noted above, relate to general levels of employment, output, and inventories, which in turn define the boundaries within which short-range capacity decisions must be made. Thus, short-term decisions essentially consist of deciding the best way to achieve desired results within the constraints resulting from long-term and intermediate-term decisions. **Short term decisions** involve scheduling jobs, workers and equipment, and the like.

The business plan establishes guidelines for the organization, taking into account the organization's strategies and policies; forecasts of demand for the organization's products or services; and economic, competitive, and political conditions. A key objective in business planning is to coordinate the intermediate plans of various organization functions, such as marketing, operations, and finance. In manufacturing companies, coordination also includes engineering and materials management. Consequently, all of these functional areas must work together to formulate the aggregate plan.

8.1.1. The Concept of Aggregation

Aggregate planning decisions are strategic decisions that define the framework within which operating decisions will be made. They are the starting point for scheduling and production

control systems. They provide input for financial plans; they involve forecasting input and demand management, and they may require changes in employment levels.

Aggregate planning is essentially a "big picture" approach to planning. Planners usually try to avoid focusing on individual products or services-unless of course the organization has only one major product or service. Instead, they focus on a group of similar products or services, or sometimes an entire product or service line.

Why do organizations need to do aggregate planning? The answer is twofold. One part is related to planning: It takes time to implement plans. For instance, if plans call for increasing the size of facilities and/or hiring (and training) new workers, that will take time. The second part is strategic: Aggregation is important because it is not possible to predict with any degree of accuracy the timing and volume of demand for individual items. So, if an organization were to "lock in" on individual items, it would lose the flexibility to respond to the market. Finally, aggregate planning is important because it can help synchronize flow throughout the supply chain; it affects costs, equipment utilization, employment levels, and customer satisfaction.

Aggregate Planning Strategies

Aggregate planning strategies can be described as **proactive, reactive, or mixed**. Proactive strategies involve demand options: They attempt to alter demand so that it matches capacity. Reactive strategies involve capacity options: They attempt to alter capacity so that it matches demand. Mixed strategies involve an element of each of these approaches.

Demand and Supply

Aggregate planners are concerned with the quantity and the timing of expected demand. If total expected demand for the planning period is much different from available capacity over that same period, the major approach of planners will be to try to achieve a balance by altering capacity, demand, or both. On the other hand, even if capacity and demand are approximately equal for the planning horizon as a whole, planners may still be faced with the problem of dealing with uneven demand within the planning interval. In some periods, expected demand may exceed projected capacity, in others expected demand may be less than projected capacity, and in some periods the two may be equal.

Inputs and Output to Aggregate Planning

Inputs

- | | |
|---|------------------------------|
| 1. Resources | 5. Overtime |
| 2. Demand forecast | 6. Inventory levels/ changes |
| 3. Policy statements on workforce changes | 7. Back orders |
| 4. Sub-contracting | 8. Costs |

Output

1. Total cost of a plan
2. Projected levels: - Inventory, output, employment, subcontracting and back ordering

8.1.3. Demand and Supply Options

Management has a wide range of decision options at its disposal for purposes of aggregate planning. Aggregate planning strategies can pertain to demand, capacity, or both. Demand strategies are intended to alter demand so that it matches capacity. Capacity strategies involve altering capacity so that it matches demand. *Mixed* strategies involve both of these approaches.

A. DEMAND OPTIONS: - Demand options include pricing, promotions, using back orders (delaying order filling), and creating new demand. The basic demand options are the following:

1. **Pricing:** - Pricing differentials are commonly used to shift demand from peak periods to off-peak periods. Some hotels, for example, offer lower rates for weekend stays, and some airlines offer lower fares for night travel. Movie theaters may offer reduced rates for matinees, and some restaurants offer "early bird specials" in an attempt to shift some of the heavier dinner demand to an earlier time that traditionally has less traffic. Some restaurants also offer smaller portions at reduced rates, and most have smaller portions and prices for children.

The smaller portions act to decrease demand. To the extent that pricing is effective, demand will be shifted so that it corresponds more closely to capacity, albeit for an opportunity cost that represents the lost profit stemming from capacity insufficient to meet demand during certain periods.

2. **Promotion:** Advertising and other forms of promotion, such as displays and direct marketing, can sometimes be very effective in shifting demand so that it conforms more closely to capacity. Obviously, timing of these efforts and knowledge of response rates and response patterns will be needed to achieve the desired results. Unlike pricing policy, there is much less control over the timing

of demand; there is always the risk that promotion can worsen the condition it was intended to improve.

3. **Back orders:** An organization can shift demand to other periods by allowing back orders. That is, orders are taken in one period and deliveries promised for a later period. The success of this approach depends on how willing customers are to wait for delivery. Moreover, the costs associated with back orders can be difficult to pin down since it would include lost sales, annoyed or disappointed customers, and perhaps additional paperwork.
4. **New demand:** Many organizations are faced with the problem of having to provide products or services for peak demand in situations where demand is very uneven. For instance, demand for bus transportation tends to be more intense during the morning and late afternoon rush hours but much lighter at other times. Creating new demand for buses at other times (e.g., trips by schools, clubs, and senior citizen groups) would make use of the excess capacity during those slack times. Manufacturing firms that experience seasonal demands for certain products (e.g., snow blowers) are sometimes able to develop a demand for a complementary product (e.g., lawn mowers, garden equipment) that makes use of the same production processes. They thereby achieve a more consistent use of labor, equipment, and facilities.

B. SUPPLY (CAPACITY) OPTION: - Supply options include hiring/laying off workers, overtime/slack time, part-time or temporary workers, inventories, and subcontractors. The basic capacity options are the following:

1. **Hire and lay off workers:** The extent to which operations are labor intensive determines the impact that changes in the workforce level will have on capacity. The resource requirements of each worker can also be a factor. For instance, if a supermarket usually has 10 of 14 checkout lines operating, an additional four checkout workers could be added. Hence, the ability to add workers is constrained at some point by other resources needed to support the workers. Conversely, there may be a lower limit on the number of workers needed to maintain a viable operation (e.g., a skeleton crew).
2. **Overtime/slack time:** Use of overtime or slack time is a less severe method for changing capacity than hiring and laying off workers, and it can be used across the board or selectively as needed. It can also be implemented more quickly than hiring and laying off and allows the firm to maintain a steady base of employees. The use of overtime can be especially attractive in dealing with seasonal demand peaks by reducing the need to hire and train people who will have to be laid off during the off-season. Overtime also permits the company to maintain a skilled workforce and employees to increase

earnings. Moreover, in situations with crews, it is often necessary to use a full crew rather than to hire one or two additional people. Thus, having the entire crew work overtime would be preferable to hiring extra people.

The use of slack when demand is less than capacity can be an important consideration. Some organizations use this time for training. It can also give workers time for problem solving and process improvement, while retraining skilled workers.

3. **Part-time workers:** In certain instances, the use of part-time workers is a viable option-much depends on the nature of the work, training and skills needed, and union agreements. Seasonal work requiring low-to-moderate job skills lends itself to part-time workers, who generally cost less than regular workers in hourly wages and fringe benefits. However, unions may regard such workers unfavorably because they typically do not pay union dues and may lessen the power of unions. Department stores, restaurants, and supermarkets make use of part-time workers. So, do parks and recreation departments, resorts, travel agencies, hotels, and other service organizations with seasonal demands. In order to be successful, these organizations must be able to hire part-time employees when they are needed.

Some companies use contract workers, also called independent contractors, to fill certain needs. Although they are not regular employees, often they work alongside regular workers. In addition to having different pay scales and no benefits, they can be added or subtracted from the workforce with greater ease than regular workers, giving companies great flexibility in adjusting the size of the workforce.

4. **Inventories:** The use of finished-goods inventories allows firms to produce goods in one period and sell or ship them in another period, although this involves holding or carrying those goods as inventory until they are needed. The cost includes not only storage costs and the cost of money tied up that could be invested elsewhere, but also the cost of insurance, obsolescence, deterioration, spoilage, breakage, and so on. In essence, inventories can be built up during periods when production capacity exceeds demand and drawn down in periods when demand exceeds production capacity.

This method is more amenable to manufacturing than to service industries since manufactured goods can be stored whereas services generally cannot. However, an analogous approach used by services is to make efforts to streamline services (e.g., standard forms) or otherwise do a portion of the service during slack periods (e.g., organize the workplace). In spite of these possibilities, services tend not to make much use of inventories to alter capacity requirements.

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5. **Subcontracting:** Subcontracting enables planners to acquire temporary capacity, although it affords less control over the output and may lead to higher costs and quality problems. The question of whether to make or buy (i.e., in manufacturing) or to perform a service or hire someone else to do the work generally depends on factors such as available capacity, relative expertise, quality considerations, cost, and the amount and stability of demand.

In some cases, a firm might choose to perform part of the work itself and let others handle the rest in order to maintain flexibility and as a hedge against loss of a subcontractor. Moreover, this gives the organization a bargaining tool in negotiations with contractors and a head start if it decides at a later date to take over the operation entirely. As an alternative to subcontracting, an organization might consider outsourcing: contracting with another organization to supply some portion of the goods or services on a regular basis.

8.1.4 Basic Strategies for Meeting Uneven Demand

As you see, managers have a wide range of decision options they can consider for achieving a balance of demand and capacity in aggregate planning. Since the options that are most suited to influencing demand fall more in the realm of marketing than in operations (with the exception of backlogging), we shall concentrate on the capacity options, which are in the realm of operations but include the use of back orders. Aggregate planners might adopt a number of strategies. Some of the more prominent ones are:

- I. Maintain a level workforce.
- II. Maintain a steady output rate.
- III. Match demand period by period.
- IV. Use a combination of decision variables.

While other strategies might be considered, these will suffice to give you a sense of how aggregate planning operates in a vast number of organizations. The first three strategies are "pure" strategies because each has a single focal point; the last strategy is "mixed" because it lacks the single focus. **Types of aggregate plans**

We categorize aggregate plans as level, chase, or hybrid plans.

- 1. Level aggregate plan:** A planning approach that produces the same quantity each time period.

-  A level aggregate plan maintains a constant workforce and produces the same amount of product in each time period of the plan.
-  Inventory and back orders are used to absorb demand fluctuations.
-  One advantage of a level production plan is workforce stability.

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- ✚ Your company sets labor and equipment capacity equal to average demand, rather than hire excess labor or buy additional tools and equipment just to meet peak demand.
 - ✚ In addition, the labor force is not subjected to varying work levels during the year, such as periods of layoff or undertime followed by periods of hiring and/or overtime.
 - ✚ The disadvantages of the level plan are the buildup of inventory and/or possible poor customer service from extensive use of back orders.
 - ✚ The level plan is often used with make-to-stock products such as stereos, kitchen appliances, and hardware.

2. Chase Aggregate Plan

Chase aggregate plan: A planning approach that varies production to meet demand each period.

- ✚ A chase aggregate plan produces exactly what is needed to satisfy demand during each period.
- ✚ The production rate changes in response to demand fluctuations. Whereas the level aggregate plan sets capacity to accommodate average demand, the chase aggregate plan sets labor and equipment capacity to satisfy demand each period.
- ✚ The advantage of the chase plan is that it minimizes finished goods holding costs.
- ✚ This may be a better option when a company produces make-to-order products such as custom cabinets, special-purpose equipment, one-of-a-kind items, or highly perishable products.
- ✚ The disadvantages are constantly changing capacity needs and the need for enough equipment to meet peak demand.
- ✚ The additional equipment needed to meet peak demand creates excess capacity in nonpeak demand periods.
- ✚ Many options for short-term capacity changes are expensive.

3. Hybrid Aggregate Plan

Hybrid aggregate plan A planning approach that uses a combination of level and chase approaches while developing the aggregate plan.

- ✚ A hybrid aggregate plan typically uses a combination of options.
- ✚ With this plan, your company might maintain a stable workforce supplemented by an inventory buildup and some overtime production to meet demand. Or the company may back-order a portion of its demand.
- ✚ Any combination of options is possible.

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- ✚ Because of the number of options, you can combine in a hybrid plan, you need to evaluate your company's current situation and limit the options you choose from.

8.1.5 Techniques of Aggregate Planning

Numerous techniques help decision makers with the task of aggregate planning. Generally, they fall into one of two categories: informal trial-and-error techniques and mathematical techniques. In practice, informal techniques are more frequently used. However, a considerable amount of research has been devoted to mathematical techniques, and even though they are not as widely used, they often serve as a basis for comparing the effectiveness of alternative techniques for aggregate planning. Thus, it will be instructive to briefly examine them as well as the informal techniques.

A general procedure for aggregate planning consists of the following steps:

1. Determine demand for each period.
2. Determine capacities (regular time, overtime, subcontracting) for each period
3. Identify company or departmental policies that are pertinent (e.g., maintain a safety stock of 5 percent of demand, maintain a reasonably stable workforce).
4. Determine unit costs for regular time, overtime, subcontracting, holding inventories, back orders, layoffs, and other relevant costs.
5. Develop alternative plans and compute the cost for each
6. If satisfactory plans emerge, select the one that best satisfies objectives. Otherwise, re-turn to step 5.

8.1.6 Aggregate Planning in Services

Aggregate planning for services takes into account projected customer demands, equipment capacities, and labor capabilities. The resulting plan is a time-phased projection of service staff requirements. Here are examples of service organizations that use aggregate planning: **Hospitals, Airlines, Financial, hospitality, transportation, and recreation services** provide a high-volume, intangible output.

Aggregate planning for these and similar services involves managing demand and planning for human resource requirements. The main goals are to accommodate peak dem. Aggregate planning for manufacturing and aggregate planning for services share similarities in some respects, but there are some important differences-related in general to the differences between manufacturing and services:

1. **Services occur when they are rendered:** Unlike manufacturing output, most services can't be inventoried. Services such as financial planning, tax counseling, and oil changes can't be stockpiled. This removes the option of building up inventories during a slow period in anticipation of future demand. On the other

hand, service capacity that goes unused is essentially wasted. Consequently, it becomes important to be able to match capacity and demand.

2. ***Demand for service can be difficult to predict:*** The volume of demand for services is often quite variable. In some situations, customers may need prompt service (e.g., police, fire, and medical emergency), while in others, they simply want prompt service and may be willing to go elsewhere if their wants are not met.
3. ***Capacity availability can be difficult to predict:*** Processing requirements for services can sometimes be quite variable, similar to the variability of work in a job shop setting. Moreover, the variety of tasks required of servers can be great, again similar to the variety of tasks in a job shop.
4. ***Labor flexibility can be an advantage in services:*** Labor often comprises a significant portion of service compared to manufacturing. That, coupled with the fact that service providers are often able to handle a fairly wide variety of service requirements, means that to some extent, planning is easier than it is in manufacturing. Of course, manufacturers recognize this advantage, and many are cross-training their employees to achieve the same flexibility.

Chapter nine

Scheduling operation

Within an organization, **scheduling** pertains to establishing the timing of the use of specific resources of that organization. It relates to the use of equipment, facilities, and human activities. Scheduling occurs in every organization, regardless of the nature of its activities.

In the decision-making hierarchy, scheduling decisions are the final step in the transformation process before actual output occurs. Many decisions about system design and operation have been made long before scheduling decisions. They include the capacity of the system, product or service design, equipment selection, selection and training of workers, and aggregate planning and master scheduling. Consequently, scheduling decisions must be made within the constraints established by many other decisions, making them fairly narrow in scope and latitude.

Effective scheduling can yield cost savings, increases in productivity, and other benefits. For example, in hospitals, effective scheduling can save lives and improve patient care. In educational institutions, it can reduce the need for expansion of facilities. In competitive environments, effective scheduling can give a company a competitive advantage in terms of customer service (shorter wait time for their orders) if its competitors are less effective with their scheduling.

Generally, the objectives of scheduling are to achieve trade-offs among conflicting goals, which include efficient utilization of staff, equipment, and facilities, and minimization of customer waiting time, inventories, and process times. This chapter covers scheduling in both manufacturing and service environments. Although the two environments have many similarities, some basic differences are important.

The Concept of Scheduling

Scheduling is the processes of determining the starting and completion times to jobs. It is the determination of when labor, equipment and facilities are needed to produce a product or provide a service. Scheduling is a time table for: performing activities, using resources, or allocating facilities. Generally, a schedule specifies the timing and sequence of production.

- ◆ Schedule must be realistic; that is, they must be capable of being achieved within the capacity limitations of the manufacturing facilities.
- ◆ Scheduling should be clearly differentiated from aggregate planning. The purpose of scheduling is to ensure that available capacity is efficiently and effectively used to achieve the organization's

objectives. The purpose of aggregate planning is to determine the resources (labor, equipment, space etc.) that should be acquired for scheduling.

Scheduling tasks are largely a function of the volume of system output. High-volume systems require approaches substantially different from those required by job shops, and project scheduling requires still different approaches. In this chapter, we will consider scheduling for high-volume systems and low-volume (job shop) scheduling.

Scheduling in High Volume Systems

Scheduling encompasses allocating workloads to specific work centers and determining the sequence in which operations are to be performed.

High-volume systems are characterized by standardized equipment and activities that provide identical or highly similar operations on customers or products as they pass through the system. The goal is to obtain a smooth rate of flow of goods or customers through the system in order to get a high utilization of labor and equipment.

High-volume systems are often referred to as flow systems; scheduling in these systems is referred to as **flow-shop scheduling**, although flow shop scheduling can also be used in medium-volume systems. Examples of high-volume products include autos, personal computers, radios and televisions, stereo equipment, toys, and appliances. In process industries, examples include petroleum refining, sugar refining, mining, waste treatment, and the manufacturing of fertilizers. Examples of services include cafeteria lines, news broadcasts, and mass inoculations. Because of the highly repetitive nature of these systems, many of the loading and sequence decisions are determined during the design of the system. The use of highly specialized tools and equipment, the arrangement of equipment, the use of specialized material-handling equipment, and the division of labor are all designed to enhance the flow of work through the system, since all items follow virtually the same sequence of operations.

A major aspect in the design of flow systems is line balancing, which concerns allocating the required tasks to workstations so that they satisfy technical (sequencing) constraints and are balanced with respect to equal work times among stations. Highly balanced systems result in the maximum utilization of equipment and personnel as well as the highest possible rate of output. In setting up flow systems, designers must consider the potential discontent of workers in connection with the specialization of job tasks in these systems; high work rates are often achieved by dividing the work into a series of relatively simple tasks assigned to different workers.

In spite of the built-in attributes of flow systems related to scheduling, a number of scheduling problems remain. One stems from the fact that few flow systems are completely devoted to a single product or service; most must handle a variety of sizes and models. Thus, an automobile manufacturer will assemble many different combinations of cars—two-door and four-door models, some with air-conditioning and some not, some with deluxe trim and others with standard trim, some with CD players, some with tinted glass, and so on. The same can be said for producers of appliances, electronic equipment, and toys.

One source of scheduling concern is possible disruptions in the system that results in less than the desired output. These can be caused by equipment failures, material shortages, accidents, and absences. In practice, it is usually impossible to increase the rate of output to compensate for these factors, mainly because flow systems are designed to operate at a given rate. Instead, strategies involving subcontracting or overtime are often required, although subcontracting on short notice is not always feasible. Sometimes work that is partly completed can be made up off the line. The reverse situation can also impose scheduling problems although these are less severe. This happens when the desired output is less than the usual rate. However, instead of slowing the ensuing rate of output, it is usually necessary to operate the system at the usual rate, but for fewer hours. For instance, a production line might operate temporarily for seven hours a day instead of eight. High-volume systems usually require automated or specialized equipment for processing and handling. Moreover, they perform best with a high, uniform output.

Scheduling in Low-Volume System

The characteristics of low-volume systems (**job shops**) are considerably different from those of high- and intermediate-volume systems. Products are made to order, and orders usually differ considerably in terms of processing requirements, materials needed, processing time, and processing sequence and setups. Because of these circumstances, job-shop scheduling is usually fairly complex. This is compounded by the impossibility of establishing firm schedules prior to receiving the actual job orders. Job-shop processing gives rise to two basic issues for schedulers: how to distribute the in requirements.

LOADING

Loading refers to the assignment of jobs to processing (work) centers. Loading decisions involve assigning specific jobs to work centers and to various machines in the work centers. In cases where a job can, be processed only by a specific center, loading presents little difficulty. However, problems arise when two or more jobs are to be processed and there are a number of work centers capable of performing the required work. In such cases, the operations manager needs some way of assigning jobs to the centers. When making

assignments, managers often seek an arrangement that will minimize processing and setup costs, minimize idle time among work centers, or minimize job completion time, depending on the situation.

Gantt Charts: Visual aids called Gantt charts are used for a variety of purposes related to loading and scheduling. They derive their name from Henry Gantt, who pioneered the use of charts for industrial scheduling in the early 1900s. Gantt charts can be used in a number of different ways, scheduling classrooms for a university and scheduling hospital operating rooms for a day. The purpose of Gantt charts is to organize and clarify the actual or intended use of resources in a time framework. Managers may use the charts for trial-and-error schedule development to get an idea of, what different arrangements would involve. Thus, a tentative surgery schedule might reveal insufficient allowance for surgery that takes longer than expected and can be revised accordingly. Use of the chart for classroom scheduling would help avoid assigning two different classes to the same room at the same time.

Two different approaches are used to load work centers, infinite loading and finite loading. **Infinite loading** assigns jobs to work centers without regard to the capacity of work center. One possible results of infinite loading are the formulation of queues in some (or all) work centers. **Finite loading** projects actual job starting and stopping times at each work center, taking into account the capabilities of each work center and the processing times of the jobs, so that capacity is not exceeded. One output of finite loading is a detailed projection of hours each work centers will operate. Schedules based on finite loading may have to be updated often, perhaps daily, due to processing delay at work centers and the addition of new jobs or cancelation of current job.

Loading can be done in various ways. **Vertical loading** refers to loading jobs at a work centers job by job, usually according to some priority criterion. Vertical loading does not consider the work centers capacity (i.e., infinite loading). In contrast, **horizontal loading** involves loading the jobs that has the highest priority on all work centers it will require, then the job with the next highest priority, and so on. Horizontal loading is based on finite loading. With infinite loading, a manager may need to make some response to overloaded work centers. Among the possible responses are shifting works to other periods or other centers, working overtime, or contracting out a portion of the work. Finite loading may reflect a fixed upper limit on capacity. For example, a bus line will have only so many buses. Hence, the decision to place into service a particular number of buses fixes capacity. Similarly, a manufacturer might have one specialized machine that it operates around the clock. Thus, it is operated at the upper limit of its capacity, so finite loading would be called for.

There are two general approaches to scheduling: **forward scheduling** and **backward scheduling**. Forward scheduling means scheduling ahead from a point in time whereas, backward scheduling means scheduling backward from a due date. Forward scheduling is used if the issue is "How long will it take to complete this job?" Backward scheduling would be used if the issue is "When is the latest the job can be started and still be completed by the due date?"

JOB SEQUENCING

Although loading decisions determine the machines or work centers that will be used to process specific jobs, they do not indicate the order in which the jobs waiting at a given work center are to be processed. **Sequencing** is concerned with determining job processing order. Sequencing decisions determine both the order in which jobs are processed at various work centers and the order in which jobs are processed at individual workstations within the work centers.

Typically, a number of jobs will be waiting for processing. **Priority rules** are simple heuristics used to select the order in which the jobs will be processed. Some of the most common are: **FCFS (first come, first served)**: Jobs are processed in the order in which they arrive at a machine or work center. **SPT (shortest processing time)**: Jobs are processed according to processing time at a machine or work center, shortest job first. **EDD (earliest due date)**: Jobs are processed according to due date, earliest due date first. **CR (critical ratio)**: Jobs are processed according to smallest ratio of time remaining until due date to processing time remaining. **S/O (slack per operation)**: Jobs are processed according to average slack time (time until due date minus remaining time to process). Compute by dividing slack time by number of remaining operations, including the current one.

Assumption of priority rules

- The set of jobs is known; no new jobs arrive after processing begins; and no jobs are canceled.
- Setup time is independent of processing sequence.
- Setup time is deterministic.
- Processing times are deterministic rather than variable.
- There will be no interruptions in processing such as machine breakdowns, accidents, or worker illness.

In effect, the priority rules pertain to static sequencing: For simplicity, it is assumed that there is no variability in either setup or processing times, or in the set of jobs. The assumptions make the scheduling

problem manageable. In practice, jobs may be delayed or canceled, and new jobs may arrive, requiring schedule revisions.

Measuring Effectiveness of Priority Rules

The effectiveness of any given sequence is frequently judged in terms of one or more performance measures. The most frequently used performance measures are:

Job flow time: This is the length of time a job is at a particular workstation or work center. It includes not only actual processing time but also any time waiting to be processed, transportation time between operations, and any waiting time related to equipment breakdowns, unavailable parts, quality problems, and so on: Job flow time is the length of time that begins when a job arrives at the shop, workstation, or work center, and ends when it leaves the shop, workstation, or work center. The average flow time for a group of jobs is equal to the total flow time for the jobs divided by the number of jobs.

Job lateness: This is the length of time the job completion date is expected to exceed the date the job was due or promised to a customer. It is the difference between the actual completion time and the due date. If we only record differences for jobs with completion times that exceed due dates, and assign zeros to jobs that are early, the term we use to refer to that is job tardiness.

Make span: Make span is the total time needed to complete a group of jobs. It is the length of time between the start of the first job in the group and the completion of the last job in the group

Average number of jobs: Jobs that are in a shop are considered to be work-in-process inventory. The average work-in-process for a group of jobs can be computed using the following formula:

$$\text{Average number of jobs} = \text{Total flow time} / \text{Make span}$$

If the jobs represent equal amounts of inventory, the average number of jobs will also reflect the average work-in-process inventory.

Scheduling in Service

An important goal in service systems is to match the flow of customers and service capabilities. An ideal situation is one that has a smooth flow of customers through the system. This would occur if each new customer arrives at the precise instant that the preceding customer's service is completed, as in a physician's office, or in air travel where the demand just equals the number of available seats. In each of these situations customer waiting time is minimized, and the service system staff and equipment would be fully utilized.

Unfortunately, the random nature of customer requests for service that generally prevails in service systems makes it nearly impossible to provide service capability that matches demand. Moreover, if service times are subject to variability-say, because of differing processing requirements-the inefficiency of the system is compounded. The inefficiencies can be reduced if arrivals can be scheduled (e.g., appointments), as in the case of doctors and dentists. However, in many situations appointments are not possible (supermarkets, gas stations, theaters, hospital emergency rooms, repair of equipment breakdowns). On waiting lines, focuses on those kinds of situations. There, the emphasis is on intermediate-term decisions related to service capacity. In this section, we will concern ourselves with short-term scheduling, in which much of the capacity of a system is essentially fixed, and the goal is to achieve a certain degree of customer service by efficient utilization of that capacity.

Scheduling in service systems may involve scheduling (1) customers, (2) the work-force, and (3) equipment. Scheduling customers often takes the form of appointment systems or reservation systems.

Appointment Systems

Appointment systems are intended to control the timing of customer arrivals in order to minimize customer waiting while achieving a high degree of capacity utilization. A doctor can use an appointment system to schedule patients' office visits during the afternoon, leaving the mornings free for hospital duties. Similarly, an attorney can schedule client meetings around court appearances. Even with appointments, however, problems can still arise due to lack of punctuality on the part of patients or clients, no-shows, and the inability to completely control the length of contact time (e.g., a dentist might run into complications in filling a tooth and have to spend additional time with a patient, thus backing up later appointments). Some of this can be avoided by trying to match the time reserved for a patient or client with the specific needs of that case rather than setting appointments at regular intervals. Even with the problems of late arrivals and no-shows, the appointment system is a tremendous improvement over random arrival.

Reservation Systems

Reservation systems are designed to enable service systems to formulate a fairly accurate estimate of the demand on the system for a given time period and to minimize customer disappointment generated by excessive waiting or inability to obtain service. Reservation systems are widely used by resorts, hotels and motels, restaurants, and some modes of transportation (e.g., airlines, car rentals). In the case of restaurants, reservations enable management to spread out or group customers so that demand matches service capabilities. Late arrivals and no-shows can disrupt the system.

Yield Management

Many companies, especially in the travel and tourism industries, operate with fixed capacities. Examples include hotels and motels, which operate with a fixed number of rooms to rent each night; airlines, which operate with a fixed number of seats to sell on any given flight; and cruise lines, which operate with a fixed number of berths to sell for any given cruise. The number of rooms, seats, or berths can be thought of as perishable inventory.

For example, unsold seats on a flight cannot be carried over to the next flight; they are lost. The same is true for hotel rooms and cruise berths. Of course, that unsold inventory does not generate income, so companies with fixed capacities must develop strategies to deal with sales.

Yield management is the application of pricing strategies to allocate capacity among various categories of demand with the goal of maximizing the revenue generated by the fixed capacity. Demand for fixed capacity usually consists of customers who make advance reservations and walk-ins. Customers who make advance reservations are typically price sensitive, while walk-ins are often price-insensitive. Companies must decide on the percentage of their limited inventory to allocate to reservations, trading off lower revenue per unit for increased certainty of sales, and how much to allocate to walk-ins, where demand is less certain but revenue per unit is higher.

The ability to predict demand is critical to the success of yield management, so forecasting plays a key role in the process. Seasonal variations are generally important, so forecasts must incorporate seasonality and plans must also be somewhat flexible to allow for ever-present random variations.

Scheduling the Workforce

Scheduling customers is demand management. Scheduling the workforce is capacity management. This approach works best when demand can be predicted with reasonable accuracy. This is often true for restaurants, theaters, rush-hour traffic, and similar instances that have repeating patterns of intensity of customer arrivals. Scheduling hospital personnel, police, and telephone operators for catalog sales, credit card companies, and mutual fund companies also comes under this heading. An additional consideration is the extent to which variations in customer demands can be met with workforce flexibility. Thus, capacity can be adjusted by having cross-trained workers who can be temporarily assigned to help out on bottleneck operations during periods of peak demand. Various constraints can affect workforce scheduling flexibility, including legal, behavioral, technical-such as workers' qualifications to perform certain operations-and budget constraints. Union contracts may provide still more constraints.

Cyclical Scheduling

In many services (e.g., hospitals, police departments, fire departments, restaurants, and supermarkets) the scheduling requirements are fairly similar: Employees must be assigned to work

shifts or time slots, and have days off, on a repeating or cyclical basis. Here is a method for determining both a schedule and the minimum number of workers needed. Generally, a basic work pattern is set (e.g., work five consecutive days, have two consecutive days off), and a list of staffing needs for the schedule cycle (usually one week) is given.

Scheduling Multiple Resources

In some situations, it is necessary to coordinate the use of more than one resource. For example, hospitals must schedule surgeons, operating room staffs, recovery room staffs, admissions, special equipment, nursing staffs, and so on. Educational institutions must schedule faculty, classrooms, audiovisual equipment, and students. The problem is further complicated by the variable nature of such systems. For example, educational institutions frequently change their course offerings, student enrollments change, and student's exhibit different course-selection patterns. Some schools and hospitals are using computer programs to assist them in devising acceptable schedules, although many appear to be using intuitive approaches with varying degrees of success.

Airlines are another example of service system that require the scheduling of multiple resources. Flight crews, aircraft, baggage handling equipment, ticket counters, gate personnel, boarding ramps, and maintenance personnel all have to be coordinated. Furthermore, government regulations on the number of hours a pilot can spend flying place an additional restriction on the system. Another interesting variable is that, unlike most systems, the flight crews and the equipment do not remain in one location. Moreover, the crew and the equipment are not usually scheduled as a single unit. Flight crews are often scheduled so that they return to their base city every two days or more often, and rest breaks must be considered.

**FINANCIAL MANAGEMENT
STUDENT HANDBOOK**

COURSE DESCRIPTION

Dear learners! The module begins with fundamental concepts, including background on the economic and financial environment, financial statements (with an emphasis on ratio analysis), corporate valuation (the time value of money), leverage analysis, capital budgeting, investment decision, cost of capital and risk analysis (portfolio theory). With this background, we go on to discuss how specific techniques and decision rules can be used to help maximize the value of the firm.

Finance is, in a real sense, the cornerstone of the free enterprise system. Therefore, good financial management is vitally important to business firms' economic health, nation, and the world. Because of its importance, corporate finance should be thoroughly understood. However, this is easier said than done in the field is relatively complex, and it is undergoing constant change in response to shifts in economic conditions. All of this makes corporate finance stimulating, exciting, challenging perplexing. We sincerely hope that Financial Management will help readers understand and solve the financial problems faced by businesses today.

The purpose of this module is to enable you to understand the financial decision-making process and to interpret the impact that financial decisions will have on value creation. The module, therefore, introduces you to the three major decision-making areas in financial management: investment, financing, and asset management decisions. We explore finance, including its frontiers, in an easy-to-understand, user-friendly manner. Although the module is designed for an introductory course in financial management, it can also be used as a reference tool.

OBJECTIVES OF THE COURSE

After studying this course, learners will be able to:

-  Explain what is meant by “financial management?” Its importance, scope & function of finance.
-  Explain why financial statement analysis planning is important to the firm and outside capital suppliers.
-  Define what is meant by “corporate valuation or the time value of money?”
-  Define, calculate, and interpret a firm's degree of operating, and financial leverage.
-  Recognize the key terms and concepts in Capital Budgeting and Investment decisions.

✚ Describe the basic concept and sources of capital associated with the cost of capital.

Chapter one

Financial management: an overview

1.1 Introduction

Dear learners! In this chapter you will learn about the overview of financial management. As you well know business organizations need finance to meet their requirements in the economic world. Any kind of business activity depends on the finance. Hence, it is called *as lifeblood of business organization*. Whether the business concerns are big or small, they need finance to fulfill their business activities.

In simple language finance is money. To start any business, we need capital. Capital is the amount of money required to start a business. The mobilization of finance is an important task for an entrepreneur therefore, finance is one of the significant factors which determine the nature and size of any enterprise. This is to be noted that identification of sources of finance from time to time to finance the assets of an enterprise is critical as it avoids the financial hardships of an enterprise. The finance is required to acquire various fixed assets and current assets.

Hence, the course Financial Management, in general presents an introduction to the theory and analytical tools essential for proper decision making in these and related areas. As a prospective finance manager, you will be introduced to the financial management process of typical firms. By learning how the financial management process works, you will establish one of the key building blocks for a successful finance career.

1.3 Meaning of Financial Management

Financial management can be defined as the “management of the finances of an organization in order to achieve the financial objectives of the organization”. In other words, financial management is concerned with the acquisition, financing, and management of assets with some overall goal in mind. Thus, the decision function of financial management can be broken down into three major areas: the capital budgeting (investment), financing, and working capital management decisions.

Financial management is a recently emerged discipline which still has no unique body of knowledge of its own, and draws heavily on economics for its theoretical concept even today. It is an area exposed for changes related with the concurrent globalization effect and increase in information technology.

Brigham (2001) described the historical perspective and the evolution of financial management as:

“When financial management emerged as a separate field of study in the early 1900s, the emphasis was on the legal aspects of mergers, the formation of new firms, and the various types of securities firms could issue to raise capital. During the depression of the 1930s the emphasis shifted to bankruptcy and reorganization, corporate liquidity, and the regulation of security markets. During the 1940s and early 1950s, it continued to be taught as a descriptive, institutional subject viewed more from the standpoint of an outsider rather than that of a manager. However, a movement toward theoretical analysis began during the late 1950s and the focus shifted to managerial decisions designed to maximize the value of the firm. The focus on value maximization continues as we begin the 21st century. However, two other trends are becoming increasingly important: (1) the globalization of business and (2) the increased use of information technology. Both of these trends provide companies with exciting new opportunities to increase profitability and reduce risks. However, these trends are also leading to increased competition and new risks.”

Setting aside the trends of change in financial management, different scholars define financial management in different technicalities, however, with similar major emphasis.

Pandey (1999) stated financial management in the form of *managerial activity* which is concerned with the planning and controlling of a firm's financial resources.

According to Solomon (1969) financial management is concerned with the efficient use of important economic resources namely, capital funds.

Phillippatus presented financial management as:

“It is concerned with managerial decisions that result in the financing and acquisition of long-term and short-term credits for the firm. As such it deals with the situation that requires selection of specific assets, the selection of specific liability as well as the problem of size and growth of an enterprise. The analysis of this decision is based on the expected inflows and outflows of funds and their effects upon managerial objectives.”

Generally, financial management can be simply defined as the decision and process of making optimal use of a firm's financial resources for the purpose of maximizing the owner's/shareholders wealth. Moreover, according to Khan and Jain (2000) it is an integral part of management and is not totally independent.

1.4 Other Discipline Related to Finance

Though finance had ceded itself from economics, it is not totally an independent field of study. It is an integral part of the firm's overall management. Finance heavily draws theories, concepts, and

techniques from related disciplines such as economics, accounting, marketing, operations, mathematics, statistics, and computer science. Among these disciplines, the field of finance is closely related to economics and accounting.

1.4.1 Finance versus Economics

Finance and economics are closely related in many aspects. First, economics is the mother field of finance. Second, the economic environment within which a firm operates influences the decisions of a financial manager. A financial manager must understand the interrelationships between the various sectors of the economy. He must also understand such economic variables as a gross domestic product, unemployment, inflation, interests, and taxes in making financial decisions.

Financial managers must also be able to use the structure of decision-making provided by economics. They must use economic theories as guidelines for their efficient financial decision making. These theories include pricing theory through the relationships between demand and supply, return analysis, profit maximization strategies, and marginal analysis. The last one, particularly, is the primary economic principle used in financial management.

Basic Differences between Finance and Economics

Finance is less concerned with theory than is economics. Finance is basically concerned with the application of theories and principles.

Finance deals with an individual firm; but economics deals with the industry and the overall level of the economic activity.

1.4.2 Finance Versus Accounting

Accounting provides financial information through financial statements. Therefore, these two fields are closely linked as accounting is an important input for financial decision-making. Besides, the accounting and finance functions generally overlap; and usually it is difficult to distinguish them. In many situations, the accounting and finance activities are within the control of the financial manager of a firm.

Basic Differences Between Finance and Accounting

Treatment of income: - in accounting income measurement is on accrual basis. Under this method revenues are recognized as earned and expenses as incurred. In finance, however, the cash method is employed to recognize the revenue and expenses.

Decision-making: - the primary function of accounting is to gather and present financial data. Finance, on the other hand, is primarily concerned with financial planning, controlling and decision-

making. The financial manager evaluates the financial statements provided by the accountant by applying additional data and then makes decisions accordingly.

Accounting is highly governed by generally accepted accounting principles.

1.4.3 Finance Versus Marketing

If you are interested in marketing, you need to know finance because, for example, marketers constantly work with budgets, and they need to understand how to get greatest payoff from marketing expenditures and programs. Analyzing costs and benefits of projects of all types is one of the most important aspects of finance, so the tools you learn in finance are vital in marketing research, the design of marketing and distribution channels, and product pricing among others. The financial manager has also to consider the impact of new product development and promotion plans in marketing areas.

1.4.4 Finance Versus Management

One of the most important areas in management is strategy. Thinking about business strategy without simultaneously thinking about financial strategy is an excellent step for disaster, and as a result, management strategists must have a very clear understanding of the financial implications of business plans.



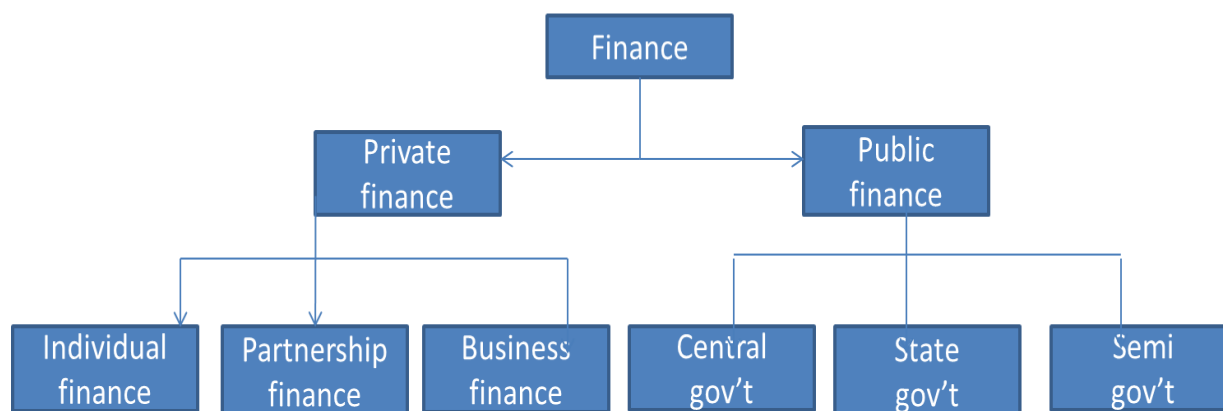
Self-check questions 1.1

1. ***Define Financial Management.***
2. How do you exactly describe the relationship between finance, economics, and accounting?

1.5 Classifications of Finance

Finance is one of the important and integral part of business concerns, hence, it plays a major role in every part of the business activities. It is used in all the area of the activities under the different names. Finance can be classified into two major parts:

Types of finance



Private Finance: includes the Individual, Firms' Business or Corporate Financial activities to meet the requirements.

Public Finance: concerns with revenue and disbursement of Government such as Central Government, State Government and Semi-Government Financial matters.

Finance consists of three interrelated areas particularly in association with the career of graduates:

Financial Market: This deals with securities market and financial institutions. For success here, one needs knowledge of valuation techniques, the factors that cause interest rate to rise and fall, the regulations to which financial institutions are subject, and the various types of financial instruments. One also needs a general knowledge of all aspects of business administrations, because the management of a financial institution involves accounting, marketing, personnel, computer system as well as financial management.

Individual/personal Finance: it focuses on the decisions made by both individual and institutional investors as they choose securities for their investment portfolios. Broadly speaking, the investment area deals with financial assets such as stock/share and bonds. Investments relate to decisions concerning stocks and bonds and include a number of activities: (1) Security analysis deals with finding the proper values of individual securities (i.e., stocks and bonds). (2) Portfolio theory deals with the best way to structure portfolios, or "baskets," of stocks and bonds. Rational investors want to

hold diversified portfolios in order to limit risks, so choosing a properly balanced portfolio is an important issue for any investor. (3) Market analysis deals with the issue of whether stock and bond markets at any given time are “too high,” “too low,” or “about right.” Behavioral finance, where investor psychology is examined in an effort to determine if stock prices have been bid up to unreasonable heights in a speculative bubble or driven down to unreasonable lows in a fit of irrational pessimism, is a part of market analysis.

Business finance or FM: This involves decisions within firms. It is the broadest of the three areas. Financial management (FM), also called corporate finance, focuses on decisions relating to how much and what types of assets to acquire, how to raise the capital needed to buy assets, and how to run the firm so as to maximize its value. The same principles apply to both for-profit and not-for-profit organizations.

Although we separate these three areas, they are closely interconnected. Banking is studied under money & capital markets, but a bank lending officer evaluating a business’ loan request must understand corporate finance to make a sound decision. Similarly, a corporate treasurer negotiating with a banker must understand banking if the treasurer is to borrow on “reasonable” terms. Moreover, a security analyst trying to determine a stock’s true value must understand corporate finance and capital markets to do his or her job. In addition, financial decisions of all types depend on the level of interest rates; so, all people in corporate finance, investments, and banking must know something about interest rates and the way they are determined.

1.6 Scope and Functions Financial Management

The scope of financial management has evolved from the traditional to the modern approach. Under the traditional approach, the major emphasis of financial management was considered in a narrower sense. Its emphasis was how corporations raise fund and it was totally having of an external orientation. The modern approach views financial management in a broader sense and provides a conceptual and analytical framework for financial decision making. According to this approach, the finance function covers both acquisitions of funds as well as their allocation. In other words, financial management is concerned with the solutions of the three major problems relating to the financial operations of a firm, corresponding to three questions of investment, financing, and dividend decisions. Thus, financial management, in the modern sense can be broken down into three major decisions as functions of finance: i) the investment decisions, ii) the financing decisions, and iii) the dividend policy decision (Khan and Jain, 2000).

1.6.1 Investment (Asset-Mix) Decisions

The investment decisions relate to the selection of assets in which funds will be invested by the firm. The assets may long term assets which yield return in the future over longer periods; and short-term assets which are coverable into cash at normal operation of the business without decrease in value, usually within a year. When the investment is made on long-term assets it is considered as capital budgeting while the other is working capital.

Capital budgeting decision is concerned with long-term assets and their compositions. Measurement of investment proposals is the major exercise of capital budgeting decision. It evaluates the business risk composition of the firm where risk and uncertainty of a certain project proposal is evaluated against a certain established standard. Moreover, it is concerned with judgment of the benefit of the firm with concept of the cost of capital.

Working capital management is concerned with the management of current assets. It is an important and integral part of financial management as a short-term survival is the prerequisite for the long-term success.

1.6.2. Financing (Capital-Mix) Decisions

Financing decision is the second important function of financial management. It is emphasized when, where and how to acquire funds to meet the firm's investment needs. The central issue, therefore, is to determine the proportion of equity and debt. The mix of debt and equity is known as the firm's capital structure. The financial manager must strive to obtain the best financing mix or the optimum capital structure for the firm. The firm's capital structure is considered to be optimum when the market value of shares is maximized. The use of debt affects the return and risk of shareholders; it may increase the return on equity funds but it always increases risk. A proper balance will have to be struck between return and risk. When the shareholder's return is maximized with minimum risk, the market value per share will be maximized and the firm's capital structure would be considered optimum. Once the financial manager is able to determine the best combination of debt and equity, he/she must raise the appropriate amount through the best available sources.

1.6.3 Dividend or Profit Allocation Decisions

Dividend decision is the third major financial decision. The emphasis is whether the firm should distribute all profits, or retain them, or distribute a portion and retain the balance. Like the debt-equity policy, the dividend policy should be determined in terms of its impact on the shareholders' value. The optimum dividend policy is one that maximized the market value of firm's shares. Thus, if shareholders are not indifferent to the firm's dividend policy, the financial manager must determine the optimum dividend-payout ratio. The financial manager should also consider the questions of dividend stability, bonus share and cash dividends in practice.

1.7 Sources of Finance

The finance required for any organization could be primarily divided into two one is long-run finance to acquire the fixed assets that are useful to the business organization over a period of time i.e. more than a year, usually we call fixed capital. The other one is short-term finance which is required to keep running the fixed assets or to make them working. This is called the working capital.

Long-term sources – The important long-term sources are common stock, preference stock bonds, loans from financial institutions and foreign capital.

Short-term sources – The short-term sources are bank loans, public deposits trade credits provisions and current liabilities.

The requirements of above nature could be financed either through external sources or internal sources if it is an existing company.

External – These are the funds drawn from outsiders. Among them the prominent are discussed below.

Share Capital – This is the primary source of finance to a corporate form of organization. It is the sale of equity or common stock and preference stock to the public. Which serves as a permanent capital to an organization. These holders will get dividend in return for their investment.

Common stock – The holders of these shares are owners of the company. They are the risk takers. They get dividend when the company earns profits, otherwise they do not get any dividend. Whatever profit is left after meeting all the expenses belongs to them. In the event of closure of the company they are the last people to get their claim.

Preference stock – Preference shares carry two preferential rights one is to get a fixed dividend at the end of each year irrespective of the profits, other one is to get back the original investment first when the company goes into liquidation.

Change par bonds – Another source of finance to a company is issue of bonds/ debentures. These holders are eligible to get fixed interest at the end of each year. The holders of these bonds do not wish to take any risk public deposits. The term is also mentioned while issuing bonds.

Public deposits – This is another mode of finance where the company will advertise and accept deposits for specified period at a fixed rate of interest.

Borrowings – The companies may borrow funds from banks, financial institutions etc for their requirements at the interest chargeable by the lender institution.

Foreign capital – The concept of liberalization is attracting many foreign companies to participate in the domestic companies. It can be either in the form of direct participation in the capital or collaboration in a project in the equity of the company and also provide loans some time.

Trade credits – The common means of short-term external finance is trade credits. Normally; every company gets its raw material and other supplies on credit basis. This is known as trade credit. This is an important source of financing.

Internal Sources – This is applicable for only those companies which are in existence. By virtue of their existence, they are in a advantageous position to generate some of the finance internally.

Retained earnings – These are the funds that are retained out of the profits for meeting future contingencies. It can be either to meet the uncertainty or future growth and expansion of business. The company would be free to utilize this source. The retained profits enable a company to withstand seasonal reactions and business fluctuations. The large accumulated savings facilitate a stable dividend policy and enhance the credit standing of the company. However, the quantum of retained earnings depend on the volume of the profits made by the company.

Provisions – Generally companies, in order to meet the legal and other obligations, create some funds for future use. These are known as provisions. They include depreciation, taxation, dividends and various current and non-current liabilities. The amount set apart in these forms would be required to be paid only on certain dates. Till then the company can use them for its own purpose. For instance, taxes payable to the government are used in the business until these are paid on due date. Therefore, though for a short-while provision would serve as a good source of internal finance.



Self-check questions 1.2

1. What are the external sources of finance?
2. List out the internal sources of finance?

1.8 Financial Markets Instruments - Institutions

1.8.1 Financial Markets

Financial markets are a place where the business houses can raise their long and short-term financial requirements. The development of financial markets indicates the development of economic system. For mobilization of savings and for rapid capital formation, healthy growth and development of these markets are crucial. These markets help promotion of investment activities, encourage entrepreneurship and development of a country. The financial markets are broadly divided as

- Capital market and
- Money market
- Capital market

1. Capital market

Capital market is defined as a place where all buyers and sellers of capital funds as well as the entire mechanism for facilitating and effecting long term funds. It provides the long-term funds that are needed for investment purpose. Thus, the capital markets are concerned with long-term finance. This also includes the institutions, facilities and arrangements for the borrowing and lending of long-term funds. Further the capital markets are divided into two categories one is primary market other one is secondary market.

- i) **Primary market** – In the primary market only new securities are issued to the public. It is a place where borrowers exchange financial securities for long-term funds. It facilitates the formation of capital. The securities may be issued directly to the individuals, institutions, through the underwriters etc.
- ii) **Secondary market** – The shares subsequent to the allotment are traded in the secondary market. Anybody can either buy or sell the securities in the market. Secondary market consists of stock exchange. In the stock exchange outstanding securities are offered for sale and purchase.

2. Money Market

Money market deals with short-term requirements of borrowers. It is concerned with the supply and demand for a commodity or service. It handles transactions in short-term government obligations, bankers' acceptances and commodity papers. In money market funds can be borrowed for short period varying from a day to a year. It is a place where the lending and borrowing of short-term funds are arranged and it comprises short-term credit instruments and individuals who participate in the lending and borrowing business.

1.8.2 Financial Instruments

There are mainly two kinds of securities namely ownership securities and loan securities. Further ownership securities are classified into two (a) common stock and (b) preference stock. These securities or instruments are being traded in capital markets.

- a. **Common stock** – It is also known as equity shares, who are the real owners of the business will enjoy the profit or loss suffered by the company.
- b. **Preferential stock** – By name these holders have two preferential rights I) to get fixed rate of dividend at the end of every year irrespective of profits / losses of the company II) to get back the investment first when the company goes into liquidation.
- c. **Bonds** – Bondholders are the money suppliers to a business unit entitled for a fixed rate of interest at the end of each year. There is stake is confined to the interest only.

1.8.3 Financial Institutions

The financial institutions include banks, development banks, investing institutions at national and international level that provide financial services to the business organizations. These financial institutions provide long-term, short-term finances and extend under writing, promotional and merchant banking services.



Self-check questions 1.3

1. Distinguish the difference between money market and capital market.
2. List out the different categories of capital market?

1.9 Objectives of Financial Management

Firm's investment and financial decisions are unavoidable and continuous. In order to make these decisions rationally, the firm must have a goal or objective (Pandey, 1999). Objectives provide a framework for optimal financial decision making. Following sections show the profit maximization and wealth maximization objectives of a firm and present the concern for the shareholders' wealth maximization. The wealth maximization is identified as theoretically logical, operationally feasible, and normative framework for guiding the financial decision making.

1.9.1 Profit Maximization

A business firm is profit-seeking organization. Hence, profit maximization is well considered to be an important objective of financial management. Profit maximization means maximizing birr income of firms. Firms produce goods and services. They may function in a market economy, or in a government-controlled economy. In market economy, prices of goods and services are determined in competitive markets. Firms in a market economy are expected to produce goods and services desired by society as efficiently as possible. In a market economy, goods and services in great demand command higher prices. This results in higher profit for firms; more of such goods and services are produced. A higher profit opportunity attracts other firms to produce such goods and services. Ultimately, with intensifying competition an equilibrium price is reached at which demand and supply match. In the case of goods and services which are not required by society, their prices and profits fall. Such goods and services are dropped out by producers in favor of more profitable opportunities.

Main aim of any kind of economic activity is earning profit. A business concern is also functioning mainly for the purpose of earning profit. Profit is the measuring techniques to understand the business efficiency of the concern. Profit maximization is also the traditional and narrow approach, which aims at, maximizes the profit of the concern. **Profit maximization consists of the following important features.**

- ✓ Profit maximization is also called as cashing per share maximization. It leads to maximize the business operation for profit maximization.
- ✓ Ultimate aim of the business concern is earning profit; hence, it considers all the possible ways to increase the profitability of the concern.

-
- ✓ Profit is the parameter of measuring the efficiency of the business concern. So it shows the entire position of the business concern.
 - ✓ Profit maximization objectives help to reduce the risk of the business.

Favorable Arguments for Profit Maximization

The following important points are in support of the profit maximization objectives of the business concern:

- i. Main aim is earning profit.
- ii. Profit is the parameter of the business operation.
- iii. Profit reduces risk of the business concern.
- iv. Profit is the main source of finance.
- v. Profitability meets the social needs also.

Unfavorable Arguments for Profit Maximization

- ii. The following important points are against the objectives of profit maximization: Profit maximization leads to exploiting workers and consumers.
- iii. Profit maximization creates immoral practices such as corrupt practice, unfair trade practice, etc.
- iv. Profit maximization objectives leads to inequalities among the stake holders such as customers, suppliers, public shareholders, etc.

In the economic theory, the behavior of firm is analyzed in terms of profit maximization. While maximizing profit, a firm either produces maximum output for a given amount of output, or uses minimum input for producing a given output. Thus, the underlying logic of profit maximization is efficiency. It is assumed to cause the efficient allocation of resources under the competitive market conditions, and profit is considered as the most appropriate measure of the firm's performance. But the profit maximization suffers from the following limitations:

i) Ambiguity in Definition The precise meaning of the profit maximization objective is not clear. Definition of the term profit is ambiguous. Does it mean short-or-long-term profit? Does it refer to profit before or after tax? Total profits or profit per share? Does it mean total operating profit or profit accruing to shareholder? These cases are difficult to specify and make the term vague.

ii) Time Value of Money The profit maximization objective does not make a distinction between returns received at different time periods. It gives no consideration to the time value of money or timing of benefits. Its values benefits received today and benefits received after a period as the same.

iii) Quality of Benefits The streams of benefits may possess different degree of certainty. The term quality refers to the degree of certainty with which benefits are expected. Two firms may have same total expected earnings, but if the earnings of one firm fluctuate considerably as compared to the other, it will be riskier. Possibly, owners of the firm would prefer smaller but assured profits to a potentially larger but less certain stream of benefits.

Generally, profit maximization criterion is inappropriate and unsuitable as an operational objective of investment, financing and dividend decisions of a firm. it is not only vague but it also ignores two important dimensions of financial analysis namely risk and time value of money.

1.9.2 Wealth Maximization

Wealth Maximization is also known as value maximization or net present worth maximization. In the current financial literatures, value maximization is almost universally accepted as an appropriate operational decision criterion for financial management decisions. It removes the technical limitations of profit maximization criterion. Its operational features satisfy all the three requirements of a suitable operational objective of financial courses of action, namely, exactness, quality of benefits and the time value of money.

The value of an asset should be viewed in terms of the benefits it can produce. The worth of a course of action can similarly be judged in terms of the value of the benefits it produces less the cost of undertaking it. A significant element in computing the value of a financial course of action is the precise estimation of the benefits associate with it. The wealth maximization criterion is based on the concept of cash flows generated by the decisions rather than accounting profit. Cash flow is precise concept with a definite connotation. Measuring benefits in terms of cash flows avoids the ambiguity associated with accounting profits. This is the first operational feature of the net present worth maximization criterion.

The second important feature of the wealth maximization criterion is that it considers both the quantity and quality of dimensions of benefits. At the same time, it also incorporates the time value of money. The operational implication of the uncertainty and timing dimensions of the benefits originates from a financial decision. Adjustments should be made in the cash flow pattern, firstly, to

incorporate risk and, secondly, to make an allowance for differences in the timing of benefits. The value of stream of cash flows is calculated by discounting its element back to the present at a discount rate. The discount rate reflects both time and risk. In applying the value maximization criterion, the time value is used in terms of worth to the owners. The discount rate that is employed is, therefore, the rate that reflects the time and risk preferences of the owners or suppliers of capital. A large discount rate is the result of higher risk and longer time period.

For the above reasons, the net present value maximization is superior to the profit maximization as an operational objective. As a decision criterion, it involves a comparison of value to cost. An action that has discounted value exceeding its cost can be said to create value.

A financial action that has a positive NPV creates wealth for shareholders and, therefore, is desirable. A financial action resulting in negative NPV should be rejected since it would destroy shareholders' wealth. Between numbers of mutually exclusive projects the one with higher NPV should be adopted. Therefore, the wealth will be maximized if this criterion is followed in making financial decisions.

For the above reasons, the net present value maximization is superior to the profit maximization as an operational objective. As a decision criterion, it involves a comparison of value to cost. An action that has discounted value exceeding its cost can be said to create value.

Favorable Arguments for Wealth Maximization

- ✓ Wealth maximization is superior to the profit maximization because the main aim of the business concern under this concept is to improve the value or wealth of the shareholders.
- ✓ Wealth maximization considers the comparison of the value to cost associated with the business concern. Total value detected from the total cost incurred for the business operation. It provides extract value of the business concern.
- ✓ Wealth maximization considers both time and risk of the business concern.
- ✓ Wealth maximization provides efficient allocation of resources.
- ✓ It ensures the economic interest of the society.

Unfavorable Arguments for Wealth Maximization

-
- (i) Wealth maximization leads to prescriptive idea of the business concern but it may not be suitable to present day business activities.
 - (ii) Wealth maximization is nothing, it is also profit maximization, it is the indirect name of the profit maximization.
 - (iii) Wealth maximization creates ownership-management controversy.
 - (iv) Management alone enjoys certain benefits.
 - (v) The ultimate aim of the wealth maximization objectives is to maximize the profit.
 - (vi) Wealth maximization can be activated only with the help of the profitable position of the business concern.

1.10 The Role of Financial Managers

A financial manager is a person who is responsible in a significant way to carry out the finance functions. A financial manager is responsible for shaping the fortunes of the enterprise, and is involved in the most vital decision of the allocation of capital. A financial manager plays a dynamic role in a modern company's development. Weighted corporate competition, technology changes, volatility in inflation and interest rates, worldwide economic uncertainty and ethical concerns over certain financial dealings must be dealt with almost daily.

The role of a financial manager changes and old ways of doing things are not good enough in a world where old ways become obsolete. Today's financial manager must have the flexibility to adapt to the changing external environment if his or her firm is to survive. The financial manager's ability to adapt to changes, raise fund, invests in asset, and manages wisely will affect the success of a firm.

Hence, a financial manager through efficient acquisition, financing and managing assets contribute to the firm in particular and to the strength and growth of an economy as a whole.

Moreover, the following specific tasks of financial staff can be stated as major responsibilities.

Forecasting and Planning The financial staff must coordinate the planning process. This means that they must interact with people from other departments as they took ahead and lay the plans that will shape the firm's future.

Major Investment and Financing Decision A successful firm usually has rapid growth in sales, which requires investments in plant, equipment, and inventory. The financial staff must help

determine the optimal sales growth rate, help decide what specific assets to acquire, and then choose the best way to finance these assets. For example, should the firm finance with debt, equity or some contributions of the two, and if debt is used, how much should be long term and how much short term?

Contribution and Control The financial staff must interact with other personnel to ensure that the firm is operating as efficiently as possible. All business decisions must have financial implications, and all managers- financial and otherwise- need to take this into account. For example, marketing decisions affect sales growth, which in turn influences investment requirements. Thus, marketing decision makers must take into account of how their actions affect and are affected by such factors as the availability of funds, inventory policies, and plant capacity utilization.

Dealing with the Financial Markets The financial staff must deal with money and capital markets. Each firm affects and is affected by the general financial markets where funds are raised, where the firm's securities are traded, and where investors either make or lose money.

Risk Management All business face risk, including natural disasters such as fires and floods, uncertainties in commodity markets, volatile interest rates and fluctuating foreign exchange rates. However, many of these risks can be reduced by purchasing insurance or by hedging in derivatives markets. The financial staff is responsible for the firm's overall risk management program, including identifying the risks that should be managed and then managing them in the most efficient manner.



Self-check questions 1.4

1. List the objectives of financial management.
2. Why profit maximization is criticized?

CHAPTER SUMMERY

Finance consists of three interrelated areas. Money and capital market, investments, and financial management are the three interrelated areas of finance. Money and capital market deals with security market and financial institutions. The investment area emphasizes on the financial assets of a firm technically speaking the shares and bonds among others. The financial management is mainly for the decision aspects.

Financial management can be defined as the process of making optimal use of financial resources for the purpose of shareholders wealth maximization. As finance is not totally independent discipline, it has interrelated features with accounting, economics, marketing and management.

The scope of financial management is modified while it passes from the traditional approach to the modern approach. Its scope was original limited to raise funds for corporate form of organizations. But under the modern approach, financial management deals with the three important decisions: investment decision, financing decision and dividend policy decision.

The investment decision is mainly for the mix of assets, long term assets and short-term assets. The financing decision concerns about the financing mix decision of firms that is the debt-equity proportions of firms. Firms pass decision on the distribution and retention of net income which is the concern of the dividend policy decision.

Goals or objectives are required for firms to make rational decisions. The profit maximization and wealth maximization are the basic criteria to make normative decisions. Since profit maximization has limitations on the ambiguity, risk and time value of money grounds, wealth maximization is considered as theoretically logical, operational feasible and provide normative framework for guiding firm's make rationale financial decisions.

Chapter two

Financial statement analysis and financial planning

2.1 Introduction

Dear reader! You have read the major financial statements in your introductory accounting courses. These financial statements are raw unless they are further processed and analyzed. Users of accounting information can get the detailed insight of financial statements when they are analyzed. Financial analysis can be conducted with the use ratio, common size statement, index, and trend analyses. Moreover, firms make financial plans based on their historical data analyzed. The chapter is, therefore, designed to show the major types of financial statement analyses and how to prepare pro-forma financial statements.

2.2 Objectives the Chapter

After the completion of the chapter, you will be able to

- Analysis the performance of organizations with the help of ratio, common size statement, trend and index analyses
- Interpret results of ratio analysis
- Understand the limitations of ratios and cares to be taken
- Use techniques of forecasting financial position and results
- Prepare projected financial statements based on historical accounting data

2.3 Financial analysis: An Introduction

In your introductory accounting courses and previous modules, you have been learned about how to construct financial statements. As you are expected to remember the financial statements include: balance sheet, income statements, cash flow statements and the shareholders' or the owner's equity statements. The information obtained in these statements is used by the management groups, creditors, investors and others to form a kind of judgment about the operating performance and financial position of a firm. Users of financial statements can get further insight about financial strengths and weaknesses of the firm if they properly analyze information reported in these

statements. The management should be particularly interested in knowing financial strengths of a firm to make their best use, to be able to spot out financial weaknesses of the firm and to take suitable corrective actions as needed. The future plans of the firm should be laid down in view of the firm's financial strengths and weaknesses. Thus, financial analysis is the starting

Financial analysis is the process of identifying the financial *strength and weakness* of a firm by properly establishing relationships between the items of financial statements for certain period(s).

Financial analysis refers to analysis of financial statements and it ***is a process of evaluating the relationships among component parts of financial statements***. The focus of financial analysis is on key figure in the financial statements and the significant relationships that exist between them. Financial analysis is used by several groups of users like managers, credit analysts, and investors.

The analysis of financial statements is designed to reveal ***the relative*** strengths and weakness of a firm. This could be achieved by comparing the analysis with other companies in the same industry, and by showing whether the firm's position has been improving or deteriorating over time. Financial analysis ***helps users*** obtain a better understanding of the firm's financial conditions and performance. It also helps users understand the numbers presented in the financial statements and ***serve as a basis for financial decisions***.

2.4 Financial Statements Analysis

Financial analysis can be undertaken by management of the firm, or by parties outside the firm such as owners, creditors, investors and so forth. ***The nature of analysis will differ depending on the purpose of the analyst***. The target of financial analysis is to produce relationships among financial data and interpret the results. The sources of data for financial analysis are accounting records, particularly balance sheet and income statement. The methods of analyzing financial statements include ratio analysis, common size statement analysis, index analysis, and trend analysis.

Financial statements include: balance sheet, income statements, cash flow statements and the shareholders' or the owners' equity statements. The information obtained in these statements is used by the management groups, creditors, investors and others to form a kind of judgment about the *operating performance and financial position* of a firm. Users of financial statements can get further insight about financial strengths and weaknesses of the firm if they properly analyze information reported in these statements. The management should be particularly interested in knowing financial

strengths of a firm to make their best use, to be able to spot out financial weaknesses of the firm and to take suitable corrective actions as needed.

The future plans of the firm should be laid down in view of the firm's financial strengths and weaknesses. Thus, financial analysis is the starting point for making plans; that is before using any sophisticated forecasting and planning procedures. Moreover, understanding the past with critical and suitable analysis is a prerequisite for anticipating the future. Hence, the discussions here followed are all about the analysis of financial statements particularly of the balance sheet and the income statement. Use the following financial statements as part of illustrations for the sections.

GLOBAL Company

Income Statement (in'000 Birr)

For the year ended December 31, 2008

Sales		Br.830
Cost of Goods Sold		<u>(539)</u>
Gross Profit on Sales		Br.291
Operating Expenses:		
Marketing Expense	Br. 91	
General and Administrative Expense	71	
Depreciation	<u>28</u>	
Total Operating Expenses		(Br.190)
Operating Income (EBIT)		Br.101
Interest Expense		<u>(20)</u>
Earnings Before tax		Br. 81
Income Tax		<u>(17)</u>
Earnings after Tax		Br.64
Dividends Paid		<u>(15)</u>
Change in Retained Earnings		<u>Br.49</u>

GLOBAL Company
Balance Sheets (in '000 Birr)
December 31, 2007 and 2008

Assets	<u>2007</u>	<u>2008</u>
Current Assets:		
Cash	Br.39	Br.44
Accounts receivable	70	78
Inventories	177	210
Prepaid expenses	<u>14</u>	<u>15</u>
Total current assets	300	347
Fixed assets:		
Gross plant and equipment	Br.759	Br.838
Accumulated depreciation	(355)	(383)
Net plant and equipment	404	455
Land	<u>70</u>	<u>70</u>
Total fixed assets	Br.474	Br.525
Patents	<u>30</u>	<u>55</u>
Total assets	<u>804</u>	<u>927</u>
Liabilities and Equity		
Current Liabilities:		
Accounts Payable	Br.61	Br.76
Income tax payable	12	17
Accrued wages and salaries	4	4
Interest payable	<u>2</u>	<u>2</u>
Total Current liabilities	Br.79	Br.99
Long-term notes payable	<u>146</u>	<u>200</u>
Total liabilities:	<u>Br.225</u>	<u>Br.299</u>
Common stock	300	300
Retained Earnings	<u>279</u>	<u>328</u>
Total Stockholders' equity	<u>579</u>	<u>628</u>

Total liabilities and equity

Br.804

Br.927

Stages in Financial Analysis

Financial analysis consists of the following three major stages.

- i. Preparation:** The preparatory steps include establishing the objectives of the analysis and assembling the financial statements and other pertinent financial data. Financial statement analysis focuses primarily on the balance sheet and the income statement. However, data from statements of retained earnings and cash flows may also be used. *So, preparation is simply objective setting and data collection.*
- ii. Computation:** This involves the *application of various tools and techniques to gain a better understanding* of the firm's financial condition and performance. Computerized financial statement analysis programs can be applied as part of this stage of financial analysis.
- iii. Evaluation and Interpretation:** Involves the determination *for the meaningfulness of the analysis and to develop conclusions, inferences, and recommendations* about the firm's performance and financial condition. This is the most important of all the three stages of financial analysis.

2.4.1 Tools and Techniques of Financial Analysis

A number of methods can be used in order to get a better understanding about a firm's financial status and operating results. The most frequently used techniques in analyzing financial statements are:

- i. Ratio Analysis
- ii. Common size Analysis
- iii. Index Analysis

2.4.1.1 Ratio Analysis

Ratio Analysis – is a mathematical relationship among money amounts in the financial statements. They standardize financial data by converting money figures in the financial statements. Ratios are usually stated in terms of times or percentages. Like any other financial analysis, a ratio analysis

helps us draw meaningful conclusions and interpretations about a firm's financial condition and performance.

Ratio analysis is a widely used tool of financial analysis. It is the systematic use of ratio to interpret financial statements so that the strength and weakness of a firm as well as its historical performance and current condition can be determined. ***The term ratio is defined as the numerical or quantitative relationship of two items or variables.***

According to Pandey (1999) ratio is defined as the indicated quotient of two mathematical expressions and the relationship between two or more things.

In financial analysis, a ratio is used as a benchmark for evaluating the financial position and performance of a firm. The absolute accounting figures reported in the financial statements do not provide a meaningful understanding of the performance and financial position of a firm. An accounting figure conveys meaning when it is related to some other relevant information (Pandey, 1999).

Moreover, for Mayes and Shank (2004) ratios are an analyst's ***microscope*** which allows users of information a better view of a firm's financial health than just looking at the raw financial data presented in financial statements. For example, it might be an impressive figure for a firm which report Br. 5,000,000 net profit at a period of time, but it is difficult to say the firm's performance is good or bad unless the net profit figure is related to the firm's investment and the total sales made among other factors.

Similarly, ***ratios help to summarize large quantities of financial data and to make qualitative judgment about the firm's, financial performance.*** For example, consider current ratio which is calculated by dividing current assets by current liabilities; the ratio indicates a relationship in terms of a *quantified* relationship between current assets and current liabilities. This relationship is an index or yardstick which permits a *qualitative judgment* to be formed about the firm's ability to meet its current obligations. It measures the firm's liquidity. The greater the ratio, the greater is the firm's liquidity and vice versa. The point that must be noted is a ratio reflecting a quantitative relationship helps to form a qualitative judgment which is the nature of all financial ratios.

2.4.1.2 Standards for Ratio Comparisons

Ratio analysis is not for the sake of the application of formula to financial data while calculating a given ratio. More important is the interpretation of the value calculated. To arrive at a proper interpretation and to make valuable judgments, a meaningful standard or basis is required and shall be established.

Standards of comparison may consist of;

-  **Past Ratios:** Ratios calculated from the past financial statements of the same firm **i.e., Longitudinal or Trend based standards**
-  **Competitors' Ratios:** Ratios of some selected firms, especially the most progressive and successful competitor, at the same point in time i.e., **Single period or Cross-section based standards**
-  **Industry Ratios:** Ratios of the industry to which the firm belongs i.e., **Industry based standards**
-  **Projected Ratios:** Ratios developed using the projected, or pro forma, financial statements of the same firm **i.e., Projected (pro forma) statements-based standards**

A) Trend Based Standards: The easiest way to evaluate the performance of a firm is to compare its present ratios with the past ratios. When financial ratios over a period of time are compared, it is known as the time series (trend) analysis. It gives an indication of the direction of change and reflects whether the firm's financial performance has improved, deteriorated or remained consistent over time. The analyst should not simply determine the change, but, more importantly, should understand why ratios changed.

B) Cross-Sectional Based Standards: Another way of comparison is to compare ratios of one firm with some selected firms in the same nature at the same point in time. This kind of comparison is known as the cross-sectional comparison. In most cases, it is more useful to compare the firm's ratios with ratios of a few carefully selected competitors.

This kind of a comparison indicates the relative financial position and performance of the firm. A firm can easily resort to such a comparison, as it is not difficult to get the published financial statements of similar firms.

C) Industry Based Standards: To determine the financial condition and performance of a firm, its ratios may be compared with average ratios of the industry of which the firm is a member. Industry

ratios are important standards in view of the fact that each industry has its characteristics which influence the financial and operating relationships. But there are certain practical difficulties in using the industry ratios.

- First, it is difficult to get average ratios for the industry.
- Second, even if industry ratios are available, they are averages of the ratios of strong and weak firms. Sometimes differences may be so wide that the average may be of little utility.
- Third, averages will be meaningless and the comparison futile if firms within the same industry widely differ in their accounting policies and practices.

If it is possible to

- Standardize the accounting data for companies in the industry and
- Eliminate extremely strong and extremely weak firms,

The industry ratios **will prove to be very useful** in evaluating the relative financial condition and performance of a firm.

D) Pro forma Statements based standards Sometimes future ratios are used as the standard of comparison. Future ratios *can be developed from projected or pro forma financial statements*. The comparison of current or past ratios with future ratios shows the firm's relative strengths and weaknesses in the past and the future. If the future ratios indicate weak financial position, corrective actions should be initiated.



Self-check questions 2.1

1. Mention some reasons for undertaking financial statement analysis.
2. What is the need for standards in ratio analysis?

2.4.1.3 Types of Ratio Analysis

Financial ratios can be grouped into different categories based on the need of analysis to be performed and the tasks to be evaluated. The interest of the users for analyzed information also matter

while classifying the types of ratio analysis. User of the information may concentrate on the liquidity position of a firm so that liquidity ratios are in demand; long-term solvency and sustenance of profitability may be in need of long-term creditors so that profitability and leverage ratios are required; similarly, owners may want to know the firm's profitability, stability of the earnings over periods and efficient and effective utilization of assets so that market value, activity and profitability ratios are requested. For the sake of convenience, however, ratios can be grouped into five basic types.

- 1) Liquidity ratios
- 2) Activity ratios
- 3) Debt/leverage ratios
- 4) Profitability ratios
- 5) Market value ratios

1. Liquidity Analysis

Liquidity refers to the firm's ability to meet its obligation in the short-run, usually one year. Liquidity ratios are generally based on the relationship between current assets and current liabilities. The most important liquidity ratios are: current ratio, acid-test ratio and cash ratio.

A. Current Ratio One of the very popular ratio, Components of current assets are cash, short term investment, receivables, inventories and pre-paid expenses. Current liabilities are those liabilities that are expected to mature usually in the next twelve months. These comprise account payables, accrued payables, loans - secured or unsecured, that are due in the next twelve months, and current other provisions.

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

The current ratio of GLOBAL Company for 2008 is:

$$\text{Current ratio} = \frac{\text{Br. 347}}{\text{Br. 99}} = 3.5\text{times or } 3.5:1$$

The general minimum norm for current ratio at international level is 2.0, but the acceptability of a value of current ratio depends on the nature of the industry in which the firm is categorized and standards established for comparisons. Assume, for example, that the industry average for GLOBAL

Company is 4.1 times. GLOBAL's current ratio is below the average of its industry, so its liquidity position is relatively weak.

As a general norm the higher the current ratio, the greater is the short-term solvency. However, in interpreting current ratio the composition of current assets must not be overlooked. A firm with a high proportion of current assets in the form of cash and receivables is more liquid than a firm with a high proportion of current assets in the form of inventories even though both firms may have the same current ratio.

A very high current ratio may imply that:

- ✓ There is excessive cash due to poor cash management
- ✓ Receivables are excess due to poor credit management
- ✓ The prevalence of excessive inventory due to lack of proper inventory management.

The result of very high current ratio is to have very high liquidity which is safety of funds for short term creditors thereby reduced risk to creditors. However, the result shows a scarification of profitability because current assets are less profitable than long term assets. On the contrary, a very low current ratio may be caused by conservative management of current assets which may be the opposite for very high current ratio.

Very low current ratio can be improved with:

- ✓ Long-term borrowing and sell of stocks to increase current assets
- ✓ Liquidate current liabilities using long-term assets.

B. Quick (Acid test) Ratio: This is a fairly stringent measure of liquidity. It is based on current assets which are highly liquid- excluding inventories and prepaid expenses. *Inventories are deemed to be the least liquid components of current assets and prepaid expenses which are not available to pay off current liabilities.* Quick assets include: cash, marketable securities, and receivables (such as notes receivable and account receivable).

The low liquid nature of inventories is resulted from:

- Many types of inventories could not be sold easily because they are partially completed items, obsolete items, special purpose items, among others, and

- The items may be sold on credit which means they become an account receivable before being converted into cash.

$$\text{Quick Ratio} = \frac{\text{Current assets} - \text{Inventory} - \text{Prepaid expenses}}{\text{Current liabilities}}$$

A quick ratio of one or greater is recommended, but as with the current ratio, an acceptable value depend on the industry group and the established standards.

The quick ratio of GLOBAL Company is:

$$\text{Quick Ratio}_{2008} = \frac{347,000 - 210,000 - 15,000}{99,000} = \frac{122,000}{99,000} = 1.23 \text{ times}$$

Therefore, Global company can cover its current liability 1.23 times without liquidating inventories and prepaid expense.

C. Cash Ratio A very short term one for which creditor may be interested in Cash ratio is defined as:

$$\text{Cash Ratio} = \frac{\text{Cash}}{\text{Current liabilities}}$$

Since cash is the most liquid asset, a financial analyst may examine cash ratio and its equivalents to current liabilities. Marketable securities are equivalent to cash; therefore, they may be included in the computation of cash ratio.

The cash ratio of GLOBAL Company is 0.44 times which is computed as follows:

$$\text{Cash ratio}_{2008} = \frac{44,000}{99,000} = 0.44 \text{ times}$$

The company can be considered as the one with small amount of cash relative to its current obligations. It can only cover 44 percent of its short-term liabilities without liquidating other current assets. The position of comparison as to the strength and weakness of the company depends upon the standards to be employed.



Self-check questions 2.2

1. What are the three type liquidity ratios?
2. Inventories are less liquid current assets and are excluded from quick ratio. Explain

2. Activity Ratios

These ratios are also called efficiency ratio, or asset management ratio, or asset utilization ratios. Utilization ratios compare the level of sales or cost of goods sold with the level of investment in various assets. *These ratios are used to measure the speed with which various assets are converted into sales revenue and assets are effectively managed.* What these ratios are intended to describe is *how efficiently, or intensively, a firm uses its assets to generate sales.* The most commonly used activity ratios inventory turnover, account receivable turnover, fixed asset turnover, and total asset turnover ratios.

A) Inventory Turnover Ratio: It measures *how quickly inventory of a firm is sold.* It indicates the efficiency of the firm in managing and selling inventories. It is calculated by dividing the cost of the goods sold by the inventory amounts. *There are arguments as to which is to be uses, either sales or cost of goods sold in the numerator.* The fact that sales are stated at market prices, while inventories are at cost, the calculated inventory turnover overstates the true turnover in cases when sales are used. Hence, the best argument is cost of goods sold is to be used unless the situation is not conducive to get the cost of goods sold in which case sale is to be implemented. The logic of comparison is ‘**apple to apple**’ so cost is to be compared with costs not market prices.

There is also debate on the use of inventory. For some average inventory is employed that is the average of ending and beginning inventories, while for the other’s ending inventory is to be taken in the denominator. For the sake simplicity, inventory at the end of a certain period is assumed.

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

The inventory turnover ratio of GLOBAL Company is

$$\text{Inventory turnover 2008} = \frac{\text{Br. 539,000}}{\text{Br. 210,000}} = 2.55 \text{ Times}$$

This implies that GLOBAL Company replaces its inventories about 2.55 times in a year. The evaluation of performance with regard to inventory turnover of a certain company depends upon the standards employed. Generally, *higher inventory turnover is considered to be good because it means that storage costs are low, but if it is too high the firm may be risking inventory outages and the loss of customers.*

B) Average Age of Inventory (AAI): It is the reciprocal of the inventory turnover. It is also called as days of inventory holdings because it is the length of time inventory is held by the firm. When the number of days in a year (say, 360) is divided by inventory turnover, days of inventory holdings derived.

$$\begin{aligned}\text{Average Age of Inventory} &= \frac{\text{Inventory}}{\text{Cost of goods Sold}} \times 360 \text{ days} \\ \text{Average Age of Inventory} &= \frac{360 \text{ days}}{\text{Inventory turnover}}\end{aligned}$$

The average age of inventories in GLOBAL Company's store is

$$\text{Average Age of Inventory}_{2008} = \frac{210,000}{539,000} \times 360 = \frac{360}{2.55} = 141 \text{ days}$$

GLOBAL Company held its inventory on average for 141 days. Whether **GLOBAL** is efficient in inventory management is a matter of comparison which is actually based on the nature of industry and other standards employed. The general principle, however, is the lower is the better without leaving the firm out of stock.

C) Account Receivable Turnover (ARTO): Firms grant credit at least to increase their sales level. It is, therefore, important to know how well the firm manages its receivables. The account receivable turnover ratio provides information about the management of account receivables. It indicates how many times account receivable is converted into cash during the year. The higher the value of account receivable turnover, the more efficient is the expected credit management. It is calculated by:

$$\text{Account Receivable Turnover} = \frac{\text{Credit sales}}{\text{Account receivable}}$$

For GLOBAL Company, the 2008 account receivable turnover ratio is (assuming all sales are credit sales):

$$\text{Account Receivable Turnover ratio} = \frac{830,000}{78,000} = 10.64 \text{ times}$$

Whether 10.64 is a good account is receivable account ratio or not is difficult to know at this point. But it is possible to say, higher is generally better. Still too higher figure of account receivable turnover ratio might indicate that the firm is delaying credit to creditworthy customers thereby losing sales. If the ratio is too low, it would suggest that the firm is having difficulty of collecting on its sales. This is particularly true if there finds that account receivables are increasing faster than sales over a prolonged period.

D) Average Collection Period (ACP) The average collection period indicates how many days it takes for a firm to collect its credit sales. It can also be defined as the average number of days for which account receivables remain outstanding. It measures the quality of account receivables since it indicates the speed of collections.

$$\text{Average Collection Period} = \frac{\text{Account Receivable}}{\text{Annual Sales}/360}$$

Note that the denominator is simply credit sales per day with the assumption of 360 days in a year. In 2008, it took GLOBAL Company an average of 33.83 days to collect credit sales.

$$\text{Average collection period} = \frac{78,0000}{830,000/360} = 33.83 \text{ days}$$

Note that this ratio provides the same information as the account receivable turnover ratio. When depicted algebraically, the relationship between average collection period and account receivable turnover ratio is show as follows.

$$\begin{aligned} \text{Account Receivable Turnover Ratio} &= 360 \div \text{Average Collection Period} \\ \text{or alternatively;} \\ \text{Average Collection period} &= 360 \div \text{Account Receivable Turnover ratio} \end{aligned}$$

Since the average collection period is the inverse of the account receivable turnover ratio, it should be apparent that the inverse criteria apply to judging this ratio. In other words, lower is usually better. The shorter the average collection period, the better is the quality of debtors, since a shorter collection period implies the prompt payment by debtors. It is advisable to compare average collection period against the firm's credit terms and policy to judge its credit and collection efficiency. For example, if the credit period granted by the firm is 30 days and its average collection period is 45 days, the comparison reveals that the firm's receivables are outstanding for a longer period than warranted by the credit period of 30 days.

An excessively longer collection period implies a very liberal and inefficient credit and collection performance. This certainly delays the collection of cash and hurts the firm's liquidity. The chances of bad debts are also increased. On the other hand, too low collection period is not necessarily favorable. Rather it may indicate a very restrictive credit and collection policy. A very restrictive credit and collection policy may avoid bad debt losses; but it also severely curtails sales.

The average collection period may help analysts in two respects:

- 1) In determining the collectability of account receivables and thus the efficiency of collection efforts, and
- 2) In ascertaining the firm's comparative strength and advantage relative to its credit policy and performance *vis -a- vis* the competitors' credit policy and performance.

E) Fixed Assets Turnover (FATO): FATO is used to measure the efficiency of a firm to utilize its investment in fixed assets. It is also described as the birr amount of sales that are generated by each birr invested in fixed assets. It is given by:

$$\text{Fixed Asset Turnover} = \frac{\text{Sales}}{\text{Net Fixed Assets}}$$

The fixed assets turnover ratio for GLOBAL Company, in 2008 is

$$\text{Fixed assets turnover} = \frac{830,000}{525,000} = 1.58 \text{ times}$$

So, GLOBAL Company generated Br.1.58 for each birr invested in fixed assets.

The firm's ability to produce a large volume of sales for a given net fixed assets is the most important aspect of its operating performance. Hence, the larger is the better for fixed assets turnover ratio.

F) Total Assets Turnover (TATO) like other ratios discussed in this section, TATO describes how efficiently a firm is using its assets to generate sales. It is the ability of a firm to generate sales from all financial resources committed to total assets. Higher total asset turnover ratio is better. It is computed as:

$$\text{Total Asset Turnover} = \frac{\text{Sales}}{\text{Total Assets}}$$

In 2008, GLOBAL Company generated 89.5 cents in sales for each birr invested in total assets.

$$\text{Total Asset Turnover} = \frac{830,000}{927,000} = 0.895 \text{ times}$$

You can interpret the total assets turnover ratio as *higher is better*. However, you should be aware that *some industries will naturally have lower turnover ratios than others*. For example, a consulting firm will have a very low investment in fixed assets, and therefore a high asset turnover ratio with other factors remaining constant. On the other hand, an electric service enterprise will have a large investment in fixed assets and probably a low asset turnover ratio. *This does not, necessarily, mean that the electric enterprise is more poorly managed than the consulting firm*. Rather, each is simply responding to the demands of their industry groups.

3. Leverage Ratios

Leverage from finance perspective refers to *multiplication of changes in profitability measures*. It is used to measure the extent to which non-owner supplied funds have been used to finance a firm's assets as compared with the funds provided by owners. It also describes the degree to which the firm uses debt in its capital structure. The amount of leverage depends on the amount of debt that a firm uses to finance its operations, so a firm which uses a lot of debt is said to be *highly leveraged*. Leverage ratios provide important information for creditors and investors. Creditors might be concerned that a firm has *too much debt and will therefore have difficulty in repaying loans*. Investors might be concerned because a larger amount of debt can *lead to a larger amount of volatility in the firm's earnings*. The most commonly stated leverage ratio types are: Debt Ratio, Debt-Equity Ratio, Time Interest Earned Ratio, Fixed Payment Coverage ratio. The first two ratios are also considered as component ratio while the last two are coverage ratios. They are component because these ratios

measure the proportion of the financing sources and they are all about the major balance sheet elements. The others are coverage ratios due to their emphasis on the ability of the firm to cover its interest and other fixed charges.

A) Total Debt Ratio (DR): It measures the proportion of total assets financed by the firm's creditors. The higher this ratio the greater is the amount of other people's money being used in an attempt to generate profit and the higher the financial costs and restrictions from creditors.

$$\text{Debt ratio} = \frac{\text{Total Debt}}{\text{Total Assets}} = \frac{\text{Total Assets} - \text{Total Equity}}{\text{Total Assets}} = 1 - \frac{\text{Total Equity}}{\text{Total Assets}}$$

The debt ratio of GLOBAL Company in 2008 shows that liabilities make up about 32.25 percent of their capital structure.

$$\text{Debt Ratio} = \frac{299,000}{927,000} = 32.25 \%$$

Creditors have supplied for GLOBAL Company about 32 cents of every birr in assets. The evaluation of the GLOBAL's situation with respect to its debt management requires comparison with industry groups and other developed standards. The generally perception, however, is too much higher ratio shows that more of the firm's assets are provided by creditors relative to owners. Under this situation the firm may face some difficulties in generating additional finance; further creditors may require higher return; and/or higher risk for existing creditors as there might not be sufficient margin of safety among others.

B) Debt Equity Ratio (DER) The Debt-equity ratio indicates the relationship between the total debt's funds provided by creditor and those provided by the owners of the firm. This ratio reflects the relative claim of creditors and shareholders against the assets of the firm. It is computed as:

$$\text{Debt Equity Ratio} = \frac{\text{Total liability}}{\text{Total equity}} = \frac{\text{Total Assets} - \text{Total equity}}{\text{Total equity}} = \frac{\text{Total assets} - 1}{\text{Total equity}}$$

The proportion of debt and equity in financing the assets of GLOBAL Company is

$$\text{Debt- Equity Ratio}_{2008} = \frac{299,000}{628,000} = 47.61 \%$$

Creditors of GLOBAL Company provided about 48 cents in financing total assets for every birr contributed by the owners.

The same implication as that of debt ratio can be stated for too higher or too lower reported debt equity ratio.

Coverage Ratios Describe the quantity of funds available to cover certain expenses. Unlike the component ratios, higher ratios are desirable in coverage ratios. Too high ratios, however, may indicate that the firm is underutilizing its debt capacity and therefore not maximizing shareholder wealth. The two most used coverage ratios are discussed in the following sections.

C) The Time Interest Earned Ratio (TIER) The interest coverage or the time interest earned ratio measures the ability of the firm to pay its interest obligations by comparing earnings before interest and taxes (EBIT) to interest expense.

$$\text{Time Interest Earned Ratio} = \frac{\text{EBIT}}{\text{Interest expense}}$$

The time interest earned ratio for GLOBAL Company for 2008 is computed as:

$$\text{Time Interest Earned Ratio} = \frac{101,000}{20,000} = 5.05 \text{ times}$$

GLOBAL Company can cover its interest 5.05 times using funds that are originally available for the payments of interest expenses.

A high ratio indicates that the firm has sufficient margin of safety to cover its interest charges and the firm's earning could decline without jeopardizing the firm's ability to make interest payments. A very low time interest earned ratio may suggest that creditors are more at risk in receiving interests due; failure to meet interest can bring legal action by creditors possibly resulting in bankruptcy; and the firm may face difficulty in raising additional funds through debt.

D) Fixed Charge Coverage Ratio (FCCR): This ratio reflects the total amount of earning available to meet all fixed payment obligations. Interest, principal payments (PP), lease payments, and preferred stock dividends (PD) are financing related fixed charges. It is computed as:

$$\text{Fixed - Charge Coverage Ratio} = \frac{\text{Earnings Before Interest and Tax} + \text{Lease Payment}}{\text{Interest} + \text{Lease payment} + \text{PD} + \text{PP}}$$

Since principal payments and preference dividends are the after-tax components, they are adjusted to the before tax amount with $1 / (1-T)$ as shown in the computational step above; where ‘T’ represents the tax rate.

4. Profitability Ratios

Profitability ratios are used to measure the *operating efficiency of a company*. In other words, they are used to evaluate the overall management effectiveness and efficiency in generating profit on sales, total assets and owners’ equity. Besides management of a company, creditors, investors and owners are interested in the profitability of the firm. Creditors want to get interest and repayment of principals regularly. Owners want to get a required rate of return on their investment. Profitability ratios are the easiest of all of the ratios to analyze. Without exception, high ratios are preferred. However, the definition of high depends on the industry in which the firm operates. For example, a 3 percent of profit margin may be quite high for a grocery while it would be terrible in software businesses. Profitability ratio includes: gross profit margin, operating profit margin, net profit margin, return on investment, and return on equity. Each of these ratios is described here under.

A) Gross Profit Margin: Measures the gross profit relative to sales. It indicates the amount of fund available to pay the firms expenses other than its cost of sales and indicates both the *efficiency of the firm’s operation and pricing policies of the firm*. It is calculated by;

$$\text{Gross Profit Margin} = \frac{\text{Gross profit}}{\text{Sales}}$$

$$\text{Gross profit margin} = \frac{291,000}{830,000} = 35\%$$

GLOBAL Company maintained 35 cents remained from each birr sales after deducting each cost of goods sold from net sale.

The gross profit margin reflects the efficiency with which management produces each unit of products. It indicates the average spread between the cost of goods sold and the sales margin. If you subtract the gross profit margin from 100 per cent, you obtain the ratio of the cost of the goods sold. A high profit margin relative to the industry average implies that the firm is able to produce or purchase at relatively lower cost.

B) Operating Profit Margin (OPM) Moving down the income statements, you can calculate the profit that remains after the firm has paid all its non- financial expenses. This profit is the operating profit. The operating profit ratio indicates how much is left over after the operating expenses. It serves as an overall measure of operating effectiveness. It is calculated as:

$$\text{Operating Profit Margin} = \frac{\text{Net Operating profit}}{\text{Sales}}$$

GLOBAL Company has an operating profit margin of about 12.2 per cent.

$$\text{Operating Profit Margin} = \frac{101,000}{830,000} = 12.2 \%$$

GLOBAL Company generated about 12 cents in operating profit per birr of net sales.

C) Net Profit Margin (NPM) Net profit margin measures the percentage of each sales birr remaining after deducting all cost and expenses. The net profit margin relates net income to sales. The net income is profit after all expenses; the net profit margin tells you the percentage of sales that remains for the shareholders of the firm.

$$\text{Net Profit Margin} = \frac{\text{Net Profit (income)}}{\text{Sales}}$$

In 2008, GLOBAL Company has net profit margin of about 8 percent.

$$\text{Net Profit Margin} = \frac{64,000}{830,000} = 8\%$$

A firm with a high net profit margin ratio would be on the advantageous *position to survive in the face of falling selling prices, rising costs of production and/or decline in demand for the product.*

D) Return on Investment (ROI) The term investment may refer to total assets or net assets. It represents pool of funds supplied by shareholders and lenders. This ratio is particularly useful since it reflects the total earning produced with the use of the total assets of the firm. It measures the overall effectiveness of management in generating profit with its available assets. It is computed as:

$$\text{Return on Total Assets} = \frac{\text{Net Income}}{\text{Total Assets}}$$

For GLOBAL Company the return on total assets is:

$$\text{Return on Total Assets} = \frac{64,000}{927,000} = 6.9\%$$

This implies that GLOBAL generates about 7 cents for every birr invested in assets.

E) Return on Equity (ROE) While total asset represents the total investment in the firm, the owners' investment, usually common stock and retained earnings, represents only a portion of the total amount. The other portion is to go for debt. Return on equity is to measure the return earned on the owners' investment. It is computed as:

$$\text{Return on Equity} = \frac{\text{Net Income}}{\text{Total Equity}}$$

The 2008 return on equity of GLOBAL Company is:

$$\text{Return on Equity} = \frac{64,000}{628,000} = 10\%$$

GLOBAL Company generates about 10 cents for every birr in shareholders equity. The comparison is left for the established standards and the industry averages. A substantially higher ROE may indicate that a firm is riskier due to high financial leverage. A very low ROE may also indicate a kind of conservative financing policy.

5. Valuation Ratios / Market/Book Value Ratios

These ratios also called Shareholder ratios. The ratios in the previous sections deal with the performance and financial condition of the firm. Those ratios provide information for managers and for creditors with little emphasis on the shareholders valuation aspects. The ratios in this section have a look at on the interest of the owners. These ratios translate the overall results of operations so that they can be compared in terms of a share of stock.

A) Earning Per Share (EPS) EPS is the amount of income earned during a period per share of common stock. It is computed as:

$$\text{Earnings per Share} = \frac{\text{Net income available for common shareholders}}{\text{Number of common shares outstanding}}$$

Assume that GLOBAL Company has a total number of common shares outstanding of 10,000. The earning per share is therefore computed as:

$$\text{EPS for 2008} = \frac{64,000}{10,000} = 6.4$$

Earnings per share does not show how much is retained and how much is distributed as dividend.

B) Dividends per Share (DPS) It is the birr amount of cash dividends paid during a period as per share of common stock. The net profits after taxes and preference dividends belong to common shareholders. But the income which they really receive is the amount of earning distributed as cash dividends. Therefore, a large number of present and potential investors may be interested in divided per share, rather than earning per share. It is computed as:

$$\text{DPS} = \frac{\text{Earnings paid to common shareholders (dividends)}}{\text{Number of ordinary shares outstanding}}$$

DPS for GLOBAL Company is $\text{DPS} = \frac{15,000}{10,000} = 1.5$

C) Dividend Payout Ratio The dividend payout ratio (or simply payout ratio) is DPS divided by the EPS. It calls also be by dividing cash dividend paid to earning for a period.

$$\text{Dividend Payout Ratio} = \frac{\text{Dividends paid common shareholders}}{\text{Earnings available for common share}} \text{ or } = \frac{\text{DPS}}{\text{EPS}}$$

Dividend payout ratio shows the percentage of earnings paid to shareholders.

For GLOBAL Company the dividend payout ratio for the year 2008 is

$$\text{Dividend Payout Ratio} = \frac{1.5}{6.4} = 23.44 \%$$

The interpretation is that GLOBAL Company paid **23.44%** of its earnings in dividends. Whether GLOBAL paid higher or lower percentage is not to be identified at this point unless standards or

industry average is revealed. In general, a very high percentage may imply that the firm is at lower growth opportunities particularly for firms which pay more than the industry average.

D) Price-Earnings Ratio: The price earnings ratio is widely used by the security analysts to value the firm's performance as expected by investors. It indicates investors' judgment or expectations about the firm's performance. It indicates the degree of confidence that investors have in the firm's future performance. The higher the price earnings ratio, the greater is the investors' confidence in the firm's future.

E) Market Values to Book Values Ratio This is the ratio of share price to book values per share.

$$\text{Price /Earnings Ratio} = \frac{\text{Market values per share}}{\text{Earnings per share}}$$

Note that book value per share is book value of shareholders' equity divided by the number of shares outstanding.

2.4.1.3 User of Ratio Analysis

Ratios of financial statements are used based on the interest of different parties. Shareholders, creditors and the management bodies have their own interest on financial performance of a firm so that these bodies develop different views towards ratio analyses.

A) Shareholders Both present and prospective shareholders are interested in the firm's current and future level of risk and return. Future earnings and stability of earnings over a period; and covariance with earnings of other companies are the interest of the shareholders. The major emphasis for the shareholders, therefore, is the profitability aspect of ratio analysis.

B) Creditors The firm's creditor may be trade creditors and/or bondholders. The trade creditors are primarily interested in the short-term liquidity of the company so that liquidity analysis is the basic concern of these parties. The bondholders or long-term creditors give emphasis on the ability of the firm to make interest and principal payments. These parties are interested on the capital structure or leverage ratios, major sources of finance and profitability over time to be assured for the sustenance of the business.

C) Management The management of a firm is concerned with aspects of the financial analysis that consider the suppliers of the capital. These bodies also use ratio analysis for the purpose of internal control with particular emphasis on the profitability of investment in various assets of the company and the efficiency of asset management.

2.4.1.4 Advantage of Ratio Analysis

- Ratios are easy to compute
- Ratios provide standards of comparison at a point in time and allow comparison to be made with industry average
- Ratios can be used to analysis corporation financial time series in order to discover trends, shifts in the trends and data outlays.
- Ratios are useful to indicate problem areas of a firm
- When combined with other tools, ratio analyses make an important contribution to the tasks of evaluating a financial performance of the firm.

2.4.1.5 Limitations and Cautions for Use of Ratio Analysis

Ratios have the following Limitations:

- A single ratio provides little information
- Ratios generally do not identify the cause of the firm's difficulties. They seldom provide answer for questions they raise
- Ratios can easily be misinterpreted. For example, a decrease in the value of ratio does not necessarily mean the something undesirable has happened.
- Very few standards exist that can be used to judge the adequacy of a ratio or set of ratios.

The following 'cautions' are fundamental while using ratios

- ❖ Financial statements being compared should be dated at the same point in time
- ❖ It is preferable to use audited financial statements for ratio analysis
- ❖ Financial data being compared should have been developed in the same way. The use of different accounting methods, for example the different inventory valuation methods and depreciation methods should be considered while calculating ratios for different firms of

probably the same nature. The use of different methods will distort the result of ratios for comparison.

- ❖ When ratios of one firm are compared with those of other's or with the ratios of the firm itself over time, distorted interpretation of the analysis may be resulted due to inflation. Inflation at times no adjustments are made, it tends older firms to appear more efficient and profitable than the newer ones.



Self-check questions 2.3

1. What is analysis of financial statements?
2. Analysis which is done by the persons within the organization is called
3. Name the financial statements
4. List the users of financial statements.

2.4.2 Common Size Statement Analysis

Common size Analysis – expresses individual financial statement accounts as a percentage of a base amount. A common size status expresses each item in the balance sheet as a percentage of total assets and each item of the income statement as a percentage of total sales. When items in financial statements are expressed as percentages of total assets and total sales, these statements are called common size statements.

In common-size statements analysis, all the balance sheet items are divided by total assets and all income statements items are divided by the total sales. Common size statements can also be used to compare firms of different sizes. Example given below is the balance sheets of AWASH Corporation for the year ended December 31, 2006 and 2007

AWASH Corporation

Balance sheet

December 31, 2006 and 2007 (in millions)

Assets

2006

2007

Current assets:

Cash	Br. 84	Br. 98
Account receivable	165	188
Inventories	<u>393</u>	<u>422</u>
Subtotal	Br.642	Br. 708

Fixed assets:

Net plant and Equipment	<u>Br.2, 731</u>	<u>Br.2, 880</u>
Total assets	<u>Br. 3,373</u>	<u>Br.3,588</u>

Liabilities and owner's Equity

Current liabilities

Account payables	Br. 312	Br.344
Note payable	<u>231</u>	<u>196</u>
Subtotal	Br.543	Br.540

Long term debts	<u>531</u>	<u>457</u>
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Total liabilities	Br. <u>1,074</u>	<u>Br.997</u>
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Owner's equity

Common stock and Paid in Capital	Br.500	Br. 550
Retained earnings	<u>1,799</u>	<u>2,041</u>

Total Owner's Equity	<u>Br.2,299</u>	<u>Br.2,591</u>
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Total liabilities and owner's equity	<u>Br. 3,373</u>	<u>Br.3,588</u>
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AWASH Corporation

Common Size Balance Sheet

December 31, 2006 and 2007 (in millions)

Assets	2006	2007	Change
Current assets			
Cash	2.5%	2.7%	+0.2%
Account receivable	4.9	5.2	+0.3
Inventories	<u>11.7</u>	<u>11.8</u>	<u>+0.1</u>

Subtotal	19.1	19.7	+0.6
Fixed assets			
Net plant and Equipment	<u>80.9</u>	<u>80.3</u>	<u>-0.6</u>
Total assets	<u>100%</u>	<u>100%</u>	<u>0%</u>
Liabilities and owner's Equity			
Current liabilities			
Account payables	9.2 %	9.6 %	+0.4%
Note payable	<u>6.8</u>	<u>5.5</u>	<u>- 1.3</u>
Sub Total	16.1 %	5.1%	-1.0%
Long term debts	<u>15.7</u>	<u>12.7</u>	<u>- 3.0</u>
Total liabilities	<u>31.8%</u>	<u>27.8%</u>	<u>-4.0%</u>
Owner's equity			
Common stock and Paid in Capital	14.8%	15.3%	+0.5%
Retained earnings	<u>53.4%</u>	<u>56.9</u>	<u>+3.5</u>
Total Owner's Equity	<u>68.2%</u>	<u>72.2%</u>	<u>+5.0</u>
Total liabilities and owner's equity	<u>100%</u>	<u>100%</u>	<u>0%</u>

N.B + Shows increase in percentage from 2006 to 2007

 - Shows decrease in percentage from 2006 to 2007

AWASH has the following income statement from which the common size income statement is presented.

AWASH Corporation
Income Statement and Common Size Income Statement
For the year ended December 31, 2007 (In millions)

Sale	Br.2311	100%
Cost of Goods sold	<u>1344</u>	<u>58.2</u>
Gross Margin	967	41.2%
Depreciation Expense	<u>276</u>	<u>11.9</u>
Income before interest and tax	691	29.9%
Interest expense	<u>141</u>	<u>6.1</u>

Taxable income	550	23.8%
Tax (34%)	<u>187</u>	<u>8.1</u>
Net income	<u>Br.363</u>	<u>15.7%</u>

2.4.3 Index Analysis

Index Analysis – expresses items in the financial statements as an index relative to the base year. All items in the base year are assumed to be 100%. Usually, this analysis is most appropriate for income statement items. Analysis of percentages for financial statements where all financial statement figures for the base year are equated to 100 percent and subsequent year financial statements items are expressed as a percentage of the base year.

Illustration Balance sheet of AWASH Corporation for the years 2003, 2004 and 2005 are given below.

AWASH Corporation			
Balance sheet			
December 31, 2003, 2004 and 2005 ('000 Birr)			
Assets	2003	2004	2005
Cash	2,507	11,310	19,648
Account receivable	70,360	85,147	11,815
Inventory	77,380	91,378	118,563
Other current asset	<u>6,316</u>	<u>6,082</u>	<u>5,891</u>
Total current assets	156,563	193,917	262,517
Net fixed assets	79,187	94,652	11,5461
Other long-term assets	<u>4,695</u>	<u>5,899</u>	<u>5,491</u>
Total assets	<u>240,445</u>	<u>294,468</u>	<u>383,469</u>
Liabilities and Shareholders' Equity			
Accounts payable	35,661	37,460	62,725
Notes Payable	20,501	14,680	17,298
Other current liabilities	11,054	8,132	15,741
Long term liabilities	<u>888</u>	<u>1,276</u>	<u>4,005</u>
Total Liabilities	<u>68,104</u>	61,548	99,769

Common stock	12,650	20,750	24,150
Additional paid in capital	37,950	70,950	87,730
Retained Earning	<u>121,741</u>	<u>141,820</u>	<u>171,820</u>
Total shareholders' Equity	<u>172,341</u>	<u>232,920</u>	<u>233,700</u>
Total liabilities and shareholders' equity	<u>240,445</u>	<u>294,468</u>	<u>383,469</u>

Required: Prepare an index analysis taking items in 2003 as a base year

AWASH Corporation

Indexed Balance sheet

December 31, 2003, 2004 and 2005 ('000)

Assets	2003	2004	2005
Cash	1.0	4.511368	7.837256
Account receivable	1.0	1.210162	0.167922
Inventory	1.0	1.180899	1.532218
Other current asset	<u>1.0</u>	<u>0.962951</u>	<u>0.932711</u>
Total current assets	<u>1.0</u>	<u>1.238588</u>	<u>1.67675</u>
Net fixed assets	1.0	1.195297	1.45808
Other long-term assets	<u>1.0</u>	1.256443	1.169542
Total assets	<u>1.0</u>	<u>1.224679</u>	<u>1.59483</u>
Liabilities and shareholders' Equity			
Accounts payable	1.0	1.050447	1.758924
Notes Payable	1.0	0.716063	0.843764
Other current liabilities	1.0	0.735661	1.424009
Long term liabilities	<u>1.0</u>	<u>1.436937</u>	<u>4.510135</u>
Total Liabilities	<u>1.0</u>	<u>0.903735</u>	<u>1.464951</u>
Common stock	1.0	1.640316	1.909091
Additional paid in capital	1.0	1.869565	2.311726
Retained Earning	<u>1.0</u>	<u>1.164932</u>	<u>1.411357</u>
Total shareholders' Equity	<u>1.0</u>	<u>1.351507</u>	<u>1.356033</u>

Total liabilities and shareholders' equity	<u>1.0</u>	<u>1.224679</u>	<u>1.59483</u>
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2.1.4 Trend Analysis

Trend analysis is used to analyze financial statements for a number of years and enable firm to project the future. It is analysis for elements of financial statements to know whether there is a tendency of change (increase or decrease) numerically for the elements of financial statements over years.

Illustration

Years	<u>2001</u>	<u>2002</u>	<u>2003</u>	<u>2004</u>
Current ratio of a firm	3.2 times	2.1 times	2.0times	1.88times

The current ratio indicates the declining tendency of liquidity position of the firm over the four years period. You can find the trend in ratios of a certain firm by taking ratios over number of periods.

2.5 Financial Planning

The management of every firm always tries to maximize efficiency and through the same process the firm goes for the maximization of wealth of the shareholders. One of the systematic approaches to bring about the intended efficiency is to have a well-designed financial planning and budgeting. In the process of financial planning, the management tries to identify the required financial gap and assess the need for external financing.

Financial planning indicates a firm's growth, performance, investment and fund requirement during a period of time. It involves the preparation of projected financial statements. It is the process of evaluating the impact of alternative investing and financing decisions; attempts to make optimal decisions, projects the consequence of these decisions for the firm in the form of financial plan and then comparing future performance against the plan.

2.5.1 Planning Process

Planning is the design of future state of an entity and effective ways to achieve stated objectives. The planning process of an enterprise involves many steps and phases. The major steps can be summarized as follows.

A) Set up Objective Objectives explain the desired state or position of an enterprise in the future. This represents the purpose for which the firm is organized. The firm must have clearly stated strategic, operational and financial objectives.

B) Make Assumptions It explains the expected conditions on which the plan is based. When one makes an objective, there exists some assumptions regarding the economic condition, political situations, and other circumstances in which the firm is working.

C) Determine Goals: Goals are operational specifications of the broad objectives with time and quantity dimensions. Goals are quantified targets to be attained within a specific period. Example The objective of a nation may be to have an economic growth for which fix a goal that is for example 5.5 percent of rate of growth in the next year.

D) Determine Strategies It lays down the foundation or the activities to be followed in attaining the objectives and goals. It specifies the way to achieve the objectives and goals.

Example objective of a firm may be to maximize return for a period of time by increasing sales. The goal may be increase sales by 10 percent at the end of the year. The firm's strategies may include reduction of selling price and/ or liberalizing credit policy among others.

E) Formulate Budget An enterprise must also have idea about the different cash inflows and outflows, the requirement of additional fund, if the plan is to be implemented. Budget is the expression of the company's plan in terms of financial terms. It is a plan explicitly stating the goals in terms of time and expected financial results for each major segments of the firm. Example cash budget, sales budget, capital budget etc.

Financial budget It is the financial expression of operating budget. It expresses the cash flows, financial positions, and operating results. The most important types of financial budgets are: Cash budget, pro-forma or projected financial statements and Capital budget. The section herein is all about pro-forma or projected financial statements.

2.5.2 Financial Statement Forecasting

It is an integral part of planning. It uses past data to estimate future financial requirements. It is future estimate of financial requirements based on some scientific and systematic approaches. It is an important activity for a wide variety of business people. Nearly all of the decisions made by financial managers are made on the basis of forecasts of one kind or another (Mayes and Shank, 2004). There

are different methods of forecasting. The percent of sales method, one of the forecasting techniques, which will be used in this sub section. The percent of sales method is the simplest method of forecasting income statements and balance sheets. The method can be used with relatively little data available.

The fundamental assumption of the percent of sales method is that many (but not all) income statement and balance sheet items maintain a constant relationship to the sales level. Moreover, the method assumes that the forecast level of sales is already known.

Sales forecast: the sales forecast generally starts with a review of sales during the past five to ten years. The development of sales forecast may consider:

- Product divisions. If firms have product divisions, sales growth is seldom the same for each division. Hence, to begin the forecasting process, divisional projections are to be made on the basis of historical growth, and then divisional forecasts are combined to produce an approximation of the firms overall sales forecast.
- The level of economic activity in each of the divisions/ branches marketing areas is forecasted. For example, the change in population growth
- The firm's probable market share in each of the divisions/ branch's distribution territories. Consideration given is to the firm's production and distribution capacity, the competitors' capacity among others.
- The exchange rate fluctuations for export-oriented firms
- The planners have to consider the effect of inflation on prices
- Advertising campaigns, promotional discounts, credit terms and the like also affect sales

Once sales have been forecasted, the forecast for future balance sheet and income statements can be undertaken.

1. Forecasting an Income Statement The level of detail that you have in an income statement will affect how many items will fluctuate with sales. The general procedure is to proceed through line by line for each component of income statement by asking the question, ***is it likely this item will change directly with sales?*** If the answer is 'yes', then calculate the percentage of sales and multiply the result by the sales forecast for the next period. If the answer is otherwise, you will take one of two actions: leave the item unchanged, or use other information to change the item.

To illustrate take the income statement of GLOBAL company for the year 2008 and prepare the pro-forma income statement for 2009. Assume that sales increases by 20 percent. The pro forma income statement is shown below with explanations.

GLOBAL Company
Pro-forma Income Statement (in' Birr)
For the year ended December 31, 2009

Sales	Br.996, 000
Cost of Goods Sold	<u>(647,400)</u>
Gross Profit on Sales	Br.348, 600
Operating Expenses:	
Marketing Expense	Br. 91,000
General and Administrative Expense	71,000
Depreciation	<u>28,000</u>
Total Operating Expenses	<u>(Br.190, 000)</u>
Operating Income (EBIT)	Br.158, 600
Interest Expense	<u>(20,000)</u>
Earnings Before tax	Br. 138,600
Income Tax (21%)	<u>(29,106)</u>
Earnings after Tax	Br. 109,494
Dividends Paid (23%)	<u>(25,184)</u>
Change in Retained Earnings	<u>Br.84,310</u>

Forecasted sales is calculated as: $830,000 + (830,000 \times 0.2)$

$$= 830,000 + 166,000 = \text{Br. } 996,000$$

Cost of goods sold has a direct relationship to sales.

$$\text{The cost of goods sold as percentage of sales} = \frac{539,000}{830,000} = 65\%$$

$$\text{Forecasted cost of goods sold} = 0.65 \times 996,000 = \text{Br. } 647,400$$

For GLOBAL Company both marketing and general administrative expenses stay constant. These expenses are conglomerates of different expenses for which some may change while the others stay constant. Depreciation expenses are not directly related to sales. The change in depreciation expense

on the pro forma income statement depends on the depreciation rate not based on sales. For GLOBAL Company there is no change in the depreciation rate.

Interest expense is not directly related to sales. It is a function of the amount and maturity structure of debt in the firm's capital structure. GLOBAL Company assumes no changes in the interest expense for the year 2009 because the same capital structure is expected for the year.

Tax depends directly on the firm's taxable income though indirectly related to sales level.

Dividend is not directly related to sales but is dependent on the retention ratio and the dividend policy of the firm.

2. Forecasting a Balance Sheet Balance sheet can be forecasted in exactly the same way as the income statement. The main difference is the common size information is not used in the case of forecasting balance sheet because the common size balance sheet is based on total assets not based on total sales. Under the percentage of sales method, the components of balance sheet are to be determined based on sales depending on their direct relationship to sales.

Like it is done in the income statement, you will move line by line for each balance sheet items to determine which items will vary with sales.

Example: take the balance sheet of GLOBAL Company for the year 2008 to prepare the pro-forma balance sheet for the year 2009. The sales forecast is reported in the pro-forma income statement. The pro-forma balance sheet and explanations for the respective components is as follows.

GLOBAL Company	
Pro-forma Balance Sheets (in Birr)	
December 31, 2009	
Assets	
Current assets	
Cash	Br. 49,800
Account Receivables	89,640
Inventories	249,000

Prepaid expenses	<u>15,000</u>
Total current assets	<u>Br. 403,440</u>
Fixed Assets:	
Gross plant and Equipment	Br. 838,000
Accumulated Depreciation	(411,000)
Net Plant and Equipment	427,000
Land	<u>70,000</u>
Total fixed assets	Br.497,000
Patent	<u>55,000</u>
Total Assets	<u>Br. 955,440*</u>
Liabilities and Shareholders' Equity	
Current Liabilities:	
Account Payable	Br. 89,640
Income tax payable	29,106
Accrued wages and salaries	4,980
Interest payable	<u>2,000</u>
Total Current Liabilities	Br.125,726
Long-term notes payable	<u>200,000</u>
Total Liabilities	<u>Br.325,726</u>
Common stock	Br.300,000
Retained Earnings	<u>412,310</u>
Total Stockholders' Equity	<u>Br. 712,310</u>
Total liabilities and Equity	<u>Br. 1,038,036*</u>

Cash The amount of cash to support a firm's sales activity will be proportional to sales. This assumption holds for most firms. If the firm sells more goods, it accumulates more cash. But the nature of cash creates some reservations on this assumption. Cash is non-earning or lower return assets so that firms seek to minimize the amount of their cash balance. Due to such reason, even though cash balance will probably change, it probably will not change by the same percentage as sales. For GLOBAL Company cash is assumed to be changed in the same proportion with sales.

Cash to sales percentage = $\frac{44,000}{880,000} = 5\%$

830,000

Hence, the forecasted cash balance: $996,000 \times 0.05 = \text{Br. } 49,800$

Account Receivable Unless the firm changes its credit policy or has changed its types of customers, account receivable will increase proportionally with sales.

For GLOBAL Company, the forecasted account receivable is:

$$\text{Account receivable to sales percentage} = \frac{78,000}{830,000} = 9\%$$

$$\text{Forecasted account receivable} = 996,000 \times 0.9 = \text{Br. } 89,640$$

Inventory As sales increase, companies generally need more inventory. Hence, the inventory of GLOBAL Company increases in the same proportion to sales.

$$\text{Inventory to sales percentage} = \frac{210,000}{830,000} = 25\%$$

$$\text{Forecasted balance of inventory} = 996,000 \times 0.25 = \text{Br. } 249,000$$

Prepaid Expenses These current assets are generally assumed not to be changed directly with increase in sales. Prepaid expenses are conglomerates of different advance payments. For GLOBAL Company Prepaid Expenses are assumed to be the same as the previous year.

Fixed Assets Even though a firm may buy and sell many pieces of fixed assets, there is no reason to believe that these actions are directly related to the level of sales. At least in the short-run plant equipment will not be changed. Furthermore, no firm builds new plants every time sales increases. But in the long-run no firm can continue to increase sale unless it eventually adds capacity. For GLOBAL Company, plant and equipment, and land stay the same as the year 2008.

Accumulated Depreciation will definitely increase, but not because of the forecasted change in sales. It will be increased by the amount depreciation expense to be reported in the forecasted periods. For GLOBAL Company accumulated depreciation increases by the amount of depreciation expenses of Br. 28,000.

Patent do not have any relationship with sales level of firms. It is also expected to be the same as the year 2008 for GLOBAL Company, disregarding the amortization expenses.

After you complete the assets section of the balance sheet, sources of financing the assets is your next part of the pro-forma balance sheet. The financing sources may be spontaneously generated and/or discretionary financing sources.

Spontaneous Sources of Financing These are the sources of financing that arise during the ordinary courses of doing business. For example, account payable, once the credit account is established with suppliers, no additional work is required to obtain credit; it just happens spontaneously when the firm makes purchases. Not all current liabilities are, however, spontaneous sources of financing. For example, short-term notes payable and long-term debts due in one year are not spontaneously emerging sources of financing.

Discretionary Sources of Financing These are financing sources which require a large effort on the part of the firm to obtain. The firm must make a conscious decision to obtain these funds. Furthermore, the firm's top-level management will use its discretion to determine the appropriate type of financing. Examples under this group are any types of bank loan, bonds, common and preferred stocks.

Now comeback to the illustration for GLOBAL Company

Accounts Payable is one of the spontaneous sources of financing for GLOBAL Company and is directly related to sales.

$$\text{Accounts payable to sales percentage} = \frac{76,000}{830,000} = 9\%$$

$$\text{Forecasted balance of accounts payable} = 996,000 \times 0.09 = \text{Br. } 89,640$$

Tax depends directly on the firm's taxable income though indirectly related to sales level.

Tax payable for GLOBAL Company is Br. 29,106 taken from the income statement.

Accrued wages and salaries are accrued liabilities representing primarily accrued expenses which are spontaneous sources of financing.

$$\text{Accrued wages and salaries to sales percentage} = \frac{4,000}{830,000} = 0.05\%$$

$$\text{Forecasted Accrued wages and salaries} = 996,000 \times 0.005 = \text{Br. } 4,980$$

Interest Payable is not directly related to sales. It is a function of the amount and maturity structure of debt in the firm's capital structure. GLOBAL Company assumes no changes in the interest payable for the year 2009 because the same capital structure is expected for the year.

Long Term Notes Payable and Common Stocks are discretionary sources of financing and are not directly related to the level of sales. For GLOBAL Company the balances of these accounts are left as they are in 2008.

Retained Earnings will increase with increase in sales but not in the same rate as sales. The new balance for retained earnings will be the old level plus the addition to retained earnings. Hence, the balance for retained earnings of GLOBAL Company is:

$$328,000 + 84,310 = \text{Br. } 412,310$$

Additional Funds Needed After you complete forecasting the balance of each balance sheet items, assets must be equal to the sum of liabilities and equity. But the sum of the projected balances will not balance. The difference between total assets and total liabilities and equity is referred to as additional fund required. The additional fund required is financed through discretionary financing sources. The fund is also called discretionary financing fund.

Deficit Discretionary Funds When the firm's forecasted assets show higher level than liabilities and equities in the pro-forma balance sheet, arrangement are made for more liabilities and/or equities to finance the level of assets needed to support the volume of sales expected. This is referred to as deficit of discretionary funds.

Surplus of Discretionary Funds If the forecast shows that there will be a higher level of liabilities and equities than assets, the firm is said to be have a surplus of discretionary funds.

Generally, the amount required to make the forecasted balance sheet balance is additional fund needed or external financial requirement.

For GLOBAL Company, the difference between total assets and total liabilities and equity is: Total assets – [Total Liabilities + Equities] = 955,440 - 1,038,036 = (Br. 82,596).

The calculation tells you that GLOBAL Company expects to have Br. 82,596 more in discretionary funds that are needed to support its forecasted level of assets. In this case, GLOBAL Company is forecasting a surplus of discretionary funds.

Self-Test Activities

1. Data for GETWAY Import-Export Enterprise are as follow.

GETWAY Import-Export Enterprise

Balance sheet

Sene 30, 2000

Assets	Liabilities and Shareholders' Equity
Cash.....Br.77, 500	Account payable.....Br. 129,000
Receivable.....336,000	Notes payable..... 84,000
Inventories..... ..241,500	Other current liabilities..... 117,000
 Total current Asset.....Br. 655,000	 Total current liabilities.....Br. 330,000
Net fixed asset.....292,000	Long-term debt.....256,500
	Common equity.....361,000
Total assets.....Br. 947,500	Total liabilities &Equity...Br.947,500

GETWAY Import-Export Enterprise

Income Statement

For the year ended Sene 30, 2000

Sales	Br. 1,607,500
Cost of goods sold:	
Materials.....Br. 717,000	
Labor.....453,000	
Heat, light and power..... 68,000	
Indirect labor.....113,000	
Depreciation.....41,500	1,392,500
Earnings before interest and taxes (EBIT).....	Br. 70,000
Interest expense.....	<u>24,500</u>
Earning before tax.....	Br. 45,000
Income tax (40%)	<u>18,200</u>
Net income.....	<u>27,300</u>

Required

Calculate the following ratios from the data given in the financial statements above for GETWAY Import-Export Enterprise.

Current ratio

Return on Total assets

Inventory turnover

Return on Equity

Total asset Turnover

Debt ratio

Net Profit Margin

2. The following hypothetical data is derived from TATU Company. Find the balance of the balance sheet accounts and sales information attached.

Debt ratio=50%

Quick ratio=0.80times

Total asset turnover = 1.5times

Receivables

Annual sales/360 = Days sales outstanding =36 days

Gross profit margin on sales = 25%

Inventory turnover ratio =5 times

Cash	_____	Accts payable	_____
Accts Receivable	_____	Long-term debt	60,000
Inventories	_____	Common stock	_____
Fixed assets	_____	Retained earnings	97,500
	—		
Total assets	Br. 300,000	Total liabilities &equity	_____
Sales	_____	Cost of goods sold	_____
	—		

Chapter three

Time value of money

3.1 Introduction

Dear reader! Firms need to determine the value of flow of cash over different periods. The cash flow may be with different intervals over periods or at lump sum amount. The value determination may also be to know the future value or the value at present of the stream of flows. The chapter on the time value of money helps you to be acquainted with the techniques to determine or calculate the values of money at present or in the future with the concept of interest.

3.2 Time Value of Money: An Introduction

The Time Value of Money Ignoring the effects of inflation, a birr today is worth more than a birr to be received a year from now. In other words, you would all prefer to receive a specific amount of money now rather than at some future date. This preference rests on the time value of money.

The term **interest** is used to describe the price charge for using money over time. If payments for the time value of money are made or accrued, interest expense is incurred; and if payments for the time value of money are received or accrued, interest revenue is realized.

Business decisions often involve receiving cash or other assets now in exchange for a promise to make payments after one or more periods. A common example is a decision to borrow money. Another important group of business decision involves investing cash now in order to receive cash, goods, or services in future periods. Inflows of birr on various future dates should not be added together as if they are of equal value and also for outflows. The future cash inflows and outflows must be restated at their present value before they are aggregated. The concept of the time value of money tells you that more distant cash inflows have a smaller present value than cash inflows to be received within a short time span.

The timing of cash receipts and payments has an important effect on the economic worth and the accounting values of both assets and liabilities. Consequently, investment and borrowing decisions should be made only after a careful analysis of the relative present values of the prospective cash inflows and outflows.

3.3 The Concept of Interest

Interest is a return earned by or the amount paid to someone who has forgone current consumption or alternative investment opportunities and rented money in a creditor relationship. Interest is the growth in a principal amount representing the fee charged for the use of money for a specified time period. Because the concept of economic earnings is periodic, you can think of return on investment in terms of a rate of return per year.

Interest can be:

1. Simple interest.
2. Compound interest.

3.3.1 Simple Interest

When we borrow money, the money borrowed or the original sum of money lent (borrowed or invested) is called the **principal**. (The principal remains fixed during the entire interest period). Interest is usually expressed as a percentage of the principal for a specified period of time which is generally a year. This percentage is termed the **interest rate**. If interest is paid on the initial amount only and not on subsequently accrued interest, it is called **simple interest**.

However, if the interest for each period is added to the principal in computing the interest for the next period, the interest is called **compound interest**. The sum of the original amount (principal) and the total interest is **the future amount** or **maturity value** or **Amount**.

Simple Interest The simple interest describes that the interest itself does not earn interest over the periods of time. The interest is earned on the principal only. The *principal* is the amount of money borrowed or invested. Under the simple interest case, the interest and the future value of a certain birr amount are computed with the use of the following formula.

Computation of the Amount of Simple Interest

Assume that a person deposits a sum of money p in a saving account or borrow a sum of money p from a lending institution. Then p is referred to as the principal. When money is borrowed a fee is charged which is called interest. Interest is usually computed as a percentage of the principal over a given period of time. Dear learners, unless otherwise stated, the interest rate is an annual rate.

Simple Interest

$$I = Prt$$

Where

I = amount of interest

P = principal

r = annual simple interest rate

t = time in years

In general, if a principal p is borrowed at a rate r , then after t years the borrower will owe the lender an amount A that will include the principal p plus the interest I . Since p is the amount that is borrowed now and A is the amount that must be paid back in the future, p is often referred to as the present value and A as the future value and is given by the following formulas respectively.

$$A = P + I$$

$$= P + Prt$$

$$A = P (1 + rt) \text{ (Simple interest future value amount)}$$

$$\frac{A}{1 + rt} = \frac{P(1 + rt)}{1 + rt}$$

$$P = \frac{A}{1 + rt} \text{ (Simple interest present value formula)}$$

Given any three of the four variables A, P, r, and t, we can solve for the fourth. At this juncture, however, you need to bear in mind that the time may be given in months, weeks or days. In simple interest formula, t must be in years, and so conversion must be made.

Example 1

Mr. Abule borrowed a principal of Birr 1000 at 8% per year simple interest. Find the future value after;

- i) two years
- ii) three months
- iii) 180 days

Solution

Given

$$P = \text{Birr } 1,000$$

$$r = 0.08 \text{ per annum } (8/100)$$

$$t = 2 \text{ years}$$

$$\begin{aligned} \text{i) } A &= P (1 + rt) \\ &= 1,000 (1 + (0.08 \times 2)) \\ &= 1,000 (1 + 0.16) \\ &= 1,000 (1.16) \\ &= 1,000 (1.16) = \underline{\underline{\text{Birr } 1,160}} \end{aligned}$$

$$\text{ii) } A = P (1 + rt)$$

Here the time period is three months. So, conversion into years must be made and hence three months corresponds to $3/12 = 0.25$ year.

$$\begin{aligned} A &= 1,000 (1 + (0.08 \times 0.25)) \\ &= 1,000 (1 + 0.02) \\ &= 1,000 (1.02) \end{aligned}$$

$$= 1,000 (1.02) = \underline{\underline{\text{Birr 1,020}}}$$

iii) $A = P (1 + rt)$

Once more again, we have to convert 180 days into years. By assuming a year to have 360 days, 180 days corresponds to $t = 180/360 = 0.5$ year. Thus, from the simple interest formula

$$\begin{aligned} A &= 1,000 (1 + (0.08 \times 0.5)) \\ &= 1,000 (1 + 0.04) \\ &= 1,000 (1.04) \\ &= 1,000 (1.04) = \underline{\underline{\text{Birr 1,040}}} \end{aligned}$$



Self-check questions 3.1

1. What rate of interest let a sum of money to double in 5 years?
2. How long does it take for a sum of money to triple at a 5% simple interest rate?
3. A saving account opened 3 months ago now has a balance of Birr 20,400. If the bank pays 8% simple interest, how much money was deposited?
4. Find the amount that an investor should deposit in a bank today if he needs Birr 20,000 in 3 months at a simple interest rate of 9%.
5. What percentage of simple interest return must a firm get on their deposit if it wants its Birr 25,000 to grow to birr 26,500 in 6 months?
6. Mr. Bitew wanted to buy a new car for his family. The cost of the car was Birr 250,000. He was in short of cash and went to his local bank and borrowed Birr 150,000 for 6 years at an annual interest rate of 12%. Find the total simple interest and the maturity value of the loan.

3.4.2 Compound Interest

Dear learners, at this juncture we need to have a look at another kind of interest computation. If the interest which is due is added to the principal at the end of each interest period, then this interest as well as the principal will earn interest during the next period. In such a case the interest is said to be compounded. The result of compounding interest is that starting with the second compounding the account earns interest on interest in addition to earning interest on principal.

The sum of the original principal and all the interest earned is the Compound Amount. The difference between compound amount and the original principal is the Compound interest.

Example 2

Suppose that Birr 1,000 is invested at 10% compound interest. This situation can be described using the following table, which shows the state of the investment year by year.

Table 4.2 Computation of Compound interest

Year	Amount on which interest is calculated	Interest earned	Cumulative amount accrued
1	Birr 1,000	10% of Birr 1,000 = 100	Birr 1,100
2	Birr 1,100	10% of Birr 1,000 = 110	Birr 1,210
3	Birr 1,210	10% of Birr 1,000 = 121	Birr 1,331
4	Birr 1,331	10% of Birr 1,000 = 133.10	Birr 1,464.10

The difference between the methods of interest computation can easily be seen by comparing the above two tables. By looking at the table, you can recognize that the amount on which simple interest is calculated is always the same, while the amount varies year after year for compound interest.

A) Amount of compound interest

If at the end of a payment period, the interest due is reinvested at the same rate, then the interest as well as the original principal will earn interest during the next payment period. Interest paid on interest reinvested is called compound interest. Since interest rates are generally quoted as annual nominal rates, the rate per compounding period is found by dividing the annual nominal rate by the number of compounding periods per year.

The rate per period is equivalent to a nominal rate of 12% compounded for various periods is given in the following table.

Table 4.3
Determin
ation of
conversion
periods

12% interest compounded	m (conversion period)	i = r/m%
Annually	1	12 or 0.12
Semi – annually	2	6 or 0.06
Quarterly	4	3 or 0.03
Monthly	12	1 or 0.01
Weekly	52	12/52 = 0.23 or 0.0023
Daily	365	12/365 = 0.033 or 0.00033

In general, if P is the principal earning interest compounded m times a year at an annual rate of r, then $i = r/m$, the rate per period, the amount A at the end of each period is

$$\begin{aligned}
 A &= P (1 + i)^1 \text{ at the end of } 1^{\text{st}} \text{ period} \\
 &= P (1 + i)^2 \text{ at the end of } 2^{\text{nd}} \text{ period} \\
 &= P (1 + i)^3 \text{ at the end of } 3^{\text{rd}} \text{ period} \\
 &= P (1 + i)^n \text{ at the end of } n^{\text{th}} \text{ period}
 \end{aligned}$$

Compound interest future value can be computed as

$$A = P (1 + r/m)^{tm} = P (1 + i)^n$$

Where

r = annual nominal rate

m = number of compounding periods per year

i = rate per compound period

n = total number of compounding periods

p = principal or present value

A = amount or future value at the end of n periods

Example 3

What are the compound amount and compound interest at the end of one year if Birr 10,000 is borrowed at 8% compound quarterly?

Solution

P = Birr 10,000 Total number of conversions = 4

r = 8% t = one year

Total number of conversion periods (m) = 4 times = quarter

$$i = \frac{r}{m} = \frac{8\%}{4} = 2\%$$

Original principal.....	Birr 10,000
Add: interest for the 1 st quarter, I = Br. 10,000 x .02.....	Br. <u>200</u>
Principal at the end of 1 st quarter.....	Br.10,200 =
Br.10,000(1.02) ¹ Add: Interest for the 2 nd quarter, 10,200 x .02.....	Br. <u>204</u>
Principal at the end of 2 nd quarter.....	Br.10,404 =
Br.10,000 (1.02) ² Add: Interest ³³⁸ for the 3 rd quarter, 10,404x.02.....	Br. <u>208.08</u>
Principal at the end of 3 rd quarter.....	Br 10 612 08 =

If we summarize the above computations, we will come up with a pattern that leads to a general formula for computing compound interest:

$$\Rightarrow 1^{\text{st}} \text{ quarter: } S = 10,000 (1.02)^1$$

$$\Rightarrow 2^{\text{nd}} \text{ quarter: } S = 10,000 (1.02) (1.02) = 10,000 (1.02)^2$$

$$\Rightarrow 3^{\text{rd}} \text{ quarter: } S = 10,000 (1.02) (1.02) (1.02) = 10,000 (1.02)^3$$

$$\Rightarrow 4^{\text{th}} \text{ quarter: } S = 10,000 (1.02) (1.02) (1.02) (1.02) = 10,000 (1.02)^4$$

Therefore, for the above question, the amount is equal to

$$\begin{aligned} A &= P (1 + I)^n \\ &= 10,000 (1.02)^4 \\ &= 10,824.3216 \text{ Birr} \end{aligned}$$

 **Compound Interest = Compound Amount - Original Principal**

$$\begin{aligned} &= 10,824.3216 - 10,000 \\ &= \text{Birr } \underline{824.3216} \end{aligned}$$

Dear learners, compound amount can also be computed using **Logarithm** as follows.

$$\begin{aligned} A &= 10,000 (1.02)^4 \\ \log A &= \log 10,000 + \log (1.02)^4 \\ &= \log 10,000 + 4 \log 1.02 \\ &= 4 + 4 (0.0086) \\ &= 4 + 0.0344 \\ \log A &= 4.0344 \\ A &= \text{antilog } 4.0344 \end{aligned}$$

$$= \underline{\underline{\text{Br.10,824.30}}}$$

☐ Rules of Logarithms

1. $\log a^a = 1$
2. $\log^{mp} = p \log m$
3. $\log^{mn} = \log^m + \log^n$
4. $\log^{m/n} = \log^m - \log^n$

EXAMPLE

Find the compound amount compound interest resulting from the investment of Birr 1000 at 6% for 10 years,

- | | |
|-----------------------------|----------------------------|
| A. Compounded annually. | F. Compounded daily |
| B. Compounded semiannually. | G. Compounded hourly |
| C. Compounded quarterly. | H. Compounded continuously |
| D. Compounded monthly. | |
| E. compounded weekly | |

Solution

A. P = Birr 1,000 t	A = p (1+i) ⁿ	
= 10 years	= 1,000 (1.06) ¹⁰	
m = 1	= Birr 1,790.85	
r = 6% A = ?	Compound interest = Compound amount - Principal	
i = 6%	= 1,790.85 - 1000	
n = 10	=	<u><u>Birr 790.85</u></u>

Solution

B. P = Birr	A = p (1+i) ⁿ	
1,000	= 1,000 (1.03) ²⁰	
r = 6%		

$$m = 2 \quad = \textbf{Birr 1,806.11}$$

$$t = 10 \quad \text{Compound interest} = \text{Compound amount} - \text{Principal}$$

years

$$i = 3\% \quad = 1,806.11 - 1000$$

$$n = 20 \quad = \textbf{Birr 806.11}$$

Solution

$$A = 1,000 (1.06)^{40}$$

$$\text{C. } P = \text{Birr 1,000} \quad = \text{Birr 1,814.02}$$

$$r = 6\% \quad \text{Compound interest} = \text{Compound amount} - \text{Principal}$$

$$m = 4 \quad = 1814.02 - 1000$$

$$t = 10 \text{ years } i = .015 \quad = \textbf{Birr 814.02}$$

$$n = 40$$

Solution**D.** P= Birr 1000 r = 6%

t = 10 years m=12

i = .005

n = 120

$$A = 1,000 (1.005)^{120}$$

$$= \underline{\text{Birr } 1,819.40}$$

$$\text{Compound interest} = \text{Compound amount} - \text{Principal}$$

$$= 1,819.40 - 1000$$

$$= \underline{\text{Birr } 819.40}$$

ion**E.** P = Birr 1000

r = 6%

t = 10 years m=365

i = .01644%

n = 3650

$$A = 1,000 (1.0001644)^{3650}$$

$$= \text{Birr } 1,822.03$$

$$\text{Compound interest} = \text{Compound amount} - \text{Principal}$$

$$= 1,822.03 - 1000$$

$$= \underline{\text{Birr } 822.03}$$

Solution**F.** P = Birr 1000 r = 6%

t = 10 years m= 8760

i = .000685%

n = 87600

$$A = 1,000 (1.00000685)^{87,600}$$

$$= \underline{\text{Birr } 1,822.12}$$

$$\text{Compound interest} = \text{Compound amount} - \text{Principal}$$

$$= 1,822.12 - 1000$$

$$= \underline{\text{Birr } 822.12}$$

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t

Solution

$$G. (1+i)^n = (1+r/m)^{mt}$$

$$f(x) = \left(1 + \frac{1}{x}\right)^x \text{ if } x \text{ approaches infinity } \left(1 + \frac{1}{x}\right)^x \text{ becomes closer to } 2.71828 = e$$

Let $m/r = x$ as $m \uparrow$ $x \uparrow$

$$\begin{aligned} \left(1 + \frac{r}{m}\right)^{mt} &= \left(1 + \frac{1}{x}\right)^{mt} \\ &= \left(1 + \frac{1}{x}\right)^{rxt} \\ r/m = 1/x = m = rx \\ &= \left[\left(1 + \frac{1}{x}\right)^x\right]^{rt} \\ &= e^{rt} \end{aligned}$$

$$A = Pe^{rt}$$

$$P = \text{Birr } 1000 \quad r = 6\%$$

$$t = 10 \text{ years } m = \text{infinite}$$

$$= 1,000 e^{0.06 \times 10} = 1000 e^0.6$$

$$= \underline{\underline{\text{Birr } 1,822.12}}$$

$$\text{Compound interest} = \text{Compound amount} - \text{Principal}$$

$$1,822.12 - 1000$$

$$= \underline{\underline{\text{Birr } 822.12}}$$

B. T
i
m
e

Computation

EXAMPLE

How many years will it take for Birr 12,000 to grow to 24,000 when interest is 7% per annum compounded annually?

Given

$$P = 12,000$$

$$A = 24,000$$

$$i = r/m = 0.07/1 \quad 0.07 \quad n = ?$$

Solution

$$A = P (1 + i)^n$$

$$24,000 = 12,000 (1 + 0.07)^n \text{ dividing both sides by 12,000, we obtain,}$$

$$2 = (1 + 0.07)^n \text{ (taking logarithm of both sides)}$$

$$\log 2 = \log 1.07^n \text{ (using log rule that } \log a^b = b \log a)$$

$\log 2 = n \log 1.07$ dividing both sides by $\log 1.07$, we obtain

$$\log 2 / \log 1.07 = n$$

$$n = \frac{0.3010}{0.0294}$$

$$= 10.24 \text{ years}$$

C. Interest Rate

EXAMPLE

Birr 2000 is deposited in an account. After one year of monthly compounding, the balance in the account is Birr 2,166. What is the annual percentage rate for this account?

Solution

$$P = \text{Birr } 2,000$$

$$A = p (1+i)^n$$

$$A = \text{Birr } 2,166$$

$$2166 = 2000 (1+i)^{12}$$

$$r = ?$$

$$1.083 = (1+i)^{12}$$

$$i = r/12$$

$$\log 1.083 = \log (1+i)^{12}$$

$$t = 1$$

$$\log 1.083 = 12 \log (1+i)$$

$$m = 12$$

$$\frac{\log 1.083}{12} = \log (1+i)$$

$$0.0028857 = \log (1+i)$$

$$\text{anti-log } 0.0028857 = 1+i$$

$$1.0066667 = 1+i$$

$$.0066667 = i$$

$$.0066667 \times 12 = r = i \times m = \mathbf{r = 8\%}$$



Self-check questions 3.2

1. How much money should be deposited in a bank paying interest at a rate of 6% compounded monthly so that at the end of 3 years accumulated amount is Birr 20,000?
2. How long will it take for Birr 40,000 to amount to Birr 70,000 if it is invested at 7% compounded quarterly?
3. A firm wishing to have enough money for a retiring manager 17 years from now decided to buy a Birr 30,000 face value zero coupon bond. If the money is worth 10% compounded annually, what should they pay for the bond?
4. How long will it take to accumulate Birr 650 if Birr 500 is invested at 10% compound quarterly?
5. Birr 2000 is deposited in an account. After one year of monthly compounding, the balance in the account is Birr 2,166. What is the annual percentage rate for this account?

3.5 Present Value

Frequently it is necessary to determine the principal P which must be invested now at a given rate of interest per conversion period in order that the compound amount “ A ” to be accumulated at the end of n conversion periods. This process is called discounting and the principal is now a discounted value of a future income A .

$A = P(1+i)^n$ dividing both sides by $(1+i)^n$ leads to

$$P = \frac{A}{(1+i)^n} = p = A(1+i)^{-n}$$

Present values of a compound amount:

$$P = A(1+i)^{-n}$$

Where:

p = principal / present value

A = compound amount (or future value)

i = interest rate per conversion period

n = total number of conversion periods

Example:

1. Find the present value of a loan that will amount to Birr 5,000 in four years if money is worth 10% compounded semiannually.

Solution.

$$A = 5,000 \text{ Birr}$$

$$t = 4 \text{ years}$$

$$m = 2$$

$$r = 10\%$$

$$P = ?$$

$$P = A (1+i)^{-n}$$

$$= 5,000 (1.05)^{-8}$$

$$= \underline{\underline{\text{Birr } 3,384.20}}$$

Present value with continuous compounding

$$\begin{aligned} \frac{A}{e^{rt}} &= \frac{Pe^{rt}}{e^{rt}} \Rightarrow P = \cancel{A} / \cancel{e^{rt}} \\ &= Ae^{-rt} \end{aligned}$$



Self-check questions 3.3

1. How much must be deposited now in an account paying 6% compounded monthly in order to have just 20,000 Birr in the account 4 years from now?
2. If money worth 14% compounded semi-annually, would it be better to discharge a debt by paying Birr 500 now or Birr 600 eighteen months from now?

3.6 Equivalent Rates

Sometimes it is helpful to convert interest rates from, for example, a compounded quarterly basis to a compounded annually basis, from a compounded quarterly basis to compounded monthly basis, etc. this is easily accomplished as long as we understand the concept of equivalent interest rates, which is defined as follows:

If at the beginning of a specified time period, the same amount of money is invested at various rates so that the resulting compound amounts are equal at the end of the time period, then the interest rates are equivalent rates.

Although we can use any length time period, we usually use a 1-year time interval. Thus, if Birr P is invested at annual rate r compounded m times a year, and another Birr P is invested at annual rate s compounded k times a year, then the rates are equivalent as long as

$$P(1 + r/m)^m = P(1 + s/k)^k$$

Dividing both sides of the above equation by P gives the equivalent rates equation which can be solved for either r or s , depending on which the unknown.

Use this equation to find equivalent rates: $(1 + r/m)^m = (1 + s/k)^k$

Example

1. What rate compounded monthly is equivalent to 8% compounded quarterly?

Solution

$$(1 + r/12)^{12} = (1 + .08/4)^4$$

$(1 + r/12)^{12} = (1.02)^4$, solving for r , we take the 12th root of each side to obtain,

$$(1 + r/12) = [(1.02)^4]^{1/12}$$

$$1 + r/12 = (1.02)^{1/3}$$

$$r/12 = (1.02)^{1/3} - 1$$

$$= 1.006622 - 1$$

$$r/12 = .006622$$

$$r = 12(.006622)$$

$$r = .079464$$

$$= 7.95\%$$

2. What nominal annual rate of interest converted monthly corresponds to 16% converted quarterly?

Solution

$$(1 + r/12)^{12} = (1 + .16/4)^4$$

$= (1.04)^4$, solving for r , we take the 12th root of each side to obtain,

$$(1 + r/12) = [(1.04)^4]^{1/12}$$

$$= (1.04)^{1/3}$$

$$\begin{aligned}
 r/12 &= (1.04)^{1/3} - 1 \\
 &= 1.013159404 - 1 \\
 r/12 &= .013159404 \\
 r &= 12(.013159404) \\
 r &= .157912845 \\
 &= \mathbf{15.79\%}
 \end{aligned}$$

Or, using logarithms

$$\begin{aligned}
 (1.04)^4 &= (1+i)^{12} \\
 \text{Log } (1.04)^4 &= \text{log } (1+i)^{12} \\
 4 \log (1.04) &= 12 \log (1+i) \\
 4 (0.017033339) &= 12 \log (1+i) \\
 0.068133357 &= 12 \log (1+i) \\
 0.0056778 &= \log 1+i \\
 \text{Antilog } .0056778 &= (1+i) \\
 1.0131594 &= 1+i \\
 .0131594 &= i \\
 r &= m \times i \\
 &= 12 \times .0131594 \\
 &= \mathbf{15.79\%}
 \end{aligned}$$

3.7 Effective Rate

Obviously, for a stated annual interest rate, the amount of interest accumulated depends upon the frequency of conversion. This is because interest which has been earned subsequently earns interest itself. When interest is compounded more than once a year, the stated annual rate is called a Nominal Rate. The effective rate corresponding to a given nominal rate r converted m times a year is the simple interest rate that would produce an equivalent amount of interest in one year. Effective rates are, therefore, the simple interest rates that would produce the same return in one year had the same principal been invested at simple interest without compounding.

If P = Principal, A = Amount, r = nominal rate, m = number of conversion periods per year, the compound interest for **one year** on principal p is,

$$I = A - P$$

$$= p (1 + r/m)^m - p$$

The effective rate of interest is $(re) = \frac{\text{Compound interest}}{\text{principal}} = \frac{I}{P}$. From the above statement:

$$I = p (1 + r/m)^m - p$$

$$= P [(1+r/m)^m - 1] \text{ Divide both sides by } p$$

$$I/p = P \left[\left(1 + \frac{r}{m} \right)^m - 1 \right]$$

$$re = \left(1 + \frac{r}{m} \right)^m - 1$$

$$= (1+i)^m - 1$$

In continuous compounding case:

$$A = Pe^{rt} \text{ for one-year } A = Pe^r$$

$$I = A - P$$

$$= Pe^r - p, \text{ multiplying both sides by } 1/p$$

$$1/p \times I = P (e^r - 1) \times 1/p$$

$$re = e^r - 1$$

Effective rates are used to compare competing interest rates offered by banks and other financial institutions.

Example:

An investor has two opportunities to invest his money. The first investment opportunity (opp A) pays 15% compounded monthly and the second investment opportunity (opp B) pays 15.2%% compounded semiannually. Which is the better investment, assuming all else is equal.

Solution

Nominal rates with different compounding periods cannot be compared directly. We must first find the effective rate of each nominal rate and then compare the effective rates to determine which investment will yield the larger return.

Effective rate for inv. opp. A

$$re = (1+r/m)^m - 1$$

Effective rate for inv. opp. B

$$re = (1+r/m)^m - 1$$

$$= (1.0125)^{12} - 1$$

$$16.075\%$$

$$= (1.076)^2 - 1$$

$$15.778\%$$

Since the effective rate for A is greater than the effective rate for B, Investment opportunity A is the preferred investment.



Self-check questions 3.4

1. A bank states that the effective interest on savings accounts that earn continuous interest is 10%. Find the nominal rate.
2. Find the effective rate of interest corresponding to a nominal rate of 9% per year compounded;
 - A. Semi-annually
 - B. Quarterly
 - C. Monthly
3. A saving account opened 3 months ago now has a balance of Birr 20,400. If the bank pays 8% simple interest, how much money was deposited?

3.8 Annuities

An annuity is a sequence of equal, periodic payments. The payments may be made weekly, monthly, quarterly, semi-annually, annually or for any fixed period of time. The time between successive payments is called the payment period for an annuity. Each payment is called periodic payment or periodic rent, and it is denoted by R . The time from the beginning of the first payment period to the end of the last period is called the term of an annuity. If payments are made at the end of each time interval, then the annuity is called an ordinary annuity. If payments are made at the beginning of the payment period, it is called an annuity due.

3.8.1 Ordinary Annuity

An ordinary annuity is a series of equal periodic payments in which each payment is made at the end of the period. In an ordinary annuity the first payment is not considered in interest calculation for the first period (because it is paid at the end of the first period for which interest is calculated) and the last

payment doesn't qualify for interest at all since the value of the annuity's computed immediately after this last payment is received.

An annuity is meant to charge an existing debt in certain number of installments inclusive of compound interest at a stipulated rate. Similarly, annuity might also be used to create a fund to redeem a liability or to replace an asset in the future without disturbing the financial position of a person. An annuity is also applicable for someone or firm to secure a regular income for one's bread or any other purpose. Finally, an annuity is a good mechanism for determining the value of the regular installments payable or receivable under a contract or to determine the present value to be set aside under a contract.

Dear learners, in this we will consider ordinary annuities that are certain and simple, with periodic payments that are equal in size in other words, we will deal with annuities that are subject to the following conditions.

- i) The terms are given by fixed time intervals
- ii) The periodic payments are equal in size
- iii) The payments are made at the end of the payment periods
- iv) The payment periods coincide with the interest conversion periods

The amount (future value) of an ordinary annuity is the sum of all payments plus all interests earned.

$$A = R \left[\frac{(1+i)^n - 1}{i} \right]$$

Example

1. What is the amount of an annuity if the size of each payment is Birr 100 payable at the end of each quarter for one year at an interest rate of 4% compounded quarterly?

Solution

Periodic payment (R) = Birr 100

Payment interval = conversion period = quarter

Nominal (annual rate), r , = 4%

Interest rate per conversion period (i) = $r/m = 4\%/4 = 1\%$

Future value of (sum of an annuity) =?

Term one year

For the above example: $A = 100 \frac{[(1.01)^4 - 1]}{0.01} = \underline{\text{Birr 406.04}}$

Compound interest = Amount - R (n)
 $= 406.04 - 100(4) \text{ existence}$
 $= \underline{\text{Birr 6.04}}$

2. A newly married couple are both working and decide to have Birr 1000 at the end of a month for a down payment on a home. The account earns 12% compound monthly. How large a down payment will they have saved in three years?

Solution

R = Birr		Compound interest = A -
1000		R(n)
t = 3	$A = R \left[\frac{(1+i)^n - 1}{i} \right]$	$= 43,076.88 - 36,000$
years.	$= 1000 \frac{[(1.01)^{36} - 1]}{0.01}$	$= \text{Birr } \underline{7,076.88}$
m = 12	$= \text{Birr } 43,076.88$	
n = 36		
r = 12%		
i = 1%		
A = ?		

3.8.2 Sinking Fund- Increasing Annuity

A Sinking fund is a fund in to which equal periodic payments are made in order to accumulate a specified amount at some point in the future. Sinking funds are generally established in order to satisfy some financial obligation or to reach some financial goal.

If the payments are to be made in the form of an ordinary annuity, then the required periodic payment into the sinking fund can be determined by reference to the formula for a mount of an ordinary annuity. That is, if

$$A = R \frac{[(1+i)^n - 1]}{i}$$

Then

$$R = \frac{A}{\frac{[(1+i)^n - 1]}{i}}$$

$$R = A \left[\frac{i}{(1+i)^n - 1} \right]$$

Example:

1. What monthly deposit will produce a balance of Birr 100,000 after 10 years? Assume that the annual percentage rate is 6% compounded monthly. What is the total amount deposited over the 10-year period?

Solution

$$\begin{aligned}
 A &= \text{Birr } 100,000 & R &= A \left[\frac{i}{(1+i)^n - 1} \right] \\
 t &= 10 \text{ years} & R &= 100,000 \left[\frac{0.005}{(1.005)^{120} - 1} \right] \\
 m &= 12 & & \\
 r &= 6\% & & = 100,000 (0.006102) \\
 R &=? & & = \text{Birr } 610.21
 \end{aligned}$$

The total amount deposited over the 10-yr period is 120 (610.21) = **Birr 73,225**. The remaining Birr **26,775** an interest.

2. XYZ Company purchased a tract of land under a purchase agreement which requires a payment of Birr 500,000 plus 5% interest compounded annually at the end of 10 years. The company plans to setup a sinking fund to accumulate the amount required to settle the land purchase debt. What should the quarterly deposit into the fund be if the account pays 15% interest, compounded quarterly?

Solution

First, we have to find the total debt (future value) at the end of ten years as

$$\begin{aligned}
 A &= P (1+i)^n & i &= mt = 1 \times 5\% = 5\% \\
 &= 500,000 (1+0.05)^{10} \\
 &= \text{Birr } 814,447.31
 \end{aligned}$$

The amount is taken as Future Value of an Ordinary annuity with $r = 15\%$ Compounded quarterly for 10 years

$$A_{40} = \text{Birr } 814,447.31 \qquad R = 814,447.31 \left[\frac{.0375}{(1.0375)^{40} - 1} \right]$$

$t = 10$ years

$m = 4$

= 9,088.80 Birr

$r = 15\%$

$i = 3.75\%$



Self-check questions 3.5

1. A person deposits Birr 500 a year for 10 years in to an account that pays 6% compounded annually. After 10 years the person transfers the money into another account that pays 8% compounded quarterly. The money is left in the second account for 8 years. What is the balance after the 18-year period?
2. An employee deposits Birr 150 at the end of each month in a credit union saving plan that pays an interest rate of 12% compounded monthly. If the monthly periodic payments have increased to Birr 200 after the 8th payment,
 - a. What is the balance after one year?
 - b. What is the amount of interest that is earned?

3.8.3 Sinking Fund Schedule

The accumulation of value in a sinking fund is illustrated by a **sinking fund schedule**.

Example

Mr. Gedion wishes to set aside semiannual payments to purchase machinery after two years (two years from now). The machinery's estimated cost is Birr 5000. Each payment earns interest at 12% compounded semiannually.

- A) Find the semiannual payment
- B) Find the total interest earned
- C) Prepare a sinking fund schedule

Solution

$A = \text{Birr} 5000$

$t = 2$ years

$$A) R = A \left[\frac{i}{(1+i)^n - 1} \right]$$

m = 2, r = 12%, i = 6% and n = 4

R = ?

$$= 5000 \left[\frac{0.06}{1.06^4 - 1} \right]$$

B) Interest = Amount - R (n)

Payment number	Payment (R)	Interest*	Total
1	Birr 1,142.96	Birr 0 = Birr 428.16	Birr 1,142.96
2	Birr 1,142.96	Birr 68.58	Birr 2,354.50
3	Birr 1,142.96	Birr 141.27	Birr 3,638.73
4	Birr 1,142.96	Birr 218.32	Birr 5000.01
		Birr 428.16	

3.8.4 Present Value of an Ordinary Annuity

The present value of an ordinary annuity is the amount of money today, which is equivalent to the sum of a series of equal payment in the future. It is the **sum of the present values of the periodic payments of an annuity**, each discounted to the beginning of an annuity. The present value represents the amount that must be invested now to purchase the payment due in the future.

In short, PV of an ordinary annuity can be computed in two ways:

- (1) Discounting all periodic payment to the beginning of the term individually.
- (2) Discounting the amount of an ordinary annuity to the beginning of the term.

$$P = R \left[\frac{1 - (1 + i)^{-n}}{i} \right]$$

Example

1. What is the PV of an annuity if the size of each payment is Birr 200 payable at the end of each quarter for one year and the interest rate is 8% compounded quarterly?

Solution

R = Birr 200

r = 8%

m = 4

t = 1yr

P = ?

Using the above formula; the PV of the former example is computed as:

R = Birr 200

$P = R \left[\frac{1 - (1 + i)^{-n}}{i} \right]$

r = 8%

i

$$\begin{aligned}
m &= 4 & &= 200 \frac{[1 - (1.02)^{-4}]}{.02} \\
t &= 1 \text{ yr} \\
P &=? & &= 200 (3.08773) \\
& & &= \text{Birr } \underline{\underline{761.55}}
\end{aligned}$$

Equivalently: Find the FV of the ordinary annuity using the formula $A = R \frac{[(1+i)^n - 1]}{i}$

$$\begin{aligned}
A &= 200 \frac{[(1.02)^4 - 1]}{.02} \\
&= \text{Birr } \underline{\underline{824.32}}
\end{aligned}$$

2. What is the cash value of a TV that can be bought for Birr 200 down payment and Birr 82 a month for 18 months, if money is worth 12% interest compounded monthly?

Solution

Cash Value = down payment + PV of an ordinary annuity

$$\begin{aligned}
&= 200 + 82 \frac{[1 - (1.01)^{-18}]}{.01} \\
&= 200 + 82(16.39827) \\
&= 200 + 1,344.658 \\
&= \text{Birr } \underline{\underline{1,544.658}}
\end{aligned}$$

3.9 Amortization- Decreasing Annuity

Amortization means retiring a debt in a given length of time by equal periodic payments that include compound interest. After the last payment, the obligation ceases to exist-it is dead-and it is said to have been amortized by the payments.

In amortization our interest is to determine the periodic payment, R, so as to amortize (retire) a debt at the end of the last payment. Solving the PV of ordinary annuity formula for R in terms of the other variables, we obtain the following amortization formula:

$$R = P \left[\frac{i}{1 - (1+i)^{-n}} \right] \quad \text{Where:}$$

R = periodic payment
P = PV of loan
i = interest rate per period

n = number of payment periods

Examples

1. Suppose you borrow Birr 5000 from a bank and agree to repay the loan in five equal installments including all interests due. The bank's interest charges are 5% compounded annually. How much should each annual payment be in order to retire the debt including the interest in 5 years?

Solution

$$\begin{aligned}PV &= \text{Birr } 5000 & R &= 5000 \left[\frac{.05}{1 - (1.05)^{-5}} \right] \\t &= 5 \text{ years} & & \\m &= 1 & & \\r &= 5\% & & \\R &=? & \text{Interest} &= (1,154.87 \times 5) - 5000 \\& & & = \underline{\underline{\text{Birr } 774.35}}\end{aligned}$$

2. At the time of retirement, a person has Birr 200,000 in an account that pays 12% compounded monthly. If he decides to withdraw equal monthly payments for 10 years, at the end of which time the account will have a zero balance, how much should he withdraw each month?

Solution

$$\begin{aligned}PV &= \text{Birr } 200,000 & R &= 200,000 \left[\frac{.01}{1 - (1.01)^{-120}} \right] \\t &= 10 \text{ years} & & \\m &= 12 & & \\r &= 12\% & & \\R &=? & & \\& & & = \underline{\underline{\text{Birr } 2,869.42}}\end{aligned}$$

3.10 Mortgage Payments

Amortization means retiring a debt in a given length of time by equal periodic payments that include compound interest. After the last payment, the obligation ceases to exist-it is dead- and it is said to have been amortized by the payments.

In atypical house purchase transaction, the home-buyer pays part of the cost in cash and borrows the remained needed, usually from a bank or a savings and loan association. The buyer amortizes the indebtedness by periodic payments over a period of time. Typically, payments are monthly and the time period is long-30 years is not unusual.

Mortgage payment and amortization are similar. The only differences are

- The time period in which the debt/loan is amortized/repaid

- The amount borrowed.
- In mortgage payments **m** is equal to 12 because the loan is repaid from monthly salary, but in amortization **m** may take other values.

In Mortgage payments we are interested in the determination of monthly payments.

Taking A = total debt

R = monthly mortgage payment

r = stated nominal rate per annum

n = 12 x t

R can be determined as follows:

$$R = A \left[\frac{i}{1 - (1+i)^{-n}} \right] \quad \text{Or} \quad R = A \left[\frac{i}{1 - (1+i)^{-n}} \right]$$

$$\text{Similarly, } A = R \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

Example:

- Mr. X purchased a house for Birr 115,000. He made a 20% down payment with the balance amortized by a 30-year mortgage at an annual interest of 12% compounded monthly.
 - What is the amount that Mr. X should pay monthly so as to retire the debt at the end of the 30th year?
 - Find the interest charged.

Solution

Selling price = Birr 115,000

r = 12%

i = 1%

Down payment (20%) 23,000

m = 12

Mortgage (A) Birr 92,000

t = 30yrs

n = 360

R = ?

$$R = A \left[\frac{i}{1 - (1+i)^{-n}} \right]$$

$$R = 92,000 \left[\frac{.01}{1 - (1.01)^{-360}} \right]$$

$$= 92,000 (.010286125)$$

$$= \text{Birr } \underline{\underline{946.32}}$$

Interest = Actual payment – Mortgage (loan)

$$= (946.32 \times 360) - 92000$$

$$= \text{Birr } 340,675.20 - 92,000$$

$$= \text{Birr } \underline{\underline{248,675.20}}$$

2. Mrs. Y purchased a house for Birr 50,000. She made an amount of down payment and pay monthly Birr 600 to retire the mortgage for 20 years at an annual interest rate of 24% compounded monthly.

Required: Find the mortgage, down payment, interest charged, and the percentage of the down payment to the selling price.

Solution

$$\text{Selling price} = \text{Birr } 50,000$$

$$\text{Mortgage (A)} = R \left[\frac{1 - (1+i)^{-n}}{i} \right]$$

$$\text{Down payment} = ?$$

$$i$$

$$\text{Mortgage (A)} = ?$$

$$= 600 \left[\frac{1 - (1.02)^{-240}}{0.02} \right]$$

$$R = \text{Birr } 600$$

$$0.02$$

$$r = 24\% \quad i = 2\%$$

$$= \text{Birr } \underline{\underline{29,741.13}}$$

$$m = 12$$

$$\text{Down payment} = \text{Selling price} - \text{mortgage}$$

$$t = 20 \quad n = 240$$

$$= 50,000 - 29,741.13$$

$$= \text{Birr } \underline{\underline{20,258.87}}$$

$$\text{Interest charged} = \text{actual payment} - \text{mortgage}$$

$$= 600 \times 240 - 29,741.13$$

$$= 144,000 - 29,741.13$$

$$= \text{Birr } \underline{\underline{114,258.87}}$$

$$\text{Percentage of down payment} = \frac{\text{Down payment}}{\text{Selling Price}} \times 100$$

$$= \frac{20,258.87}{50,000} \times 100 = \underline{\underline{40.52\%}}$$



Self-check questions 3.6

1. What is the cash value of a car that can be bought for Birr 200 down payment and Birr 82 a month for 18 months, if money is worth 12% interest compounded monthly?
2. How much should you deposit in an account paying 6% compounded quarterly in order to be able to withdraw Birr 1000 every 3 months for the next 3 years?
3. What is the present value of an annuity of seven payments of Birr 1000 each made at the end of each quarter with an interest rate of 12% compounded monthly?
4. At the time of retirement, a person has Birr 200,000 in an account that pays 12% compounded monthly. If he decides to withdraw equal monthly payments for 10 years, at the end of which time the account will have a zero balance, how much should he withdraw each month?
5. Mrs. Almaz purchased a house for Birr 50,000. She made an amount of down payment and pay monthly Birr 600 to retire the mortgage for 20 years at an annual interest rate of 24% compounded monthly.

Required: Find the mortgage, down payment, the interest charged, and the percentage of the down payment to the selling price.

Chapter four

Security valuation and the cost of capital

4.1 Introduction

Dear learners! As you have seen in the previous accounting courses, the value of an asset is determined based on its cost (historical cost). That means all the necessary expenditures incurred from the time the asset is acquired until it is placed in operation will be the cost of the asset. However, in financial management, the value of an asset is quite different. Since finance is interested more on decision making rather than recording, the value of an asset is determined before it is purchased. The purpose is to decide whether to acquire or not to acquire the asset. Therefore, here the historical cost cannot be used as the value of the asset. Rather, the value of the asset is determined by valuation.

Valuation is the process of determining the worth of any asset whose value is obtained from future cash flows. The value of any asset in finance is the present value of all future cash flows. It is expected to provide over the relevant time period. This value is called **intrinsic value**. The intrinsic value of an asset is determined based on three basic inputs: cash flows (returns), time pattern of the returns, and the discount rate. The value of an asset is, therefore, determined by discounting the expected cash flows to their present value. To determine the present value, we use a discount rate appropriate based on the asset's risk.

4.2 Bond Valuation

Governments and corporations borrow money by selling bonds to investors. The money they collect when the bond is issued, or sold to the public, is the amount of the loan. In return, they agree to make specified payments to the bondholders, who are the lenders. When you own a bond, you generally receive a fixed interest payment each year until the bond matures. This payment is known as the coupon because most bonds used to have coupons that the investors clipped off and mailed to the bond issuer to claim the interest payment. At maturity, the debt is repaid: the borrower pays the

bondholder

the

bond's face value (equivalently, its par value).

Bond is a long-term contract under which a borrower agrees to make payments of interest and principal on specific dates to the holder of the bond. And also, bond is a long-term debt instrument or security issued by businesses and governmental units to raise large sums of money. Investment in a bond provides two types of cash flows

- i. *The periodic interest payment by the issuing party and=COUPON PAYMENT*
- ii. *The price paid to the investor upon maturity=MV*

The first, i.e., the interest payment is based on the par value of the bond and the coupon interest rate. The par value is the face value of the bond which will be paid to the investor upon maturity. Par value is also called maturity value.

In order to value a bond, you must understand the following.

4.3.1 Features of Bonds

- ❖ **Par value:** It is the amount or value stated on the face of the bond. It represents the amount of the firm borrows and promises to repay at the time of maturity. It can be any denomination.
- ❖ **Coupon Rate of Interest:** A bond carries a specific interest rate, which is called the coupon rate. The interest payable to the bondholder is simply par value of bond multiplied by the coupon rate.
- ❖ **Maturity period:** Every bond will have maturity period. On completion of the maturity period the principal amount has to be repaid as per the agreed terms while issuing such bonds call provision. Sometimes bonds may be issued under a provision that the business unit will have an option to pay back the bond amount before the **maturity period**. These are known as callable bonds.
- ❖ **The intrinsic value of a bond** is equal to the present value of its expected cash flows. The coupon interest payments and principal payments are known and the present value is determined by discounting these future payments from the issuer at an appropriate discount rate or market yield.
- ❖ **Zero-Coupon Bonds.** Corporations sometimes issue zero-coupon bonds. In this case, investors receive face value at the maturity date but do not receive a regular coupon payment. In other words, the bond has a coupon rate of zero. You learned how to value such bonds earlier.

These bonds are issued at prices considerably below face value, and the investor's return comes from the difference between the purchase price and the payment of face value at maturity.

- ❖ **Floating-Rate Bonds.** Sometimes the coupon rate can change over time. For example, floating-rate bonds make coupon payments that are tied to some measure of current market rates. The rate might be reset once a year to the current Treasury bill rate plus 2 percent. So, if the Treasury bill rate at the start of the year is 6 percent, the bond's coupon rate over the next year would set at 8 percent. This arrangement means that the bond's coupon rate always approximates current market interest rates.
- ❖ **Convertible Bonds.** If you buy a convertible bond, you can choose later to exchange it for a specified number of shares of common stock. For example, a convertible bond that is issued at par value of \$1,000 may be convertible into 50 shares of the firm's stock. Because convertible bonds offer the opportunity to participate in any price appreciation of the company's stock, investors will accept lower interest rates on convertible bonds.
- ❖ **Put Bond** - allows the holder to force the issuer to buy back the bond at a stated price.
- ❖ **The Call Provision:** A **call provision** allows the company to repurchase or "call" part or all of the bond issue at stated prices over a specific period. Corporate bonds are usually callable. The difference between the call price and the stated value is the **call premium**.
- ❖ **The Indenture:** is the written agreement between the corporation (the borrower) and its creditors. It is sometimes referred to as the *deed of trust*. Usually, a trustee (a bank, perhaps) is appointed by the corporation to represent the bondholders. The trust company must make sure the terms of the indenture are obeyed, manage the sinking fund, and represent the bondholders in default—that is, if the company defaults on its payments to them.

Investors have many choices when investing in Bonds, but bonds are classified into four main types: treasury, corporate, municipal, and foreign.

4.3.2 Types of Bonds

1. **Treasury Bonds:** sometimes referred to as government bonds, are issued by the federal government. It is reasonable to assume that the federal government will make good on its promised payments, so these bonds have no default risk. However, Treasury bond prices decline when interest rates rise, so they are not free of risks.

2. **Corporate Bonds:** as the name implies, are issued by corporations. Unlike Treasury bonds, corporate bonds are exposed to default risk- if the issuing company gets into trouble, it may be unable to make the promised interest and principal payments. Different corporate bonds have different levels of default risk, depending on the issuing company's characteristics and on the terms of the specific bond. Default risk is often referred to as "credit risk". The larger the default or the credit risk, the higher the interest rate the issuer must pay.
3. **Municipal Bonds:** or "minis," are issued by state and local governments. Like corporate bonds, munis have default risk. However, munis offer one major advantage over all other bonds. The interest earned on most municipal bonds is exempted for federal taxes, and also from state taxes if the holder is the resident of the issuing state. Consequently, municipal bonds carry interest rates that are considerably lower than those on corporate bonds with the same default risk.
4. **Foreign Bonds:** are issued by foreign governments or foreign corporations. Foreign corporate bonds are, of course, exposed of default risk, and so are sometimes foreign government bonds. An additional risk exists if the bonds are denominated in a currency other than that of the investor's home currency.

4.3.3 Basic Bond Valuation Model

The value of a bond is the present value of the periodic interest payments plus the present value of the par value. The value of a bond can be computed using the following equation:

$$B_0 = I (PVIFA\ k_d, n) + M(PVIF\ k_d, n) \quad \text{Where:}$$

B_0 = the value of the bond

I = interest paid each period = Par Value x Coupon interest rate

K_d = the appropriate interest rate on the bond

n = The number of periods before the bond matures

M = the par value of the bond

$(PVIFA\ k_d, n)$ = The present value interest factor for an annuity at interest rate of k_d per period for n

$$\text{periods} = \frac{1 - \frac{1}{(1 + k_d)^n}}{k_d}$$

$(PVIF_{k_d, n})$ = The present value interest factor at interest rate of k_d per period for n periods =

$$\frac{1}{(1 + k_d)^n}$$

Notice that we have used k_d instead of i . This is because, generally, in financial management **k** labels rate of return and the subscript **d** denotes debt security. So k_d labels the rate of return on a debt security.

Illustration: Tebaber Corporation has a Br. 1,000 par value bond with an 8% coupon interest rate outstanding. Interest is paid semiannually and the bond has 12 years remaining to its maturity date.

Required: What is the value of the bond if the required return on the bond is 8%?

Solution:

Given: $M = \text{Br. } 1,000$; $k_d = 8\%$ per year or 4% ($8\% \div 2$) per semiannual period; $I = \text{Br. } 40$ ($\text{Br. } 1,000 \times 4\%$); $n = 24$ semiannual periods (12×2); $B_0 = ?$

$$\begin{aligned} B_0 &= I (\text{PVIFA } k_d, n) + M (\text{PVIF } k_d, n) \\ &= \text{Br. } 40 (\text{PVIFA}_{4\%, 24}) + \text{Br. } 1,000 (\text{PVIF}_{4\%, 24}) \\ &= \text{Br. } 40 (15.2470) + \text{Br. } 1,000 (0.3901) \\ &= \underline{\underline{\text{Br. } 1,000}} \end{aligned}$$

If the appropriate discount rate (k_d) remains constant at 8% (4% per semiannual period), the value of the bond will not be changed. It will remain Br. 1,000. Suppose the appropriate discount rate is 8% 2 years from now, what would be the value of the bond?

Solution: now n is reduced to $20[24 - (2 \times 2)]$

$$\begin{aligned} B_0 &= \text{Br. } 40 (\text{PVIFA}_{4\%, 20}) + \text{Br. } 1,000 (\text{PVIF}_{4\%, 20}) \\ &= \text{Br. } 40 (13.5903) + \text{Br. } 1,000 (0.4564) \\ &= \underline{\underline{\text{Br. } 1,000}} \end{aligned}$$

Suppose the interest rate in the economy when Tebaber's bonds were issued was 6% rather than 8%, what would be the value of the bond? Since Tebaber's bond now will be paying more interest than do other bonds in the market, the company's bond will be selling at a larger price. Such bonds which are selling more than their par value are called premium bonds. Here, k_d is 6% (3% per semiannual payment), but other things are not changed. So

$$\begin{aligned} B_0 &= \text{Br. } 40 (\text{PVIFA}_{3\%, 24}) + \text{Br. } 1,000 (\text{PVIF } 3\%, 24) \\ &= \text{Br. } 40 (16.9355) + \text{Br. } 1,000 (0.4919) \\ &= \underline{\underline{\text{Br. } 1,169.32}} \end{aligned}$$

So, when the market interest rate (k_d) is less than the coupon interest rate, the value of a bond is always larger than the par value. An investor by deciding to invest his money on Tebaber's bond, he will receive a 1% (4% - 3%) more interest payment than he would receive if he invested somewhere else. This allows the investor to receive Br. 10 [Br. 1,000 x (4% - 3%)] more every semiannual period. As a result, the investor would be willing to give more price to the bond. The additional price is the present value of each Br. 10 he is going to receive for the next 24 semiannual periods. Therefore, the value of a premium bond can also be computed as:

$$\begin{aligned} B_o &= \text{Br. } 1,000 + \text{Br. } 10 (\text{PVIFA}3\%, 24) \\ &= \text{Br. } 1,000 + \text{Br. } 10 (16.9355) \\ &= \underline{\underline{\text{Br. } 1,169.36^*}} \end{aligned}$$

* The previous value was Br. 1,169.32. The difference is due to rounding problem.

Assuming the interest rate remains constant at 6% for the next 11 years (12 periods), what would happen to Tebaber's bond?

$$\begin{aligned} B_o &= \text{Br. } 40 (\text{PVIFA}3\%, 22) + \text{Br. } 1,000 (\text{PVIF}3\%, 22) \\ &= \underline{\underline{\text{Br. } 1,159.38}} \end{aligned}$$

Thus, the value of the bond would fall from Br. 1,169.32 to Br. 1,159.38. If you calculate the value of the bond at other future dates, the price would continue to fall as the maturity date approaches.

Had the interest rate (k_d) was 10% when Tebaber's bond was selling, the value of the bond would be:

$$\begin{aligned} B_o &= \text{Br. } 40 (\text{PVIFA}5\%, 24) + \text{Br. } 1,000 (\text{PVIF}5\%, 24) \\ &= \text{Br. } 40 (13.7986) + \text{Br. } 1,000 (0.3101) \\ &= \underline{\underline{\text{Br. } 862.04}} \end{aligned}$$

Since Tebaber's bond now will be paying less interest than do other bonds in the market, they are selling at a smaller price (discount bond).

If the interest rate remains constant at 10% for the next 11 years (22 periods), the value of Tebaber Berta's bond would be Br. 868.32. Thus, the value of the bond will have risen from Br. 862.04 to Br. 868.32. If you further calculate the value of the bond at other future dates, the price would continue to rise as the maturity date approaches.

We see a relation developing between the coupon rate, the yield, and the value of a debt security:

- ➡ If the coupon rate is more than the yield, the security is worth more than its maturity value—it sells at a premium and it is called **premium bond**.
- ➡ If the coupon rate is less than the yield, the security is less than its maturity value—it sells at a discount and it is called **discount bond**.
- ➡ If the coupon rate is equal to the yield, the security is valued at its maturity value and it is called **par value**.

Generally, the relationship among a Bond's price and its coupon rate, current yield and yield to Maturity

- ➡ Bond Price < Face Value: Coupon rate < Current yield < YTM
- ➡ Bond Price = Face Value: Coupon rate = Current yield = YTM
- ➡ Bond Price > Face Value: Coupon rate > Current yield > YTM

4.3.4 Interest Rate on a Bond

So far, we have been seeing how to determine the value of a bond if we are given the par value, the coupon interest rate, the number of periods, and the interest rate on the bond. Next, we shall discuss on how to find the interest rate on a bond, i.e., k_d if we are given the value of the bond. We will consider yield to maturity and yield to call.

Yield to Maturity (YTM) is the rate of return investors earn if they buy a bond at a specific price B_0 and hold it until maturity. The approximate YTM can be found using the following approximation formula:

$$\text{Approximate YTM} = \frac{I + \frac{M - B_0}{n}}{\frac{M + B_0}{2}}$$

Example: Zebra Company has a Br. 1,000 par value, 10% coupon interest rate, and 15 years to maturity. The bond is currently selling at Br. 1,090. Compute the YTM.

Solution:

Given: $M = \text{Br. } 1,000$; $I = \text{Br. } 100$ ($\text{Br. } 1,000 \times 10\%$); $n = 15$; $B_0 = \text{Br. } 1,090$; $\text{YTM} = ?$

$$\text{Approximate YTM} = \frac{\text{Br.}100 + \frac{\text{Br.}1,000 - 1,090}{15}}{\frac{\text{Br.}1,000 + \text{Br.}1,090}{2}} = 9\%$$

If an investor buys Zebra's bond at Br. 1,090 and holds it for 15 years, the approximate yield or rate of return per year is 9%.



Self-check questions 4.1

1. When is a bond selling at premium, at par value, and at discount?
2. ABC Corporation issued a new series of bonds on January 1, 1985. The bonds were sold at their par of Br. 1,000, have 12% coupon, and mature in 20 years. Coupon payments are made quarterly. What was the price of the bond on December 31, 1989, assuming that the level of interest rate had fallen to 8%?
3. The Salem Company bond currently sells for Br. 955, has a 12% coupon interest rate and Br. 1,000 par value, pays interest annually, and has 15 years to maturity. Calculate the yield to maturity on this bond.

Preferred stock is a type of equity security that provides its owners with limited or fixed claims on a corporation's income and assets. Investment in a preferred stock provides a single cash flow, i.e., constant periodic dividend payments. Preferred stock has similarities to both a bond and a common stock. As to similarities to a bond, preferred dividends are fixed in amount and are like interest payments. As to a common stock, the preferred dividends are paid for an indefinite time period.

4.4.1.1 Preferred Stock Valuation Model

The value of a preferred stock is the present value of all future preferred dividends it is expected to provide over an infinite time horizon. Most preferred stocks entitle their owners to regular and fixed dividend payments. If the payments last forever, the issue is a perpetuity. Since dividends from preference shares are assumed to be perpetual payments, the intrinsic value of such shares will be estimated from the following equations.

$$V_{ps} = \frac{C}{(1+k)} + \frac{C}{(1+k)^2} + \frac{Cn}{(1+k)^n}$$

V_p = Value of perpetual today

C = Constant dividends received

Kps = Required rate of return appropriate

Therefore, the value of a preferred stock is found by the following formula:

$$VPS = \frac{Dps}{Kps}$$

Where:

Vps = Value of the preferred stock

Dps = Preferred stock dividends

Kps = The required rate of return on the preferred stock

Example: Abebe wishes to estimate the value of its outstanding preferred stock. The preferred issue has a Br. 80 par value and pays an annual dividend of Br. 6.40 per share. Similar-risk preferred stocks are currently earning a 9.3% annual rate of return. What is the value of the outstanding preferred stock?

Solution:

Given: Dps = Br. 6.40; Kps = 9.3%; Vps = ?

So, the Br. 6.40 annual dividend an investor receives for an infinite year is equal to today's Br. 68.82 if the required rate of return is 9.3%.

4.4.1.2 Rate of Return on a Preferred Stock

To evaluate the worthiness of investment in a preferred stock in comparison to other investment opportunities, we should be able to compute the rate of return on a preferred stock. If we know the current price of a preferred stock and its dividend, we can compute the expected rate of return on the preferred stock. This can be done using the following formula:

$$Kps = \frac{Dps}{Vps}$$

Where

Kps = The expected rate of return on the preferred stock

Dps = Preferred stock dividends

Vps = Value or current price of the preferred stock

Example: A preferred stock pays an annual dividend of Br. 9 and the current market price is Br. 81. Compute the required rate of return from the preferred stock.

Solution:

Given: Dps = Br. 9; Vps = Br. 81; Kps = ?

For an investor to invest Br. 81 in this preferred stock and to receive an annual dividend of Br. 9, his minimum required rate of return is 11.11%.

4.4.2 Common Stock Valuation

The value of a share of common stock is the present value of the common stock's dividend expected over an infinite time horizon. The value of a share of common stock is also equal to the sum of the present value of the expected dividends and the present value of the expected selling price of the stock. The selling price in turn will depend on the dividends to be received by the purchasing party.

To understand the value of a common stock we should keep in mind two points. First, the dividends are expected for an infinite time period. Second, the dividends are not constant. Therefore, the value of a common stock is found by summing the present values of annual dividends.

$$P_0 = \frac{D_1}{(1+ks)^1} + \frac{D_2}{(1+ks)^2} + \dots + \frac{D_\infty}{(1+ks)^\infty}$$

Where:

P_0 = Value of the common stock at time zero (as of today)

$D_1, D_2, \dots, D_\infty$ = Per share dividend expected at the end of each year

Ks = the required rate of return on the common stock.

The common stock valuation equation can be simplified by redefining each year's dividend. The dividends are defined in terms of anticipated dividends growth. Generally, there are three cases accordingly. These are:

- i) Zero growth common stock,
- ii) Constant growth common stock, and
- iii) Variable growth common stock.

Hence, common stock valuation approaches are developed under each of the above dividend growth models. Next sections will discuss each model one by one.

i) Zero Growth Stock

A zero-growth stock is a common stock whose future dividends are not expected to grow at all. The expected growth rate (g) is zero. This is the simplest model to common stock valuation. It assumes a

constant, non-growing annual dividend. So here the annual dividends are all equal. That is $D_1 = D_2 = \dots = D_\infty = D$.

A common stock with zero growth rate is a security that is expected to provide a fixed dividend each year. Hence, a zero-growth common stock is a perpetuity. Therefore, the value of a zero-growth stock is given as:

$$P_0 = \frac{D}{K_s}$$

Example: The most recent common stock dividend of Shalom Manufacturing Corporation was Br. 3.60 per share. Due to the firm's maturity as well as stable sales and earnings, the dividends are expected to remain at the current level of the foreseeable future.

Required: Determine the value of Shalom's common stock for an investor whose required return is 12%.

Solution:

Given: $D = \text{Br. } 3.60$; $K_s = 12\%$; $P_0 = ?$

The maximum price the investor would be willing to pay for a share of Shalom's common stock is Br. 30 for he to receive a Br. 3.60 annual dividend for an indefinite year.

II) Constant Growth Stock

Constant growth stock is a common stock whose future dividends are expected to grow at a constant dividend growth rate (g). It is sometimes called normal growth stock. The constant (normal) growth common stock valuation model is the most widely cited approach to common stock valuation.

The value of a constant growth stock is the present value of the expected future dividends growing at a constant rate of g . Here the value can be found by using the following formula:

$$P_0 = \frac{D_1}{K_s - g}; K_s > g \text{ Where:}$$

D_1 = The expected dividend at the end of year 1.

g = The expected growth rate in dividends.

$D_1 = D_0(1+g)$, where D_0 is the most recent dividend. Similarly, $D_2 = D_1(1+g)$ and so on. To find the value of a common stock (constant growth) at one year, first, find the expected dividend at the end of next year.

Example: Zeila Motor Corporation's common stock currently pays an annual dividend of Br. 5.40 per share. The dividends are expected to grow at a constant annual rate of 5% to infinity. Estimate the value of Zeila's common stock if the required return is 12%.

Solution:

Given: $D_0 = \text{Br. } 5.40$; $g = 5\%$; $K_s = 12\%$; $P_0 = ?$

For an investor to receive an annual dividend of Br. 5.40 growing at 5% constantly to infinity, the maximum price he would pay today is Br. 81.

If we are given the value of a constant growth stock, the most recent dividend, the expected dividend growth rate, we can compute the expected rate of return as follows.

$$K_s = \frac{D_1}{P_0} + g \quad \text{Where;}$$

K_s = The expected rate of return on a constant growth stock

D_1/P_0 = Expected dividend yield.

g = Expected dividend growth rate = capital gains yield.

Example: Assume the above example except that you are given the value of common stock of Br. 81 instead of the required return. Compute the expected rate of return?

iii) Variable Growth Stock

Variable growth stock is a stock whose dividends are expected to grow at variable or non-constant rates. The model of common stock valuation that allows for a change in the dividend growth rate is called Variable (Non constant) Growth Model. It sometimes is also called supernormal growth model.

The value of a share of variable growth stock is determined by following 4 procedures;

1. Find the value of the dividends at the end of each year during the initial growth period.
2. Find the present values of the dividends found in step 1.
3. Find the value of the stock at the end of the initial growth period
4. Add the present value of the dividends found in step 2 and the present value of the value of the stock found in step 3 to determine the value of the stock at time zero, i.e. P_0 .

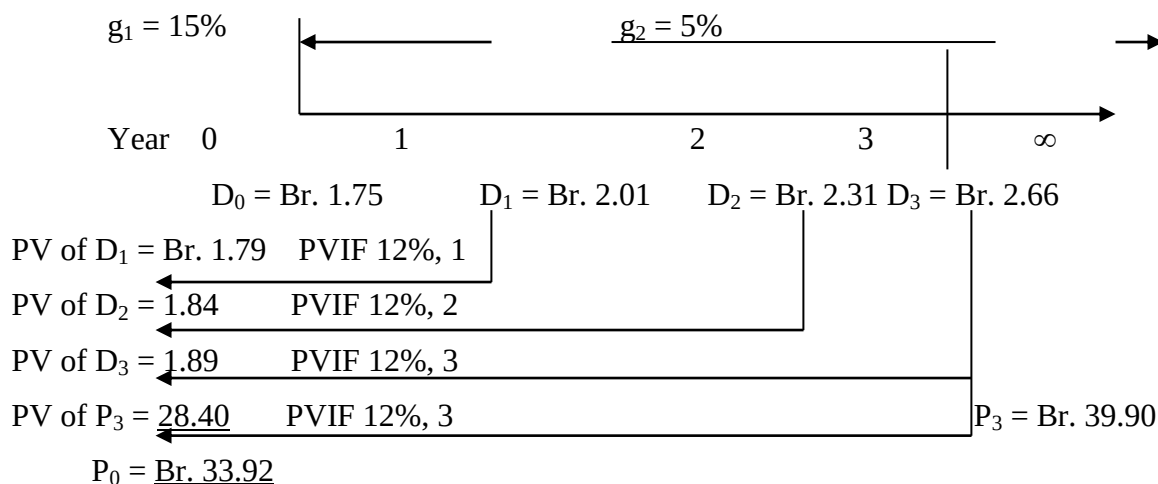
Example: Addis Company's most recent annual dividend, which was paid yesterday, was Br. 1.75 per share. The dividends are expected to experience a 15% annual growth rate for the next 3 years. By the end of 3 years growth rate will slow to 5% per year to infinity.

Stockholders require a return of 12% on Addis' stock

Required: Calculate the value of the stock today.

Solution:

Given: $D_0 = \text{Br. } 1.75$; $g_1 = 15\%$ for 3 years; $g_2 = 5\%$ from year 3 to infinity; $k_s = 12\%$; $p_0 = ?$



$$D_1 = D_0 (1 + g_1) = \text{Br. } 1.75 (1.15) = \text{Br. } 2.01$$

$$D_2 = D_1 (1 + g_1) = \text{Br. } 2.01 (1.15) = \text{Br. } 2.31$$

$$D_3 = D_2 (1 + g_1) = \text{Br. } 2.31 (1.15) = \text{Br. } 2.66$$

$$P_3 = \frac{D_4}{k_s - g_2} = \frac{D_3(1 + g_2)}{k_s - g_2} = \frac{\text{Br. } 2.66(1.05)}{0.12 - 0.05} = \text{Br. } 39.90$$

Therefore, the value of Addis Company's common stock today is Br. 33.92



Self-check questions 4.2

1. You are considering the purchase of Zemen company common stock that paid dividend of Br. 2 yesterday. You expect the dividend to grow at the rate of 5% per year for the next 3 years, and, if you buy the stock, you plan to hold it for 3 years and then sell it. Calculate the value of the common stock if your required rate of return is 15%
2. Melat computers Incorporated is experiencing a period of rapid growth. Earnings and dividends are expected to grow at 15% rate during the next 2 years, at 13% in the third year, and at a constant rate of 6% thereafter. Melat's last dividends was Br. 1.15, and the required rate of return on the stock is 12%. Calculate the value of the stock today.
3. Can we compute the value of a common stock whose future dividends are expected to decline at a constant rate?

4.5 Capital Structure

Capital structure is defined as the relative amount of permanent short-term debt, long-term debt, preferred stock, and common equity used to finance a firm. The optimal capital structure occurs at the point at which the cost of capital is minimized and firm value is maximized.

Capital structure is defined as the amount of permanent short-term debt, long-term debt, preferred stock, and common equity used to finance a firm. In contrast, **financial structure** refers to the amount of total current liabilities, long-term debt, preferred stock, and common equity used to finance a firm. Thus, capital structure is part of the financial structure, representing the permanent sources of the firm's financing. The emphasis of capital structure analysis is on the firm's long-range **target capital structure**, that is, the capital structure at which the firm ultimately plans to operate. For most companies, the current and target capital structures are virtually identical, and calculating the target structure is a straightforward process. Occasionally, however, companies find it necessary to change from their current capital structure to a different target. The reasons for such a change may involve a change in the company's asset mix (and a resulting change in its risk) or an increase in competition that may imply more risk.

4.5.1 Assumptions of Capital Structure Analysis

The analysis that follows is based on some important assumptions. First, it is assumed that a firm's investment policy is held constant when we examine the effects of capital structure changes on firm value and particularly on the value of common stock. This assumption means that the level and variability of operating income (EBIT) is not expected to change as changes in capital structure are contemplated. Therefore, capital structure changes affect only the distribution of the operating income between the claims of debt holders, preferred stockholders, and common stockholders.

By assuming a constant investment policy, we also assume that the investments undertaken by the firm do not materially change the debt capacity of the firm. This assumption does not always hold in practice, but for the overwhelming majority of investment projects, it is a realistic assumption that also helps us focus on the key determinants of an optimal capital structure.

Two elements of risk are primary considerations in the capital structure decision: the business risk and the financial risk of a firm. Financial risk is discussed in the following section.

i) Business Risk

Business risk refers to the variability or uncertainty of a firm's operating income (EBIT). Many factors influence a firm's business risk (holding constant the effects of all other important factors), including

- ✓ *The variability of sales volumes over the business cycle.*
- ✓ *The variability of selling prices.*
- ✓ *The variability of costs*
- ✓ *Existence of market power*
- ✓ *The extent of product diversification*
- ✓ *The level and rate of growth*
- ✓ *The degree of operating leverage (DOL).*

ii) Financial Risk

Financial risk refers to the additional variability of earnings per share *and* the increased probability of insolvency that arises when a firm uses fixed-cost sources of funds, such as debt and preferred stock, in its capital structure.⁴ (Insolvency occurs when a firm is unable to meet contractual financial obligations—such as interest and principal payments on debt, payments on accounts payable, and income taxes—as they come due.) Fixed capital costs represent contractual obligations a company must meet regardless of the EBIT level.

The use of increasing amounts of debt and preferred stock raises the firm's fixed financial costs; this, in turn, increases the level of EBIT that the firm must earn in order to meet its financial obligations and remain in business. The reason a firm accepts the risk of fixed cost financing is to increase the possible returns to stockholders.

4.5.2 Capital Structure Theory

Capital Structure Without a Corporate Income Tax

In 1958, two prominent financial researchers, Franco Modigliani and Merton Miller (MM), showed that, under certain assumptions, a firm's overall cost of capital, and therefore its value, is *independent* of capital structure.⁸ In particular, assume that the following *perfect* capital market conditions exist

- ❖ There are no transaction costs for buying and selling securities.
- ❖ A sufficient number of buyers and sellers exists in the market, so no single investor can have a significant influence on security prices.

- ❖ Relevant information is readily available to all investors and is costless to obtain.
- ❖ All investors can borrow or lend at the same rate.

MM also assumed that all investors are rational and have homogeneous expectations of a firm's earnings. Additionally, firms operating under similar conditions are assumed to face the same degree of business risk. This assumption is called the *homogeneous risk class assumption*. Finally, MM assumed that there are no income taxes. MM later relaxed this no tax assumption. The results of the tax case follow after the no-tax case discussion.

In the no-tax MM case, the cost of debt and the overall cost of capital are constant regardless of a firm's financial leverage position, measured as the firm's debt-to-equity ratio, B/E . As a firm increases its relative debt level, the cost of equity capital, k_e , increases, reflecting the higher return requirement of stockholders due to the increased risk imposed by the additional debt. The increased cost of equity capital exactly offsets the benefit of the lower cost of debt, k_d , so that the overall cost of capital does not change with changes in capital structure.

MM support their theory by arguing that a process of arbitrage will prevent otherwise equivalent firms from having different market values simply because of capital structure differences.

Arbitrage is the process of simultaneously buying and selling the same or equivalent securities in different markets to take advantage of price differences and make a profit. Arbitrage transactions are risk-free. For example, suppose two firms in the same industry differed only in that one was *levered* (that is, it had some debt in its capital structure) and the other was *unlevered* (that is, it had no debt in its capital structure). If the MM theory did not hold, the unlevered firm could increase its market value by simply adding debt to its capital structure. However, in a perfect capital market without transactions costs, MM argue that investors would not reward the firm for increasing its debt. Stockholders could change *their* own financial debt–equity structure without cost to receive an equal return. Therefore, stockholders would not increase their opinion of the market value of an unlevered firm just because it took on some debt.

The MM argument is based on the *arbitrage* process. If one of two unlevered firms with identical business risk took on some debt and the MM theory did not hold, its value should increase and, therefore, so would the value of its stock. MM suggest that under these circumstances, investors will sell the overpriced stock of the levered firm. They then can use an arbitrage process of borrowing, buying the unlevered firm's stock, and investing the excess funds elsewhere. Through these costless

transactions, investors can increase their return without increasing their risk. Hence, they have substituted their own personal financial leverage for corporate leverage. MM argue that this arbitrage process will continue until the selling of the levered firm's stock drives down its price to the point where it is equal to the unlevered firm's stock price, which has been driven up due to increased buying.

The arbitrage process occurs so rapidly that the market values of the levered and unlevered firms are equal. Therefore, MM conclude that *the market value of a firm is independent of its capital structure in perfect capital markets with no income taxes.*

Four primary factors influence the capital structure decision

- A. Business risk**, or the riskiness inherent in the firm's operations if it used no debt. The greater the firm's business risk, the lower its optimal debt ratio.
- B. The firm's tax position.** A major reason for using debt is that interest is deductible, which lowers the effective cost of debt. However, if much of a firm's income is already sheltered from taxes by depreciation tax shields or tax loss carry-forwards, its tax ratio will be low, so debt will not be as advantageous as it would be to a firm with a higher effective tax rate.
- C. Financial flexibility, or the ability to raise capital on reasonable terms under adverse conditions.** Corporate treasurers know that a steady supply of capital is necessary for stable operations, which is vital for long-run success. They also know that when money is tight in the economy, or when a firm is experiencing operating difficulties, suppliers of capital prefer to provide funds to companies with strong balance sheet. Therefore, both the potential further need for funds and the consequences of a fund's shortage have a major influence on the target capital structure – the greater the probable future need for capital, and the worse the consequences of a capital budget, the stronger the balance sheet should be.
- D. Managerial conservatism or aggressiveness.** Some managers are more aggressive than others, hence some firms are more inclined to use debt in an effort to boost profits. This factor does not affect, the optimal, or value- maximizing, capital structure, but it does influence the target capital structure.

These four points largely determine the target capital structure, but operating conditions can cause the actual capital structure to vary from the target.

4.6 Cost of Capital

The cost of capital is the minimum rate of return that a firm must earn in order to satisfy the overall rate of return required by its investors. It is also the minimum rate of return a firm must earn on its invested capital to maintain the value of the firm unchanged. The second definition considers the cost of capital as a break-even rate.

If a firm's actual rate of return exceeds its cost of capital, the value of the firm would increase. If on the other hand, the cost of capital is not earned, the firm's market value will decrease. So, the cost of capital is the rate of return that is just sufficient to leave the price of the firm's common stock unchanged. The cost of capital serves as a discount rate when a firm evaluates an investment proposal.

4.6.1 Measuring the Specific Cost of Capital

The cost of capital for any particular capital source or security issue is called the specific cost of capital. It is also called individual cost of capital or component cost of capital. Each type of capital contained the capital structure of a firm include:

1. Debt
2. Preferred stock
3. Common stock
4. Retained earnings

Two important points you should bear in mind about the specific cost of capital. One is that it is computed on an after-tax basis. Meaning, if there would be any tax implication on the individual source of capital, it should be considered. In almost all circumstances, the tax implication is only on debt sources of finance. The second point is that the specific cost of capital is expressed as an annual percentage or rate like 6%, 9%, or 10%. The cost of capital is not stated in terms of birrs.

4.6.1.1 The cost of debt

This is the minimum rate of return required by suppliers of debt. The relevant specific cost of debt is the after-tax cost of new debt. Generally, debt is the cheapest source of finance to a firm and, hence, the cost of debt is the lowest specific cost of capital. There are two basic explanations for this. First, debt suppliers, generally, assume the lowest risk among all suppliers of capital. They receive interest payments before preferred and common dividends are paid. Since they assume the smallest risk, their

return is the lowest. Their lowest return would be the lowest cost of capital to the firm. Second, raising capital through debt sources entails interest expense. The interest expense in turn reduces the firm's income which ultimately would cause tax payment to be reduced. So, raising money in the form of debt results in the smallest tax burden, and finally, the firm's cost of debt would be the lowest.

Debt sources of finance may take several forms like bonds, promissory notes, bank loans. Here, for our convenience we consider bond issue to illustrate the cost of debt.

Computing the cost of new bond issue involves three steps:

1. Determine the net proceeds from the sale of each bond

$$NPd = Pd - f$$

Where:

NPd = The net proceeds from the sale of each bond

Pd = The market price of the bond

f = Flotation costs

2. Compute the effective before tax cost of the bond using the following approximation formula:

$$Kd = \frac{I + \frac{M - Bo}{n}}{\frac{M + Bo}{2}}$$

Where:

Kd = The effective before tax cost of debt

I = Annual interest payment

Pn = The par value of the bond

n = Length of the holding period of the bond in years.

3. Compute the after-tax cost of debt

$$Kdt = Kd (1 - t)$$

Where:

Kdt = The after-tax cost of debt

t = The marginal tax rate

Example: Currently, Abyssinia Industrial Group is planning to sell 15-year, Br. 1,000 par-value bonds that carry a 12% annual coupon interest rate. As a result of lower current interest rates, Abyssinia bonds can be sold for Br. 1,010 each. Flotation costs of Br. 30 per bond will be incurred in the process of issuing the bonds. The firm's marginal tax rate is 40%.

Required: Calculate the after-tax cost of Abyssinia's new bond issue:

Solution:

Given: $P_n = \text{Br. } 1,000$; $I = \text{Br. } 120$ ($\text{Br. } 1,000 \times 12\%$); $n = 15$; $P_d = \text{Br. } 1,010$; $f = \text{Br. } 30$;

$t = 40\%$; $K_{dt} = ?$

Then apply the three steps:

$$1. \text{ NPd} = \text{Br. } 1,010 - \text{Br. } 30 = \text{Br. } 980$$

$$2. \text{ Kd} = \frac{\text{Br. } 120 + \frac{\text{Br. } 1,000 - \text{Br. } 980}{15}}{\frac{\text{Br. } 1,000 + \text{Br. } 980}{2}} = 12.26\%$$

$$3. \text{ Kdt} = 12.26\% (1 - 40\%) = \underline{7.36\%}$$

Therefore, the after – tax cost of Abyssinia's new bond issue is 7.36%. That is, Abyssinia should be able to earn a minimum of 7.36% to satisfy bondholders. Otherwise, the firm's value will decline.

4.6.1.2 The cost of preferred stock

The cost of preferred stock is the minimum rate of return a firm must earn in order to satisfy the required rate of return of the firm's preferred stock investors. It is also the minimum rate of return a firm's preferred stock investors require if they are to purchase the firm's preferred stock.

When a firm raises capital by issuing new preferred stock, it is expected to pay fixed number of dividends to the preferred stockholders. So, it is the dividend payment that is the cost of the preferred stock to the firm stated as an annual rate.

The cost of a new preferred stock issue can be computed by following two steps:

1. Determine the net proceeds from the sale of each preferred stock.

$$\text{NPpf} = \text{Ppf} - f$$

Where:

NPpf = Net proceeds from the sale of each preferred stock

Ppf = Market price of the preferred stock

f = Flotation costs

2. Compute the cost of preferred stock issue

$$Kps = \frac{Dps}{NPpf}$$

Where:

Kps = The cost of preferred stock

DPs = The pre share annual dividend on the preferred stock

Example: Sefa Computer Systems Company has just issued preferred stock. The stock has 12% annual dividend and Br. 100 par value and was sold at 102% of the par value. In addition, flotation costs of Br. 2.50 per share must be paid. Calculate the cost of the preferred stock.

Solution:

Given: Pps = Br. 102 (Br. 100 x 102%); Dps = Br. 12 (Br 100 x 12%); f = Br. 2.50;

Kps =?

Then apply the two steps:

i) $NPpf = Br. 102 - Br. 2.50 = Br. 99.50$

ii) $Kps = \frac{Br. 12}{Br. 99.50} = 12.06\%$

Therefore, Sefa Company should be able to earn a minimum of 12.06% on any investment financed by the new preferred stock issue. Otherwise, the firm's value will decrease.

4.6.1.3 The cost of common stock

The cost of common stock is the minimum rate of return that a firm must earn for its common stockholders in order to maintain the value of the firm. A firm does not make explicit commitment to pay dividends to common stockholders. However, when common stockholders invest their money in a corporation, they expect returns in the form of dividends. Therefore, common stocks implicitly involve a return in terms of the dividends expected by investors and hence, they carry cost.

Generally, common stock dividends are paid after interest and preferred dividends are paid. As a result, common stock investors assume the maximum risk in corporate investment. They compensate

the maximum risk by requiring the highest return. This highest return expected by common stockholders make common stock the most expensive source of capital.

The cost of common stock can be computed using the constant growth valuation model.

$$K_s = \frac{D_1}{NPo} + g$$

NPo

Where:

K_s = The cost of new common stock issue

D_1 = The expected dividend payment at the end of next year

NPo = Net proceeds from the sale of each common stock

g = The expected annual dividends growth rate

The net proceeds from the sale of each common stock (NPo) is computed as follows:

$$NPo = Po - f$$

Where:

Po = The current market price of the common stock

f = flotation costs

Example: An issue of common stock is sold to investors for Br. 20 per share. The issuing corporation incurs a selling expense of Br. 1 per share. The current dividend is Br. 1.50 per share and it is expected to grow at 6% annual rate. Compute the specific cost of this common stock issue.

Solution

Given: $Po = \text{Br. } 20$; $Do = \text{Br. } 1.50$; $g = 6\%$; $f = \text{Br. } 1$; $K_s = ?$

Then apply the two steps:

1. $NPo = \text{Br. } 20 - \text{Br. } 1 = \text{Br. } 19$

2. $K_s = \frac{D_1}{Npo} + g = \frac{\text{Br. } 1.50 (1.06)}{\text{Br. } 19} = \underline{14.37\%}$

Therefore, the firm should be able to earn a minimum return of 14.37% on investments that are financed by the new common stock issue.

4.6.1.4 The Cost of Retained Earnings

Retained earnings represent profits available for common stockholders that the corporation chooses to reinvest in itself rather than payout as dividends. Retained earnings are not securities like stocks and bonds and hence do not have market price that can be used to compute costs of capital.

The cost of retained earnings is the rate of return a corporation's common stockholders expect the corporation to earn on their reinvested earnings, at least equal to the rate earned on the outstanding common stock. Therefore, the specific cost of capital of retained earnings is equated with the specific cost of common stock. However, flotation costs are not involved in the case of retained earnings.

Computing the cost of retained earnings involves just a single procedure of applying the following formula:

$$K_r = \frac{D_1}{P_0} + g$$

Where:

K_r = The cost of retained earnings

D_1 = The expected dividends payment at the end of next year

P_0 = The current market price of the firm's common stock

g = The expected annual dividend growth rate.

Example: Zeila Auto Spare Parts Manufacturing company expects to pay a common stock dividend of Br. 2.50 per share during the next 12 months. The firm's current common stock price is Br. 50 per share and the expected dividend growth rate is 7%. A flotation cost of Br. 3 is involved to sale a share of common stock.

Required: Compute the cost of retained earnings

Solution

Given: P_0 = Br. 50; D_1 = Br. 2.50; g = 7%; K_r = ?

Then apply the formula:

$$K_r = \frac{D_1}{P_0} + g = \frac{\text{Br. 2.50}}{\text{Br. 50}} + 7\% = 12\%$$



Self-check questions 4.3

1. Ayenew Company's financing plans for next year include the sale of long-term bonds with a 10% coupon. The company believes it can sell the bonds at a price that will provide a yield to maturity of 12% to investors. If its marginal tax rate is 35%, what is Ayenew's after-tax cost of debt?
2. Sattelite Share Company plans to sale preferred stock with par value of Br. 50 per share. The issue is expected to pay quarterly dividends of Br. 1.25 per share and to have flotation costs of 6% of the par value. The preferred stock sells at 95% of its par.

Required: Calculate the cost of preferred stock to Satellite Share Company.

3. ABC Corporation's share of common stock is currently selling at Br. 75. The firm's projected dividend per share during the next year is Br. 3.38 and the expected dividend growth rate is 8%. Because of competitive nature of the market a Br. 3 per share underpricing is necessary. In addition, the sale of new common stock involves underwriting fee of Br. 0.60 per share and other flotation costs of Br. 0.90 per share.

Required: Calculate the cost of common stock for Repentance Corporation.

4. Zequala Textiles Share Company wishes to measure its cost of retained earnings. The firm's stock is currently selling for Br. 57.50. The firm expects to pay Br. 3.40 dividend at the end of the year. The expected dividend growth rate is 8%.

Required: Determine the cost of retained earnings.

The firm's capital structure is composed of debt, preferred stock, common stock, and retained earnings. Each capital source accounts to some portion of the total finance. But the percentage contribution of one source is usually different from another. So, we must compute the weighted average cost of capital rather than the simple average.

The weighted average cost of capital (WACC) is the weighted average of the individual costs of debt, preferred stock and common equity (common stock and retained earnings). It is also called the composite cost of capital.

If the weights of the component capital sources are all given, the weighted average cost of capital can be computed as:

$$WACC = W_d K_d + W_{ps} K_{ps} + W_{ce} K_s$$

Where:

WACC = The weighted average cost of capital

Wd = The weight of debt

Wps = The weight of preferred stock

Wce = The weight of common equity

Kdt = The after – tax cost of debt

Kps = The cost of preferred stock

Ks = The cost of common equity

The WACC is found by weighting the cost of each specific type of capital by its proportion in the firm's capital structure. Weights of the individual capital sources can be calculated based on their book value or market value.

To illustrate the computation of the WACC, look at the following example.

Muna Tools Manufacturing Company's financial manager wants to compute the firm's weighted average cost of capital. The book and market values of the amounts as well as specific after-tax costs are shown in the following table for each source of capital.

Source of capital	Book value	Market value	Specific cost
Debt	Br. 1,050,000	Br. 1,000,000	5.3%
Preferred stock	84,000	125,000	12.0
Common equity	966,000	1,375,000	16.0
Total	<u>Br. 2,100,000</u>	<u>Br. 2,500,000</u>	

Required: Calculate the firm's weighted average cost of capital using:

- book value weights
- market value weights

Solution:

a. Total book value = Br. 2,100,000

Wd = $\frac{\text{Br. 1,050,000}}{\text{Br. 2,100,000}} = 0.5$; Wps = $\frac{\text{Br. 84,000}}{\text{Br. 2,100,000}} = 0.04$; Wce = $\frac{\text{Br. 966,000}}{\text{Br. 2,100,000}} = 0.46$

Br. 2,100,000

Br. 2,100,000

Br. 2,100,000

$$\begin{aligned} \text{WACC} &= W_d K_d + W_p K_p + W_c K_c \\ &= 0.5 (5.3\%) + 0.04 (12.0\%) + 0.46 (16.0\%) \\ &= 2.65\% + 0.48\% + 7.36\% \\ &= \underline{10.49\%} \end{aligned}$$

The minimum rate of return on all projects should be 10.49%. Meaning, Muna should accept all projects so long as they earn a return greater than or equal to 10.49%

b. Total Market value = Br. 2,500,000

$$W_d = \frac{\text{Br. 1,000,000}}{\text{Br. 2,500,000}} = 0.4; W_p = \frac{\text{Br. 125,000}}{\text{Br. 2,500,000}} = 0.05; W_c = \frac{\text{Br. 1,375,000}}{\text{Br. 2,500,000}} = 0.55$$

Br. 2,500,000

Br. 2,500,000

Br. 2,500,000

$$\begin{aligned} \text{WACC} &= 0.4 (5.3\%) + 0.05 (12.0\%) + 0.55 (16.0\%) \\ &= 2.12\% + 0.60\% + 8.80\% \\ &= \underline{11.52\%} \end{aligned}$$

If the market value weights are used, Muna should accept all projects with a minimum rate of return of 11.52%



Self-check questions 4.4

1. On January 1, 2002, the total assets of Zway share company were Br. 54 million. There was no short-term debt. The firm's optimal capital structure is given below.

Long-term debt

Br. 27,000,000

Common equity

27,000,000

Total liabilities and equity

Br. 54,000,000

New bonds will have a 10% coupon rate and will be sold at Par. Common stock currently has a market price of Br. 60 and can be sold with a flotation cost of Br. 6 per share. Dividend yield is estimated to be 4% and the expected dividend growth rate is 8%

Required: Calculate:

- A. The cost of debt assuming a 40% marginal corporate tax rate
- B. The cost of common equity (50% common stock and 50% retained earnings)
- C. The weighted average cost of capital

4.8 Marginal Cost of Capital (MCC)

As a firm tries to have more new capital, the cost of each birr will rise at some point. Thus, the marginal cost of capital (MCC) is the cost of obtaining additional new capital. Technically speaking, the MCC is the weighted average cost of the last birr of new capital obtained. So, the concept of marginal cost of capital is discussed in the context of the weighted average cost of capital.

As a firm raises larger and larger amounts of capital, the weighted average cost of capital also rises. But the question would be at what point the firm's costs of debt, preferred stock, and common equity as well as WACC increase?

The first point, therefore, in computing the MCC is to determine the breaking points where the cost of capital will increase.

The technical aspects of the MCC can be better understood using an example.

Example: The target capital structure of Shala Corporation and other pertinent data are given below.

Long-term debt -----	40%	cost of preferred stock (Kps) = 12.06%
Preferred stock -----	10%	cost of retained earnings (Kr) = 14%
Common equity -----	50%	cost of common stock (Ks) = 15%

Shala Corporation has Br. 900,000 available retained earnings. But when the firm fully utilizes its retained earnings, it must use the more expensive new common stock financing to meet its equity needs. In addition, the firm expects that it can borrow up to Br. 1,200,000 of debt at 7.3% after-tax cost. Additional debt will have an after-tax cost of 9.1%.

Required

1. What is the breaking point associated with the;
 - a. Exhausting of retained earnings?
 - b. Increment of debt between Br. 0 to Br. 1,200,000?
2. Determine the ranges of total new financing where the WACC will rise
3. Calculate the WACC for each range of finance.

Solutions

- 1) a. Breaking point (BP) common equity = $\frac{\text{Br. } 900,000}{50\%} = \text{Br. } 1,800,000$

$$\text{b. Breaking point (BP) long-term debt} = \frac{\text{Br. } 1,200,000}{40\%} = \text{Br. } 3,000,000$$

The breaking points computed above can be interpreted as:

Shala can meet its equity needs using retained earnings until its total finance need is Br. 1,800,000. But when total capital required is more than Br. 1,800,000, its equity needs should be met with common stock. Similarly, until the firm's total finance need reaches Br. 3,000,000, shala can raise any debt at 7.3% cost. Any further finance need beyond Br. 3,000,000 will cause the cost of debt to rise to 9.1%.

2) There are three ranges of finance that could be identified on the basis of the breaking points:

1st Range: Br. 0 to Br. 1,800,000,

2nd Range: Br. 1,800,000 to Br. 3,000,000, and

3rd Range: Br. 3,000,000 and above

$$\begin{aligned} 3) \text{ WACC (1}^{\text{st}} \text{ range)} &= 0.40 (7.3\%) + 0.10 (12.06\%) + 0.50 (14\%) \\ &= 2.92\% + 1.21\% + 7.00\% \\ &= \underline{11.13\%} \end{aligned}$$

$$\begin{aligned} \text{WACC (2}^{\text{nd}} \text{ range)} &= 0.40 (7.3\%) + 0.10 (12.06\%) + 0.50 (15\%) \\ &= 2.92\% + 1.21\% + 7.50\% \\ &= \underline{11.63\%} \end{aligned}$$

$$\begin{aligned} \text{WACC (3}^{\text{rd}} \text{ range)} &= 0.40 (9.1\%) + 0.10 (12.06\%) + 0.50 (15\%) \\ &= 3.64\% + 1.21\% + 7.50\% \\ &= \underline{12.35\%} \end{aligned}$$

Chapter five

Capital budgeting and investment decisions

5.1 Introduction

Dear learner! This chapter deals with the financial management of the assets on a firm 's balance sheet and looks at some widely used capital budgeting decision models, discussing and illustrating their relative strengths and weaknesses. The cash flow procedure, capital budgeting criteria and the time value of money procedures which was developed in Chapter 3 provide the basis for making capital expenditure decisions

Capital budgeting is the process of planning for purchases of assets whose returns are expected to continue beyond one year (beyond one operating period). A capital expenditure is a cash outlay that is expected to generate a flow of future cash benefits lasting longer than one year. It is distinguished from a normal operating expenditure, which is expected to result in cash benefits during the coming 1-year period (during the coming operating period). Capital expenditures are important to a firm both because they require sizable cash outlays and because they have a long-range impact on the firm's performance.

A firm's capital expenditures affect its future profitability and, when taken together, essentially plot the company's future direction by determining which products will be produced, which markets will be entered, where production facilities will be located, and what type of technology will be used. Capital expenditure decision making is important for another reason as well. Specifically, it is often difficult, if not impossible, to reverse a major capital expenditure without incurring considerable additional expense. For these reasons, a firm's management should establish a number of definite procedures to follow when analyzing capital expenditure projects. Choosing from among such projects is the objective of capital budgeting models.

5.2. Importance of Capital Budgeting Decisions

A. Long-term Implications of Capital Budgeting

A capital budgeting decision has its effect over a long-time span and inevitably affects the company's future cost structure and growth. A wrong decision can prove disastrous for the long-term survival of firm. On the other hand, lack of investment in asset would influence the competitive position of the firm. So, the capital budgeting decisions determine the future destiny of the company.

B. Involvement of large amount of funds in Capital Budgeting

Capital budgeting decisions need substantial amount of capital outlay. This underlines the need for thoughtful, wise and correct decisions as an incorrect decision would not only result in losses but also prevent the firm from earning profit from other investments which could not be undertaken.

C. Irreversible decisions in Capital Budgeting

Capital budgeting decisions in most of the cases are irreversible because it is difficult to find a market for such assets. The only way out will be scrapping the capital assets so acquired and incur heavy losses.

D. Risk and uncertainty in Capital budgeting

Capital budgeting decision is surrounded by great number of uncertainties. Investment is present and investment is future. The future is uncertain and full of risks. Longer the period of project, greater may be the risk and uncertainty. The estimates about cost, revenues and profits may not come true.

E. Difficult to make decision in Capital budgeting

Capital budgeting decision making is a difficult and complicated exercise for the management. These decisions require an overall assessment of future events which are uncertain. It is really a marathon

job to estimate the future benefits and cost correctly in quantitative terms subject to the uncertainties caused by economic-political social and technological factors.

F. Large and Heavy Investment

The proper planning of investments is necessary since all the proposals are requiring large and heavy investment. Most of the companies are taking decisions with great care because of finance as key factor.

G. Long term Effect on Profitability

Capital expenditures have great impact on business profitability in the long run. If the expenditures are incurred only after preparing capital budget properly, there is a possibility of increasing profitability of the firm.

H. Complicacies of Investment Decisions

Generally, the long-term investment proposals have more complicated in nature. Moreover, purchase of fixed assets is a continuous process. Hence, the management should understand the complexities connected with each project.

5.4 Classifications of Projects

A firm usually encounters several different types of projects when making capital expenditure decisions such as;

Independent Projects: an independent project is one whose acceptance or rejection does not directly eliminate other projects from consideration. In the absence of a constraint on the availability of funds, more than one project could be adopted if they meet minimum investment criteria.

Mutually Exclusive Projects: a mutually exclusive project is one whose acceptance excludes the acceptance of one or more alternative proposals. Because two mutually exclusive projects have the capacity to perform the same function for a firm, only one should be chosen.

Contingent Projects: a contingent project is one whose acceptance is dependent on the adoption of one or more other projects. When a firm is considering contingent projects, it is best to consider together all projects that are dependent on one another and treat them as a single project for purposes of evaluation.

Basic Framework for Capital Budgeting

Briefly, the capital investment decision process consists of four important steps, actually the last step will not be discussed here:

- i. Generating capital investment project proposals
- ii. Estimating cash flows
- iii. Evaluating alternatives and selecting projects to be implemented
- iv. Reviewing a projects performance after it has been implemented, and post-auditing its performance after its termination

i. Generating Capital Investment Project Proposals

Ideas for new capital investments can come from many sources, both inside and outside a firm. Proposals may originate at all levels of the organization-from factory workers up to the board of directors. Most large and medium-size firms allocate the responsibility for identifying and analyzing capital expenditures to specific staff groups. These groups can include cost accounting, industrial engineering, marketing research, research and development, and corporate planning. In most firms, systematic procedures are established to assist in the search and analysis steps. For example, many firms provide detailed forms that the originator of a capital expenditure proposal must complete. These forms normally request information on the projects initial cost, the revenues it is expected to generate, and how it will affect the firm's overall operating expenses. These data are then channeled to a reviewer or group of reviewers at a higher level in the firm for analysis and possible acceptance or rejection. Where a proposal goes for review often depends on how the particular project is classified. The capital expenditures noted above can be regrouped as projects generated by growth opportunities, projects generated by cost reduction opportunities, and projects generated to meet legal requirements and health and safety standards.

ii. Estimating Cash Flows

The capital budgeting process is concerned primarily with the estimation of the cash flows associated with a project, not just the project's contribution to accounting profits. Typically, a capital expenditure requires an initial cash outflow, termed the net investment. Thus, it is important to measure a project's performance in terms of the net (operating) cash flows it is expected to generate over a number of future years.

Project's cash flows are expected to exhibit normal (conventional) pattern. If we represent the cash outflows with the minus "-" sign and the cash inflows with the plus "+" sign, then conventional cash

flow pattern under capital budgeting is defined as the time series of cash flows that contains only one change in sign. For example, if an investment alternative has one cash outflow followed by three cash inflows, it can be represented as -, +, +, + and is considered to exhibit conventional cash flow patterns. In the process of capital budgeting, therefore, we assume investment alternatives exhibit conventional cash flow patterns as indicated above. Evaluating an investment alternative that violates this assumption can become very difficult, if not impossible. Regardless of whether a project's cash flows are expected to be normal (conventional) or non-normal, certain basic principles should be applied during their estimation, including the following:

a. Cash flows should be measured on an incremental basis.

In other words, the cash flow stream for a particular project should be estimated from the perspective of how the entire cash flow stream of the firm will be affected if the project is adopted as compared with how the stream will be affected if the project is not adopted. Therefore, all changes in the firm's revenue stream, cost stream, and tax stream that would result from the acceptance of the project should be included in the analysis. In contrast, cash flows that would not be changed by the investment should be disregarded.

b. Cash flows should be measured on an after-tax basis.

Because the initial investment made on a project requires the outlay of after-tax cash dollars, the returns from the project should be measured in the same units, namely, after-tax cash flows.

c. Consider all indirect (Incidental) effects in the cash flow calculations.

In addition to the direct cash flows of the project, all its incidental effects on the rest of the firm must be considered. The project may enhance the profitability of some of the existing activities of the firm because it has a complementary relationship with them or it may detract from the profitability of some of the existing activities of the firm because it has a competitive relationship with them. Such incidental effects should be taken into account.

Examples:

Complementary Projects: Establishing Drug Store in a Hospital
Competitive Projects: Textile Factory plus Leather Processing Factory

d. Sunk costs should not be considered when evaluating a project

A sunk cost is an outlay that has already been made (or committed to be made). Because sunk costs cannot be recovered, they should not be considered in the decision to accept or reject a project. Sunk

costs cannot be recovered and hence, are not relevant or sunk costs are not differential, i.e. they do not vary among alternatives.

e. The value of resources used in a project should be measured in terms of their opportunity costs.

Opportunity costs of resources (assets) are the cash flows those resources could generate if they are not used in the project under consideration. The opportunity cost of a resource is the benefit forgone that would have been derived from it by putting it to its best alternative use. That is, the resources used by the project might have been rented out, sold, or required elsewhere in the firm. The opportunity cost is the best-forgone option that arises for using scarce resources.



Self-check questions 5.1

1. Describe independent projects, mutually exclusive projects and contingent projects.
2. Identify and describe the principles of cash flow estimation.

iii. Relevant Cash Flows

To evaluate investment opportunities, financial managers must determine the relevant cash flows associated with the project. These are the incremental cash outflows (investment) and inflows (return). The incremental cash flows represent the additional cash flows- outflows or inflows- expected to result from a proposed capital expenditure. The cash flows of any project may include three basic components: (1) an initial investment, (2) operating cash inflows, and (3) terminal cash flow. All projects- whether for expansion, replacement or renewal, or some other purpose-have the first two components. Some, however, lack the final component, terminal cash flow.

A) Net Investment (NINV)

The net investment (NINV) in a project is defined as the project's initial net cash outlay, that is, the outlay at time (period) 0. It is calculated using the steps on the following page.

The tax consequences can influence the net investment of a project. These tax effects are the treatment of gains and losses from asset sales in the case of replacement decisions. If a project generates additional revenues and the company extends credit to its customers, an additional initial investment in accounts receivable is required. Moreover, if additional inventories are necessary to

generate the increased revenues, then an additional initial investment in inventory is required, too. This increase in initial working capital-that is, cash, accounts receivable, and inventories-should be calculated net of any automatic increases in current liabilities, such as accounts payable or wages and taxes payable, that occur because of the project. As a general rule, replacement projects require little or no net working capital increase. Expansion projects, on the other hand, normally require investments in additional net working capital.

Steps involved in net investment (NINV) determination of a project

- Step 1. The new project cost plus any installation and shipping costs associated with acquiring the asset and putting it into service¹
- PLUS
- Step 2. Any increases in net working capital initially required as a result of the new Investment
- MINUS
- Step 3. The net proceeds from the sale of existing assets when the investment is a replacement decision²
- PLUS, or MINUS
- Step 4. The taxes associated with the sale of the existing assets and/or the purchase of the new assets³
- EQUALS
- The net investment (NINV).

¹The asset cost plus installation and shipping costs form the basis upon which depreciation is computed.

²This is normally computed as the actual salvage value of the asset being replaced less any costs associated with physically removing or selling it.

³These taxes include any taxes associated with disposal of the old asset. The total tax effect may be either positive or negative; this is why it is either added to or subtracted from the new project cost.

B) Net (Operating) Cash Flows

Capital investment projects are expected to generate after-tax cash flow streams after the initial net investment has been made. The process of estimating incremental cash flows associated with a specific project is an important part of the capital budgeting process. Capital budgeting is concerned primarily with the after-tax net (operating) cash flows (NCF) of a particular project, or change in cash inflows minus change in cash outflows. For any year during the life of a project, these may be defined as the change in operating earnings after taxes, ΔOEAT , plus the change in depreciation, ΔDep , minus the change in the net working capital investment required by the firm to support the project, NWC:

$$\text{NCF} = \Delta\text{OEAT} + \Delta\text{Dep} - \Delta\text{NWC} \dots \dots \dots \mathbf{5.1}$$

The term change (Δ) refers to the difference in cash (and noncash) flows with and without adoption of the investment project.

Operating earnings after tax, OEAT, differs from earnings after tax, EAT, because OEAT does not consider interest expenses in its calculation. Net cash flows, NCF, as used for capital budgeting purposes, do not normally consider financing charges, such as interest, because these financing charges will be reflected in the cost of capital that is used to discount a project's cash flows. To include financing charges in NCF would result in double-counting these costs.

In years when a firm must increase its investment in net working capital (NWC) associated with a particular project, this increased investment in NWC reduces NCF. Normally, however, at the end of the life of a project, the NWC investment accumulated over the life of the project is recovered (for example, as inventories are sold and accounts receivable are collected). Thus, NWC is negative (a reduction). Because of the minus sign before NWC in Equation 5.1, the effect of a decline in the net working capital investments is an increase in NCF.

Depreciation is the systematic allocation of the cost of an asset with an economic life in excess of one year. Although depreciation is not a cash flow, it does affect a firm's after-tax cash flows by reducing reported earnings and thereby reducing taxes paid by a firm. If a firm's depreciation increases in a particular year as a result of adopting a project, after-tax net cash flow in that year will increase, all other things being equal.

After-tax net cash flow also considers changes in a firm's investment in net working capital. Changes in net working capital can occur as part of the net investment at time 0 or at any time during the life of the project. If a firm increases its accounts receivable, for example, in a particular year without increasing its current liabilities as a result of adopting a specific project, after-tax net cash flow in that year will decrease, all other things being equal. On the other hand, a reduction in a firm's investment in net working capital during a given year results in an increase in the firm's NCF for that year.

From basic accounting definitions, the change in operating earnings after taxes (ΔOEAT) in Equation 5.1 is equal to the change in operating earnings before taxes (ΔOEBT) times $(1 - T)$ where T is the marginal corporate income tax rate. Furthermore, OEBT is equal to the change in revenues (ΔR) minus the change in operating costs (ΔO) minus the change in depreciation (ΔDep). Substituting these terms in Equation 5.1 yields the following definition of net cash flow:

$$\text{NCF} = (\Delta R - \Delta O - \Delta\text{Dep}) (1 - T) + \Delta\text{Dep} - \Delta\text{NWC} \dots \dots \dots \mathbf{5.2}$$

Equation 5.2 can be further extended into an operational definition of NCF by defining

ΔR as $R_w - R_{wo}$, ΔO as $O_w - O_{wo}$, and ΔDep as $\text{Dep}_w - \text{Dep}_{wo}$

Where;

R_{wo} = Revenues of the firm without the project

R_w = Revenues of the firm with the project

O_{wo} = Operating costs exclusive of depreciation for the firm without the project
 O_w = Operating costs exclusive of depreciation for the firm with the project
 Dep_{wo} = Depreciation charges for the firm without the project

Dep_w = Depreciation charges for the firm with the project

The definition given in Equation 5.2 can be rewritten as follows:

$$\text{NCF} = [(R_w - R_{wo}) - (O_w - O_{wo}) - (\text{Dep}_w - \text{Dep}_{wo})] (1-T) + (\text{Dep}_w - \text{Dep}_{wo}) - \text{NWC} \dots 5.3$$

In the final year of a project's economic life, Equation 5.2 must be modified to reflect recovery of the incremental after-tax salvage value of the asset(s).

Summary of Net Cash Flow Calculations and Equations

		Equation
Calculation	5.1 & 5.2	5.3
Change in revenue	ΔR	$(R_w - R_{wo})$
Less: Change in operating costs	$-\Delta O$	$-(O_w - O_{wo})$
Less: Change in depreciation	$-\Delta \text{Dep}$	$-(\text{Dep}_w - \text{Dep}_{wo})$
Equals: Change in operating earnings before taxes	ΔOEBT	ΔOEBT
Less: Taxes	$-T(\Delta \text{OEBT})$	$-T(\Delta \text{OEBT})$
Equals: Change in operating earnings after taxes	ΔOEAT	ΔOEAT
Plus: Change in depreciation	$+\Delta \text{Dep}$	$+(\text{Dep}_w - \text{Dep}_{wo})$
Less: Change in net working capital	$-\Delta \text{NWC}$	$-(\text{NWC}_w - \text{NWC}_{wo})$
<i>Equals: Net cash flow</i>	<i>NCF</i>	<i>NCF</i>

C) Terminal cash flow

When the project is terminated, any assets directly related to the project will be sold and any change in the net working capital (from the initial investment) will be recaptured. The general assumption is that the asset purchased for the project lasts exactly the same as the duration of the project. What if a firm decides to sell the asset before the asset reaches the end of its life span? In that case, the firm will

have to determine the difference between the asset's sale price and the then-existing undepreciated tax book value, and the firm will be taxed based on that difference. The following points give us detail information about after-tax salvage value and recovery of net working capital.

Recovery of after-tax salvage value

Whenever an asset that has been depreciated is sold, there is potential tax consequences that may affect the after-tax net proceeds received from the asset sale. These tax consequences are important when estimating the after-tax salvage value to be received at the end of the economic life of any project. As discussed earlier, the tax consequences of asset sales are also important when calculating the net investment required in a replacement investment project. There are four cases that need to be considered.

Case 1: Sale of an asset for its book value

Normally if an asset is sold for an amount exactly equal to its tax book value, there will be no tax consequences as there will be no gain or loss on the sale. Tax book value is equal to the installed cost of the asset minus accumulated tax depreciation.

Case 2: Sale of an asset for less than its book value

If a Company disposes of an asset for an amount less than the asset's book value, there is a loss on the sale and thus this loss may be treated as an operating loss or an offset to operating income. This operating loss effectively reduces the company's taxes by an amount equal to the loss times the company's marginal tax rate. For example, Arba Minch Textile sells for Br. 20,000 an asset having a tax book value of Br. 50,000; Arba Minch Textile incurs a Br. 30,000 pretax loss. Assume that the company's earnings before taxes is Br. 100,000 (before consideration of the operating loss from the disposal of the asset). Taxes on these earnings are Br. 100,000 times the company's marginal (40 percent) tax rate, or Br. 40,000. Because of the operating loss incurred by selling the asset for Br. 20,000, the company's taxable income is reduced to Br. 70,000 and the taxes decline to Br. 28,000 (40 percent of Br. 70,000). The Br. 12,000 difference in taxes is equal to the tax loss on the sale of the old asset times the company's marginal tax rate (Br. 30,000 X 40%).

Case 3: Sale of an Asset for More than Its Book Value but Less than Its Original Cost

In this case the amount equal to the book value may be treated as a tax-free cash flow while the remainder which is in excess of the book value may be taxed at the same rate as that applied to the

operating income. If Arba Minch Textile sells the asset for Br. 60,000, Br. 10,000 more than the current tax book value, Br. 50,000 of this amount constitutes a tax-free cash inflow, and the remaining Br. 10,000 is taxed as operating income. As a result, the firm's taxes increase by Br. 4,000, or the amount of the gain times the firm's marginal tax rate (Br. 10,000 X 40%).

Case 4: Sale of an Asset for More than Its Original Cost

In this case part of the gain may be treated as ordinary income and part may be treated as capital gain. The part treated as ordinary income may be equal to the difference between the original cost and the current tax book value. The capital gain portion may be the amount in excess of the original asset cost.

If Arba Minch Textile sells the asset for Br. 120,000 (assuming an original asset cost of Br. 110,000), part of the gain from the sale is treated as ordinary income and part is treated as a long-term capital gain. The gain receiving ordinary income treatment is equal to the difference between the original asset cost and the current tax book value, or Br. 60,000 (Br. 110,000 – Br. 50,000). The capital gain portion is the amount in excess of the original asset cost, or Br. 10,000. Both ordinary income and capital gains are taxed at the same corporate rate.

Recovery of Net Working Capital

At the end of a project's life, all net working capital additions required over the project's life are recovered—not just the initial net working capital outlay occurring at time 0. Hence, the total accumulated net working capital is normally recovered in the last year of the project. This decrease in net working capital in the last year of the project increases the net cash flow for that year, all other things being equal. Of course, no tax consequences are associated with the recovery of NWC.

Illustrative Example 1: Expansion Project

A project that requires a firm to invest funds in additional assets in order to increase sales (or reduce costs) is called an expansion project.

Suppose the TLC Yogurt Company has decided to capitalize on the exercise fad and plans to open an exercise facility in conjunction with its main yogurt and health foods store. To get the project under way, the company will rent additional space adjacent to its current store. The equipment required for the facility will cost \$50,000. Shipping and installation charges for the equipment are expected to total \$5,000. This equipment will be depreciated on a straight-line basis over its 5-year economic life

to an estimated salvage value of \$0. In order to open the exercise facility, TLC estimates that it will have to add about \$7,000 initially to its net working capital in the form of additional inventories of exercise supplies, cash, and accounts receivable for its exercise customers (less accounts payable).

During the first year of operations, TLC expects its total revenues (from yogurt sales and exercise services) to increase by \$50,000 above the level that would have prevailed without the exercise facility addition. These incremental revenues are expected to grow to \$60,000 in year 2, \$75,000 in year 3, decline to \$60,000 in year 4, and decline again to \$45,000 during the fifth and final year of the project's life. The company's incremental operating costs associated with the exercise facility, including the rental of the facility, are expected to total \$25,000 during the first year and increase at a rate of 6 percent per year over the 5-year project life. Depreciation will be \$11,000 per year (\$55,000 installed cost, assuming no salvage value, divided by the 5-year economic life). TLC has a marginal tax rate of 40 percent. In addition, TLC expects that it will have to add about \$5,000 per year to its net working capital in years 1, 2, and 3 and nothing in years 4 and 5. At the end of the project, the total accumulated net working capital required by the project will be recovered.

Required: Calculate the Net Investment

First, TLC must make a cash outlay of \$50,000 to pay for the facility equipment. In addition, it must pay \$5,000 in cash to cover the costs of shipping and installation of the equipment. Finally, TLC must invest \$7,000 in initial net working capital to get the project under way. The net investment (NINV) required at time 0 would be:

Net Investment Calculation for TLC	
Purchase price of exercise equipment	\$50,000
Plus, Shipping and installation	5,000
Plus, Initial net working capital required	<u>7,000</u>
Equals Net investment (NINV)	<u>\$62,000</u>

Calculate Annual Net Cash Flows

The first-year net cash flows can be computed by substituting $\Delta R = \$50,000$, $\Delta O = \$25,000$,

$\Delta \text{Dep} = \$11,000$, $T = 0.40$ and $\Delta \text{NWC} = \$5,000$ into Equation 5.2.

$$\text{NCF}_1 = (\$50,000 - \$25,000 - \$11,000)(1 - 0.40) + \$11,000 - \$5,000 = \$14,400$$

For the second year NCF calculation; $\Delta R = \$60,000$, $\Delta O = \$25,000(1 + 0.06)^1 = \$26,500$,

$\Delta\text{Dep} = \$11,000$, $T = 0.40$, and $\Delta\text{NWC} = \$5,000$,

$$\text{NCF}_2 = (\$60,000 - \$26,500 - \$11,000)(1 - 0.40) + \$11,000 - \$5,000 = \$19,500$$

For the third year NCF calculation; $\Delta R = \$75,000$, $\Delta O = \$25,000(1 + 0.06)^2 = \$28,090$,

$\Delta\text{Dep} = \$11,000$, $T = 0.40$, and $\Delta\text{NWC} = \$5,000$,

$$\text{NCF}_3 = (\$75,000 - \$28,090 - \$11,000)(1 - 0.40) + \$11,000 - \$5,000 = \$27,546$$

For the fourth year NCF calculation; $\Delta R = \$60,000$, $\Delta O = \$25,000(1 + 0.06)^3 = \$29,775$,

$\Delta\text{Dep} = \$11,000$, $T = 0.40$ and $\Delta\text{NWC} = \$5,000$

$$\text{NCF}_4 = (\$60,000 - \$29,775 - \$11,000)(1 - 0.40) + \$11,000 = \$22,535$$

Finally, in the fifth year, TLC will recover its working capital investment of \$22,000, i.e., \$7,000 (Year 0) and \$5,000 (Years 1, 2, and 3). Therefore, $\Delta\text{NWC} = -\$22,000$, $\Delta R = \$45,000$ and $\Delta O = \$25,000(1 + 0.06)^4 = \$31,562$, $\Delta\text{Dep} = \$11,000$, and $T = 0.40$

$$\text{NCF}_5 = (\$45,000 - \$31,562 - \$11,000)(1 - 0.40) + \$11,000 - (-\$22,000) = \$34,463$$

The calculations of annual cash flows are summarized in Table 5.1. Note that since the equipment was fully depreciated and had no salvage value at the end of the fifth year, the after-tax salvage value was \$0.

Estimated Cash Flows for TLC Investment Project

Year	Net Investment and Net Cash Flows	
0	-\$62,000	Net Investment
1	14,400	Net Cash Flows
2	19,500	
3	27,546	
4	22,535	
5	34,463	

Table 5.1 Calculation of Annual Net Cash Flows for TLC Exercise Facility					
	Year 1	Year 2	Year 3	Year 4	Year 5
Change in revenues (ΔR)	\$50,000	\$60,000	\$75,000	\$60,000	\$45,000
Minus Change in operating costs (ΔO)	25,000	26,500	28,090	29,775	31,562
Minus Change in depreciation (ΔDep)	<u>11,000</u>	<u>11,000</u>	<u>11,000</u>	<u>11,000</u>	<u>11,000</u>
Equals Change in operating earnings before tax (ΔOEBT)	\$14,000	\$22,500	\$35,910	\$19,225	\$2,438

Minus Change in taxes (40%) (T)	<u>5,600</u>	<u>9,000</u>	<u>14,364</u>	<u>7,690</u>	<u>975</u>
Equals Change in operating earnings after tax (Δ OEAT)	\$8,400	\$13,500	\$21,546	\$11,535	\$1,463
Plus, Change in depreciation (Δ Dep)	11,000	11,000	11,000	11,000	11,000
Minus Change in net working capital (Δ NWC)	5,000	5,000	5,000	0	— 22,000
Plus, After-tax salvage	—	—	—	—	0
Equals Net cash flow (NCF)	<u>\$14,400</u> <u>0</u>	<u>\$19,500</u> <u>0</u>	<u>\$27,546</u> <u>6</u>	<u>\$22,535</u> <u>5</u>	<u>\$34,463</u> <u>3</u>

Illustrative Example 2: Replacement Project

Replacement project involve retiring one asset and replacing it with a more efficient asset.

Suppose Briggs & Stratton purchased an automated drill press 10 years ago that had an estimated economic life of 20 years. The drill presses originally cost \$150,000 and has been fully depreciated, leaving a current book value of \$0. The actual market value of this drill press is \$40,000. The company is considering replacing the drill press with a new one costing \$190,000. Shipping and installation charges will add an additional \$10,000 to the cost. The new machine would be depreciated to zero on a straight-line basis. The new machine is expected to have a 10-year economic life, and its actual salvage value at the end of the 10-year period is estimated to be \$25,000. Briggs & Stratton's current marginal tax rate is 40 percent.

Suppose Briggs & Stratton expects annual revenues during the project's first year to increase from \$70,000 to \$85,000 if the new drill press is purchased. (This might occur because the new press is faster than the old one and can meet the increasing demands for more work.) After the first year, revenues from the new project are expected to increase at a rate of \$2,000 a year for the remainder of the project life.

Assume further that while the old drill press required two operators, the new drill press is more automated and needs only one, thereby reducing annual operating costs from \$40,000 to \$20,000 during the project's first year. After the first year, annual operating costs of the new drill press are expected to increase by \$1,000 a year over the remaining life of the project. The old machine is fully depreciated, whereas the new machine will be depreciated on a straight-line basis. The marginal tax rate of 40 percent applies. Assume also that the company's net working capital does not change as a result of replacing the drill press.

Required: a) Calculate the Net Investment

In this case, no initial incremental net working capital is required and the net proceeds received from the sale of the old drill press have to be adjusted for taxes. Because the old drill press is sold for \$40,000, the gain from this sale is treated as a recapture of depreciation and thus taxed as ordinary income.

Net Investment Calculation for Briggs & Stratton	
Cost of new drill press	\$190,000
Plus, Shipping and installation charges	<u>10,000</u>
Equals Installed cost	\$200,000
Plus, Increase in initial net working capital	0
Minus Proceeds from sale of old drill press	<u>40,000</u>
Equals Net investment before taxes	\$160,000
Plus, Tax on gain from sale of old drill press (40% X \$40,000)	<u>16,000</u>
Equals Net investment (NINV)	<u>\$176,000</u>

b) Calculate Annual Net Cash Flows

The first-year net cash flow resulting from the purchase of the new drill press can be computed by substituting $R_w = \$85,000$, $R_{wo} = \$70,000$, $O_w = \$20,000$, $O_{wo} = \$40,000$, $Dep_w = \$20,000$ (= $\$200,000/10$), $Dep_{wo} = \$0$, $T = 0.40$ and $NWC = \$0$ into Equation 5.3 as follows:

$$NCF1 = [(\$85,000 - \$70,000) - (\$20,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$29,000$$

For NCF2 calculation, $R_w = \$87,000$, $O_w = \$21,000$, therefore NCF2 would be

$$NCF2 = [(\$87,000 - \$70,000) - (\$21,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$29,600$$

$$NCF3 = [(\$89,000 - \$70,000) - (\$22,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$30,200$$

$$NCF4 = [(\$91,000 - \$70,000) - (\$23,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$30,800$$

$$NCF5 = [(\$93,000 - \$70,000) - (\$24,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$31,400$$

$$NCF6 = [(\$95,000 - \$70,000) - (\$25,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$32,200$$

$$\text{NCF7} = [(\$97,000 - \$70,000) - (\$26,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$32,600$$

$$\text{NCF8} = [(\$99,000 - \$70,000) - (\$27,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$33,200$$

$$\text{NCF9} = [(\$101,000 - \$70,000) - (\$28,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 = \$33,800$$

Finally, in year 10, the \$25,000 estimated salvage from the new drill press must be added along with its associated tax effects. This \$25,000 salvage is treated as ordinary income because it represents a recapture of depreciation for tax purposes. The incremental after-tax salvage value of the asset would be calculated using the following formula.

$$\text{Salvage value} - [(\text{salvage value} - \text{book value}) \times \text{tax rate}] = \$25,000 - [(\$25,000 - 0) \times 40\%] = \$15,000$$

$$\text{NCF10} = [(\$103,000 - \$70,000) - (\$29,000 - \$40,000) - (\$20,000 - \$0)] (1 - 0.4) + (\$20,000 - \$0) - \$0 + \$15,000 = \$49,400$$

Annual Net Cash Flow Worksheet for the Briggs & Stratton is available in Table A.

Summary Project Cash Flows for Briggs & Stratton

Year	Net Investment and Net Cash Flows
0	-\$176,000
1	29,000
2	29,600
3	30,200
4	30,800
5	31,400
6	32,000
7	32,600
8	33,200
9	33,800
10	49,400

Problems in Cash Flow Estimation

Because project cash flows occur in the future, there are varying degrees of uncertainty about the value of these flows. Therefore, it is difficult to predict the actual cash flows of a project. The capital budgeting process assumes that the decision maker is able to estimate cash flows accurately enough that these estimates can be used in project evaluation and selection. For this assumption to be realistic, a project proposal should be based on inputs from marketing managers regarding revenue estimates and inputs from the production and engineering staffs regarding costs and achievable levels of performance. Objective inputs from these sources can help reduce the uncertainty associated with cash flow estimation.

5.5 Decision Models for Evaluating Alternatives

This subsection of the chapter looks at some widely used capital budgeting decision models. When combined with the cash flow procedures developed earlier and the time value of money procedures developed in Chapter 3, these models provide the basis for making capital expenditure decisions. The long-term investment (capital budgeting) decision has a significant effect on the value of the firm. Here you will see various investment decision criteria in light of the wealth maximization objective of the firm.

Four criteria are commonly used for evaluating and selecting investment projects.

-  Net present value (NPV)
-  Profitability index (PI)
-  Internal rate of return (IRR)
-  Payback (PB) period

5.5.1 Net present value (NPV)

The net present value rule is the primary decision-making rule used throughout the practice of financial management. The net present value—that is, the present value of the expected future cash flows minus the initial outlay of an investment made by a firm represents the contribution of that investment to the value of the firm and, accordingly, to the wealth of the firm's shareholders. The net present value method is also sometimes called the discounted cash flow (DCF) technique. The cash flows are discounted at the firm's required rate of return; that is, its cost of capital. A firm's cost of capital is defined as its minimum acceptable rate of return for projects of average risk. The net present value of a project may be expressed as:

$$\mathbf{NPV = PVNCF - NINV \dots\dots\dots 5.4}$$

Where; NPV is the net present value;

PVNCF is the present value of net (operating) cash flows and

NINV, the net investment.

For a series of uneven net (operating) cash flows, the net present value of a project may be calculated as:

$$\mathbf{NPV = \sum_{t=1}^n \frac{NCF_t}{(1+k)^t} - NINV \dots\dots\dots 5.5}$$

OR

$$\mathbf{NPV = \sum_{t=1}^n NCF_t \times PVIF_{k,t} - NINV \dots\dots\dots 5.6}$$

Where; NCF_t is the net (operating) cash flow in year t, n is the expected project life (years),

k is the cost of capital, and

PVIF_{k,t} is the present value interest factor.

The net (operating) cash flow in the final year (n) of the project, NCF_n, includes any salvage value remaining at the end of the project's life.

If all the net (operating) cash flows are equal over the life of the project, that is, an annuity NCF = NCF₁ = NCF₂ = . . . = NCF_n, then Equation 5.7 can be expressed as follows:

$$\mathbf{NPV = NCF \times PVIFA_{k,n} - NINV \dots\dots\dots 5.7}$$

where PVIFA_{k,n} is the present value of annuity interest factor (in the third Table).

Alternatively, you can use present value of ordinary annuity formula for computing the NPV of a project with equal stream of cash flows using equation 5.8.

$$\mathbf{NPV = NCF \left[\frac{1 - \frac{1}{(1+k)^n}}{k} \right] - NINV \dots\dots\dots 5.8}$$

The annual net cash flows for normal projects are usually positive after the initial net investment. Occasionally, however, one or more of the expected net cash flows over the life of a project may be

negative. When this occurs, positive numbers are used for years having positive net cash flows (net inflows), and negative numbers are used for years having negative net cash flows (net outflows).

Illustrative Example 3

Suppose Ace Lumber is considering two projects, A and B, having net investments and net cash flows as shown below. Assume that cash flows are received at the end of each year and the cost of capital of 14%.

	Project A		Project B
	Net Cash Flow After Taxes		Net Cash Flow After Taxes
Year	NCF _t		NCF _t
1	\$12,500		\$5,000
2	12,500		10,000
3	12,500		15,000
4	12,500		15,000
5	12,500		25,000
6	12,500		30,000
	Net investment (NINV) = \$50,000		Net investment (NINV) = \$50,000

Required: Compute the NPV of the projects and make the appropriate decision

In project A the annual net cash flows are uniform so that you can use the ordinary annuity formula or Present Value of an Annuity Interest Factor (PVIFA) table (equation 5.7 or 5.8).

Present value of an annuity of \$12,500 for n = 6 years at k = 14%:

$$PVNCF = \$12,500 (PVIFA_{0.14, 6}) = \$12,500 (3.889) = \$ 48,613$$

$$\text{Less: Net investment (NINV)} \quad \underline{-\$50,000}$$

$$\text{Net present value (NPV)} \quad \underline{\underline{- \$ 1,387}}$$

OR

$$\begin{aligned}
 NPV &= \$12,500 \left[\frac{1 - \frac{1}{(1 + 0.14)^6}}{0.14} \right] - \$50,000 \\
 &= \$12,500 \left(1 - \frac{0.456}{0.14} \right) - \$50,000 \\
 &= \$12,500(3.889) - \$50,000 \\
 &= \$48,613 - \$50,000
 \end{aligned}$$

$$= -\$1,1387$$

In project B the annual net cash flows are uneven; therefore, you can't use both the ordinary annuity formula and Present Value of an Annuity Interest Factor (PVIFA) table. The Present Value of an uneven cash flow stream is found as the sum of the Present Values of the individual cash flows of the stream (equation 5.5 and 5.6).

Present value			
Year	NCF _t	PVIF _{0.14, t}	NCF _t X PVIF _{0.14, t}
1	\$5,000	0.877	\$4,385
2	10,000	0.769	7,690
3	15,000	0.675	10,125
4	15,000	0.592	8,880
5	25,000	0.519	12,975
6	30,000	0.456	<u>13,680</u>
PVNCF			\$57,735
Less: Net investment (NINV)			<u>50,000</u>
Net present value (NPV)			<u>\$7,735</u>

Decision Rule: In general, a project should be accepted if its net present value is greater than or equal to zero and rejected if its net present value is less than zero. This is so because a positive net present value in principle translates directly into increases in stock price and increases in shareholder's wealth. In the Ace Lumber example, Project A would be rejected because it has a negative net present value, and Project B would be accepted because it has a positive net present value. If two or more mutually exclusive investments have positive net present values, the project having the largest net present value is the one selected.

Advantages of Net Present Value (NPV)

- ✚ The net present value method is consistent with the goal of shareholder wealth maximization.
- ✚ The net present value approach considers both the magnitude and the timing of cash flows over a project's entire expected life.

- ✚ When the firm undertakes a new project, the firm's value is increased by the net present value of the new project; therefore, NPV follows value additivity principle.
- ✚ Indicates whether a proposed project will yield the rate of return required by the firm's investors which is represented by the cost of capital.

Disadvantages of Net Present Value (NPV)

- ✚ Does not give visibility into how long a project will take to generate a positive NPV i.e, NPV rule tells us to accept all investments where the NPV is greater than zero, however, the measure doesn't tell us when a positive NPV is achieved.
- ✚ The model assumes that capital is abundant - that is there is no capital rationing. If resources are scarce, then the analyst has to look carefully at not just the NPV for each project they are evaluating, but also the size of the investment itself.



Self-check questions 5.2

1. List the advantages and disadvantages of NPV model.
2. Assuming 15% as the cost of capital for ABC investment project determine the NPV of the firm.

5.5.2 Profitability index (PI)

The profitability index (PI), or benefit–cost ratio, is the ratio of the present value of expected net cash flows over the life of a project (PVNCF) to the net investment NINV. It is expressed as follows:

$$PI = \frac{PVNCF}{NINV}$$

Example: Assuming a 14 percent cost of capital compute the profitability index for Projects A and B in previous example and pass the appropriate accept-reject decision.

$$PI_A = \frac{\$48,613}{\$50,000} = 0.97$$

$$PI_B = \frac{\$57,735}{\$50,000} = 1.15$$

The profitability index is interpreted as the present value return for each dollar of initial investment. In comparison, the net present value approach measures the total present value dollar return.

Decision Rule: A project whose profitability index is greater than or equal to 1 is considered acceptable, whereas a project having a profitability index less than 1 is considered unacceptable. When a project has a profitability index equal to 1, the present value of the net cash flows is exactly equal to the net investment. Thus, the project has a net present value of zero, meaning that it is expected to earn the investor's required rate of return and nothing more.

In the case of Ace Lumber, Project B is acceptable, whereas Project A is not. When two or more independent projects with normal cash flows are considered, the profitability index, net present value, and internal rate of return approaches all will yield identical accept–reject signals.

When dealing with mutually exclusive investments, conflicts may arise between the net present value and the profitability index criteria. This is most likely to occur if the alternative projects require significantly different net investments.

Consider, for example, the following information on Projects J and K.

	Project J	Project K
Present value of net cash flows(A)	\$25,000	\$14,000
Less Net investment(B)	<u>20,000</u>	<u>10,000</u>
Net present value (NPV)	\$5,000	\$4,000
profitability index(A/B)	1.25	1.4

According to the net present value criterion, Project J would be preferred because of its larger net present value. According to the profitability index criterion, Project K would be preferred.

When a conflict arises, the final decision must be made on the basis of other factors. For example, if a firm has no constraint on the funds available to it for capital investment—that is, no capital rationing—the net present value approach is preferred because it will select the projects that are expected to generate the largest total dollar increase in the firm's wealth and, by extension, maximize shareholder wealth. If, however, the firm is in a capital rationing situation and capital budgeting is being done for only one period, the profitability index approach may be preferred because it will indicate which projects will maximize the returns per dollar of investment—an appropriate objective when a funds constraint exists. If the firm makes capital budgeting decisions for more than one period in the future, it is usually necessary to use some kind of programming model.

Advantages of profitability index (PI)

- ✚ Tells whether an investment increases the firm's value.
- ✚ Considers all cash flows of the project.
- ✚ Considers the time value of money.
- ✚ Considers the risk of future cash flows (through the cost of capital).
- ✚ Useful in ranking and selecting projects when capital is rationed.

Disadvantages of profitability index (PI)

- ✚ Requires an estimate of the cost of capital in order to calculate the profitability index.
- ✚ May not give the correct decision when used to compare mutually exclusive projects.



Self-check questions 5.3

1. When and why profitability index yields conflicting results between or among projects.
2. Assuming 15% as the cost of capital for ABC investment project determine the PI of the firm.

5.5.3 Internal Rate of Return

The internal rate of return is defined as the discount rate that equates the present value of the net cash flows from a project with the present value of the net investment or the internal rate of return is the discount rate that causes a project's net present value to equal zero. A project's internal rate of return can be determined by means of the following equation.

$$\sum_{t=1}^n \frac{NCF_t}{(1+r)^t} = NINV \quad \text{or} \quad \sum_{t=1}^n NCF_t \times PVIF_{r,t} = NINV \dots \dots \dots 5.9$$

Where;

$NCF_t / (1 + r)^t$ is the present value of net (operating) cash flows in period t discounted at the rate r,

NINV is the net investment in the project, r is the internal rate of return, and

$PVIF_{r,t}$ is the present value interest factor.

Subtracting the net investment, NINV, from both sides of Equation 5.9 yields the following:

$$\sum_{t=1}^n \frac{NCF_t}{(1+k)^t} - NINV = 0$$

This is essentially the same equation as that used in the net present value method (Equation 5.5).

The only difference is that in the net present value approach a discount rate, k , is specified and the net present value is computed, whereas in the internal rate of return method the discount rate, r , which causes the project net present value to equal zero, is the unknown.

If all the net (operating) cash flows are equal over the life of the project, that is, an annuity $NCF = NCF_1 = NCF_2 = \dots = NCF_n$, then Equation 5.10 can be expressed as follows:

$$NCF \times PVIFA_{r,n} = NINV \dots\dots\dots 5.10$$

Where; $PVIFA_{r,n}$ is the present value of an annuity factor

Decision Rule: Generally, the internal rate of return method indicates that a project whose internal rate of return is greater than or equal to the firm's cost of capital should be accepted, whereas a project whose internal rate of return is less than the firm's cost of capital should be rejected. The cost of capital, in the context of the IRR, is a hurdle rate- the minimum acceptable rate of return.

When two independent projects are considered under conditions of no capital rationing, the net present value and internal rate of return techniques result in the same accept– reject decision. When two or more mutually exclusive projects are being considered, it is generally preferable to accept the project having the highest internal rate of return as long as it is greater than or equal to the cost of capital.

Example: Compute the internal rate of return for Ace Lumber's Projects A and B.

Because Project A is an annuity, its internal rate of return may be computed directly with the aid of a PVIFA table. Substituting $NCF = \$12,500$, $NINV = \$50,000$, $n = 6$ into Equation 5.10 yields:

$$\$12,500 \times PVIFA_{r,6} = \$50,000,$$

$$PVIFA_{r,6} = \$50,000 \div \$12,500 = 4.000$$

Referring to Table C and reading across the table for $n = 6$, it can be seen that the interest factor of 4.000 occurs near 13 percent, where it is 3.998. Thus, the internal rate of return for Project A is about 13 percent.

The internal rate of return for Project B is more difficult to calculate because the project is expected to yield uneven cash flows. In this case, the internal rate of return is computed with the help of a financial calculator or through trial and error. In order to solve IRR with the trial and error process, we would “plug” in different estimates for the IRR, work through the calculations, and determine if we have found the rate that causes NPV to equal \$0.

$$-NINV + \frac{NCF_1}{(1 + IRR)^1} + \frac{NCF_2}{(1 + IRR)^2} + \cdots + \frac{CFN_n}{(1 + IRR)^n} = 0$$

The calculation of the project’s IRR does not depend upon the required rate of return. The IRR is compared to the required rate of return to determine whether to accept or reject the project. Also, if a project’s NPV is positive, its IRR will exceed the required rate of return. If a project’s NPV is negative, its IRR will be below the required rate of return. If the IRR to be calculated is the form of fraction or decimal linear interpolation formula can be used.

$$IRR = Base\ Factor + \frac{Positive\ NPV}{Difference\ between\ Positive\ and\ Negative\ NPVs} \times Difference\ in\ \%$$

Since Ace lumber’s project B NPV computed at 14% cost of capital is positive, the IRR exceeds the required rate of return. So, let us try at 18%

$$-\$50,000 + \frac{5,000}{(1.18)^1} + \frac{10,000}{(1.18)^2} + \frac{15,000}{(1.18)^3} + \frac{15,000}{(1.18)^4} + \frac{25,000}{(1.18)^5} + \frac{30,000}{(1.18)^6}$$

$$-\$50,000 + \$4,237.3 + \$7,181.8 + \$9,129.5 + \$7,736.8 + \$10,927.7 + \$11,112.9$$

$$NPV = -\$50,000 + \$50,326 = \$326$$

Still the NPV is positive so let us increase the rate further by 1%

$$-\$50,000 + \frac{5,000}{(1.19)^1} + \frac{10,000}{(1.19)^2} + \frac{15,000}{(1.19)^3} + \frac{15,000}{(1.19)^4} + \frac{25,000}{(1.19)^5} + \frac{30,000}{(1.19)^6}$$

$$-\$50,000 + \$4,201.7 + \$7,061.6 + \$8,901.2 + \$7,480.0 + \$10,476.2 + \$10,564.3$$

$$NPV = -\$50,000 + \$48,685 = -\$1,315$$

At 19% the computed NPV is negative therefore the IRR lies between 18% and 19%, therefore,

through linear interpolation the approximate IRR can be determined.

$$IRR = \text{Base Factor} + \frac{\text{Positive NPV}}{\text{Difference between Positive and Negative NPVs}} \times \text{Difference in \%}$$

$$IRR = 18\% + \frac{\$326}{\$326 - (-\$1,315)} \times 1\%$$

$$= 18\% + \frac{\$326}{\$1,641} \times 1\%$$

$$= 18\% + 0.2$$

$$= 18.2\%$$

Ace lumber's project B IRR = 18.2%

Advantages of IRR

- ✚ The internal rate of return method is used widely in industry by firms that employ present value-based capital budgeting techniques, the fact is that, some people feel more comfortable dealing with the concept of a project's percentage rate of return than with its dollar amount of net present value.
- ✚ Like the net present value approach, the internal rate of return technique takes into account both the magnitude and the timing of cash flows over the entire life of a project in measuring the project's economic desirability.
- ✚ Considers all cash flows of the project.
- ✚ Considers the risk of future cash flows through the cost of capital.

Disadvantages of IRR

- ✚ The possible existence of multiple internal rates of return is basic problem IRR model. Recall that a normal project has an initial cash outlay or outlays (net investment) followed by a stream of positive net cash flows. If for some reason—such as large abandonment costs at the end of a project's life or a major shutdown and rebuilding of a facility sometime during its life—the initial net investment is followed by one or more positive net cash flows (inflows) that then are followed by a negative cash flow, it is possible to obtain more than one internal rate of return.



Self-check questions 5.4

1. Discuss the advantage and disadvantage of IRR
2. Assuming 15% as the cost of capital for ABC investment project determine the

5.5.4 Payback (PB) period

Payback period is commonly used to evaluate proposed investments. The payback period is the amount of time required for the firm to recover its initial investment in a project, as calculated from cash inflows. One can also compute a discounted payback period, where the net cash inflows are discounted at the firm's cost of capital in determining the number of years required to recover the net investment in a project. In the case of an annuity, the payback period can be found by dividing the initial investment or net investment by the annual net cash inflows (equation 5.11). For a mixed stream of cash inflows, the yearly cash inflows must be accumulated until the initial investment is recovered. That means, if the payback period constitutes fraction periods the equation 5.12 could be used. Although popular, the payback period is generally viewed as an unsophisticated capital budgeting technique, because it does not explicitly consider the time value of money and ignores the post pay back cash flows.

$$PB = \frac{\text{Net Investment}}{\text{Annual net cash inflows}} \dots\dots\dots 5.11$$

$$PB = \text{Years before full recovery of NINV} + \frac{\text{unrecovered NINV at start of year}}{\text{cash inflow during the year}} \dots\dots\dots 5.12$$

Note: in case of discounted cash flow model the denominators should be discounted cash flows i.e. discounted annual Cash Inflow (5.11) and discounted cash inflow during the year (5.12).

Decision rule: when the payback period is used to make accept–reject decisions, the following decision rule or criteria apply:

- ✓ If the payback period is less than the maximum acceptable payback period, accept the project.
- ✓ If the payback period is greater than the maximum acceptable payback period, reject the project.

The length of the maximum acceptable payback period is determined by management. This value is set subjectively on the basis of a number of factors, including the type of project (expansion, replacement or renewal, other), the perceived risk of the project, and the perceived relationship between the payback period and the share value. It is simply a value that management feels, on average, will result in value-creating investment decisions.

Example: Compute the undiscounted payback period and discounted payback period for Ace Lumber's Projects A and B.

Project A				
	Undiscounted		Discounted (at 14%)	
	Cash	Cumulative	Cash	Cumulative
Year(t)	Inflow	Cash Inflow	Inflow	Cash Inflow
1	\$12,500	\$12,500	\$10,963	\$10,963
2	12,500	25,000	9,612.50	20,575.00
3	12,500	37,500	8,437.50	29,012.50
4	12,500	50,000	7,400.00	36,412.50
5	12,500	62,500	6,487.50	42,900.00
6	12,500	75,000	5,700.00	48,600.00
				Years before full recovery
PB=	<u>Net Investment</u>		PB = Years before full +	of NINV
	Annual Cash Inflow		recovery of NINV	Discounted Cash inflow
=	<u>50,000</u>			during the year
	12,500		=	<u>Undefined</u>
=	<u>4.0 years</u>			

Project B				
	Undiscounted		Discounted (at 14%)	
	Cash	Cumulative	Cash	Cumulative
Year(t)	Inflow	Cash Inflow	Inflow	Cash Inflow
1	\$5,000	\$5,000	\$4,385	\$4,385
2	10,000	15,000	7,690	12,075
3	15,000	30,000	10,125	22,200
4	15,000	45,000	8,880	31,080
5	25,000	70,000	12,975	44,055
6	30,000	100,000	13,680	57,735
PB=	4+ <u>(50,000-45,000)</u>		PB=	5 + <u>(50,000 – 44,055)</u>

	25,000		13,680
=	<u>4.2 years</u>	=	<u>5.43 years</u>

In project A, the undiscounted PB period is 4.0 years and the discounted PB period is undefined. This occurs because the NPV of the project is negative, that is, the discounted cash inflows are less than the net investment. In project B, the undiscounted PB period is 4.2 and B the discounted PB period is 5.43 years.

Advantages of Payback period

- ✚ A more valid justification for the use of the payback method is that it gives some indication of a project's desirability from a liquidity perspective because it measures the time required for a firm to recover its initial investment in a project.

Disadvantages of Payback period

- ✚ The undiscounted payback method gives equal weight to all cash inflows within the payback period, regardless of when they occur during the period. In other words, the technique ignores the time value of money
- ✚ Payback methods (both discounted and undiscounted) essentially ignore cash flows occurring after the payback period. Payback figures are biased against long-term projects and can be misleading.
- ✚ Payback provides no objective criterion for decision making that is consistent with shareholder wealth maximization. The payback methods (both discounted and undiscounted) may reject projects with positive net present values. The choice of an acceptable payback period is largely a subjective one, and different people using essentially identical data may make different accept–reject decisions about a project.



Self-check questions 5.4

1. Girma Electronics is considering the purchase of testing equipment that will cost birr 500,000 to replace old equipment. Assume the new machine will generate after-tax savings of birr 250,000 per year over the next four years.
 - a) What is the discount payback period of the investment if the firm has a 15% cost of capital?
 - b) What is the NPV of the investment if Girma Electronics has a 15% cost of capital?
2. You are asked to evaluate two projects for Adventures Club, Inc. Using the net present value method combined with the profitability index approach so which project would you select? Use a discount rate of 12 percent.

Project X (Br. 10,000 investment)		Project Y (Br. 22,000 investment)	
Year	Cash Flow	Year	Cash Flow
1.....	\$4,000	1	\$10,800
2.....	5,000	2	9,600
3.....	4,200	3	6,000
4.....	3,600	4	7,000

Leverage analysis

6.2 Introduction

Dear learner! This chapter discusses leverage and capital-structure concepts and techniques and how the firm can use them to create the best capital structure. Leverage involves the use of fixed costs to magnify returns. Its use in the capital structure of the firm has the potential to increase its return and risk. Leverage and capital structure are closely related concepts that are linked to capital budgeting decisions through the cost of capital. These concepts can be used to minimize the firm's cost of capital and maximize its owners' wealth.

6.3 The Concept of Leverage

In a business context, however, **leverage** refers to the use of fixed costs in an attempt to increase (or lever up) profitability. In this chapter, we explore the principles of both **operating leverage** and **financial leverage**. The former is due to fixed operating costs associated with the production of goods or services, whereas the latter is due to the existence of fixed financing costs – in particular, interest on debt. Both types of leverage affect the level and variability of the firm's after-tax earnings, and hence the firm's overall risk and return.

Leverage results from the use of fixed-cost assets or funds to magnify returns to risk, whereas decreases in leverage result in decreased return and risk. The amount of leverage in the firm's capital structure—the mix of long-term debt and equity maintained by the firm—can significantly affect its value by affecting return and risk. Unlike some causes of risk, management has almost complete control over the risk introduced through the use of leverage. Because of its effect on value, the financial manager must understand how to measure and evaluate leverage, particularly when making capital structure decisions.

The three basic types of leverage can best be defined in the following section:

- ✚ **Operating leverage** is concerned with the relationship between the firm's sales revenue and its earnings before interest and taxes, or EBIT. (EBIT is a descriptive label for operating profits.)
- ✚ **Financial leverage** is concerned with the relationship between the firm's EBIT and its common stock earnings per share (EPS).
- ✚ **Total leverage** is concerned with the relationship between the firm's sales revenue and EPS.



Self-check questions 6.1

1. What is Leverage? Explain
2. Discuss the different types of leverage.

6.4 Operating Leverage

Operating leverage is present any time a firm has fixed operating costs – regardless of volume. In the long run, of course, all costs are variable. Consequently, our analysis necessarily involves the short

run. We incur fixed operating costs in the hope that sales volume will produce revenues more than sufficient to cover all fixed and variable operating costs. One of the more dramatic examples of an effect of operating leverage is the airline industry, where a large proportion of total operating costs is fixed. Beyond a certain break-even load factor, each additional passenger essentially represents straight operating profit (earnings before interest and taxes, or EBIT) to the airline.

It is essential to note that fixed operating costs do not vary as volume changes. These costs include such things as depreciation of buildings and equipment, insurance, part of the overall utility bills, and part of the cost of management. On the other hand, variable operating costs vary directly with the level of output. These costs include raw materials, direct labor costs, part of the overall utility bills, direct selling commissions, and certain parts of general and administrative expenses.

One interesting potential effect caused by the presence of fixed operating costs (operating leverage) is that a change in the volume of sales results in a more than proportional change in operating profit (or loss).

A high degree of operating leverage implies that a relatively small change in sales results in a relatively large change in EBIT, net operating profits after taxes (NOPAT), and return on invested capital (ROIC). Other things held constant, the higher a firm's fixed costs, the greater its operating leverage. Higher fixed costs are generally associated with

(1) highly automated, capital intensive firms; (2) businesses that employ highly skilled workers who must be retained and paid even when sales are low; and (3) firms with high product development costs that must be maintained to complete ongoing R&D projects.

Dear learner, we will examine the three types of leverage concepts in detail in sections that follow. First, though, we will look at breakeven analysis, which lays the foundation for leverage concepts by demonstrating the effects of fixed costs on the firm's operations.

6.4.1 Break-Even Analysis

Breakeven analysis, sometimes called cost-volume-profit analysis, is used by the firm

-  To determine the level of operations necessary to cover all operating costs Indicates the level of operations and
-  To evaluate the profitability associated with various levels of sales. The firm's operating breakeven point is the level of sales necessary to cover all operating costs. At that point,

earnings before interest and taxes equal \$0. Or the operating break-even point occurs when earnings before interest and taxes (EBIT) equal zero.

$$EBIT = PQ - VQ - F = 0$$

The first step in finding the operating breakeven point is to divide the cost of goods sold and operating expenses into fixed and variable operating costs. Fixed costs are a function of time, not sales volume, and are typically contractual; rent, for example, is a fixed cost. Variable costs vary directly with sales and are a function of volume, not time; shipping costs, for example, are a variable cost.

The Algebraic Approach

Using the following variables, we can recast the operating portion of the firm's income statement given in Table 6.1 into the algebraic representation shown in below:

P = sale price per unit

Q = sales quantity in units

$(P - V)$ = unit contribution margin

FC = fixed operating cost per period

VC = variable operating cost per unit

Rewriting the algebraic calculations in Table 6.1 as a formula for earnings before interest and taxes yields Equation 6.1:

$$EBIT = (P \times Q) - FC - (VC \times Q)$$

Simplifying Equation 4.1 yields

$$EBIT = Q \times (P - VC) - FC \quad (4.2)$$

As noted above, the operating breakeven point is the level of sales at which all fixed and variable operating costs are covered—the level at which EBIT equals \$0. Setting EBIT equal to \$0 and solving Equation 6.2 for Q yield:

At the break-even point (Q_{BE}), EBIT is zero. Therefore, $0 = Q_{BE} (P - V) - FC$

Thus, the break-even (quantity) point is equal to fixed costs divided by the unit contribution margin ($P - V$).

$$Q_{BE} = \frac{FC}{P - VC} \quad (6.3)$$

Q_{BE} is the firm's operating breakeven point.

Assume that Cheryl's Posters, a small poster retailer, has fixed operating costs of \$2,500, its sale price per unit (poster) is \$10, and its variable operating cost per unit is \$5. Applying Equation 4.3 to these data yields

$$Q_{BE} = \frac{\$2,500}{\$10 - \$5} = 500 \text{ units}$$

At sales of 500 units, the firm's EBIT should just equal \$0. The firm will have positive EBIT for sales greater than 500 units and negative EBIT, or a loss, for sales less than 500 units. We can confirm this by substituting values above and below 500 units, along with the other values given, into Equation 6.1.

Break-Even (Sales) Point: Calculating a break-even point on the basis of dollar sales instead of units is often useful. Sometimes, as in the case of a firm that sells multiple products, it would be impossible, for example, to come up with a meaningful break-even point in total units for a firm such as General Electric, but a break-even point based on sales revenues could easily be imagined. When determining a general break-even point for a multi-product firm, we assume that sales of each product are a constant proportion of the firm's total sales.

Recognizing that at the break-even (sales) point the firm is just able to cover its fixed and variable operating costs, we turn to the following formula:

$$S_{BE} = FC + VC_{BE} \quad (6.4)$$

Where;

S_{BE} = break-even sales revenues

FC = fixed costs

VC_{BE} = total variable costs at the break-even point

Unfortunately, we are now faced with a single equation containing two unknowns – S_{BE} and VC_{BE} . Such an equation is unsolvable. Luckily, there is a trick that we can use in order to turn Eq. (6.4) into a single equation with a single unknown. First, we need to rewrite Eq. (6.4) as follows:

$$S_{BE} = FC + (VC_{BE} / S_{BE}) S_{BE} \quad (6.5)$$

Because the relationship between total variable costs and sales is assumed constant in linear break-even analysis, we can replace the ratio (VC_{BE}/S_{BE}) with the ratio of total variable costs to sales (VC/S) for any level of sales. For example, we can use the total variable costs and sales figures from the firm's most recent income statement to produce a suitable (VC/S) ratio. In short, after replacing the ratio (VC_{BE}/S_{BE}) with the generic ratio (VC/S) in Eq. (6.5), we get:

$$S_{BE} = FC / [1 - (VC/S)] \quad S_{BE} [1 - (VC/S)] = FC$$

$$S_{BE} = FC / [1 - (VC/S)] \quad (6.6)$$

To illustrate, consider a firm that produces a high-quality child's bicycle helmet that sells for \$50 a unit. The company has annual fixed operating costs of \$100,000, and variable operating costs are \$25 a unit regardless of the volume sold.

The break-even (quantity) point is equal to fixed costs divided by the unit contribution margin $(P - V)$. In our example,

$$Q_{BE} = \$100,000 / (\$50 - \$25) = 4,000 \text{ units}$$

For our example bicycle-helmet manufacturing firm, the ratio of total variable costs to sales is 0.50 regardless of sales volume. Therefore, using Eq. (4.6) to solve for the break-even (sales) point, we get $S_{BE} = \$100,000 / [1 - 0.50] = \$200,000$

At \$50 a unit, this \$200,000 break-even (sales) point is consistent with the 4,000-unit break-even (quantity) point determined earlier [i.e., $(4,000) (\$50) = \$200,000$]. (See figure 6.1 below)

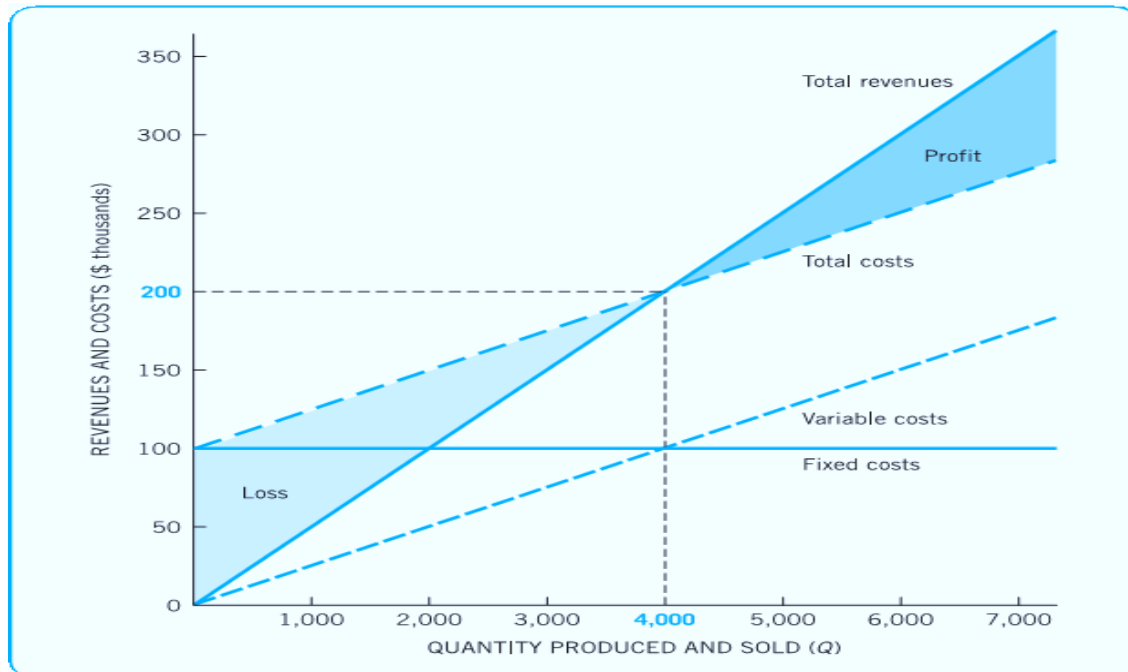


Figure 6.1: Break-even chart with the break-even point expressed in units and sales dollars

The intersection of the total costs line with the total revenues line determines the break-even point. The break-even point is the sales volume required for total revenues to equal total operating costs or for operating profit to equal zero. In Figure 6.1 this break-even point is 4,000 units of output (or \$200,000 in sales). Mathematically, we find this point (in units) by first noting that operating profit (EBIT) equals total revenues minus variable



Self-check questions 6.2

1. Explain the potential effect caused by the presence of fixed operating costs.
2. Break-even analysis also called *cost/volume/profit (C/V/P) analysis*. Explain.
3. Stallings Specialty Paint Company has fixed operating costs of \$3 million a year. Variable operating costs are \$1.75 per half pint of paint produced, and the average selling price is \$2 per half pint.
 - a) What is the annual operating break-even point in half pints (Q_{BE})? In dollars of sales (S_{BE})?
 - b) If variable operating costs decline to \$1.68 per half pint, what would happen to the operating break-even point (Q_{BE})?

6.5 Measuring the Degree of Operating Leverage (DOL)

Earlier, we said that one potential effect of operating leverage is that a change in the volume of sales results in a more than proportional change in operating profit (or loss). A quantitative measure of this sensitivity of a firm's operating profit to a change in the firm's sales is called the degree of operating leverage (DOL). The degree of operating leverage of a firm at a particular level of output (or sales) is simply the percentage change in operating profit over the percentage change in output (or sales) that causes the change in profits. Thus, it can be derived using the following equation:

$$DOL = \frac{\text{Percentage change in operating profit (EBIT)}}{\text{Percentage change in output or sales}} \quad (6.7)$$

Whenever the percentage change in EBIT resulting from a given percentage change in sales is greater than the percentage change in sales, operating leverage exists. This means that as long as DOL is greater than 1, there is operating leverage.

It is often difficult to work directly with Eq. (6.7) to solve for the DOL at a particular level of sales because an anticipated percentage change in EBIT (the numerator in the equation) will not be observable from historical data. Thus, although Eq. (6.7) is crucial for defining and understanding DOL, a few simple alternative formulas derived from Eq. (6.7) are more useful for actually computing DOL values:

$$DOL \text{ at base sales level } Q = \frac{Q(P - VC)}{Q(P - VC) - FC} \quad (6.8)$$

$$DOL_S \text{ dollar of sales} = \frac{EBIT + FC}{EBIT} \quad (6.9)$$

Example: Substituting Q 1,000, P \$10, VC \$5 and FC \$2,500 into Equation 6.8 yields the following result:

$$DOL \text{ at 1,000 units} = \frac{1,000 (\$10 - \$5)}{1,000 (\$10 - \$5) - \$2,500} = 2.0$$

Fixed Costs and Operating Leverage

Changes in fixed operating costs affect operating leverage significantly. Firms sometimes can incur fixed operating costs rather than variable operating costs and at other times may be able to substitute one type of cost for the other. For example, a firm could make fixed-dollar lease payments rather than payments equal to a specified

percentage of sales. Or it could compensate sales representatives with a fixed salary and bonus rather than on a pure percent-of-sales commission basis. The effects of changes in fixed operating costs on operating leverage can best be illustrated by continuing our example.

Example: Assume that Cheryl's Posters exchanges a portion of its variable operating costs for fixed operating costs by eliminating sales commissions and increasing sales salaries. This exchange results in a reduction in the variable operating cost per unit from \$5 to \$4.50 and an increase in the fixed operating costs from \$2,500 to \$3,000. Although the EBIT of \$2,500 at the 1,000-unit sales level is the same as before the shift in operating cost structure. This shows that the firm has increased its operating leverage by shifting to greater fixed operating costs.

$$\begin{aligned} DOL \text{ at } 1,000 \text{ units} &= 1,000 (\$10 - \$4.50) / 1,000 (\$10 - \$4.50) - \$3,000 \\ &= 2.2 \end{aligned}$$

Comparing this value to the DOL of 2.0 before the shift to more fixed costs makes it clear that the higher the firm's fixed operating costs relative to variable operating costs, the greater the degree of operating leverage.

DOL and Business Risk

It is important to recognize that the degree of operating leverage is only one component of the overall business risk of the firm. The other principal factors giving rise to business risk are variability or uncertainty of sales and production costs. The firm's degree of operating leverage magnifies the impact of these other factors on the variability of operating profits. However, the degree of operating leverage itself is not the source of the variability.

A high DOL means nothing if the firm maintains constant sales and a constant cost structure. Likewise, it would be a mistake to treat the degree of operating leverage of the firm as a synonym for its business risk. Because of the underlying variability of sales and production costs, however, the degree of operating leverage will magnify the variability of operating profits, and hence the company's business risk.



Self-check questions 6.3

1. Gahlon Gearing, Ltd., has a DOL of 2 at its current production and sales level of 10,000 units. The resulting operating income figure is \$1,000.
 - a. If sales are expected to increase by 20 percent from the current 10,000-unit sales position, what would be the resulting operating profit figure?
 - b. At the company's new sales position of 12,000 units, what is the firm's –new DOL figure?

6.6 Financial Leverage

As stated earlier, a company can finance its investments by debt and equity. The company may also use preference capital. The rate of interest on debt is fixed irrespective of the company's rate of return on assets. The company has a legal binding to pay interest on debt. The rate of preference dividend is also fixed; but preference dividends are paid when the company earns profits. The ordinary shareholders are entitled to the residual income.

Financial leverage involves the use of fixed cost financing. Interestingly, financial leverage is acquired by choice, but operating leverage sometimes is not. The amount of operating leverage (the amount of fixed operating costs) employed by a firm is sometimes dictated by the physical requirements of the firm's operations. For example, a steel mill by way of its heavy investment in plant and equipment will have a large fixed operating cost component consisting of depreciation. Financial leverage, on the other hand, is always a choice item. No firm is required to have any long-term debt or preferred stock financing. Firms can, instead, finance operations and capital expenditures from internal sources and the issuance of common stock.

The use of the fixed-charges sources of funds, such as debt and preference capital along with the owners' equity in the capital structure, is described as financial leverage or

gearing or trading on equity. The use of the term trading on equity is derived from the fact that it is the owner's equity that is used as a basis to raise debt; that is, the equity that is traded upon.

Financial leverage results from the presence of fixed financial costs in the firm's financial leverage income stream. It is leverage as the potential use of fixed financial costs to magnify the effects of changes in earnings before interest and taxes on the firm's earnings per share.

The two fixed financial costs that may be found on the firm's income statement are (1) interest on debt and (2) preferred stock dividends. These charges must be paid regardless of the amount of EBIT available to pay them.

The effect of financial leverage is such that an increase in the firm's EBIT results in a more-than-proportional increase in the firm's earnings per share, whereas a decrease in the firm's EBIT results in a more-than-proportional decrease in EPS.

Financial leverage is employed in the hope of increasing the return to common shareholders. Favorable or positive leverage is said to occur when the firm uses funds obtained at a fixed cost (funds obtained by issuing debt with a fixed interest rate or preferred stock with a constant dividend rate) to earn more than the fixed financing costs paid. Any profits left after meeting fixed financing costs then belong to common shareholders. Unfavorable or negative leverage occurs when the firm does not earn as much as the fixed financing costs.

The favorability of financial leverage, or trading on the equity¹ as it is sometimes called, is judged in terms of the effect that it has on earnings per share to the common shareholders. In effect, financial leverage is the second step in a two-step profit-magnification process. In step one, operating leverage magnifies the effect of changes in sales on changes in operating profit. In step two, the financial manager has the option of using financial leverage to further magnify the effect of any resulting changes in operating profit on changes in earnings per share.



Self-check questions 6.4

1. What is financial Leverage?
2. What is the impact of financial leverage on stockholders?
3. What is known as the indifference point in EBIT-EPS analysis?
4. Differentiate between degree of operating leverage (DOL) and degree of financial leverage (DFL).

EBIT-EPS Break-Even, or Indifference, Analysis

Calculation of Earnings per Share: To illustrate an EBIT-EPS break-even analysis of financial leverage, suppose that Cherokee Tire Company with long-term financing of \$10 million, consisting entirely of common stock equity, wishes to raise another \$5 million for expansion through one of three possible financing plans. The company may gain additional financing with a new issue of (1) all common stock, (2) all debt at 12 percent interest, or (3) all preferred stock with an 11 percent dividend. Present annual earnings before interest and taxes (EBIT) are \$1.5 million but with expansion are expected to rise to \$2.7 million.

The income tax rate is 40 percent, and 200,000 shares of common stock are now outstanding. Common stock can be sold at \$50 per share under the first financing option, which translates into 100,000 additional shares of stock.

To determine the EBIT-EPS break-even, or indifference, points among the various financing alternatives, we begin by calculating earnings per share, EPS, for some hypothetical level of EBIT using the following formula:

$$EPS = \frac{(EBIT - I)(1 - t) - PD}{NS} \quad (6.10)$$

Where; I = annual interest paid

PD = annual preferred dividend paid t = corporate tax rate

NS = number of shares of common stock outstanding

Suppose we wish to know what earnings per share would be under the three-alternative additional-financing plans if EBIT were \$2.7 million. The calculations are shown in Table 6.2. Note that interest on debt is deducted before taxes, whereas preferred stock

dividends are deducted after taxes. As a result, earnings available to common shareholders (EACS) are higher under the debt alternative than they are under the preferred stock alternative, despite the fact that the interest rate on debt is higher than the preferred stock dividend rate.

Table 6.2: Calculations of earnings per share under three additional financing alternatives

	COMMON STOCK	DEBT	PREFERRED STOCK
Earnings before interest and taxes (<i>EBIT</i>)	\$2,700,000	\$2,700,000	\$2,700,000
Interest (<i>I</i>)	—	600,000	—
Earnings before taxes (<i>EBT</i>)	\$2,700,000	\$2,100,000	\$2,700,000
Income taxes ($EBT \times t$)	1,080,000	840,000	1,080,000
Earnings after taxes (<i>EAT</i>)	\$1,620,000	\$1,260,000	\$1,620,000
Preferred stock dividends (<i>PD</i>)	—	—	550,000
Earnings available to common shareholders (<i>EACS</i>)	\$1,620,000	\$1,260,000	\$1,070,000
Number of shares of common stock outstanding (<i>NS</i>)	300,000	200,000	200,000
Earnings per share (<i>EPS</i>)	\$5.40	\$6.30	\$5.35

EBIT-EPS Chart: Given the information in Table 6.2, we are able to construct an EBIT-EPS break-even chart similar to the one for operating leverage. On the horizontal axis we plot earnings before interest and taxes, and on the vertical axis we plot earnings per share.

For each financing alternative, we must draw a straight line to reflect EPS for all possible levels of EBIT. Because two points determine a straight line, we need two data points for each financing alternative. The first is the EPS calculated for some hypothetical level of EBIT. For the expected \$2.7 million level of EBIT, we see in Table 6.2 that earnings per share are \$5.40, \$6.30, and \$5.35 for the common stock, debt, and preferred stock financing alternatives. We simply plot these earnings per share levels to correspond with the \$2.7 million level of EBIT. Technically, it does not matter which hypothetical level of EBIT we choose for calculating EPS. On good graph paper one EBIT level is as good as

the next. However, it does seem to make common sense to choose the most likely, or expected, EBIT level rather than some level not too likely to occur.

The second data point – chosen chiefly because of its ease of calculation – is where EPS is zero. This is simply the EBIT necessary to cover all fixed financial costs for a particular financing plan, and it is plotted on the horizontal axis. We can make use of Equation (6.10) to determine the horizontal axis intercept under each alternative. We simply set the numerator in the equation equal to zero and solve for EBIT. For the common stock alternative, we have:

$$0 = (EBIT - 1) (1 - t) - PD \quad (6.11)$$

$$(EBIT - 0)(1 - 0.40) - 0$$

$$(EBIT) (0.60)$$

$$EBIT = \frac{0}{0.60} = 0$$

Notice that there are no fixed financing costs whatsoever (on either old or new financing). Therefore, EPS equals zero at zero EBIT. For the debt alternative we have:

$$0 = (EBIT - 1) (1 - t) - PD$$

$$(EBIT - \$600,000) (1 - 0.40) - 0$$

$$(EBIT) (0.60) - \$360,000$$

$$EBIT = \frac{\$360,000}{0.60} = \$600,000$$

Thus, the after-tax interest charge divided by 1 minus the tax rate gives us the EBIT necessary to cover these interest payments. In short, we must have \$600,000 to cover interest charges, so \$600,000 becomes the horizontal axis intercept. Finally, for the preferred stock alternative we have:

$$0 = (EBIT - 1) (1 - t) - PD$$

$$(EBIT - 0) (1 - 0.40) - \$550,000$$

$$(EBIT) (0.60) - \$550,000$$

$$EBIT = \frac{\$550,000}{0.60} = \$916,667$$

We divide total annual preferred dividends by 1 minus the tax rate to obtain the EBIT necessary to cover these dividends. Thus, we need \$916,667 in EBIT to cover \$550,000 in preferred stock dividends, assuming a 40 percent tax rate. Again, preferred dividends are deducted after taxes, so it takes more in before-tax earnings to cover them than it does to cover interest. Given the horizontal axis intercepts and earnings per share for some hypothetical level of EBIT (like the —expected (EBIT), we draw a straight line through each set of data points. The break-even, or indifference, chart for Cherokee Tire Company is shown in Figure 6.2.

We see from Figure 6.2 that the earnings per share indifference point between the debt and common stock additional-financing alternatives is \$1.8 million in EBIT. If EBIT is below that point, the common stock alternative will provide higher earnings per share. Above that point the debt alternative produces higher earnings per share.

The indifference point between the preferred stock and the common stock alternative is \$2.75 million in EBIT. Above that point, the preferred stock alternative produces more favorable earnings per share. Below that point, the common stock alternative leads to higher earnings per share. Note that, there is no indifference point between the debt and preferred stock alternatives. The debt alternative dominates for all levels of EBIT and by a constant amount of earnings per share, namely 95 cents.

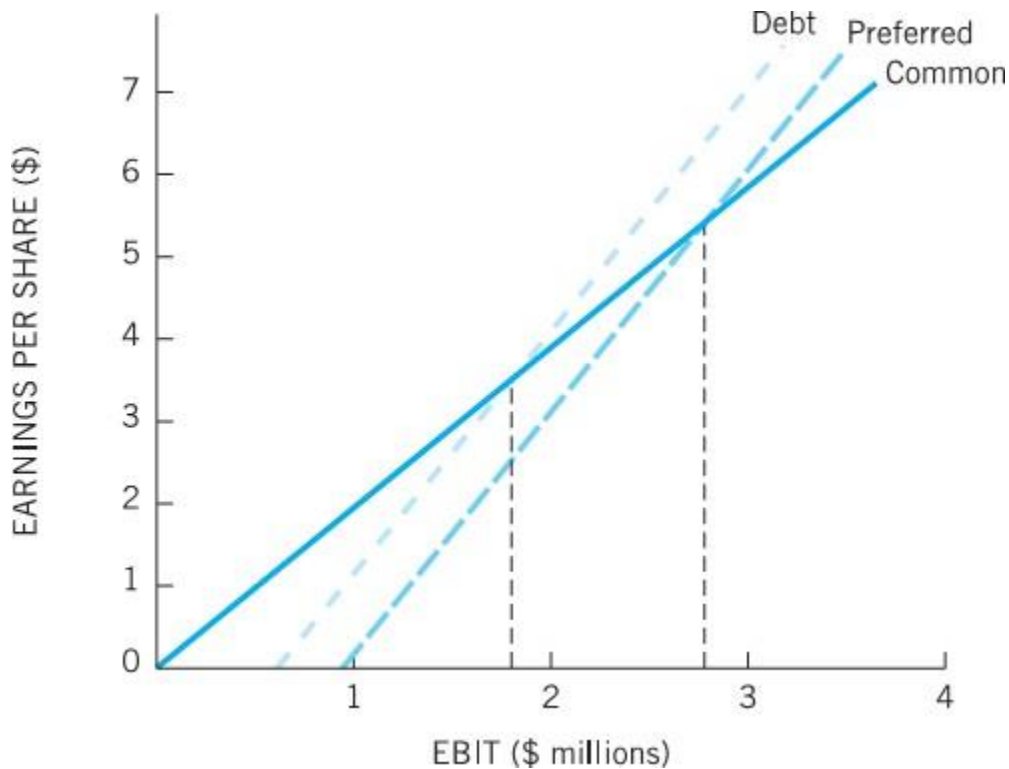


Figure 6.2: EBIT-EPS break-even, or indifference, chart for three additional- financing alternatives.

Indifference Point Determined Mathematically: The indifference point between two alternative financing methods can be determined mathematically by first using Equation (6.12) to express EPS for each alternative and then setting these expressions equal to each other as follows;

$$= \frac{(EBIT_{1,2} - I_1)(1 - t) - PD_1}{NS_1} = \frac{(EBIT_{1,2} - I_2)(1 - t) - PD_2}{NS_2}$$

Where; $EBIT_{1,2}$ = EBIT indifference point between the two alternative financing methods that we are concerned with – in this case, methods 1 and 2

I_1, I_2 = annual interest paid under financing methods 1 and 2

PD_1, PD_2 = annual preferred stock dividend paid under financing methods 1 & 2

t = corporate tax rate

NS_1, NS_2 = number of shares of common stock to be outstanding under financing methods 1 and 2

Suppose that we wish to determine the indifference point between the common stock and debt-financing alternatives in our example. We would have:

$$= \frac{\text{Common Stock}}{(EBIT_{1,2} - 0)(1 - 0.40) - 0} \quad \frac{\text{Debt}}{(EBIT_{1,2} - \$600,000)(1 - 0.40) - 0}$$

$$= \frac{300,000}{200,000}$$

Cross multiplying and rearranging, we obtain

$$\begin{aligned} (EBIT_{1,2})(0.60)(200,000) &= (EBIT_{1,2})(0.60)(300,000) - (0.60)(\$600,000)(300,000) \\ (EBIT_{1,2})(600,000) &= \$108,000,000,000 \\ EBIT_{1,2} &= \$1,800,000 \end{aligned}$$

The EBIT-EPS indifference point, where earnings per share for the two methods of financing are the same, is \$1.8 million. This amount can be verified graphically in Figure 6.2. Thus, indifference points can be determined both graphically and mathematically.

Effect on Risk: So far, our concern with EBIT-EPS analysis has been only with what happens to the return to common shareholders as measured by earnings per share. We have seen in our example that, if EBIT is above \$1.8 million, debt financing is the preferred alternative from the standpoint of earnings per share. We know from our earlier discussion, however, that the impact on expected return is only one side of the coin. The other side is the effect that financial leverage has on risk. An EBIT-EPS chart does not permit a precise analysis of risk.

Nevertheless, certain generalizations are possible. For one thing, the financial manager should compare the indifference point between two alternatives, such as debt financing versus common stock financing, with the most likely level of EBIT. The higher the expected level of EBIT, assuming that it exceeds the indifference point, the stronger the case that can be made for debt financing, all other things the same.

In addition, the financial manager should assess the likelihood of future EBITs actually falling below the indifference point. As before, our estimate of expected EBIT is \$2.7 million. Given the business risk of the company and the resulting possible fluctuations in EBIT, the financial manager should assess the probability of EBITs falling below \$1.8 million. If the probability is negligible, the use of the debt alternative will be supported. On the other hand, if EBIT is presently only slightly above the indifference point and the probability of EBITs falling below this point is high; the financial manager may conclude that the debt alternative is too risky.



Self-check questions 6.5

1. Assuming EBIT of \$120,000 and a Tax rate of 50% for raising equity investment of \$125,000, the firm will sell 12,500 shares and pay \$56,250 interest on a debt of \$375,000 at 15 percent.
 - a) What is the amount of EPS?
2. Kyle Corporation is comparing two different capital structures, an all-equity plan (Plan 1) and a levered plan (Plan II). Under Plan I, Kyle would have 900,000 shares of stock outstanding. Under Plan II, there would be 650,000 shares of stock outstanding and \$10 million in debts outstanding. The interest rate on the debt is 10 percent, and there are no taxes.
 - a) If EBIT is \$11.5 million, which plan will result in the higher EPS?
 - b) If EBIT is \$5 million, which plan will result in the higher EPS?
 - c) What is the break-even EBIT?

The degree of financial leverage (DFL) is the numerical measure of the firm's financial leverage. Computing it is much like computing the degree of operating (DOL) leverage. A quantitative measure of the sensitivity of a firm's earnings per share to a change in the firm's operating profit is called the degree of financial leverage (DFL). The degree of financial leverage at a particular level of operating profit is simply the percentage change in earnings per share over the percentage change in operating profit that causes the change in earnings per share. Thus, the following equation presents one approach for obtaining the DFL.

$$DFL = \frac{\text{percentage change in EPS}}{\text{percentage change in EBIT}} \quad (6.13)$$

Whenever the percentage change in EPS resulting from a given percentage change in EBIT is greater than the percentage change in EBIT, financial leverage exists. This means that whenever DFL is greater than 1, there is financial leverage.

Whereas Equation (6.13) is useful for defining DFL, a simple alternative formula derived from Equation (6.13) is more useful for actually computing DFL values:

$$DFL = \frac{EBIT}{EBIT - 1 - [PD/(1-t)]} \quad (6.14)$$

Equation (6.14) states that DFL at a particular level of operating profit is calculated by dividing operating profit by the dollar difference between operating profit and the amount of before-tax operating profit necessary to cover total fixed financing costs. (Remember, it takes more in before-tax earnings to cover preferred dividends than it does to cover interest: hence we need to divide preferred dividends by 1 minus the tax rate in our formula.) For our example firm, using the debt-financing alternative at \$2.7 million in EBIT, we have

$$DFL \text{ at EBIT of } \$2.7 \text{ million} = \frac{\$2,700,000}{\$2,700,000 - \$600,000} = 1.29$$

For the preferred stock financing alternative, the degree of financial leverage is:

$$DFL \text{ at EBIT of } \$2.7 \text{ million} = \frac{\$2,700,000}{\$2,700,000 - \$550,000/0.60} = 1.51$$

Interestingly, although the stated fixed cost involved with the preferred stock financing alternative is lower than that for the debt alternative (\$550,000 versus \$600,000), the DFL is greater under the preferred stock option than under the debt option. This is because of the tax deductibility of interest and the non-deductibility of preferred dividends. It is often argued that preferred stock financing is of less risk than debt financing for the issuing firm. With regard to the risk of **cash insolvency**, this is probably true. But the DFL tells us that the relative variability of EPS will be greater under the preferred stock financing arrangement, everything else being equal. This discussion



Self-check questions 6.6

1. Take the case of Bright ways Ltd. when EBIT increases from \$120,000 to \$160,000, EPS increases from \$1.65 to \$2.45. when it employs 50 percent debt and pays interest

naturally leads us to the topic of financial risk and its relationship to the degree of financial leverage.

6.8 Total Leverage

When financial leverage is combined with operating leverage, the result is referred to as ***total (or combined) leverage***. The effect of combining financial and operating leverage is a two-step magnification of any change in sales into a larger relative change in earnings per share. A quantitative measure of this total sensitivity of a firm's earnings per share to a change in the firm's sales is called ***the degree of total leverage (DTL)***.

6.8.1 Degree of Total Leverage (DTL)

The degree of total leverage of a firm at a particular level of output (or sales) is equal to the percentage change in earnings per share over the percentage change in output (or sales) that causes the change in earnings per share. Thus,

$$\text{Degree of total leverage (DTL) at (or Dollars) of output (or sales) = } \frac{\% \Delta \text{ in EPS}}{\% \Delta \text{ in output or sales}} \quad (6.15)$$

Computationally, we can make use of the fact that the degree of total leverage is simply the product of the degree of operating leverage and the degree of financial leverage as follows:

$$\begin{aligned} DTL, Q \text{ units (or } S \text{ dollars)} &= \\ DTL, Q \text{ units (or } S \text{ dollars)} \times DFL, EBIT \text{ of } X \text{ dollars} &\quad (6.16) \end{aligned}$$

In addition, multiplying alternative *DOLs*, Equation (6.8) and (6.9), by *DFL*, Equation (6.14), gives us;

$$\text{Degree of total leverage (DTL) at } Q \text{ units} = \frac{Q(P - V)}{Q - P - V - FC - 1 - \left[\frac{PD}{1 - t} \right]}$$

$$\text{Degree of total leverage at } S \text{ dollars of sales} = \frac{EBIT + FC}{EBIT - 1 - \left[\frac{PD}{1 - t} \right]} \quad (6.17)$$

These alternative equations tell us that for a particular firm the greater the *before-tax* financial costs, the greater the degree of total leverage over what it would be in the absence of financial leverage.

Suppose that our bicycle-helmet manufacturing firm used to illustrate operating leverage has \$200,000 in debt at 8 percent interest. Recall that the selling price is \$50 a unit, variable operating costs are \$25 a unit, and annual fixed operating costs are \$100,000.

Assume that the tax rate is 40 percent, and that we wish to determine the degree of total leverage at 8,000 units of production and sales. Therefore, using Eq. (6.16), we have

$$\begin{aligned} \text{Degree of total leverage (DTL) at } Q \text{ units} &= \frac{8,000(50 - 25)}{8,000 - 50 - 25 - 100,000 - 16,000} \\ &= 2.38 \end{aligned}$$

Thus, a 10 percent increase in the number of units produced and sold would result in a 2.38 percent increase in earnings per share. Stating the degree of total leverage for our example firm in terms of the product of its degree of operating leverage times its degree of financial leverage, we get:

$$DOL \text{ at } 8,000 \text{ units} \times DFL \text{ at } 8,000 \text{ units} = DTL \text{ at } 8,000 \text{ units}$$

$$\frac{8,000(50 - 25)}{8,000(50 - 25) - \$100,000} \times \frac{\$100,000}{\$100,000 - 16,000} = 2.38$$

$$2.00 \times 1.19 = 2.38$$

In the absence of financial leverage, our firm's degree of total leverage would have been equal to its degree of operating leverage for a value of 2 (remember, DFL for a firm with no financial leverage equals 1). We see, however, that the firm's financial leverage magnifies its DOL figure by a factor of 1.19 to produce a degree of total leverage equal to 2.38.

6.8.2 DTL and Total Firm Risk

Operating leverage and financial leverage can be combined in a number of different ways to obtain a desirable degree of total leverage and level of total firm risk. High business risk can be offset with low financial risk and vice versa. The proper overall level of firm risk involves a trade-off between total firm risk and expected return. This trade-off must be made in keeping with the objective of maximizing shareholder value. The discussion, so far, is meant to show how certain tools can be employed to provide information on the two types of leverage –operating and financial – and their combined effect.



Self-check questions 6.7

2. XYZ computer cable manufacturer, expects sales of 20,000 units at \$ 5 per unit in the coming year and must meet the following obligations: variable operating costs of \$2 per unit, fixed operating costs of \$10,000, the interest of \$20,000, and preferred stock dividends of \$12,000. The firm is in the 40% tax bracket and has 5,000 shares of common stock outstanding.
 - a. Calculate DTL at 20,000 units.